



ADDIS ABABA
**SCIENCE AND
TECHNOLOGY**
UNIVERSITY
UNIVERSITY FOR INDUSTRY

Student Handbook

2023/2024





“University for Industry”

ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY

OFFICE OF THE ACADEMIC PROGRAM DIRECTORATE

STUDENT HANDBOOK

November, 2023

Addis Ababa, Ethiopia

Contents

1. General Information.....	1
1.1 Introduction.....	1
1.2 Site/location	2
1.3 University Vision and Mission.....	4
1.3.1 Vision.....	4
1.3.2 Mission	4
1.4 AASTU Organization	4
2. Academic Programs and Academic Guide.....	5
2.1 The University Registrar and Alumni Directorate.....	5
2.2 Welcome & Orientation Session	5
2.3 University Life	6
2.4 Academic Calendar.....	6
2.5 Admission	7
2.5.1 Undergraduate Study	7
2.5.2 Admission to Graduate Program (GP).....	8
2.6 Registration.....	9
2.6.1 Course Regular Registration	9
2.6.2 Late Registration	9
2.7 Identification Card.....	9
2.8 Add and Drop.....	10
2.9 Withdrawing & Clearing.....	10
2.10 Readmission.....	10
2.10.1 Readmission for Students in Good Academic Standing	11
2.10.2 Readmission for Academically Dismissed Students.....	11
2.11 Assessments and Evaluation.....	12
2.12 Class Attendance	13
2.13 Semester Load.....	13
2.14 The Grading System	15
2.15 NO GRADE (NG).....	15
2.16 Academic Status of Undergraduate Students	16
2.17 Discretionary Probation.....	18
2.18 Repeating Courses	19
2.19 Make-up Examinations.....	19
2.20 Re- Examinations	21
2.21 Returning & Remarking of Exam Papers	21
2.22 Transfer of Students.....	22
2.23 Course Exemption	23
2.24 Graduation Requirements	24
2.25 Credentials	24

2.25.1 Transcripts.....	24
2.25.2 Degrees	25
2.25.3 Tuition and Other Fees.....	25
2.26 Fast-Track Program	26
2.26.1 Eligibility	26
2.26.2 Application and Admission Procedures	26
2.26.3 General Rules of FTP	27
2.26.4 Requirements for Staying in FTP	27
2.27 Other Services	28
2.27.1 Social Services	28
2.27.2 Counseling Service.....	28
2.28 Undergraduate Degree Programs.....	29
2.29 Freshman and Pre-engineering program	29
2.29.1 Program Courses Synopsis.....	30
3. College of Engineering	35
3.1 Bachelor of Science Degree in Architecture	36
3.1.1 Introduction.....	36
3.1.2 Program Educational Objectives	37
3.1.3 Program Education Outcome (PEO)	37
3.1.4 Graduation Requirements	39
3.1.5 Course Category and list of Courses	40
3.1.6 Program Curriculum Structure: Bachelor of Science in Architecture	44
3.1.7 Program Courses Synopsis	49
3.1.8 Introduction.....	65
3.1.9 Program Educational Objectives	66
3.1.10 Program Outcomes (POs)	66
3.1.11 Graduation Requirements.....	69
3.1.12 Course Category and list of Courses	69
3.1.13 Program Curriculum Structure: Bachelor of Science in Civil Engineering	73
3.1.14 Program Courses Synopsis.....	78
3.2 Bachelor of Science Degree in Mining Engineering	90
3.2.1 Introduction.....	90
3.2.2 Program Educational Objectives	91
3.2.3 Program Outcome.....	92
3.2.4 Graduation Requirement.....	93
3.2.5 Course Category and list of Courses	94
3.2.6 Program Curriculum Structure: Bachelor of Science in Mining Engineering	96
3.2.7 Program Courses Synopsis	102
3.3 Bachelor of Science Degree in Chemical Engineering.....	111
3.3.1 Introduction.....	111
3.3.2 Program Education Objective (PEO).....	112

3.3.3 Program Outcomes (PO).....	113
3.3.4 Graduation Requirements	114
3.3.5 List of courses and category.....	114
3.3.6 Program Curriculum Structure: Bachelor of Science in Chemical Engineering	118
3.3.7 Program Courses Synopsis	124
3.4 Bachelor of Science Degree in Environmental Engineering.....	135
3.4.1 Introduction.....	135
3.4.2 Program Education Objective (PEO).....	137
3.4.3 Program Outcomes (PO):.....	137
3.4.4 Graduation Requirements	139
3.4.5 List of Courses and Category	139
3.4.6 Program Curriculum Structure: Bachelor of Science in Environmental Engineering	143
3.4.7 Program Courses Synopsis	149
3.5 Bachelor of Science Degree in Electrical and Computer Engineering	164
3.5.1 Introduction.....	164
3.5.2 Program Education Objective (PEO).....	164
3.5.3 Program Outcome (PO)	165
3.5.4 Graduation Requirement.....	166
3.5.5 Program Curriculum Structure: Bachelor of Science in Electrical and Computer Engineering.....	173
3.5.6 Program Courses Synopsis	184
3.6 Bachelor of Science Degree in Electromechanical Engineering	193
3.6.1 Introduction.....	193
3.6.2 Program Educational Objectives (PEO)	195
3.6.3 Program Outcomes (PO).....	196
3.6.4 Graduation Requirements.....	197
3.6.5 List of Courses and Category	198
3.6.6 Program Curriculum Structure: Bachelor of Electromechanical engineering.....	200
3.6.7 Program Courses Synopsis	205
3.7 Bachelor of Science Degree in Mechanical Engineering.....	218
3.7.1 Introduction.....	218
3.7.2 Program Education Objective (PEO).....	218
3.7.3 Program Outcomes (PO):.....	218
3.7.4 Graduation Requirements	220
3.7.5 List of Courses and Category	221
3.7.6 Program Curriculum Structure: Bachelor of Science in Mechanical Engineering	225
3.7.7 Program Courses Synopsis	235
3.8 Bachelor of Science Degree in Software Engineering.....	249
3.8.1 Introduction.....	249
3.8.2 Program Education Objective (PEO).....	250
3.8.3 Program Outcomes (PO):.....	251
3.8.4 Graduation Requirements	252

3.8.5 List of Courses and Category	253
3.8.6 Program Curriculum Structure: Bachelor of Science in Software Engineering	257
3.8.7 Program Courses Synopsis	261
Common Course Synopsis	261
4. College of Natural and Applied Sciences	275
4.1 Bachelor of Science Degree in Food Sciences and Applied Nutrition	275
4.1.1 Introduction.....	275
4.1.2 Program Education Objective (PEO).....	276
4.1.3 Program Outcomes (PO).....	277
4.1.4 Graduation Requirements	277
4.1.5 List of Courses and Category	278
4.1.6 Program Curriculum Structure: Bachelor of FSAN	280
4.1.7 Program Courses Synopsis	284
4.2 Bachelor of Science Degree in Geology.....	296
4.2.1 Introduction.....	296
4.2.2 Program Education Objective (PEO).....	296
4.2.3 Program Outcomes (PO):.....	297
4.2.4 Graduation Requirements	298
4.2.5 List of Courses and Category.....	298
4.2.6 Program Curriculum Structure: Bachelor of geology.....	301
4.2.7 Program Courses Synopsis	305
4.3 Bachelor of Science Degree in Biotechnology.....	320
4.3.1 Introduction.....	320
4.3.2 Program Education Objectives (PEO)	321
4.3.3 Program Learning Outcomes (PO):.....	321
4.3.4 Graduation Requirements	322
4.3.5 Course Category and list of Courses	322
4.3.6 Course Breakdown for Bachelor of Science in Biotechnology.....	325
4.3.5 Program Courses Synopsis	329
Supportive Module.....	344
4.4 Bachelor of Science Degree in Industrial Chemistry.....	346
4.4.1 Introduction.....	346
4.4.2 Program Education Objectives (PEO)	347
4.4.3 Program outcome (PO).....	347
4.4.4 Graduation Requirements	348
4.4.5 Program Curriculum Structure: Bachelor of Industrial Chemistry.....	352
4.4.6 Program Courses Synopsis	355

FOREWORD

This handbook is intended to provide students with brief and readily available information with emphasis on academic rules and the regulations of the university that matters most of academic success of students and the programs that is offered in AASTU. These academic rules, regulations and information are, by and large, extracted from the Senate Legislation which governs the whole academic exercise of the University and the curricula in AASTU.

Hence, for your informed decision, a successful and healthy university life, the office of the academic program directorate highly recommends you to read and comprehend the essence of this student handbook as soon as you start university life and classes. The book will be available at all University libraries, student council of the university and other bodies of the university as needed. You can also access from the university web site: www.aastu.edu.et. However, we proudly assure you that the student handbook is not the only available source of information. The whole university community, in particular, the student affairs directorate, the academic program directorate office, implementation of women and social affair inclusiveness office, the Office of the Registrar and alumni, the Colleges, Departments and most importantly your instructors and advisors are there to serve and nurture you.

Please feel free to consult any of the bodies for any range of information you need on time. As members of the university community, we are all one family to support and help each other with love and respect for one motto- education for change. One more important point is that, AASTU is gender sensitive and particularly works hard to create conducive and comfortable academic environment for female students particularly through the implementation of women and social affair inclusiveness office. Therefore, we advise female students to work closely with the gender office to let not exist any social and cultural barrier that limit your potential academic wisdom exercise.

MESSAGE FROM THE PRESIDENT



Dear students, welcome to Addis Ababa Science and Technology University (AASTU). For the new students who are joining us this academic year with high performance in national and university entrance exams, congratulations!

Established in 2011, AASTU is one of the two science and technology universities in the country. Since its establishment, AASTU has been working to meet the needs of the country in the transformation of its economy focusing in the fields of applied sciences, ICT, and engineering. These disciplines will remain the university's major strength.

AASTU strives to make your campus environment conducive for education, innovation and living. Each one of you will meet students from different parts of the country which will give you an opportunity to learn each other's languages and cultures. That will be an important experience to work and live in a multicultural society in your future life.

AASTU has a vision of becoming "an internationally recognized Ethiopian Hub of science and technology with strong national commitment and significant continental impact by 2030"

In the course of making AASTU a real Technology University, we are embarking on technology enhanced teaching and learning including E-learning and E-library. To access these platforms and for all official communications we have created institutional email for all our students. The institutional email will allow our students to access free access educational software and cloud space.

The foundation for an exemplary professional competence and ethics in your later work life will be built on your hard work and discipline during your university life. I therefore would like to advise you to avoid practices and behaviors that may derail you from the noble professional objectives you have set to achieve as you decided to join AASTU; moreover I encourage you to read and abide by the 'Discipline guideline for students of higher learning Institutions' which is available at our website ww.aastu.edu.et.

I wish you all the best in your studies!

Dereje Engida (PhD, CEng MIMechE)

AASTU President

1. General Information

1.1 Introduction

Addis Ababa Science and Technology University (here after, AASTU), is one of the new public universities of the country established to play as a forefront changing actor in the technological transformation of the country by creating strong linkage with industries. As it was stated in the Five-Year Growth and Transformation Plan (2010 - 2015 G.C), the establishment of well-institutionalized and strong science and technology universities and institutes of technology will serve as a cornerstone to build an economically developed and industrialized state of Ethiopia. As a result, AASTU was founded in 2011 under the Directive of the Council of Ministers No. 216/ 2011 as well as amended by regulation numbers 314/2014 by admitting the first batch (2000 students) in November 2011.

Since 2015, Addis Ababa Science and Technology University was following nationally harmonized undergraduate programs curricula. But, by now it has developed the new undergraduate curricula to meet the requirements of accreditation, which consists of continuous quality improvements (CQIs).

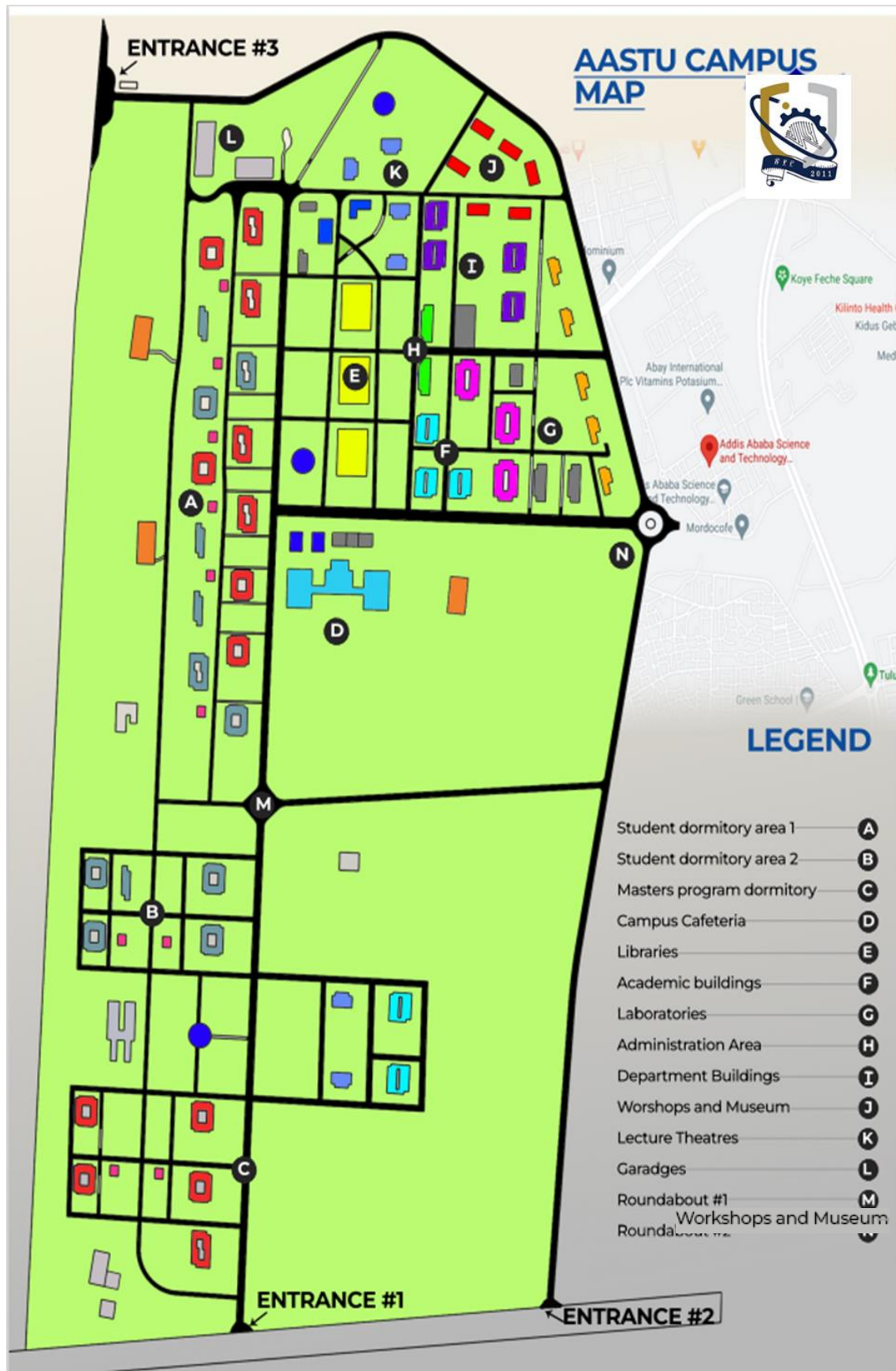
In January 2016, the Ministry of Education (Education Strategy Center) developed a concept note to reform the education sector in accordance with the national vision and national development goals. Hence, one of the strategic plans proposed in the Ethiopian education road map has included common courses, which account a total of 40 credit hours as a national requirement that led to the development of fundamental changes to the Ethiopian educational system.

In line with the given strategic direction of Ethiopian development, Addis Ababa Science and Technology University has a mission to be a problem solver of the industry, leading in the nation research, and delivering world-class education. To meet this mission, the university has given special attention to strengthen the academic sector by working towards accreditation of all undergraduate programs.

As a result of the above reasons, the university planned to revise the entire undergraduate program curricula from the accreditation point of views. The goal of accreditation is to ensure the education provided by higher education to an acceptable level of quality. Therefore, this curriculum framework is developed based on the requirements of the Washington accord and ABET for engineering program and applied sciences programs accreditation respectively.

1.2 Site/location

Addis Ababa Science and Technology University is established in accordance with Article 49 (3) of Higher Education Proclamation No. 650/ 2009 of the Federal Democratic Republic of Ethiopia and then transformed to be a new Science and Technology University in accordance with the Council of Ministers regulation No. 314/2014; in line with its massive program to expand higher education as a key element for technological transformation of the country. It is located, about 3.2 km North of Turinesh-Bejing Hospital, in wereda nine (Kilinto site), Akaki-kality sub-city of the Addis Ababa city administration.



1.3 University Vision and Mission

1.3.1 Vision

To be an internationally recognized Ethiopian Hub of science and technology with strong national commitment and significant continental impact by 2030.

1.3.2 Mission

- Delivering world-class education and training in strategically prioritized science and technology disciplines based on national economic demand.
- Conducting problem-solving applied researches to support the productivity and competitiveness of industries.
- Serving as a center for knowledge and technological adaptation, innovation and transfer.
- Building the technical and managerial capabilities of industries.
- Building a national hub of science and technology.

1.4 AASTU Organization

Governing and advisory bodies of the University consist of:

- a) Board;
- b) President;
- c) Senate;
- d) Managing Council;
- e) University Council;
- f) University Advisory Council;
- g) College Council;
- h) Academic Unit Managing Council;
- i) Department Assembly;
- j) Advisory, Standing or Ad-Hoc committee's

2. Academic Programs and Academic Guide

2.1 The University Registrar and Alumni Directorate

The university Registrar and alumni directorate is dedicated to the principles of service, applying academic policies and procedures of the university. The directorate performs the essential roles of supporting, facilitating and promoting the academic mission of the university by connecting student to the faculty, and a curriculum from enrollment to graduation and beyond. The office provides the organizational structure and technology to ensure a sound and efficient management of students' record.

The office of the registrar and alumni directorate ensures the integrity, accuracy, and security of all academic records of current and former students; facilitate effective student registration and enrollment; builds secure student data files and sets policy and procedure for their responsible use. The registrar office supervises the processes for the transfer of credits, graduation and certification of degrees, issuance of grade reports, official transcripts, diplomas, and related credentials. It counsels and advises students, faculty, and staff on academic matters; and interprets and enforces academic policies and regulations of the university.

With this brief overview of the responsibilities of the directorate, we are glad to inform you that there is one record officer in charge of your department that you have to communicate for all the services and advise you seek from the directorate. When necessary, the record officer shall communicate with the associate registrar and even the registrar to resolve your request as soon as possible. We are deeply committed to providing personal attention and efficient service to our students. You have to feel at ease to contact the office of the directorate by email, by telephone, or in person.

2.2 Welcome & Orientation Session

The University delivers welcome and orientation session for freshmen (new in coming) students at the beginning of every academic year. The Orientation focuses on university life, academic programs, academic rules, rights and duties of students, disciplinary issues, gender etc. Hence, students need to attend this orientation session as firsthand information about AASTU.

2.3 University Life

For most of you, University life is your first experience to live away from family. As a result, some of you may feel that the University student life such as loneliness, peer-group interaction etc is challenging.

Indeed, it is not easy to miss family comfort and cope up to a new and complex university social life. However, sooner or later you will adopt the new environment to be independent and self-supporting grown person, just the same way a one year child manages to stand by her/his self and start walking. In this regard, AASTU is committed to facilitate and support you through all of its organs not only to adapt University life but also to fully utilize the resources set to create a responsible and capable citizen.

Helpful Ten Commandments of University Life that you have to Exercise are:

- ❖ Respect university rules and regulations;
- ❖ Be respectful but never fail to insist for your rights appropriately;
- ❖ Never miss a class: be it lecture, tutorial, or lab session;
- ❖ Plan to study and exploit all resources to make your dream true;
- ❖ Never fail to consult and use relevant information to make intelligent decision;
- ❖ Never to be manipulated for any decision by a friend or peer group;
- ❖ Plan to entertain and take part in social events such as co-curricular activities;
- ❖ Believe in that the university personnel are here to serve you;
- ❖ Never fail to share your problem with appropriate body of the university as well as your family;
- ❖ Never to be frustrated and give up.

2.4 Academic Calendar

The university academic calendar is a time schedule of the major academic activities of the university that students and the university community in general should be aware of. In its regular program, the university shall have a two-semester calendar a minimum of sixteen and a maximum of eighteen weeks of which two weeks of exam) each , running from September to

June. In Continuing Education Program (CEP), there are two semesters, same with the regular program and a Kiremt term of 8- 9 weeks running from July to first week of September.

The university academic calendar, among other things, shall provide a schedule for admission, readmission and registration; adding and dropping of courses/modules, examination periods, beginning and end of semesters, inter- semester breaks, last date for reporting grade reports to the registrar, instructor evaluation weeks, the regular meetings of the Senate, etc.

2.5 Admission

2.5.1 Undergraduate Study

Admission to Regular Program

Admissions to undergraduate programs of any institution from high school colleges shall be based on completion of the high school education and obtaining the necessary pass marks in the Ethiopian Higher Education Entrance Examination (EHEE) or equivalent academic achievements from foreign countries as well as the science and technology university entrance examination to be set by the ministry of education (MOE) and /or AASTU as the case may be.

Admission to Continuing Education Program

Admissions to all undergraduate continuing education programs shall be processed through the academic units, in consultation with the university registrar and CEP, according to policies set for such admissions in line with ministry of education.

Relevant information including student handbook, academic calendar, readmission form etc... shall be posted at the university web site.

Other Admission Cases

The university in certain circumstances may admit students jointly with other affiliated universities or institutes. Such admission may be processed at any time of the academic calendar as deemed appropriate. Students may also be admitted to more than one program at the same time. However, criteria for such special admissions to individual programs shall be developed

and recommended by college academic council (CAC) and endorsed by the admission and enrollment committee (AEC).

2.5.2 Admission to Graduate Program (GP)

General Provisions on Admission

A candidate who has a high scholastic standing, whose professional experience has been superior and who has received good recommendations and passed the graduate admission test shall be given priority for admission. Moreover, priorities in admission may be granted to government sponsors as per the pressing need of the government.

The minimum intake of a master program shall be five students. However, the optimum (minimum and maximum) number of students in the different graduate programs shall be determined by the respective unit taking into consideration the number of staff, availability of space and demand for training.

The academic unit may, from time to time, include additional admission requirements in conformity with university rules and regulations. These may be of general application or may pertain to specific programs of Graduate Programs.

Admission Requirements for Graduate Programs

- The applicant must have completed the academic requirements for the Bachelor's Degree in the specific/related field(s) of study with a minimum CGPA of 2.00 for all gender at the university or any other recognized institution of higher learning as recommended in the respective curriculum.
- The applicant must meet satisfactorily the selection criteria which may include the entrance examination to be administered by the academic units concerned graduate admission test. Foreign applicants may, inline of sitting for entrance examination, submit results of GRE/TOEFL/IELTS or equivalent, language proficiency examinations for applicants whose undergraduate medium of instruction is not English etc.
- An applicant for admission to a Ph.D. program must have a Master's degree by thesis from an accredited institution of higher learning rated at least "good". However, in cases where applicant is Master's degree without thesis work or rated below "good", such

applicants may be provisionally admitted to Ph.D. program provided that the applicant shall complete preliminary thesis, rated at least very good, as a requirement to be a full-fledged Ph.D. student. DGC/CAC shall decide other equivalent requirements for admission to Ph.D. programs.

2.6 Registration

2.6.1 Course Regular Registration

- Normally, students register for courses using formal procedures, electronic or manually by the registrar office during registration days specified in the academic calendar of the year in person for every course to be taken in each regular, evening or kiremt semesters.
- Students are required to present clearance paper for semester I registration of every new academic year.
- Students are also required to register in each semester. A student who fails to register in a given semester is considered as a “dropout”.
- Students can only claim course grade records for courses they register officially through registrar office

2.6.2 Late Registration

There is usually a one-day registration period after the regular registration closes during the regular, evening or Kiremt semester. Late registration is subject to penalty.

2.7 Identification Card

A student registering at the University is issued a non-transferable identification card with a unique and permanent registration number. The card, which is a very valuable document and requires a careful handling, is a means to access services and facilities in the University. The identification card is required every time the student requests for any University services.

Identification cards are issued to all new students free of charge. Lost ID card shall be replaced after a report produced from the Zonal Police office which is subjected to ID replacement fee charge. The ID cards must be immediately returned to the registrar office when student discontinues or completes his/her studies (Dismissal, Withdrawal or graduation).

2.8 Add and Drop

In consultation with the academic advisor or department head, students can add and/or drop courses within the specified add and drop time in each semester. All add and drop forms must be filled out carefully and submitted to the Registrar office on time.

Any registered course shall have grades at the end of the semester. Not submitted grade shall count an F grade immediately after the end of the next semester registration.

2.9 Withdrawing & Clearing

Students withdrawing from the University must consult their academic advisors and explain why they wish to drop out or withdraw. Students withdrawing from the University for an Inadequate Reason will be denied readmission. Any student who wants to withdraw should complete withdrawal form on time. **The University denies readmission request if an applicant fails to withdraw officially on time.** Students should fulfill the proper clearance procedure at the end of each academic year or Kiremt session or upon graduation or when they withdraw or drop out.

Every student must collect two copies of the clearance form from the Registrar Office. After, properly completing such form and getting them signed by the appropriate offices, the student must submit one copy to the registrar office, and keep the other with him/her. No transcript, degree and letter of recommendation will be issued to a person who failed to appropriately withdraw.

2.10 Readmission

- Students, who withdraw officially from the University, can request for readmission into the College/Department they withdrew from. They cannot apply to join a different College/Department. Academic dismissals who meet the minimum cut-off points for readmission or withdrawals in good academic status shall be eligible for consideration.
- Readmission application form can be taken from the registrar office or accessed from the website of the University. Simply fill out, sign and submit to the Registrar office.
- Readmission is subject to availability of space, facilities and the necessary budget.

2.10.1 Readmission for Students in Good Academic Standing

- A student who, for reasons beyond control, discontinues studies may apply for re-admission into the same program for any number of times provided that the maximum duration of stay in the program, counted from the date of first enrolment, has not expired or will not have expired before completion of the program following re-admission. Such readmission shall be to the same semester that the student discontinued studies.
- Re-admission of students in good standing is subject to availability of facilities and necessary budgetary allocations.
- In those cases where there had been curricular changes in a program, the academic unit may require the readmitted student to take new courses that have replaced those taken and passed prior. However, where there is an equivalent course such shall be done for the student.
- Upon re-admission of a student in good standing, the academic unit shall supply a list of courses from which the student will be exempted and of new courses substituting phased out courses to be taken.

2.10.2 Readmission for Academically Dismissed Students

A student may be readmitted after dismissal provided all of the following are fulfilled:

- The dismissal was not due to acts of violation that resulted in his/her termination with the university.
- The student may apply for re-admission at least after one semester following withdrawal from the University except for CDEP students who may apply for readmission at any time. However, a student may be re-admitted in a semester when there are more courses the student is eligible to register for.
- Space is available in the program and there exists likelihood that the student will raise their grade point to the required level after removing any deficiencies which should not take more than one year.
- The maximum duration of stay in the program has not expired or is not likely to expire before the completion of the remaining courses of study.
- A dismissed regular student, who is re-admitted and allowed to repeat a course or courses in a given semester, may be dismissed indefinitely, for not attaining good academic

standing upon determination of status. However, such students may be admitted as new student into the CEP as per the rules and regulations of CEP program.

A cut-off SANG/SGPA and/or a CANG/CGPA for readmission of an academically dismissed student shall be as follows:

- A first year student, dismissed at the end of the first semester with a SGPA of not less than 1.25 shall be readmitted.
- A first year student dismissed at the end of the second semester with a CGPA of not less than 1.67 shall be readmitted.
- A second year and above student dismissed at any semester with a CGPA of not less than 1.75 shall be readmitted.
- Any readmitted student repeats courses in which the student scored “C-“, “D” or “F” grades with the approval of the student’s academic advisor/Head of academic unit.

Readmission procedures

- Students collect readmission application form from the Admissions Office of the Registrar or accessed from the website of AASTU at specified times indicated in the Academic Calendar.
- The filled out form is then returned to the Admission office of the Registrar or mailed electronically. The Office forwards the applications to the appropriate faculty or college. The academic commission of the College acts on the application and decisions are communicated to the Office of the Registrar.
- The Office of the Registrar informs all readmission applicants whether they have been accepted or rejected preferably by e-mail or text message. Hence, don’t miss to provide your e-mail address and cell-phone number when you fill out the readmission application form.

2.11 Assessments and Evaluation

- Student learning shall be assessed on a variety of ways /continuous assessment in the form of tests, assignments, presentations, etc. to determine the final grade earned.

- The final grade is obtained based on the sum of continuous assessment and summative assessment (final exam).
- The weight of the continuous assessment should be out of 50% comprising at least five assessments weighting 5% -15% each in at least three different modalities. While the final exam shall be evaluated out of 50%. A student should score at least 40% in the final exam to pass.

2.12 Class Attendance

A student is required to attend all lecture, laboratory and practical sessions as well as field work. In situations where a student, because of reasons beyond his/her control, fails to attend some classes, a minimum of 80% attendance is required if he/she is to earn credit in a given course. A student who has missed more than 20% attendance will be given a grade of IA (Incomplete Attendance) and will be required to provide acceptable reasons for his/her failure to attend classes.

- If a student's incomplete attendance is proven to have been for valid reasons, his/her registration for the course will be cancelled and he/she shall be permitted to do the course afresh.
- If a student's incomplete attendance was due to reasons that were not valid, the IA grade will be changed to an "F" grade at the end of the sixth week of his/her next enrolment in the program.

2.13 Semester Load

i. Regular Program

A semester load is the number of credits' a student is supposed to take in a semester.

- The normal semester load is 17-19 credit hours or 28-31 ECTS. However, a curriculum may have 20 cr. Hrs. per semester twice except the final year.
- In exceptional cases, as per decision by CAC, graduating class students may be allowed to take a maximum of 22 credits or 35 ECTS.

- A minimum credit or ECTS hours for full-time regular students shall be 15 or 25 per semester, respectively.
- The maximum load for part time students is 9 credit hours or 16 ECTS and the minimum is 10 ECTS or 6 Credit Hours.

A student may be especially allowed by the CAC to take up to 22 credit hours or 35 ECTS when:

- The student secures a GPA of 2.5 and above upon completion of an equivalent course work corresponding to one academic year; and the student is at graduation year after passing all courses taken up to date.
- Graduation failures and readmitted students may take below the minimum semester load. However, status shall only be determined by at least the minimum load.

ii. Continuing Education program

The normal load in evening & weekend programs shall be 8 to 12 credit hours or 14 to 18 ECTS or per semester.

- A student may be especially allowed by the CAC to take up to 15 credit hours or 22 ECTS or when:
- The student secures a GPA of 2.5 and above upon completion of an equivalent course work corresponding to one academic year; and the student is at graduation year after passing all courses taken up to date.
- The normal load for evening and weekend students in a Kiremt semester shall be 12 to 14 ECTS or 6 to 8 credit hours.
- The normal semester load in the continuing Education for Kiremt students shall be 20 to 22 ECTS or 13 to 15 credit hours.
- For the purpose of determining academic status of Kiremt students, one Kiremt session is counted as a Kiremt Semester, but three Kiremt semesters are counted to one regular academic year. The Semester Average Number Grade (SANG) or SGPA calculated at the end of each Kiremt Semester, in conjunction with Cumulative Student Status, shall be used to determine the academic status of a student.

2.14 The Grading System

Final grades of students both in the undergraduate and postgraduate studies are on letter grade system. There are some courses that use P (pass) and F (fail) grades.

The letter grade for undergraduate program is defined as shown in the table.

Raw Mark interval (100%)	Corresponding Letter Grade	Corresponding fixed number Grade	Status Description
[90,100]	A+	4.0	Excellent
[85,90)	A	4.0	
[80,85)	A-	3.75	
[75,80)	B+	3.5	Very Good
[70,75)	B	3.0	
[65,70)	B-	2.75	Good
[60,65)	C+	2.5	
[50,60)	C	2.0	Satisfactory
[45,50)	C-	1.75	Unsatisfactory
[40,45)	D	1.0	Very Poor
[0,40)	F	0	Fail

Grade point on specific course is a product of credit hours and points of a given letter grade. The total grade points are divided by the number of credit hours attempted in order to determine the grade point average (GPA).

2.15 NO GRADE (NG)

In case where a student does not have full examination records, the instructor shall record “NG” for No Grade. A student with “NG” grade has to immediately communicate with the course instructor or Department Head as deemed necessary for appropriate procedure to resolve the problem. All “NG” s will be changed to one or another of the following within two weeks from registration day(s).

- To an “I” (incomplete) by the CAC in consultation with the instructor concerned for a student who because of illness or of other reasons beyond his/her control, fails to complete the course; or
- To an “F” by the CAC in consultation with the instructor concerned for a student who fail to justify for not taking the final exam. Measures related to incomplete test or an assignment is determined by the instructor; or
- To a ”W” (withdrawn) by the department chair/Instructor for a student who has formally withdrawn from the program, or
- To a “DO” (dropout) by the department, chair/Instructor for a student who has not formally withdrawn from the program in accordance with the withdrawal procedures set forth by the University.
- “NG” grade not addressed as per the procedures listed above within two weeks from the registration of the next semester shall be changed to “F” grade by the Registrar Office & status determined accordingly.
- Neither “W”, “DO” nor shall “I” play any part in the computation of the semester grade point average;

2.16 Academic Status of Undergraduate Students

The following requirements apply to all undergraduate degree students:

- Student who earned a 2.0 (C)” grade or above in all courses of a semester shall be in good standing.
- A student cannot register a course for more than three times in anyway.
- Any consecutive warning leads to Academic Dismissal.

How GPA is calculated (Example)

Course Title	CHr	Grade	Grade point
Course1	4	A	16
Course2	3	A-	11.25
Course3	2	B	6
Course4	5	C+	12.5
Course5	3	F	0
Course6	4	D	4
Total	21		49.75

VLG = value of letter grade: A+ =4, A=4, A- =3.75, B+=3.5, B-=2.75, C+ = 2.5, C=2.0, C- =1.75, D= 1, F=0

Gpt = VLG*CHr; SGPA = SGpt/SCHr= 49.75/21= **2.37** and CGPA= CGpt/CCHr

A cut-off SANG/SGPA and/or a CANG/CGPA for status determination of all students shall be as follows:

First Year Students

- A student who, at the end of the first semester, receives an SGPA/SANG of less than 1.50 shall be dismissed for academic reasons.
- A student, who, at the end of the first semester, scores a semester grade point average (SGPA/SANG) of 1.50 up to 1.74 both inclusive, shall be warned.
- If a student, who has been put on warning during the first semester, fails to maintain an SGPA/SANG of 1.75 or a CGPA/CANG of 2.00.

During the second semester of the first academic year, he is subject to dismissal unless otherwise put on probation at the discretion of the academic commission.

- Student, who, at the end of the second semester, receives either an SGPA/SANG of less than 1.75 or a CGPA/CANG of less than 2.00 shall be warned.
- However, If the student's SGPA/SANG is less than 1.75 and his CGPA/CANG is less than 2.00, the student is subject to dismissal.

- If a student fails to achieve a SGPA/SANG of 1.00 at any semester, s/he is subject to dismissal.

Second year and above students

- A student, who fails to achieve either SGPA/SANG of at least 1.75 and/or a CGPA/CANG of at least 2.00 at any semester, shall be warned by his dean.
- However, a student who fails to achieve a SGPA/SANG of 1.00 at any semester is subject to dismissal.
- A student, who for a second consecutive semester scores either an SGPA of less than 1.75 or fails to maintain a CGPA/CANG of 2.00, is subject to dismissal unless put on probation. A student who is subject to dismissal cannot claim probation as a matter of right.

Academic status determination summary Table

Status	Year I		Year II above
	Sem I	Sem II	
Warning	$1.5 \leq \text{SGPA} < 1.75$	$\text{SGPA} < 1.75$ or $\text{CGPA} < 2.0$	$\text{SGPA} < 1.75$ or $\text{CGPA} < 2.0$ Or $\text{SGPA} < 1.75$ & $\text{CGPA} < 2.0$
Dismissal	$\text{SGPA} < 1.5$	Consecutive warning or $\text{SGPA} < 1.75$ & $\text{CGPA} < 2.0$ or $\text{SGPA} < 1.0$	Consecutive warning or $\text{SGPA} < 1.0$
Pass	Otherwise		

2.17 Discretionary Probation

Students subject to dismissal for academic reasons will be dismissed unless placed on probation.

Probation is a privilege that may be granted after a student has exhausted all other privileges such as readmission. No student subject to dismissal may expect probation as a matter of right.

The decision to permit a student to continue his/her studies on probation must be based on individual consideration of his/her case and a conclusion that, in view of all the relevant circumstances, there are valid reasons to believe that the student can raise himself/herself

academically to the required levels of achievement after one semester on probation. Such valid reasons should be spelled out and documented in Academic Commission minutes for each student placed on probation.

2.18 Repeating Courses

A readmitted student repeats a course or several courses with “C-“, “D” and “F” grade to remove academic deficiency at the discretion of the Academic commission.

- A student who obtains an “F” grade in a course may repeat the course in consultation with his/her academic advisor until the “F” grade is removed whenever it is offered.
- A student who has been allowed to repeat a course based on an original “D” or “C-“ grade cannot repeat the course more than twice unless it is required to remove the student’s academic deficiency for graduation.
- The final grade of any repeated course shall be maintained regardless of the initial one. Previous grade of repeated courses should be shown as cancelled on the transcript to indicate that the course has been repeated; and the new grade shall be included in the computation of the SANG/SGPA in the new reporting period.
- Students repeating courses must register for the course and carry out all academic activities pertaining to the course.
- No course is repeated for the purpose of raising CGPA.

2.19 Make-up Examinations

A student unable to take a final examination for reasons beyond his/her control, such as hospitalization or accidents, and therefore given an “I” grade for a course by the AC may be allowed to take make-up examination in the course.

Procedure:

- Any such student or a person representing the student must submit in writing with valid and documented reasons for not taking the final examination to the head of his/her academic department after the date of administration of the exam.
- A student allowed to take a makeup examination will register for and take the examination with other students when the final examination in the course is next administered
- Students allowed to take make-up examination must apply and register for the examination at least one month (the dates are as announced by the office of the Registrar in the academic calendar) before the final examination in the course is scheduled to be administered.
- A student who has three or more “ I “ grades in a semester and is allowed to take makeup examination must, irrespective of his/her academic status, withdraw from the University for academic reasons and apply to take the makeup examination. No GPA will be calculated and no academic status will be determined for such a student until the results of the makeup examination are known.
- A student who has less than three (one or two) “ I “ grades in a semester and who is allowed to take a makeup examination may continue his/her studies in a subsequent semester if the student is in good academic standing. For such a student GPA will be calculated and his/her academic status will be determined based on grades obtained and excluding courses in which the “I“ grades are reregistered. Such a student will also apply for and take makeup examination in incomplete courses.
- For a student who has less than three “ I “ grades but is not in good academic standing shall withdraw from the University and apply to take the makeup examination. The grades earned through the make-up examination are used to compute the GPA of the student in the semester in which the make-up examination is given.
- Any “I” grade not removed within a year as per procedures specified above will be automatically converted into an “F “grade.

2.20 Re- Examinations

Re-examinations are allowed under the following conditions:

- A student in any year who may not have a possibility of repeating a course(s) due to discontinuity of an academic program and/or a graduating student.

The following shall be the conditions under which re-examination is entitled:

- The courses are those that the student had taken during the foregoing semesters of the program;
- The maximum number of courses shall be two for a graduating student and a maximum of four for a student with phase out courses.
- During the student's tenure at the University, a student can sit for re-examination twice in the same course.
- Re-examination is administered within 2-3 weeks after the commencement of the subsequent semester.

2.21 Returning & Remarking of Exam Papers

A student who is aggrieved by the grade obtained in a course shall have the right to petition for remarking their exam paper.

- The student has the right to see the results of their examination papers.
- At the end of each semester, a period for remarking examination papers is specified in the academic calendar.
- When a student need for remarking, he/she will take a copy of an application form from the office of their respective Departments.
- A completed copy of the application form and exam paper will be submitted to the head of the Department responsible for offering the course.
- The Department head, in consultation with the Dean, shall take actions for investigation when a student requests for remarking. For details of implementation procedures see the

Senate Legislation. The petition shall be formally submitted to the head of academic unit within the time frame set in the Academic Calendar.

- The maximum number of courses that the student can apply for remarking shall be only two at any one time. However, if the remarking result proved that the complaint of the student was correct, the remarking shall not be counted against the student in which case the student may apply for additional remarking as the case may be even after the time frame set in the Academic Calendar.
- A grade obtained in remarking shall be final even if it is lower than the one previously obtained.

2.22 Transfer of Students

Regular programs

- Transfer from one program to another is not allowed. However, transfer between programs may be allowed under special circumstances up on request made by respective college AC and the approval of AAVP.
- Students seeking transfer from other Science and Technology universities should have stayed in their department or program of study of the previous University for at least one semester and at the time of request;
 - (a) Transfer students shall have accomplished less than 50% of the credit/ECTS required for the program.
 - (b) Must be in good academic standing with a minimum CGPA of 3.0.
 - (c) Successfully accomplished each module taken and required for status determination.

Continuing Education Program (CEP)

Transfers from program to program or from one academic unit to another within CEP is not allowed. However exchange between students is allowed only once.

Students in CEP may transfer to regular programs as a private students upon approval by the concerned SC/DC and endorsed by AEC provided they meet the conditions set forth below.

Transfer to the regular study program is to enable CEP students to complete studies quicker and will not change their admission classification from CEP to a regular program and does not entail waiver of tuition fee.

- Applicants must have accumulated 75% of the total credits required for graduation.
- Outstanding students with a minimum Cumulative Average Grade point of 3.75 and/or students who would complete their studies within one academic year can apply for transfer and selection will be made on a competitive basis. However, such students may not exceed 2% of the total student enrolled in the batch of the program of study.
- Transfer shall take place within the same programs only.
- Applicants who meet the requirements stated above should fill in forms prepared for this purpose by the Registrar and submit them to the academic unit.
- Applications shall be submitted before registration in the program and the transfer should be completed within two weeks after beginning of classes.
- Academic unit may accept or reject transfer requests on the basis of availability of place and facilities on a competitive basis. Such decisions shall be approved by the relevant SC/DC and forward to the Registrar.
- All transferred students shall pay all fees applicable to CEP. Under exceptional conditions, such students may be allowed to take courses offered in evening, Kiremt or regular programs as their situation warrants such an arrangement and when approved by the SC/DC.
- Transfer from other Universities to AASTU's CEP programs shall be allowed and handled as per the provisions for regular programs.

2.23 Course Exemption

- Students who have attended equivalent level accredited higher institution and earned Degree/Diploma may request for course(s) exemption upon submission of official records to the Registrar and a copy to concerned department.
- The concerned Department carefully examines each previously attended course in terms of its depth, coverage and credit hours before suggesting its corresponding course for exemption.

- Courses may only be exempted if the student has taken the course in the last five years.
- A student is expected to attend courses of at least 60% of the total credit requirement of the department in order to earn a degree from the university.
- The Department submits a list of courses from which the student will be exempted within two months from the date of registration or admission of the student to office of the Registrar where endorsement of exemption list of courses and official notification to the student is made.

2.24 Graduation Requirements

The following constitute requirements for Graduation:

- All the required courses/modules and the minimum credit hours set by the respective academic unit should be satisfied, except to phase in and phase out program.
- A Cumulative Grade Point Average (CGPA)/ CANG of 2.00 must be obtained;
- A cumulative Grade Point Average /CANG/ of 2.0 in major area courses;
- No "F" grade in any course/module taken for undergraduate program
- Pass in exit exam
- Students, who fail to graduate due to less CGPA than the required or due to "F" grade and have no more chance to upgrade their CGPA or to remove their "F" may be given certificate of attendance with their transcript stated as "graduate failure".

2.25 Credentials

2.25.1 Transcripts

Transcripts are records of students' academic performance. They are the most valuable private and sensitive documents. Utmost care is taken in their recording, storing and issuance. The following is a set of regulation governing issuance of transcripts.

- No student records are shown or given to a third party without the written consent of the student. The University may make discretionary exceptions to this.

- No transcripts will be issued for requests made through third parties, that is, representatives, friends or relatives, agencies, etc. In exceptional cases transcripts are given to third parties if the third party meets the following conditions:
- The third party must carry a power of attorney, and the third party does not demand that the transcripts be given to him/her.
- All students who have dropped out, withdrawn or graduated from the University must present an official clearance sheet to get transcripts and other services. Third parties of such students must present clearance sheets of students they represent. The official clearance form is an evidence for the students that he/she clears from any financial and other obligations to the university.
- Transcripts given by the Registrar Office are of two kinds: student copy and official copy. Official copies carry the Registrar's seal and signature and are directly sent to institutions or organizations upon the request and/or the consent of the individual. Student copy, Official transcript to local and foreign destination shall be charged.
- Transcripts are processed and issued or sent on regular or express service bases.
- Student grade reports shall not be issued for graduating class students.

2.25.2 Degrees

Degrees are issued by the Registrar's Office. Following graduation or approval to graduate by each College Academic Commission, a student is, upon presentation of clearance papers, issued temporary certificates of completion. Temporary certificate is later on replaced by the original diplomas. The concerned individual has to pay the required charges for the original degrees. Degrees are given to the individual and not to a third party. Students who fail to collect their Degrees shall be subject to penalty fee. Degrees are issued when candidates produce clearance of their responsibilities. Replacement Degree is also possible upon official announcement of the lost Degree through public media designated for such purpose.

2.25.3 Tuition and Other Fees

- All Ethiopian students in the Regular Undergraduate University program are not charged for admission application and regular registration. However, they are charged for tuition and boarding as well as food services under the cost sharing system, either in advance or after their graduation depending on the policy of the MoE.

- Foreign students joining the regular programs and all students joining the evening program are subject to tuition fees. Tuition and other fees shall be decided by the University and the details will be made available at the Office of the Registrar or the Office of the Continuing and Distance Education Directorate Director.

2.26 Fast-Track Program

Starting from 2008 E.C. Addis Ababa Science and Technology University is admitting outstanding students through specially designed entrance exam from all over the country. To encourage these students, the University has implemented a Fast-Track Program (FTP) in its Legislation to enable these gifted students complete and receive their bachelor's degrees on time while simultaneously making substantial progress in Master's level studies. This guideline will be used to screen the applicants and meet the degree requirements that will ensure the success of the students by maintaining the quality of their education. Once applicants have been screened and selected to participate in this program, they will complete a rigorous and carefully selected set of organized advanced undergraduate and graduate courses. These courses may be used to satisfy the undergraduate and graduate degree requirements. The guidelines contained in this program set an upper limit on the number of courses that a student might count toward both the bachelors and the Master's degree. In no case may the dual counted courses exceed the maximum specified below. All departments must ensure that only outstanding and academically capable students will be allowed to pursue the proposed course of study.

2.26.1 Eligibility

Students who meet the following conditions are eligible to apply for the FTP:

Undergraduate students with a CGPA of at least 3.5 for male and 3.4 for female and disabilities obtained in at least four semesters for science and in at least 6 semesters for engineering programs are eligible to apply for Master's programs fast-track system.

2.26.2 Application and Admission Procedures

Eligible students will complete FTP application form and submit to the department which is running the graduate program.

The DGC shall prepare evaluation criteria to assess the applicant's critical thinking skill (30%), analytical skill (20%), research potential (30%), and writing skill (20%). The evaluation must include standard comprehensive exam and an interview. In addition, applicants for FTP shall pass in Graduate Admission Test (GAT) as determined by MoE to be considered for admission.

The DGC will make the final decision to admit or deny the student to Fast-track Program. The evaluation result along with the application form of the selected students shall be forwarded to the College's Academic Council (CAC) and the CAC endorses the decision of the DGC.

2.26.3 General Rules of FTP

Departments are responsible to identify core courses which are mandatory to be taken by students enrolled under fast track programs. Departments are responsible to identify courses under each undergraduate program which can potentially be substituted by advanced MSc courses. In this case, dual credit count can be considered for both programs for maximum of 9 (nine) credit hours. The Master's level courses, which are counted as dual credit, taken by the Fast Track student will appear on both the undergraduate and graduate transcripts. These Master's level courses will count in the cumulative earned hours and CGPA for both undergraduate and graduate careers assisting the fast track student towards degree completion at the bachelor's and Master's level. For Fast-Track students, credit can be counted by exam only (self-study) for a maximum of two undergraduate courses. The College/Departments shall plan appropriate time for fast track students to take internship. If the student takes the internship in summer season, the minimum required time for the internship shall be three months. The Fast-Track student can take maximum of 25 credit hours per semester.

2.26.4 Requirements for Staying in FTP

A student must maintain an overall GPA of at least 3.25 in undergraduate courses. A student is expected to get grades of B or better in graduate courses. If the student earns a grade less than B in a graduate course before earning the undergraduate degree, the student will lose eligibility to remain in FTP. If, at any time, a student fails to satisfy the requirements in 2.26.3 and 2.26.4, the student will be removed from FTP. Any graduate credits earned will be applied only to the undergraduate degree, and none of the other benefits of studentship in FTP will apply.

Students dismissed from FTP will be eligible to apply to the graduate programs of the University as a regular applicant. All graduate courses taken during FTP will not be considered if the student is admitted as a regular graduate program. Request to be unrolled from the FTP program is allowed within 2 weeks period after admission. Termination in the mid-semester is not allowed. All FTP privileges will be waived for students who are unrolled from FTP.

2.27 Other Services

2.27.1 Social Services

The University facilitated for the provision of several social services in the University campus to make your life easy. These include healthcare, restaurants, cafés, shops such as barber, beauty salon, photocopy and stationery and mini supermarkets. In this regard, the university management will continue to exert its effort in extending necessary social and financial services in order to meet your satisfaction.

2.27.2 Counseling Service

Gender, Guidance and Counseling, and HIV counseling services are established at office level to provide you support and counseling service in educational, social, psychological and university life issues at large. Please do not hesitate to contact these offices when deemed necessary.

2.28 Undergraduate Degree Programs

College of Engineering

- Bachelor of science degree in Architecture
- Bachelor of science degree in Civil Engineering
- Bachelor of science degree in Mining Engineering
- Bachelor of science degree in Chemical Engineering
- Bachelor of science degree in Environmental Engineering
- Bachelor of science degree in Electromechanical Engineering
- Bachelor of science degree in Electrical and Computer Engineering
- Bachelor of science degree in Mechanical Engineering
- Bachelor of science degree in Software Engineering

College of Natural and Applied Sciences

- Bachelor of science degree in Biotechnology with Honors
- Bachelor of science degree in Geology with Honors
- Bachelor of science degree in Industrial Chemistry with Honors
- Bachelor of science degree in Food Science and Applied nutrition with Honors

2.29 Freshman and Pre-engineering program

Freshman courses

S. No	Course Title	Course code	ESTS	Cr. Hr	Category	Remark
1	Entrepreneurship	Entr1006	5	3	Common Course	
2	Communicative English Language Skills I	FLEn 1003	5	3(2+1)	Common Course	
3	Geography of Ethiopian and The horn	GeEs 1005	5	3	Common Course	
4	Mathematics for Natural	Math 1007	5	3(2+3)	Common Course	

	Science					
5	Logic and Critical Thinking	Phil 1009	5	3	Common Course	
6	General Psychology	Psyc 1011	5	3	Common Course	
7	Social Anthropology	Anth1002	3	2	Common Course	
8	General Physics	Phys 1001	5	3(2+1)	Common Course	
9	Physical Fitness	SpSc 1013	2	P/F	Common Course	
10	Communicative English Language Skills II	FLEn 1004	5	3(2+1)	Common Course	
11	Introduction to Emerging Technology	EmTe 1008	5	3	Common Course	
12	General Chemistry	Chem 1006	5	3(2+1)	Common Course	
13	Global Trend	GLTr 2011	4	2	Common Course	
14	Inclusiveness	Incl 1010	3	2	Common Course	
15	Moral and Civic Education	MCiE 1012	3	2	Common Course	
16	Economics	Econ 2009	5	3	Common Course	
17	History of Ethiopia and the horn	Hist. 2002	5	3	Common Course	
18	Applied Mathematics IA	Math1014	7	4(3+3)	Common Course	
19	Applied Mathematics IB	Math1014	7	4(3+3)	Common Course	
20	Applied Mathematics IIB	Math2007	7	4(3+3)	Common Course	

2.29.1 Program Courses Synopsis

Course Code: Phil1012

Course Name: **Logic and Critical Thinking**

Credit Hour: 3

Prerequisite: No

Logic and Critical Thinking is an inquiry that takes arguments as its basic objects of investigation and it is an exercise, a habit, a manner of perception and reasoning that has principles of rationality as its fulcrum, and dynamically involves various reasoning skills that ought to be human approach to issues and events of life. Logic and Critical Thinking enables to construct one's own sound argument and to evaluate the arguments of other's and evaluate arguments' validity, strengths and weaknesses.

Course Code: Psych 1011

Course Name: General Psychology

Credit Hour: 3

Prerequisite: No

The course encompasses the fundamental concepts and principles of psychology and psychological processes which have immense applications to human life and to develop life skills based on the theories and principles of psychology.

Course Code: **FLEn 1011**

Course Name: **Communicative English Language Skills I**

Credit Hour: 3

Prerequisite: No

This course will cover specific language aspects such as developing basic functions of English language skills such as reading, listening, writing, speaking, vocabulary and grammar.

Course Code: GeES 1005

Course Name: **Geography of Ethiopia and the Horn**

Credit Hour: 3

Prerequisite: No

This course covers a brief description on the location, shape and size of Ethiopia as well as basic skills of reading map, the physical background and natural resource endowment of Ethiopia and the Horn which includes its geology and mineral resources, topography, climate, drainage and water resources, soil, fauna and flora. It also deals with the demographic characteristics of the country and its implications on economic development.

Course Code: Math 1001

Course Name: **Mathematics (For natural science)**

Credit Hour: 3

Prerequisite: No

This course covers the basic concepts of logic and set theory, the real and complex number systems, Mathematical induction, least upper bound and greatest lower bound, functions and types of functions, polynomial and rational functions, logarithmic and exponential functions, trigonometric functions, hyperbolic functions and their graphs and analytic geometry.

Course Code: **SpSc 1001**

Course Name: **Physical Fitness**

Credit Hour: 2

Prerequisite: No

This course covers: - Concepts of physical fitness, the health benefits of physical activity, making well-informed food choices, health related components of fitness and principles of exercise prescription and assessment of fitness components.

Course Code: Phys1011

Course Name: **General Physics**

Credit Hour: 3

Prerequisite: No

This algebra based introductory course is designed to enable students to learn the basic concepts, principles and applications of elementary physics. The topics include vectors, Kinematics & Dynamics of Particles, Fluids Mechanics, Heat and Thermodynamics, Oscillations, waves and optics, Electromagnetism & Electronics, Cross Cutting Applications of Physics.

Course Code: FLEn 1012

Course Name: **Communicative English Language Skills II**

Credit Hour: 3

Prerequisite: Communicative English Language Skills I

Communicative English Language Skills II Module is a continuation of Communicative English Language skills I Module, and it mainly aims to provide first year University students proficiency with speaking, listening, reading and writing skills.

Course Code: MCiE1012

Course Name: **Moral and Civic Education**

Credit Hour: 2

Prerequisite:

This course generally covers basic understanding of civics and ethics, approaches of ethical decision-making, moral judgment, state and government, Constitution, Democracy and Human Rights. In so doing students will have basic knowledge of their duties and rights besides the understanding of theories of state, morality and global issues.

Course Code: SNIE 1002

Course Name: **Inclusiveness**

Credit Hour: 2

Prerequisite: None

The course will provide concepts of disabilities/vulnerabilities and inclusiveness, application of assessment strategies for service provision, skills of demonstrating inclusive culture and introduce the existing national and international legal frameworks. In addition, the course will offer techniques on establishing partnership with stakeholders and apply inclusiveness for peace, democracy and development.

Course Code: Anth 1002

Course Name: **Social Anthropology**

Credit Hour: 2

Prerequisite: None

This course as a freshman course gives an understanding of different key areas of anthropological inquiry: family, means of production, political organizations, social organization, language, religion, and gender.

Course Code: Econ 1011

Course Name: Introduction to Economics

Credit Hour: 3

Prerequisite: None

This course is an introductory course covering basic principles and issues of economics. Broadly speaking, the course has two components, microeconomics and macroeconomics. In microeconomics, the focus is on the way in which individual economic agents – workers, consumers, households and firms – make decisions. This part begins with a discussions and applications of the concepts of demand & supply; theory of the consumer; theory of producer; and market structures. While in macroeconomics, the study involves the study of the economy as a whole, especially issues related to macroeconomic goals, national income account and its measurement, interest rates macroeconomic problems such as unemployment, deficit and inflation, growth and policy instruments.

Course Code: GLTr2001

Course Name: **Global Trend**

Credit Hour: 2

Prerequisite: None

The course aims to equip students with the basics of international relations, foreign policy and diplomacy, introduction to international political economy, globalization and regionalism, and major contemporary global trends.

Course Code: HiES2010

Course Name: History of Ethiopia and the Horn

Credit Hour: 3

Prerequisite: None

This course describes why history is important, how history is studied and introduces the region Ethiopia and the Horn. It treats human evolution, Neolithic Revolution, settlement patterns as well as religion and religious processes in Ethiopia and the Horn. Based on these historical backgrounds, the course describes states, external contacts, economic formations and achievement in terms of architecture, writing, calendar, and others to the end of the 13th century. Historical processes including states formation and power rivalry, trade, external relation, threats and major battles, centralization and modernization attempts, Italian occupation, and socio-economic conditions from 1800 to 1941 makes central position in the modern history of the region.

Course Code: EmTe-1012

Course Name: **Introduction to Emerging Technologies**

Credit Hour: 3

Prerequisite: None

This course will enable students to explore current breakthrough technologies in the areas of Artificial Intelligence, Internet of Things and Augmented Reality that have emerged over the past few years. Besides helping learners become literate in emerging technologies, the course will prepare them to use technology in their respective professional preparations.

Course Code: Entr.1002

Course Name: **Entrepreneurship**

Credit Hour: 2

Prerequisite: None

This course is designed to encourage students to start their own business and to acquaint them with the peculiar challenges and management decisions faced by owners of small business. It will develop such entrepreneurial skills as: identifying business opportunities; initiating, financing, and developing new venture business plans. It also addresses issues of small business, such as: legal aspects, financing, costing, locating, personnel, marketing, competition, sources of funding, and constituency services.

Course Code: Math1002

Course Name: Applied Mathematics I

Credit Hour: 4

Prerequisite: None

The course is designed to develop students' confidence with mathematical concepts, use of mathematical skills and techniques in a wide range of contexts specifically problem solving and abstract thinking. It covers basic elements of vectors, vector spaces, matrices, determinants, solving systems of linear equations, concepts and applications of differential and integral calculus of one variable.

Course Code: Econ2009

Course Name: **Economics**

Credit Hour: 3

Prerequisite: None

This course is an introductory course covering basic principles and issues of economics. Broadly speaking, the course has two components, microeconomics and macroeconomics. In microeconomics, the focus is on the way in which individual economic agents – workers, consumers, households and firms – make decisions. This part begins with a discussions and applications of the concepts of demand & supply; theory of the consumer; theory of producer; and market structures. While in macroeconomics, the study involves the study of the economy as a whole, especially issues related to macroeconomic goals, national income account and its measurement, interest rates macroeconomic problems such as unemployment, deficit and inflation, growth and policy instruments.

3. College of Engineering

- Bachelor of science Degree in Architecture
- Bachelor of science Degree in Civil engineering
- Bachelor of science Degree in Mining engineering
- Bachelor of Science Degree in Chemical Engineering
- Bachelor of Science Degree in Environmental Engineering
- Bachelor of Science Degree in Electrical and Computer Engineering
- Bachelor of Science Degree in Electromechanical Engineering
- Bachelor of Science Degree in Mechanical Engineering
- Bachelor of Science Degree in Software Engineering

3.1 Bachelor of Science Degree in Architecture

3.1.1 Introduction

Architecture is the art and science of designing and constructing buildings. A wider definition would include within its scope the design of the total built environment, from the macro-level of town planning, urban design, and landscape to the micro-level of furniture and product design. Architecture equally importantly, also refers to the product of such a design. The practice of Architecture is the act of planning and architectural designing, structural conceptualization, specifying, supervising and giving general administration and responsible direction to the erection, enlargement or alterations of buildings and built environments.

As a corollary to the first purpose mentioned, architectural education should also aspire to improve the quality of architectural output in general and architectural practice in particular. In addition to educating well-equipped graduates, this can be done by research (i.e. constantly expanding architectural knowledge), by setting good examples, by publishing and by fostering contacts with practices.

Architectural design in engineering structures or any part thereof; the scientific, aesthetic, and orderly coordination of all the processes which enter into the production of a complete building or structure, performed through the medium of unbiased preliminary studies of plans, consultations, specifications, conferences, evaluations, investigations, contract documents and oral advice and directions. Architecture is one of the key professions involved in shaping the built environment and urban space. Architectural education that prepares architects for a professional life should therefore, be seen, at least in the following contexts and the specific objectives about them:

- Social, cultural, political contexts
- Professional, technological, industrial contexts
- The World: local, global, ecological contexts
- Academic contexts: including science and knowledge in general.
- International contexts

Architecture is an interdisciplinary field that comprises several major components: humanities, social and physical sciences, technology and the creative arts. The future of architecture also depends on an understanding and assimilation of the achievements of other disciplines and professions.

Although, like other branches of professional education, it is primarily assigned the task of educating future architects, architectural education is, generally, part of the university system and must adjust to and benefit from the traditions of the university, such as doing research and seeing the professional practice within the context of society and science.

3.1.2 Program Educational Objectives

The Department of Architecture aims to:

- Train Architects who are capable of practicing in architectural projects,
- Impart knowledge and skill in the field of architecture and produce qualified architects to satisfy the manpower needs of the country,
- Advance knowledge in the fields of architecture through research,
- Render all services, considered appropriate and best handled by the Department to the community of the region,
- Participate in the study and preservation of the architectural heritage of Ethiopia.

3.1.3 Program Education Outcome (PEO)

The Program Education Outcome (PEO) describe accomplishments that program graduates are expected to attain within five years duration after graduation. Graduates will have applied their expertise to contemporary problem solving, be engaged professionally, and have continued to learn and adapt, and have contributed to their organizations through leadership and teamwork. More specifically, the objectives of the Bachelor of Science Degree in Architecture are to engage in critical thinking, acquire the knowledge, skills, and techniques needed for the genesis, nurturing, development and feasibility of ideas and, finally, master a wide range of media necessary to express and communicate them adequately after they graduate from Addis Ababa Science and Technology University from architecture department within five years.

These criteria seek to evaluate the outcomes of architecture programs and student work within their unique institutional, regional, national, international, and professional contexts, while encouraging innovative approaches to architecture education and professional preparation.

Table Program Education Outcome (PEO)

PEO/ PC	Statement
PEO/PC -1	Career Paths- the program must ensure that students understand the paths to becoming licensed as an architect in the country and the range of available career opportunities that utilize the discipline's skills and knowledge.
PEO/PC -2	Design. Our graduates must be prepared to engage in design activity as a multi-stage process aimed at addressing increasingly complex problems, engaging a diverse consultancy, and providing value and an improved future. The architecture program is centered on creative and critical thinking. This will be accomplished by having strong learning and design studio cultures. It will reinforce through a strong connection between design studio and non-design studio courses. The program instills in students the role of the design process in shaping the built environment and conveys the methods by which design processes integrate multiple factors, in different settings and scales of development, from buildings to cities.
PEO/PC-3	Ecological Knowledge and Responsibility- the program instills in students a holistic understanding of the dynamic between built and natural environments, enabling future architects to mitigate climate change responsibly by leveraging ecological, advanced building performance, adaptation, and resilience principles in their work and advocacy activities.
PEO/PC -4	History and Theory- the program ensures that students understand the histories and theories of architecture and urbanism, framed by diverse social, cultural, economic, and political forces, nationally and globally.
PEO/PC -5	Research and Innovation —the program prepares students to engage and

	participate in architectural research to test and evaluate innovations in the field.
PEO/PC -6	Leadership and Collaboration —the program ensures that students understand approaches to leadership in multidisciplinary teams, diverse stakeholder constituents, and dynamic physical and social contexts, and learn how to apply effective collaboration skills to solve complex problems.
PEO/PC-7	Learning and Teaching Culture —the program fosters and ensures a positive and respectful environment that encourages optimism, respect, sharing, engagement, and innovation among its faculty, students, administration, and staff.
PEO/PC-8	Social Equity and Inclusion —the program furthers and deepens students' understanding of diverse cultural and social contexts and helps them translate that understanding into built environments that equitably support and include people of different backgrounds, resources, and abilities.

A program demonstrates how its curriculum, structure, and other experiences address the NAAB criteria based on (see for our PC reference detail see appendix 1).

3.1.4 Graduation Requirements

The graduation requirement for all undergraduate programs should satisfy the following minimum requirements as stated in the university's senate legislation in July 2017, Article 109.

- All the required courses/Courses and the minimum credit hours set in the program curriculum by the respective academic unit should be satisfied, except to phase in and phase out the program.
- A cumulative grade point average CGPA of 2.00 must be obtained;
- A cumulative grade point average CGPA of 2.00 in major area courses;
- No "F" grade in any course/ Course taken for the undergraduate program;
- Pass the exit exam
- The score pass mark for all courses which have Pass/Fail grade.

3.1.5 Course Category and list of Courses

Table Distribution of credit hours for all course categories

Category		Total Cr. hr	Percentage (%)
1	Core Course (major and supportive course)	152	80
2	Core Elective/focus area course	3	1.60
3	University requirement (Core)	3	1.60
Core Course Total Cr. hr		158	83.16
4	National Requirement	32	16.84
Total Cr. Hr		190	100

Table List of the Courses

S. No	Code	Course Name	Cr. Hr	L	F	P	Pre request	Column3
1	Phys 1001	General Physics	3	2	3	0	None	Year 1 Semester 1
2	FLEn 1003	Communicative English Language Skills I	3	2	3	0	None	Year 1 Semester 1
3	GeEs 1005	Geography of Ethiopian and The horn	3	2	3	0	None	Year 1 Semester 1
4	Math 1007	Mathematics for Natural Science	3	2	3	0	None	Year 1 Semester 1
5	Phil 1009	Logic and Critical Thinking	3	1	0	3	None	Year 1 Semester 1
6	Psyc 1011	General Psychology	3	2	3	0	None	Year 1 Semester 1
7	SpSc 1013	Physical Fitness	(P/F)	1	3	0	None	Year 1 Semester 1
8	Anth1002	Social Anthropology	2	2	3	0	None	Year 1 Semester 2
9	FLEn 1004	Communicative English Language Skills II	3	2	1	2	FLEn 1003	Year 1 Semester 2
10	MCiE 1012	Moral and Civic Education	2	3	3	0	None	Year 1 Semester 2
11	Math 1014	Applied Mathematics I for Engineering	4	1	0	3	None	Year 1 Semester 2

S. No	Code	Course Name	Cr. Hr	L	T	P	Pre request	Column3
12	Entr1106	Entrepreneurship	3	2	3	0	None	Year 1 Semester 2
13	EmTe1108	Introduction to Emerging Technology	3	2	3	0	None	Year 1 Semester 2
14	Incl 1010	Inclusiveness	2	2	0	3	None	Year 1 Semester 2
15	ARCH2101	Basic Design	3	1	3	3	None	Year 2 Semester 1
16	ARCH2103	Building Materials and Construction I	3	1	3	3	None	Year 2 Semester 1
17	Econ2009	Economics	3	3	0	0	None	Year 2 Semester 1
18	GLTr2011	Global Trend	2	2	0	0	None	Year 2 Semester 1
19	ARCH2105	Drawing (Geometric Descriptive and Drafting)	3	1	3	3	None	Year 2 Semester 1
20	ARCH2107	Graphics Communication Skills I(Sketching I)	2	1	0	3	None	Year 2 Semester 1
21	ARCH2109	Theory and Design of Structures I (Engineering Mechanics)	3	2	3	0	None	Year 2 Semester 1
22	ARCH2102	Basic Architectural Design	3	1	0	6	ARCH2101	Year 2 Semester 2
23	ARCH2104	Building Materials and Construction II	3	1	3	3	ARCH2103	Year 2 Semester 2
24	ARCH2108	Graphics Communication Skills II (Sketching II, Painting II)	2	1	0	3	ARCH2105	Year 2 Semester 2
25	ARCH2106	History of Architecture I	3	2	3	0	None	Year 2 Semester 2
26	Hist 2002	History of Ethiopia and the horn	3	3	0	0	None	Year 2 Semester 2
27	ARCH2110	Theory and Design of Structures II (Strength of Materials)	3	2	3	0	ARCH2109	Year 2 Semester 2
28	ARCH3111	Architectural Design I	4	1	0	9	ARCH2102	Year 3 Semester 1
29	ARCH3113	Building Materials and	3	1	3	3	ARCH2104	Year 3 Semester 1

S. No	Code	Course Name	Cr. Hr	L	T	P	Pre request	Column3
		Construction III						
30	ARCH3115	History of Architecture II	3	2	3	0	ARCH2106	Year 2 Semester 1
31	ARCH3128	Ethiopian History of Architecture	2	1	3	0	None	Year 3 Semester 2
32	ARCH3126	Introduction to Environmental Planning	2	1	0	3	None	Year 3 Semester 1
33	ARCH2112	Model Making Technique	2	0	3	3	None	Year 3 Semester 1
34	ARCH3121	Theory and Design of Structures III	3	2	3	0	ARCH2110	Year 3 Semester 1
35	ARCH3117	Visual & History of Arts	3	2	0	3		Year 3 Semester 1
36	ARCH3120	Architectural Design II	4	1	0	9	ARCH3111	Year 3 Semester 2
37	ARCH3114	Architectural Sciences I (Water and sewage)	2	1	0	3	None	Year 3 Semester 2
38	ARCH3116	General Building Heritage	2	1	0	3	None	Year 3 Semester 2
39	ARCH3119	Graphics Communication Skills III (Professional CAD)	3	1	3	3	ARCH2108	Year 3 Semester 1
40	ARCH3118	Landscape Design	3	2	0	3	None	Year 3 Semester 2
41	ARCH3122	Professional Practice I	2	1	0	3	None	Year 3 Semester 2
42	ARCH3124	Surveying	3	1	3	3	None	Year 3 Semester 2
43	ARCH3130	Internship-I	3	0	0	9	None	Year 3 Semester 3
44	ARCH4125	Architectural Sciences II (heating, cooling, ventilation)	2	1	3	0	None	Year 4 Semester 1
45	ETP4115	integrated Engineering Team Project	3	1	0	3	None	Year 4 Semester 1
46	ARCH4129	Building Workshop I (Masonry)	1	0	0	3	None	Year 4 Semester 1
47	ARCH4131	Integrated Design Project I	5	1	0	12	ARCH3120	Year 4 Semester 1
48	ARCH4133	Introduction to Urban	3	2	0	3	None	Year 4 Semester 1

S. No	Code	Course Name	Cr. Hr	L	T	P	Pre request	Column3
		Planning						
49	ARCH4135	Professional Practice II	2	1	0	3	ARCH3122	Year 4 Semester 1
50	ARCH4123	Theory of Architecture I	3	2	3	0	None	Year 4 Semester 1
51	ARCH4142	Appropriate Building Technology	2	1	0	3	None	Year 4 Semester 2
52	ARCH4144	Architectural Sciences III (Light & Energy)	2	1	0	3	None	Year 4 Semester 2
53	ARCH4132	Basic Urban Design	3	1	3	3	ARCH4133	Year 4 Semester 2
54	ARCH4134	Building Workshop II (Carpentry)	1	0	0	3	None	Year 4 Semester 2
55	COMP4136	Computer Programming	3	2	3	0	None	Year 4 Semester 2
56	ARCH4138	Integrated Design Project II	5	1	0	12	ARCH4131	Year 4 Semester 2
57	ARCH4140	Theory of Architecture II	3	2	3	0	ARCH4137	Year 4 Semester 2
58	ARCH4146	Internship-II	3	0	0	9	None	Year 4 Semester 3
59	ARCH5137	Architectural Science IV (Acoustics)	2	1	0	3	None	Year 5 Semester 1
60	ARCH5135	Building Information Modeling (BIM)	3	1	3	3	None	Year 5 Semester 1
61	ARCH5141	Final Year Project (Research)	P/F	0	3	3	ARCH4138	Year 5 Semester 1
62	ARCH5143	Housing & Inner-City Redevelopment	3	1	3	3	ARCH4132	Year 5 Semester 1
63	ARCH5145	Integrated Design Project III	5	1	0	12	ARCH4138	Year 5 Semester 1
64	ARCH5152	Urban Sociology	2	2	0	0	None	Year 5 Semester 1
65	ARCH5150	Final Year Project (Studio)	6	0	0	18	ARCH5141	Year 5 Semester 2
66	ARCH5148	Ecological Architecture & Urbanism	3	2	0	3	None	Year 5 Semester 2
67	ARCH5149	Interior Design	2	1	3	3	None	Year 5 Semester 2
68	ARCH5152	Construction Management	3	2	0	3	None	Year 5 Semester 2

S. No	Code	Course Name	Cr. Hr	L	T	P	Pre request	Column3
		Elective	3	0	0	0	None	Year 5 Semester 2

Table Core Elective Courses

No	Course code	Course title	Cr. Hr	L	P	T	Pre-requisite
69	ARCH5254	Advanced Building Structures	3	1	3	3	None
70	ARCH5251	Advanced Landscape Architecture	3	1	0	3	ARCH3119
71	ARCH5252	Advanced Urban Design	3	1	3	3	ARCH4132
73	ARCH5253	Architecture Heritage and Conservation	3	2	0	3	None
Total Credits			12	5	6	12	

3.1.6 Program Curriculum Structure: Bachelor of Science in Architecture

Year I, Semester I

S. No	Course Title	Course code	ESTS	Cr. Hr	Category
1	General Physics	Phys1001	5	3	Common Course
2	Communicative English Language Skills I	FLEn1003	5	3	Common Course
3	Geography of Ethiopian and The horn	GeEs1005	5	3	Common Course
4	Mathematics for Natural Science	Math1007	5	3	Common Course
5	Logic and Critical Thinking	Phil1009	5	3	Common Course
6	General Psychology	Psyc1011	5	3	Common Course
7	Physical Fitness	SpSc1013	2 C.H	(P/F)	Common Course
	Total			18	

Year I, Semester II

S. No	Course Title	Course code	ESTS	Cr. Hr	Category
1	Social Anthropology	Anth1002	3	2	Common Course
2	Communicative English Language Skills II	FLEn1004	5	3	Common Course
3	Entrepreneurship	Entr1006	5	3	Common Course
4	Introduction to Emerging Technology	EmT1008	5	3	Common Course
5	Inclusiveness	Incl1010	3	2	Common Course
6	Moral and Civic Education	MCiE1012	3	2	Common Course
7	Applied Mathematics I for Engineering	Math1014	7	4	Common Course
	Total			19	

Year 2 Semester 1			Credits		Contact hours		
S. No	Code	Course Name	Cr. Hr	L	T	P	Pre-requisite
1	ARCH2101	Basic Design	3	1	3	3	None
2	ARCH2103	Building Materials and Construction I	3	1	3	3	None
3	Econ2009	Economics	3	3	0	0	None
4	GLTr2011	Global Trend	2	2	0	0	None
5	ARCH2105	Drawing (Descriptive Geometry and Drafting)	3	1	3	3	None
6	ARCH2107	Graphics Communication Skills I (Sketching I)	2	1	0	3	None
7	ARCH2109	Theory and Design of Structures I (Engineering Mechanics)	3	2	0	3	None
Total Sum			19	6	12	15	

Year 2 Semester 2			Credits		Contact hours		
S. No	Code	Course Name	Cr. Hr	L	T	P	Pre-requisite
1	ARCH2102	Basic Architectural Design	3	1	0	6	ARCH2101
2	ARCH2104	Building Materials and Construction II	3	1	3	3	ARCH2103
3	ARCH2108	Graphics Communication Skills II (Sketching II and Painting II)	2	1	0	3	ARCH2107
4	ARCH2106	History of Architecture I	3	2	3	0	None
5	Hist2002	History of Ethiopia and the horn	3	3	0	0	None
6	ARCH2110	Theory and Design of Structures II (Strength of Materials)	3	2	3	0	ARCH2109
7	ARCH2112	Model Making Technique	2	0	3	3	None
Total Sum			19	7	6	21	

Year 3 Semester 1			Credits		Contact hours		
S. No	Code	Course Name	Cr. Hr	L	T	P	Pre-requisite
1	ARCH3111	Architectural Design I	4	1	0	9	ARCH2102
2	ARCH3113	Building Materials and Construction III	3	1	3	3	ARCH2104
3	ARCH3115	History of Architecture II	3	2	3	0	ARCH2106
4	ARCH3119	Graphics Communication Skills III (Professional CAD)	3	1	3	3	ARCH2108
5	ARCH3121	Theory and Design of Structures III	3	2	0	3	ARCH2110
6	ARCH3117	Visual & History of Arts	3	2	0	3	None
Total Sum			19	7	9	18	

Year 3 Semester 2			Credits		Contact hours		
S. No	Code	Course Name	Cr. Hr	L	T	P	Pre-requisite
1	ARCH3120	Architectural Design II	4	1	0	9	ARCH3111
2	ARCH3114	Architectural Sciences I (Water and sewage)	2	1	0	3	None
3	ARCH3116	General Building Heritage	2	1	0	3	None
4	ARCH3118	Landscape Design	3	2	0	3	None
6	ARCH3122	Professional Practice I	2	1	0	3	None
7	ARCH3124	Surveying	3	1	3	3	None
5	ARCH3128	Ethiopian History of Architecture	2	2	0	0	None
7	ARCH3126	Introduction to Environmental Planning	2	1	0	3	None
Total Sum			20	6	12	15	

Year 4 Semester 1			Credits		Contact hours		
S. No	Code	Course Name	Cr. Hr	L	T	P	Pre-requisite
1	ARCH4125	Architectural Sciences II (heating, cooling, ventilation)	2	1	3	0	None
2	IETP4115	Integrated Engineering Team Project	3	1	3	3	None
3	ARCH4129	Building Workshop I (Masonry)	1	0	0	3	None
4	ARCH4131	Integrated Design Project I	5	1	0	12	ARCH3120
5	ARCH4133	Introduction to Urban Planning	3	2	0	3	None
6	ARCH4127	Professional Practice II	2	1	0	3	ARCH3122
7	ARCH4123	Theory of Architecture I	3	2	3	0	None
Total Sum			19	6	15	15	

Year 4 Semester 2			Credits		Contact hours		
S. No	Code	Course Name	Cr. Hr	L	T	P	Pre-requisite
1	ARCH4142	Appropriate Building Technology	2	1	0	3	None
2	ARCH4144	Architectural Sciences III (Light & Energy)	2	1	0	3	None
3	ARCH4132	Basic Urban Design	3	1	3	3	ARCH4133
4	ARCH4134	Building Workshop II (Carpentry)	1	0	0	3	None
5	COMP4136	Computer Programming	3	2	3	0	None
6	ARCH4138	Integrated Design Project II	5	1	0	12	ARCH4131
7	ARCH4140	Theory of Architecture II	3	2	3	0	ARCH4137
Total Sum			19	5	15	15	

Year 4 Semester 3			Credits		Contact hours		
S. No	Code	Course Name	Cr. Hr	L	T	P	Pre-requisite
1	ARCH4146	Internship-II	3	0	9	0	ARCH3130
Total Sum			3	0	9	0	

Year 5 Semester 1			Credits		Contact hours		
S. No	Code	Course Name	Cr. Hr	L	T	P	Pre-requisite
1	ARCH5137	Architectural Science IV (Acoustics)	2	1	0	3	None
2	ARCH5141	Final Year Project (Research)	P/F	0	3	3	ARCH4138
3	ARCH5143	Housing & Inner-City Redevelopment	3	1	3	3	ARCH4132
4	ARCH5145	Integrated Design Project III	5	1	0	12	ARCH4138
5	ARCH5135	Building Information Modeling (BIM)	3	1	3	3	None
6	ARCH5139	Urban Sociology	2	2	0	0	None
7	ARCH5149	Interior Design	2	1	0	3	ARCH3120
Total Sum			17	7	12	15	

Year 5 Semester 2			Credits		Contact hours		
S. No	Code	Course Name	Cr. Hr	L	T	P	Pre-requisite
1	ARCH5150	Final Year Project (Studio)	6	0	0	18	All Major Design Courses and ARCH5141
2	ARCH5148	Ecological Architecture & Urbanism	3	2	0	3	None
3	ARCH5152	Construction Management	3	2	0	3	None
4		Elective	3	0	0	0	
Total Sum			15	4	3	24	

Table Core Elective Courses

No	Course code	Course title	Cr. Hr	L	P	T	Pre-requisite
1	ARCH5254	Advanced Building Structures	3	1	3	3	None
2	ARCH5251	Advanced Landscape Architecture	3	1	3	3	ARCH3119
3	ARCH5252	Advanced Urban Design	3	1	3	3	ARCH4132
4	ARCH5253	Architecture Heritage and Conservation	3	2	0	3	None
Total Credits				12	5	6	12

Conversion rate: from Cr.hr to ECTS is given as, 1ECTS=1.67Cr.hr.

3.1.7 Program Courses Synopsis

Course Code: Arch2101

Course Name: Basic Design

Credit Hour: 3

Prerequisite: None

This course will introduce basic elements of design with their qualities to explore their application for spatial development. It also introduces the basic spatial qualities related to composition, transformation, arrangement and pattern

Course Code: Arch2102

Course Name: Basic Architectural Design

Credit Hour: 3

Prerequisite: Arch2101

This course will introduce a context in architecture, urban scale, urban cartography, a review on traditional city to modern city. Architectural space qualities, spatial assemblage, elements and principles of architecture. Identifying the steps of an architectural design process and exploring the basic elements and principles of architecture by integrating to create spatial qualities.

Course Code: Arch3111

Course Name: Architectural Design I

Credit Hour: 4

Prerequisite: Arch2102

This course will introduce architectural design process through the development of program for living functional space for a single family to address the material, contextual, and technological aspects of architecture.

Course Code: Arch3120

Course Name: Architectural Design II

Credit Hour: 4

Prerequisite: Arch3111

The course will introduce a basic understanding on the material, and construction aspects of architecture from previous course, this course focus on the social, cultural, economic, contextual aspects in design and development of architectural projects and developments for multi-family residential unit [apartment, community building]. In addition, student is able to develop a multi- housing unit using all the knowledge and skill they have developed from year to now. However, the overall theme of the semester focuses on cultural identity and thus, all the exercises and the final semester major project would be in predilection for both Ethiopian and African cultural entities and taste, and contextual understanding of the life style.

Course Code: Arch4131

Course Name: Integrated Design Project I

Credit Hour: 5

Prerequisite: Arch3120

The main objective of this course is to empower students to develop a skill and knowledge of various interdependence factors of architecture, construction and structure. It uses a process of analysis, and synthesis in proposing design solutions for existing spatial and structural problems of buildings. This comprehensive course

demands that students work in teams integrating constructional structural and environmental systems in the design and documentation of a large and complex building. Students research building type and systems precedents and their resulting impact on built form, analyze material properties, specify component building systems and apply codes and standards to fulfill technical, programmatic and aesthetic needs.

Course Code: Arch4138

Course Name: Integrated Design Project II

Credit Hour: 5

Prerequisite: ARCH4131 Integrated Design Project I

The course aims to focus on creating community within the city. Students investigate socio-cultural and environmental aspects of the city as they relate to architecture. The studio includes exploring history, theory and principles of architecture by integrating with urban and landscape design for the making of architecture and urban form .

Course Code: Arch5145

Course Name: Integrated Design Project III

Credit Hour: 5

Prerequisite: ARCH4138 Integrated Design Project II

Bioclimatic architecture is the design of buildings and spaces based on the local climate to provide thermal and visual comfort, using solar energy and other environmental sources. Basic natural elements of bioclimatic design are passive solar systems which are incorporated into buildings utilizing the sun and other environmental sources like air movement, vegetation, water evaporation, and shading for heating, cooling and to provide day lighting. Develop skills in architectural design through the exploration of technology, materials, structures, construction, and urban modification integral with programmatic detail within a building design initiated in response to environmental studies, building science, HVAC & other Services.

Course Code: Arch5141

Course Name: Final Year Project (Research)

Credit Hour: P/F

Prerequisite: ARCH4138 : Integrated Design Project II and All Major Courses

The course of Final Year Project work is a comprehensive Architectural Design project which necessitates the coordination of other fields in the building design process. It comprises program preparation and design of buildings containing spatial, functional and social aspect.

Course Code: Arch5150

Course Name: Final Year Project (Studio)

Credit Hour: 6

Prerequisite: ARCH5141 Final Year Project (Research) and All Major Courses

The course Final Year Project is a comprehensive architectural design project that necessitates the coordination of other fields in the building design process. It comprises program preparation and design of buildings containing spatial, functional and social aspect based on the research output in the previous semester.

Course Code: Arch5149

Course Name: Interior Design

Credit Hour: 3

Prerequisite: All Architectural Design Courses

The course aims to introduce the students to the theoretical principles and practical problems related to the interior design as a form of management of the spaces inside a building (as apartments, offices, commercials, etc.) Furthermore, the course will provide the tools and techniques in creating a place that is practical, functional and fulfils the philosophical and aesthetic needs of the occupants.

Course Code: Arch3118

Course Name: Landscape Design

Credit Hour: 3

Prerequisite:

This studio introduces students to the programmatic, artistic, and technical aspects of land form and topographic adjustments to accommodate human use. Topics include pedestrian and vehicular circulation, conservation and addition of plant materials, movement of water, recreation use, and creation of views. Sculptural land forms will be emphasized through the use of topographic plans, sections, and contour models

Course Code: Arch5251

Course Name: Advanced Landscape Architecture

Credit Hour: 3

Prerequisite:

In this course, student will be offered in close consultation with urban design and planning staff of the Addis Ababa City Administration, to explore the potential improvements and impacts of increased mixed-use and housing development in central city neighborhoods.

Course Code: Arch2107

Course Name: Graphics Communication Skills I (Sketching I)

Credit Hour: 2

Prerequisite:

This course aims at developing the skills needed for documenting designs using drawings and for performing graphical analysis of two-dimensional and three-dimensional problems. The course is open to all interested students. The class designed for beginner of architecture students. Previous skill or any formal or informal class is necessary. This course introduces the student to the process of visual communication using basic drawing techniques and concepts. Students will develop sound observational skills through visualization using a variety of mark-making tools. Students will learn the fundamentals of drawing (line weight, proportion, rendering and perspective techniques) and understand how it applies to design development.

Course Code: Arch2108

Course Name: Graphics Communication Skills II (Sketching and Painting II)

Credit Hour: 3

Prerequisite: ARCH2107 Graphics Communication Skills I (Sketching I)

Ability to utilize mainly manual and instruments to explore, develop, define and communicate a design proposal. An introduction to the nature and vocabulary of graphical expression used in construction drawings, details, and sketches to include, architectural, structural, civil, mechanical, electrical, disciplines. Students develop an appreciation for the importance of effective graphical documentation and interpret drawings in terms of form, size, distance, quantity and interrelation of elements. Emphasis placed on effective sketched, verbal, and written expression of drawing interpretations to audiences not familiar with construction drawings. Advanced free hand drawing experiences continue the study of form and structure begun in communication skills I. In addition, students are encouraged to develop individual expression in a variety of graphic media. Drawing as a means of developing graphic ideas is stressed

Course Code: Arch2105

Course Name: Drawing (Descriptive Geometry and Drafting)

Credit Hour: 3

Prerequisite

The course deals with physical space. Descriptive geometry is essentially the technique of accurately representing objects by means of drawings and of solving graphically all problems related to their forms and position. It also provides the theoretical basis for technical drawing. Descriptive geometry deals specifically with the graphical representation on a plane (the drawing surface) of the basic geometrical elements and solution of space problems connected with their representation.

Drafting and design involves the creation of drawings and models used to build structures. Architectural drafting and design is one of several career options related to architecture and engineering.

This course aims at developing the skills needed for documenting design using drawing and for performing graphical analysis of two and three dimensional problems. Which is to provide Fundamental knowledge of drafting and engineering drawings, to train students in detail about drafting convention, equipment's and geometrical Construction, to expose students to methods and practice to use drawing scale and drafting of a house and also to train students how they use different types of projection in developing of surface.

Course Code: Arch3117

Course Name: Visual & History of Arts

Credit Hour: 2

Prerequisite

This course deals with the world history of art; it pays close attention to the social, political, and religious function of the arts created by these cultural traditions, what they might have meant to their original audiences, and the specific cultural contexts that influenced their creation. The class will specifically deal with the art of Neolithic Europe, ancient Mediterranean art, and through the development Byzantine Art, the middle Ages through the Gothic Period, and The Art of Italy (13th-16thc). Additionally it also gives high emphasis on modern art and postmodern Art and Ethiopian traditional art. In the practical part of this course, students acquire special knowledge of the human figure and anatomy. A variety of media and methods of graphic representation in watercolor might be explore. Perceptual skills, as well as cognitive aspects of drawing the human form, will be cover on the study. Live models, both clothed and nude as sources for study.

Course Code: Arch3119

Course Name: Graphics Communication Skills III (Professional CAD)

Credit Hour: 3

Prerequisite ARCH3108 Graphics Communication Skills II (Sketching II and Painting II)

This course will lead the way for students for a more complex usage of professional CAD and animation software for a more realistic presentation of their projects. The course will lead the way for students for a more complex usage of professional CAD, animation, and programming software's for a more realistic presentation of their projects. This course introduces students to the history, theory, practices, institutions, and impact of modern communications media. We will examine both print and non-print media and address the media's impact on society, how audiences respond to media, how people produce and consume media, and media industries and careers. Students will apply what they learn by composing in multiple mediums and other means of communication integral to contemporary communications.

Course Code: ARCH3122

Course Name: Professional Practice I

Credit Hour: 2

Prerequisite

Examination of organizational and management theories and practices for delivering professional design services. Includes a historic overview of the profession and a review of the architect's responsibilities from the pre-contract phase through cost estimating and specifications to construction. Application of computer technology in preparing specifications.

Course Code: ARCH4135

Course Name: Professional Practice II

Credit Hour: 2

Prerequisite : Professional Practice I

The course introduces the main issues of rule and law in the organizational structural and management of the Architects office. The contract administration and management related to professional code of ethics, legal aspects of professional liability in the law of contract offer acceptance and discharge of contracts, contract administration will be presented. Examination of organizational and management theories and practices for delivering professional design services. Includes a historic overview of the profession and a review of the architect's responsibilities from the pre-contract phase through cost estimating and specifications to construction. Application of computer technology in preparing specifications.

Course Code: ARCH3130

Course Name: Internship - I

Credit Hour: 3

Prerequisite : ARCH3120 Architectural design II

The internship program has Hands-on training in practical skills typical for the construction industry.

Course Code: ARCH4146

Course Name: Internship-II

Credit Hour: 3

Prerequisite Internship - I

The students are expected to exercise full-time position for the entire summer term as an intern in all sizes of architectural firms, design departments of public bodies or non-profit agencies in related fields. In one or several of the following fields (depending on the type of industry and company profile) Design and aiming at developing special skills related to the industrial practice. The internship programs will hands-on training in practical skills typical for the construction industry.

Course Code: ARCH2106

Course Name: History of Architecture - I

Credit Hour: 3

Prerequisite

Introducing the study of the chronological record of events (as affecting a nation or people), based on a critical examination of architectural source materials and usually presenting an explanation of their causes from ancient periods to the present day.

Analytical study of past architectural antecedents will aid in acquainting student's "with our architectural heritage and arousing interest in postulating architectural solutions that are responsive to the built environment of this civilization.

Emphasis is placed upon the understanding of the technological, sociological, aesthetic, and artistic influences which determine our built environment.

This subject provides an outline of the history of architecture and urbanism, from the first societies to the present. Students analyze buildings and the built environment as the products of culture and in relation to the special problems of architectural design and the history of architecture, with an urbanist perspective that stresses the cultural and political context from which building arises. The course develops critical tools for the analysis and appreciation of architecture, for its role in the intellectual environment in which we conduct our lives. (this one added)

Course Code: ARCH3115

Course Name: History of Architecture - II

Credit Hour: 3

Prerequisite : History of Architecture - I

Introducing the study of the chronological record of events (as affecting a nation or people), based on a critical examination of architectural source materials and usually presenting an explanation of their causes from ancient periods to the present day. Analytical study of past architectural antecedents will aid in acquainting students' "with our architectural heritage and arousing interest in postulating architectural solutions that are responsive to the built environment of this civilization. Provides an outline of the history of architecture and urbanism, from the first societies to the present. Students analyze buildings and the built environment as the products of culture and in relation to the special problems of architectural design and the history of architecture, with an urbanist perspective that stresses the cultural and political context from which building arises. The course develops critical tools for the analysis and appreciation of architecture, for its role in the intellectual environment in which we conduct our lives.(this one added)

Course Code: ARCH3128

Course Name: Ethiopian History of Architecture

Credit Hour: 3

Prerequisite : History of Architecture - II

This course introduces students to the development of vernacular, historic and modern architecture and settlements including heritage sites in Ethiopia and Horn of Africa. Introducing the study of the chronological record of events (as affecting a nation or people), based on development of vernacular, historic and modern architecture and settlements in Ethiopia. Analytical study of past architectural antecedents will aid in acquainting students "with Ethiopian architectural heritage and arousing interest in postulating architectural solutions. It introduces students to the development of vernacular, historic and modern architecture and settlements, and heritage sites in Ethiopia.

Course Code: ARCH4123

Course Name: Theory of Architecture I

Credit Hour: 3

Prerequisite

The course aims to define what constitutes "theory" in Architecture, how it relates to other types of Architectural writing such as criticism and history, and especially how it relates to the intellectual context and built works of its day, as well as to theories that came before and after it. Students are expected to influential critical and creative themes in modern Architecture and able to make Analysis of the dimensions of Art & Architecture, Determents of Architectural space & form, Typology & design; Structure, Technology & materials; Culture, Society & economy; Climate & topography. Introduces students to influential critical and creative themes in modern Architecture.

Help students understand discourses and practices, the questions and contexts that they engage, and their implications for contemporary thinking and design.

Course Code: ARCH4140

Course Name: Theory of Architecture II

Credit Hour: 3

Prerequisite Theory of Architecture I

Students are expected to influential critical and creative themes in modern Architecture with focus on 20th century Architecture: theory and practice; historiography and scientific writing; Interpretation in Architectural history and criticism; Ideology & symbolism; semantic explanations; universalism versus regionalism; modernism versus Postmodernism; de-constructivism; theories of contemporary architecture and urbanism

Course Code: ARCH3116

Course Name: General Building Heritage

Credit Hour: 2

Prerequisite

The course is designed to study different conventions and practices derived from these theories and apply Architectural heritage documentation and recording, and inventory techniques

Course Code: ARCH5253

Course Name: Architectural Heritage and Conservation

Credit Hour: 3

Prerequisite

This course has the purpose to introduce the students to practical problems by both reading various documents related to the preservation of historical cities, streets, and buildings and analyzing their real application cases.

Course Code: ARCH2103

Course Name: Building Materials and Construction I

Credit Hour: 3

Prerequisite

It introduces the basic principles and practices of building constructions.

Course Code: ARCH2104

Course Name: Building Materials and Construction II

Credit Hour: 3

Prerequisite Building Materials and Construction I

It focuses on details and systems of all main building parts as well as industrialized building components as appropriate building and give the direction of future developments in construction systems.

Course Code: ARCH3113

Course Name: Building Materials and Construction III

Credit Hour: 3

Prerequisite Building Materials and Construction II

Introducing alternative materials and alternative construction methods as well as to show advanced construction work of digital building models.

Course Code: ARCH5254

Course Name: **Advanced Building Structures – (Elective)**

Credit Hour: 3

Prerequisite

The course is designed to provide various types of loading on structures and intensify knowledge and planning skills concerning issues in building science & space.

Course Code: ARCH4129

Course Name: Building workshop I(Masonry)

Credit Hour: 1

Prerequisite

To provide a practical understanding of building materials, construction techniques and structures through practical experience.

Course Code: ARCH4134

Course Name: Building workshop II(carpentry)

Credit Hour: 1

Prerequisite

To provide a practical understanding of building materials, construction techniques and structures through practical experience.

Course Code: ARCH4142

Course Name: Appropriate Building Technology

Credit Hour: 2

Prerequisite

Introduces building processes and tools that are appropriate to the climate, socio-economic conditions and natural resources of an area, and which contribute to sustainable development.

Course Code: ARCH5147

Course Name: Construction Management

Credit Hour: 3

Prerequisite

The course provides an overview of the actual situations in a construction work and how can be handled by acquiring knowledge in the scope of construction.

Course Code: ARCH3114

Course Name: Architectural Sciences I (Water And Sewage)

Credit Hour: 2

Prerequisite

This course provides a theoretical knowledge and application of water supply and liquid waste management system of a building and its surrounding.

Course Code: ARCH4125

Course Name: Architectural Sciences II (Heating, Cooling, Ventilation)

Credit Hour: 2

Prerequisite

This course intends to discuss basic principles of heat, heat transfer, air movement and the application of this principle in developing a building system to control the indoor bioclimatic comfort.

Course Code: ARCH4144

Course Name: Architectural Sciences III (Light & Energy)

Credit Hour: 2

Prerequisite

The course discusses scientific principles of light in relation to building interiors and exterior design

Course Code: ARCH5137

Course Name: Architectural Science IV (Acoustics)

Credit Hour: 2

Prerequisite

The course intends to study Principles of sound production, propagation and transmutation of sound level to apply in creating safe and comfortable acoustical environment.

Course Code: ARCH2109

Course Name: Theory and Design of Structure – I

Credit Hour: 3

Prerequisite

This course gives an introduction to structural building systems. It provides an understanding of basic structural engineering.

Knowledge of structural systems, understanding the internal and external force systems, material properties and cross sectional properties.

Course Code: ARCH2110

Course Name: Theory and Design of Structure – II (Strength of Materials)

Credit Hour: 3

Prerequisite Theory and Design of Structure – I

This course provides an introduction to the analysis of determinate and indeterminate structure. This knowledge is very essential for an engineer to enable him in designing all types of structures.

Course Code: ARCH3121

Course Name: Theory and Design of Structure - III

Credit Hour: 3

Prerequisite Theory and Design of Structure – II (Strength of Materials)

The main objective of the course work is to make students understand core concept of design of concrete structures for a specific purpose under service loading conditions.

Course Code: ARCH3124

Course Name: Surveying

Credit Hour: 3

Prerequisite

The aim of the course is to equip the students with a theoretical and practical understanding of surveying in Architectural field. Subject areas include taping, tape corrections, leveling, angle measurements, traversing, traverse adjustments, contouring, mapping, Setting out and proper use and care of surveying instruments.

Course Code: Comp4136

Course Name: Computer Programming

Credit Hour: 3

Prerequisite

A programming language is a formal language comprising a set of instructions that produce various kinds of output. As programming involves activities such as analysis, developing understanding, generating algorithms, verification of requirements of algorithms including their correctness and resources consumption, and coding of algorithms, this course teaches students to program using a target programming language.

As an introduction, this course gives students an overview of the different concepts of programming and problem solving strategies such as Variables, Reserved words, Syntax Diagram, Constant declarations, Data types, Conditional statements, Loops, Error handling and functions. For each topic, some coding guidelines will be discussed.

Course Code: ARCH4127

Course Name: Building Information Modeling (BIM)

Credit Hour: 3

Prerequisite ARCH3118 Graphics Communication Skill III(Professional CAD)

The course is concerned specifically with the utilization of Building Information Modeling (BIM) technology. The course will be focusing on the processes involved in developing a full 3D design object model, not for the purpose

of visualization alone, but more importantly as a tool for understanding and documenting how a proposed building design fits together and how it will perform during use.

Course Code: ARCH3119

Course Name: Model Making Technique

Credit Hour: 2

Prerequisite

The aim of the course is to equip the students with a theoretical and practical understanding of model making techniques, mainly the selection of materials, typology of model building, connection and masking techniques, preparation of appropriate miniature and scaled building and landscaping materials.

Course Code: ARCH3117

Course Name: Introduction to Environmental Planning

Credit Hour: 2

Prerequisite

The objective of this course is to provide an introduction to the ideas and information necessary to integrate environmental viability and sustainable development with other primary concerns namely, equity, healthy communities and economic development. The course will explore the historical roots of current trends in environmental planning; examine theories that have developed recently to encourage environmental planning, environmental regulation and systems of production; and review the status of some of the basic methods and processes of environmental planning. The focus will be on providing an overview of the major concepts, actors, methods and policies used in the decision-making context, while emphasizing practical considerations and local examples.

Course Code: ARCH4133

Course Name: Introduction to Urban Planning

Credit Hour: 3

Prerequisite

This course has relevance of the subject that deals with settlement formation and growth as a response to social, economic, political and cultural needs. It designed to give students firsthand information about urban areas (the city), its development, structure and institutions. It introduces notions of urban areas, urban planning and its relationship with other disciplines. It also discusses the role of urban planning, process of planning, functional sphere of planning, urban structure and its formation, history of urban planning and theories of urban planning.

Course Code: ARCH4135

Course Name: Ecological Architecture & Urbanism

Credit Hour: 3

Prerequisite

Through lectures, studio problems, research projects, and discussion, this course will explore the challenge and potential incorporating ecological factors in urban contexts. The course focuses on the interaction of landscape science (hydrology, geology, etc.) with the necessities and mechanisms of the human environment (urban design, transportation, economics, etc.). Lectures and research projects will particularly emphasize innovative approaches.

Course Code: ARCH4132

Course Name: Basic Urban Design

Credit Hour: 3

Prerequisite ARCH4133 Introduction to Urban Planning

The course aims to explain urban design as a tool of sustainable urban development and quality of life, fill the gap between Architecture and urban planning. This course introduces the history and theory that informs contemporary issues in urban design. It examines the evolving structure of the modern city and the ways in which theoretical ideas and design strategies impact the urban landscape. The importance of environmental and social responsibility and the role of the architect in the urban design process are stressed, while at the same time considering form and spatial qualities of buildings in the urban environment.

Course Code: ARCH5252

Course Name: Advanced Urban Design

Credit Hour: 3

Prerequisite

The aim of this course is to enable student explain contemporary and contextual urban city problems in the country in order to invent practical solutions with the available technology and advanced techniques of analysis

Course Code: ARCH5141

Course Name: Housing and Inner-City Redevelopment

Credit Hour: 3

Prerequisite ARCH4132 Basic Urban Design

The course will introduce the meaning of housing, what it is and does for the community in terms of different space philosophies in cities, towns, districts and neighborhoods. It is based on studio project, with strategies and scenarios of on-site relocation and nearby-relocation of a target population in urban upgrading. It gives the relationship between housing and inner city redevelopment on seminars presentation and practical studio work.

Course Code: ARCH5149

Course Name: Urban Sociology

Credit Hour: 2

Prerequisite ARCH4132 Basic Urban Design

This course is designed to give students a holistic understanding of the issues of social institutions, behavior of the society and to introduce students to a key concept of Urban Sociology to spaces particular to urban development in the Ethiopian context while giving the experiences of other developing countries.

Course Code: ETP4115

Course Name: Integrated Engineering Team Project

Credit Hour: 3

Prerequisite

This is a multidisciplinary team integrated engineering project in connection with a special engineering problem and under the guidance of a faculty member. The project synopsis consists of literature review, design, project management, business acumen, multidisciplinary team work, entrepreneurship

Bachelor of Science Degree in Civil Engineering

3.1.8 Introduction

Engineering is an art of converting scientific knowledge, principles, theories and other facts in to useful practical applications for the benefit of mankind. Civil engineering is a branch of engineering dealing with the design, construction and maintenance of the physical and naturally built environment including highways, buildings, bridges, tunnels, waterworks, harbors, etc. It was defined to distinguish non-military engineering from military engineering

Civil Engineering as defined by the London based Institute of Civil Engineers (ICE) is a great art, on which the wealth and well-being of the whole society depends. Its essential feature, as distinct from science and the arts, is the exercise of imagination to develop products, processes and people needed to create a sustainable physical and natural built environment. It requires a broad understanding of scientific principles, knowledge of materials and the art of analysis and synthesis. It also requires research, team working, and leadership and business skills. A Civil Engineer is someone who practices all or part of this art.

Civil engineers are primarily responsible for the planning, design and construction of infrastructure which includes major buildings, bridges, dams, pipelines, sewage and water treatment plants, and various transportation systems and facilities. In order to provide workable, durable, and affordable solutions to society's infrastructure needs, civil engineers must develop an understanding of the physical laws that govern the actions of nature and their environmental forces, and the behavior of natural and man-made materials. It is not surprising, therefore, that basic research on mechanics of solids and fluids was initially conducted by civil engineers working on solutions to practical problems.

The importance of a sound knowledge based on these subjects is likely to increase in the future as civil engineers are called upon to build in more hostile and delicate environments, to handle new materials, and to preserve natural resources. With thorough knowledge of both the principles of construction and the possible environmental consequences of a structure, the Civil engineer's expertise is one that is essential to our present civilization and one that will become ever more valuable in the future.

3.1.9 Program Educational Objectives

In Ethiopia, Civil Engineers are leaders in the conceptualization, design, construction, and maintenance of the infrastructures on which the society depends. Civil Engineers build and maintain bridges, highways, railways, tunnels, airports, dams, water treatment and distribution systems and large buildings, along with many other structures. Civil engineers work on environmental projects, such as ecological restoration, waste containment, and soil remediation sites or design of a safe and efficient transportation system.

As society evolves, the solutions to Civil Engineering problems are no longer exclusively technical issues. Instead, they require consideration of demographic trends, human aspirations, laws of supply and demand, and in general, social, economic and political factors. The civil engineers of the future will have to develop a better appreciation and understanding of these subjects to assume their rightful place in society.

3.1.10 Program Outcomes (POs)

The practice of Civil Engineering includes the provision of professional services in connection with identification of problems of existing infrastructure and elaboration of technically and economically feasible concepts for their solution, construction supervision, control and approval of contractors' documents and settlement of claims and disputes. The minimum standards for the BSc in Civil Engineering program at the completion of their degree program are expressed in the following minimum set of Program Learning Outcomes (PLOs). These program outcomes are statements on what students shall know, understand, and perform upon completing their course and/or program of study.

This program is aimed at creating well-qualified Civil Engineers with adequate knowledge in the area of structural, highway, geotechnical and water resources and who can be actively engaged in the planning, development and management of Civil Engineering projects.

Generally, the trainees will be equipped with the knowledge that enables them to:

- Undertake project identification, pre-feasibility and feasibility study and detail design of Civil Engineering works.

- Prepare complete contract documents and terms of references for Civil Engineering projects Plan, manage, monitor and evaluate the operation and maintenance of Civil Engineering works.
- Renovate and rehabilitate existing Civil Engineering works.

Specifically, graduates of the program will:

- Be knowledgeable of the historical context, the state-of-the-art, and emerging issues in the field of Civil Engineering and its role in contemporary society.
- Demonstrate critical reasoning and requisite quantitative skills to identify, formulate and resolve Civil Engineering problems, and to create designs that reflect economic environmental, and social sensitivities.
- Display a systems viewpoint, critical thinking, effective communication and interpersonal skills, a spirit of curiosity, and conduct reflecting a professional and ethical manner.
- Exhibit a commitment to lifelong learning and professional development, involvement in professional activity and public service, and achievement of professional licensure.
- Reflect a broad intellectual training for success in multidisciplinary professional practice, in Civil Engineering or diverse related careers, and toward achieving leadership roles in industry, government, and academia.

Table Program Outcomes/Program Learning Outcomes of Civil Engineering program

Engineering knowledge	PLO1: Ability to apply knowledge of mathematics, natural science, engineering fundamentals and Civil Engineering specialization to the solution of complex Civil Engineering problems.
Problem Analysis	PO2: Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
Design/development of solutions	PO3: Design solutions for complex Civil Engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health, and safety, cultural, societal and environmental considerations.

Investigation	PO4: Conduct investigations of complex problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
Modern tool usage	PO5: Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modelling, to complex engineering problems, with an understanding of the limitations.
The engineer and society	PO6: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to Civil Engineering practices and solutions to complex engineering problems.
Environment and sustainability	PO7: Understand and evaluate the sustainability and impact of Civil Engineering work in the solution of complex engineering problems in societal and environmental contexts.
Ethics	PO8: Apply ethical principles and commit to professional ethics and responsibilities and norms of Civil Engineering practice.
Individual and teamwork	PO9: Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
Communication	PO10: Communicate effectively on complex engineering activities with the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations and give and receive clear instructions.
Project management and finance	PO11: Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work as a member and leader in a team, to manage projects and in multi-disciplinary environments.
Lifelong learning	PO12: Recognize the need for, and have the preparation and ability to engage in, independent and life-long learning in the broadest context of technological change. (Technological Megatrends: BIM, Remote Sensing, Virtual /Augment Reality, AI, Smart Sensor, IoT)

3.1.11 Graduation Requirements

A student is required to take courses that will bring the total 190 credit hours for graduation with the Degrees of Bachelor of Science in Civil Engineering. Student must take and pass all the required courses to satisfy the requirement for graduation.

All the required courses and the minimum credit hours set in the program curriculum by the respective academic unit should be satisfied, except to phase in and phase out program.

- A cumulative grade point average CGPA of 2.00 must be obtained;
- A cumulative grade point average CGPA of 2.00 in major area courses;
- No “F” grade in any course taken for undergraduate program;
- Pass the exit exam
- Score pass mark for all courses which have Pass/Fail grade.

Other requirements are same as those of the University graduation requirements.

3.1.12 Course Category and list of Courses

Table: Distribution of credit hours for all course categories

S.No	Category	Total Cr. hr	Percentage (%)
1	Core Course (major and supportive course)	150	78.95
2	Core Elective/focus area course	2	1.05
3	University requirement (Core)	3	1.58
Core Course Total Cr. Hr		155	81.58
4	National Requirement	35	18.42
Total Cr. Hr		190	100.00

Table: List of Core/Compulsory Courses

Core/Compulsory Courses			
S. No	Course Name	Code	CH
1	Entrepreneurship for Engineers	Entr1106	3
2	General Physics	Phys1001	3
3	Applied Mathematics I for Engineering	Math1014	4

4	Applied Mathematics II	Math2007	4
5	Engineering Drawing	MEng2001	3
6	Introduction to Computer Programming	Comp2003	3
7	Engineering Mechanics I (Statics)	CEng2005	3
8	Engineering Mechanics II (Dynamics)	MEng2102	3
9	Emerging Technologies for Engineers	EmTe1012	3
10	Strength of Materials	CEng2104	4
11	Hydraulics	CEng2106	4
12	Engineering Surveying I	CEng2108	3
13	General Workshop Practice	CEng2110	1
14	Engineering Geology	CEng2112	2
15	Transport Engineering	CEng3101	3
16	Soil Mechanics I	CEng3103	3
17	Theory of Structures I	CEng3105	3
18	Open Channel Hydraulics	CEng3107	3
19	Construction Materials	CEng3109	2
20	Engineering Surveying II	CEng3111	3
21	Highway Engineering I	CEng3102	3
22	Soil Mechanics II	CEng3104	3
23	Theory of Structures II	CEng3106	3
24	Engineering Hydrology	CEng3108	3
25	Building Construction	CEng3110	3
26	Numerical Methods	CEng3112	2
27	Computer Aided Drafting (CAD)	CEng3114	2
28	Highway Engineering II	CEng4103	3
29	Environmental Engineering	EnEng4105	2
30	Reinforced Concrete Structures I	CEng4107	3
31	Hydraulic Structures I	CEng4109	3
32	Specification & Quantity Survey	CEng4111	3
33	Fundamental of Architecture	CEng4113	2
34	Construction Equipment	CEng4102	2
35	Technical Report Writing & Research Methodology	CEng4104	1
36	Foundation Engineering I	CEng4106	3

37	Reinforced Concrete Structures II	CEng4108	3
38	Hydraulic Structures II	CEng4110	3
39	Water Supply and Urban Drainage	CEng4112	3
40	Procurement and Contract Administration	CEng4114	2
41	Internship Practice	CEng4116	6
42	BSc thesis I (Proposal Preparation)	CEng5101	P/F
43	Integrated Civil Engineering Design	CEng5103	3
44	Railway Engineering	CEng5105	2
45	Foundation Engineering II	CEng5107	3
46	Structural Design	CEng5109	3
47	Steel & Timber Structures	CEng5111	3
48	Irrigation Engineering	CEng5113	2
49	Engineering Economics	CEng5115	2
50	BSc thesis II (Main Research)	CEng5102	6
51	Construction Management	CEng5106	2
52	Fundamental of Bridge Design	CEng5112	3
53	Waste Water and Solid Waste Treatment	CEng5114	3
Total Sum			150

Table: List of Core/Elective Courses

2- Core Elective Courses			
S. No	Course Name	Course Code	Cr.Hr.
1	Highway Engineering III	CEng5204	2
2	Tunneling	CEng5206	2
3	Reinforced Concrete structures III	CEng5210	2
4	Water Resource Development	CEng5212	2
5	Geographic Information System (GiS)	CEng5214	2
6	Building Information Modelling (BIM)	CEng5216	2
Total Sum			2

Table: University Requirement Course

3- University Requirement			
S. No	Course Name	Course Code	Cr.Hr.
1	Integrated Engineering Team Project	IETP 4115	3
Total			3

Table: National Requirement Common Courses

4-National Requirement Common Courses			
S. No	Course Name	Course Code	Cr.Hr.
1	Communicative English Language Skills I	FLEn 1003	3
2	Communicative English Language skills II	FLEn 1004	3
3	Geography of Ethiopia and the Horn	GeES1005	3
4	Mathematics for Natural Sciences	Math1007	3
5	Logic and Critical thinking	Phil1009	3
6	General psychology	Psyc1011	3
7	Physical Fitness	SpSc1013	P/F
8	Social Anthropology	Anth1002	2
9	Inclusiveness	IncL1010	2
10	Moral and Civic Education	MCiE1012	2
11	Economics	Econ2009	3
12	Global Trend	GLTr 2011	2
13	History of Ethiopia and the Horn	Hist 2002	3
14	Probability and Statistics	Stat 3027	3
Total Sum			35
Grand Total			190

3.1.13 Program Curriculum Structure: Bachelor of Science in Civil Engineering*Table: Course Break Down Per Semester for Regular Civil Engineering Department*

1 st Year Semester I				CH	Contact hours		
S. No	Course Name	Code	Pre-Requisite		L	Tu	LP
1	General Physics	Phys1001	None	3	2	3	0
2	Communicative English Language Skills I	FLEn1003	None	3	2	3	0
3	Geography of Ethiopia and the Horn	GeES1005	None	3	2	3	0
4	Mathematics for Natural Sciences	Math1007	None	3	2	3	0
5	Logic and Critical thinking	Phil1009	None	3	2	3	0
6	General psychology	Psyc1011	None	3	2	3	0
7	Physical Fitness	SpSc1013	None	P/F	1	0	3
Total Sum				18	13	18	3

1 st Year Semester II				CH	Contact hours		
S. No	Course Name	Code	Pre-Requisite		L	TU	LP
1	Social Anthropology	Anth1002	None	2	2	0	0
2	Communicative English Language skills II	FLEn1004	FLEn1003	3	2	3	0
3	Entrepreneurship for Engineers	Entr1106	None	3	2	3	0
4	Emerging Technologies for Engineers	EmTe1108	None	3	2	0	3
5	Inclusiveness	Incl1010	None	2	2	0	0
6	Moral and Civic Education	MCiE1012	None	2	2	0	0
7	Applied Mathematics I for Engineering	Math1014	None	4	3	3	0
Total Sum				19	15	9	3

2 nd Year Semester I (Pre-Engineering)				CH	Contact hours		
S. No	Course Name	Code	Pre-Requisite		L	TU	LP
1	Engineering Drawing	MEng2001	None	3	1	1	5
2	Introduction to Computer Programming	Comp2003	None	3	2	0	3
3	Engineering Mechanics I (Statics)	CEng2005	Phys1001	3	2	3	0
4	Applied Mathematics II	Math2007	Math1014	4	3	3	0
5	Economics	Econ2009	None	3	2	3	0
6	Global Trend	GLTr2011	None	2	2	0	0
Total Sum				18	12	10	8

2 nd Year Semester II				CH	Contact hours		
S. No	Course Name	Code	Pre-Requisite		L	TU	LP
1	Engineering Mechanics II (Dynamics)	MEng2102	CEng2005	3	2	3	0
2	Strength of Materials	CEng2104	CEng2005	4	3	3	0
3	Hydraulics	CEng2106	CEng2005	4	2	3	3
4	Engineering Surveying I	CEng2108	Math1007	3	2	0	3
5	General Workshop Practice	CEng2110	None	1	1	0	2
6	Engineering Geology	CEng2112	None	2	1	1	2
7	History of Ethiopia and the Horn	Hist2002	None	3	2	3	0
Total Sum				20	13	13	10

3 rd Year Semester I				CH	Contact hours		
S. No	Course Name	Code	Pre-Requisite		L	TU	LP
1	Transport Engineering	CEng3101	None	3	2	3	0
2	Soil Mechanics I	CEng3103	CEng2104, CEng2106	3	2	2	1
3	Theory of Structures I	CEng3105	CEng2104	3	2	3	0
4	Open Channel Hydraulics	CEng3107	CEng2106	3	2	2	1
5	Construction Materials	CEng3109	None	2	1	2	1
6	Engineering Surveying II	CEng3111	CEng2108	3	2	0	3
7	Probability and Statistics	Stat3027	None	3	2	3	0
Total Sum				20	13	15	6

3 rd Year Semester II				CH	Contact hours		
S. No	Course Name	Code	Pre-Requisite		L	TU	LP
1	Highway Engineering I	CEng3102	CEng3101, CEng3111	3	2	3	0
2	Soil Mechanics II	CEng3104	CEng3103	3	2	2	1
3	Theory of Structures II	CEng3106	CEng3105	3	2	3	0
4	Engineering Hydrology	CEng3108	CEng3107	3	2	3	0
5	Building Construction	CEng3110	CEng3109	3	2	3	0
6	Numerical Methods	CEng3112	Math2007	2	1	3	0
7	Computer Aided Drafting (CAD)	CEng3114	MEng2001	2	1	0	3
Total Sum				19	12	17	4

4 th Year Semester I				CH	Contact hours		
S. No	Course Name	Code	Pre-Requisite		L	TU	LP
1	Highway Engineering II	CEng4103	CEng3102	3	2	2	1
2	Environmental Engineering	EnEng4105	None	2	2	0	2
3	Reinforced Concrete Structures I	CEng4107	CEng3106	3	2	3	0
4	Hydraulic Structures I	CEng4109	CEng3108, CEng3107	3	2	3	0
5	Specification and Quantity Surveying	CEng4111	CEng3110, CEng3114	3	2	3	0
6	Fundamental of Architecture	CEng4115	CEng3114	2	1	0	3
7	Integrated Engineering Team Project	IETP4115	None	3	1	0	5
Total Sum				19	12	11	11

4 th Year Semester II				CH	Contact hours		
S. No	Course Name	Code	Pre-Requisite		L	TU	LP
1	Construction Equipment	CEng4102	CEng3109	2	2	2	0
2	Technical Report Writing & Research Methodology	CEng4104	FLEn1004	1	1	2	0
3	Foundation Engineering I	CEng4106	CEng3104, CEng4107	3	2	3	0
4	Reinforced Concrete Structures II	CEng4108	CEng4107	3	2	3	0
5	Hydraulic Structures II	CEng4110	CEng4109	3	2	3	0
6	Water Supply and Urban Drainage	CEng4112	CEng3108	3	2	2	1
7	Procurement and Contract Administration	CEng4114	CEng4111	2	1	3	0
8	Engineering Economics	CEng4116	Econ2009	2	1	3	0
Total Sum				19	13	21	1

4 th Year Semester II & III (Summer)				CH	Contact hours		
S. No	Course Name	Code	Pre-Requisite		L	TU	LP
1	Internship Practice	CEng4118	CEng4103, CEng4106, CEng4108, CEng4110, CEng4111	6	1	0	15
Total Sum				6	1	0	15

5 th Year Semester I				CH	Contact hours		
S. No	Course Name	Code	Pre-Requisite		L	TU	LP
1	Final year project I	CEng5101	CEng4118	P/F	1	0	5
2	Integrated Civil Engineering Design	CEng5103	CEng4118	3	0	0	6
3	Railway Engineering	CEng5105	CEng3101	2	2	2	0
4	Foundation Engineering II	CEng5107	CEng4106	3	2	3	0
5	Structural Design	CEng5109	CEng4108	3	2	3	0
6	Steel and Timber Structures	CEng5111	CEng3106	3	2	2	1
7	Irrigation Engineering	CEng5113	CEng4110	2	2	2	0
Total Sum				16	11	12	12

5 th Year Semester II				CH	Contact hours		
S. No	Course Name	Code	Pre-Requisite		L	TU	LP
1	Final Year Project II	CEng5102	CEng5101	6	0	6	6
2	Construction Management	CEng5106	CEng4116, CEng4102	2	2	2	0
3	Fundamental of Bridge Design	CEng5112	CEng4110, CEng4108	3	2	3	0
4	Waste Water and Solid Waste Treatment	CEng5114	CEng4112	3	2	2	1
5	Elective I	CEngxxxx	CEngxxxx	2	2	2	0
Total Sum				16	8	15	7

Elective Courses

2-Core Elective Courses				CH	Contact hours		
S. No	Course Name	Code	Pre-Requisite		L	TU	LP
1	Reinforced Concrete structures III	CEng5210	CEng4108	2	2	2	0
2	Highway Engineering III	CEng5204	CEng4103	2	2	2	0
3	Water Resource Development	CEng5212	CEng3108, CEng4110	2	2	2	0
4	Tunneling	CEng5206	CEng3104	2	2	2	0
5	Geographic Information System (GIS)	CEng5214	CEng3111	2	2	0	2
6	Building Information Modelling (BIM)	CEng5216	CEng4115, CEng4108	2	1	0	3

3.1.14 Program Courses Synopsis

Course Name: **Engineering mechanics - I (Statics)**

Course Code: CEng2005

Credit Hour: 3

Prerequisite: Phy-1001 General Physics

Engineering mechanics (Statics) is the application of mechanics (one of the three branches of Physics) to solve problems involving common engineering elements

Course Name: **Engineering Mechanics II (Dynamics)**

Course Code: **MEng 2102**

Credit Hour: 3

Prerequisite: **CEng2015 , Engineering Mechanics I (Statics)**

This course covers the theory and applications of engineering mechanics, Select appropriate coordinate systems for physical systems and analyze motion variables such as position, velocity, and acceleration and Conduct kinematic analysis for the velocity & acceleration of moving bodies.

Course Name: **Applied Mathematics I for Engineering**

Course Code: **Math-1014**

Credit Hour: 4

Prerequisite: **No**

This course covers basic elements of vectors, vector spaces, matrices, determinants, solving systems of linear equations, concepts and applications of differential and integral calculus of one variable

Course Name: **Applied Mathematics II**

Course Code: **Mat2007**

Credit Hour: 4

Prerequisite: **Applied Mathematics I for Engineering (Math 1014)**

This course covers basic elements of sequence and series, power series, differential calculus of several variables and multiple integral concepts and their applications

Course Name: **Probability and Statistics**

Course Code: **Stat 3027**

Credit Hour: 3

Prerequisite: **No**

This course covers basic elements of statistics, probability, central tendencies, variation, and types of sampling, concepts and applications of estimation and hypothesis testing, regression analysis.

Course Name: **Numerical Methods**

Course Code: **CEng3112**

Credit Hour: 3

Prerequisite: Math 2051 and computer programming

The course aims at introducing students in finding numerical solutions to problems for which analytical solutions either do not exist or are not readily or cheaply obtainable. It enables students to apply linear algebra and calculus. It also aims to help student develop programming skills.

Course Name: **Emerging Technologies for Engineers**

Course Code: **EmTel108**

Credit Hour: 3

Prerequisite: **No**

This course will enable students to explore current breakthrough technologies in the areas of Artificial Intelligence, Internet of Things and Augmented Reality that have emerged over the past few years. Besides helping learners become literate in emerging technologies, the course will prepare them to use technology in their respective professional preparations

Course Name: **Engineering Drawing**

Course Code: **MEng 2001**

Credit Hour: 3

Prerequisite: No

The course Engineering Drawing covers introduction and importance of engineering drawing, drawing instruments, drawing standards and conventions; theories of projection: types and systems of projections, projections of points and lines; multi-view drawings; pictorial drawing; auxiliary views and sectional view

Course Name: **General Workshop Practice**

Course Code: **CEng2110**

Credit Hour: 1

Prerequisite: No

This course is mainly designed to impart students to a hands-on exercises and practices on plumbing, masonry works, concrete mixing, wood work , metal work, finishing and electrical installation and observe construction site.

Course Name: **Computer Programming**

Course Code: **Comp 2003**

Credit Hour: 3

Prerequisite: No

A programming language is a formal language comprising a set of instructions that produce various kinds of output. As programming involves activities such as analysis, developing understanding, generating algorithms, verification of requirements of algorithms including their correctness and resources consumption, and coding of algorithms, this course teaches students to program using a target programming language. As an introduction, this course gives students an overview of the different concepts of programming and problem solving strategies such as Variables, Reserved words, Syntax Diagram, Constant declarations, Data types, Conditional statements, Loops, Error handling and functions. For each topic, some coding guidelines will be discussed.

Course Name: **Computer Aided Drafting (CAD)**

Course Code: **CEng3114**

Credit Hour: 2

Prerequisite: **MEng 2001**

Students will learn the Application Software for Civil Engineering. This course will lead the way for students for a more complex usage of professional CAD and animation software for a more realistic presentation of their projects. The course will lead the way for students for a more complex usage of professional CAD, animation, and programming software's for a more realistic presentation of their projects.

Course Name: **Fundamental of Architecture,**

Course Code: **CEng4115**

Credit Hour: 2

Prerequisite: **CEng3114**

- To understand the fundamentals of construction planning and design procedures, and site selection.
- To develop skills and knowledge in the preparation of working drawings.
- To understand the concepts of various components of a low-rise building and their construction methods.
- To acquire a thorough understanding of the basics of framed structures, shell and dome structures and prefabricated building systems.

Course Name: **Engineering Surveying I**

Course Code: **CEng3111**

Credit Hour: 3

Prerequisite: Math 2007 (Mathematics for Natural Sciences)

This course is designed to introduce Civil Engineering students to know basic principles of surveying, accurately measure distances and angles using high precision and up-to-date surveying equipment.

Course Name: **Engineering Surveying II**

Course Code: **CEng2108**

Credit Hour: 3

Prerequisite: CEng2108 (Engineering Surveying I)

Triangulation; Trilateration; Resection; GPS Surveying; Topographic Surveying; Topographic Mapping; Horizontal Curve; Vertical Curve; Curve Setting Out; Introduction to Photogrammetry

Course Name: **Strength of Material**

Course Code: **CEng2104**

Credit Hour: 4

Prerequisite: **CEng2103 Engineering Mechanics I**

This course covers the Engineering mechanics (Statics) and the application of mechanics (one of the three branches of Physics) to solve problems involving common engineering elements

Course Name: **Theory of Structure I**

Course Code: **CEng3106**

Credit Hour: 3

Prerequisite: CEng2104, Strength of Material

This course covers the different types of structural systems, loads and methods of analysis for statically determinate structures

Course Name: **Theory of Structure II**

Course Code: **CEng3105**

Credit Hour: 3

Prerequisite: Theory of Structure I , CEng3105

This course introduces the analysis of indeterminate structural systems common in Civil Engineering

Course Name: **Reinforced Concrete Structure I**

Course Code: **CEng4107**

Credit Hour: 3 (5CP)

Prerequisite: CEng3106 , Theory of Structure II

This course induces the structural analysis and design of reinforced concrete structural members subjected to gravity load.

Course Name: **Reinforced Concrete Structure II**

Course Code: **CEng4108**

Credit Hour: 3

Prerequisite: **Reinforced Concrete Structures I (CEng4107)**

This course induces the structural analysis and design of reinforced concrete of structural members.

Course Name: **Steel and Timber Structures**

Course Code: CEng5111

Credit Hour: 3 (5CP)

Prerequisite: CEng3106

This course induces the structural analysis with design of steel and timber members subjected to gravity and lateral loading along with the connection details

Course Name: **Fundamentals of Bridge Design**

Course Code: CEng5112

Credit Hour: 3 (5CP)

Prerequisite: CEng4110, CEng4108

To identify the different types of bridges: arch, suspension, truss, beam, cantilever and cable stayed. Learning various loads acting on bridges.

Course Name: **Structural Design**

Course Code: **CEng5109**

Credit Hour: 3

Prerequisite: CEng4108, Reinforced Concrete Structures II

This course induces the plastic analysis of frame structures, calculation lateral load on structure and design of detailing and connections.

Course Name: **Engineering Geology**

Course Code: **CEng2112**

Credit Hour: 2

Prerequisite: No

To provide an introduction to the geotechnical significance of earth materials, rock defects, structural geology, geomorphology, hydrogeology, active tectonics, earthquakes, volcanism, erosion and mass movement in the Civil Engineering practice

Course Name: **Soil Mechanics I**

Course Code: **CEng3103**

Credit Hour: 3

Prerequisite: Hydraulics (**CEng2106**) & Strength of Materials (**CEng2104**)

In this course, students will be introduced to the basic concepts of the physical properties of soils as a civil engineering material and the fundamental principles of soil mechanics.

Course Name: **Soil Mechanics II**

Course Code: **CEng3104**

Credit Hour: 3

Prerequisite: Soil Mechanics I (**CEng3103**)

To equip students with the basic inputs required for the design and construction of geotechnical structures.

Course Name: **Foundation Engineering I**

Course Code: **CEng4106**

Credit Hour: 3

Prerequisite: **Soil Mechanics II (CEng3104), Reinforced Concrete Structure I (CEng4107)**

This course covers the basic concepts of logic and set theory, the real and complex number systems, Mathematical induction, least upper bound and greatest lower bound, functions and types of functions, polynomial and rational functions, logarithmic and exponential functions, trigonometric functions, hyperbolic functions and their graphs and analytic geometry.

Course Name: **Foundation Engineering II**

Course Code: **CEng5107**

Credit Hour: 3

Prerequisite: Foundation Engineering I (**CEng4106**)

To design and construction of deep foundations, marine structures, foundations on expansive soils, and understand environmental issues related to geotechnical engineering.

Course Name: **Hydraulics**

Course Code: **CEng2106**

Credit Hour: **4 cr hr (6CP)**

Prerequisite: **Engineering Mechanics I (Statics) CEng2005**

The course provides students with basic principles of hydraulics for design of dams and its appurtenant structures

Course Name: **Open Channel Hydraulics**

Course Code: **CEng3107**

Credit Hour: 3 cr. Hr

Prerequisite: **Hydraulics (CEng2106)**

The course provides students with basic hydraulic principles that are required to solve open channel flows.

Course Name: **Hydraulic Structures I**

Course Code: **CEng4109**

Credit Hour: 3(5CP)

Prerequisite: **Engineering Hydrology (CEng3108), Open Channel Hydraulics (CEng3107)**

The course provides students with basic principles of design of dams and its appurtenant structures

Course Name: **Hydraulic Structures II**

Course Code: **CEng4110**

Credit Hour: 3

Prerequisite: **Hydraulic Structures- I (CEng4109)**

Provide students with principles of river Engineering, design of flood protection structures, and design of river bank protection structures. Design of diversion structures are taught in the course. Design of Conveyance and Cross drainage structures, drops/falls and Culvert and Bridges

Course Name: **Irrigation Engineering**

Course Code: **CEng5113**

Credit Hour: 2Cr.hr(4CP)

Prerequisite: **Hydraulic Structures II (CEng4110)**

Provide students with the basic principles of irrigation design, selection of irrigation systems, determination of the quality of water for irrigation and finally assessment of the feasibility of irrigation schemes

Course Name: **Water Supply and Urban Drainage**

Course Code: **CEng4112**

Credit Hour: 3

Prerequisite: **Engineering Hydrology (CEng3108)**

To provide students with fundamental skill on planning and design of water supply and urban drainage systems.

Course Name: **Waste Water and Solid Waste Treatment**

Course Code: CEng5114

Credit Hour: **3cr. hr (5CP)**

Prerequisite: **Water Supply and Urban Drainage, CEng4112**

Students will learn the basic methods for industrial and municipal wastewater treatment facilities and about the processes involved; they will learn the basic design of wastewater treatment facilities.

Course Name: **Transport Engineering**

Course Code: **CEng3101**

Credit Hour: 3

Prerequisite: **No**

- ✓ Introduction to transport systems: Highways, railways, airways, and waterways.
- ✓ Transport planning: elements of transport planning, urban transport planning, and evaluating transport alternatives.
- ✓ Driver, pedestrian, vehicle, and road characteristics.
- ✓ Traffic Engineering studies: spot speed studies, volume studies, travel time and delay studies parking studies and road traffic safety.
- ✓ Fundamental principles of traffic flow: traffic flow elements, and flow-density relationships.
- ✓ Traffic-control devices and systems: traffic signs, pavement markings and islands, and traffic signals.
- ✓ Highway capacity: level of service, capacity of highways, and intersections.

Course Name: **Highway Engineering- I**

Course Code: **CEng3102**

Credit Hour: 3

Prerequisite: **Engineering Surveying II (CEng3111) and Transport Engineering (CEng3101)**

- ✓ Introduction to Highway Engineering,
- ✓ Highway Geometric Design Factors,
- ✓ Geometric Design of Highway Alignment,
- ✓ Highway Drainage Design & drainage Structures,
- ✓ Earth work quantity & Mass Haul Diagram,

- ✓ Introduction to Intersection, Road Terminal & Interchanges
- ✓ Introduction to geometric design of Airfields

Course Name: **Railway Engineering**

Course Code: **CEng5105**

Credit Hour: **3 Cr. Hr.**

Prerequisite: **Transport Engineering (CEng3101)**

The course will be an introduction to the railway engineering and rail infrastructures and their impacts on the society and on the environment. The course will focus on alignment, track geometry, superstructure and substructure components, switches, Railway planning and capacity, railway signaling control system together with operation of railway.

Course Name: **Technical Report Writing & Research Methodology**

Course Code: **CEng4104**

Credit Hour: **1**

Prerequisite: **FLEn 1004 (Communicative English Language skills II)**

The course provides students to Prepare a standardized Proposal for their research. The topic for the thesis is assigned by the department and can be connected to any of the major subjects already taught. The subject of the assignment preferably considers the needs of the country

Course Name: **Internship Practice**

Course Code: **CEng4118**

Credit Hour: **6 cr. Hr**

Prerequisite: **CEng4103, CEng4106, CEng4108, CEng4110, CEng4111**

- The Internship Program creates the opportunity for the students to familiarize themselves with the community/industrial practices.
- The objective of the Internship is to supplement the intern's general theoretical knowledge that has been gained through academic course work with a practical work experience
- To evaluate students by companies or any organization how much they understand the practical knowledge during the intern ship.

Course Name: **Integrated Civil Engineering Design**

Course Code: **CEng5103**

Credit Hour: **3**

Prerequisite: **Internship (CEng4118)**

An integrated project will assist the students in consolidating their Knowledge acquired in various previous modules. The students will apply their knowledge in solving practical and real-time projects in due time.

Students will perform a comprehensive design project using a team approach requiring interaction with practitioner's development of a team project report and a formal presentation

The main aim of this project work will let students to acquire knowledge and skills of planning and management of Civil Engineering projects. In addition to this, the objective of the course is to equip students with effective report writing skills and research methodologies. Students shall develop solid technical report and paper/thesis writing skills, analysis and data interpretation techniques, and research methodologies

Course Name: **Integrated Engineering Team Project**

Course Code: **IETP4015**

Credit Hour: **3**

Prerequisite: Comp 2003 , Entr 1006

This is a multidisciplinary team integrated engineering project in connection with a special engineering problem and under the guidance of a faculty member. The project synopsis consists of literature review, design, project management, business acumen, multidisciplinary team work, entrepreneurship

Course Name: **Construction Material**

Course Code: **CEng3109**

Credit Hour: **3**

Prerequisite: No

Introduces the production, nature and characteristics of different construction materials and identify them with respect to their suitability to different Engineering structures. Understand the physical and chemical property of construction materials under different conditions having basic skill how to mix and produce concrete

Course Name: **Building Construction**

Course Code: **CEng3110**

Credit Hour: **3**

Prerequisite: CEng3109

The course introduces students with the different types of buildings, their components and methods of construction. The overall building processes beginning from site works will be covered

Course Name: **Specification and Quantity Survey,**

Course Code: **CEng4111**

Credit Hour: 3cr. hr

Prerequisite: CEng3110 ,CEng3114

Students will gain knowledge in the legal aspects of contracts and bidding; types of construction documents including bonds; interpretation of technical building specifications and their application to selection and installation of materials, equipment and systems

Course Name: **Procurement and Contract Administration**

Course Code: **CEng4114**

Credit Hour: 3

Prerequisite: **Specification and Quantity Survey (CEng4111)**

Students will gain knowledge in the legal aspects of contracts and bidding; types of construction documents including bonds; interpretation of technical building specifications and their application to selection and installation of materials, equipment and systems

Course Name: **Engineering Economics ,**

Course Code: CEng4116

Credit Hour: 2cr hr

Prerequisite: Econ-2009

Expressing the basic concepts of engineering economics and identifying the time value of money, the concepts behind benefit-cost analyze. Understanding the concept of depreciation. Determining design projects on the basis of the cost implications when compared with the expected benefits.

Course Name: **Construction Equipment,**

Course Code: **CEng4102**

Credit Hour: 2cr hr

Prerequisite: CEng3109

At the end of this course, students will be able to have knowledge on equipment and properly select earthmoving, excavation, lifting and other type of equipment with safety requirements. Moreover, students will understand the economic aspects of construction equipment

Course Name: **Construction Management**

Course Code: CEng5106

Credit Hour: 3Cr.hr (5ECTS)

Prerequisite: CEng5115, CEng4102

Time Management, Cost Management, Quality Assurance and Control, Risk Management, Safety and Health in Construction

Course Name: **Final Year Project I (Proposal Preparation)**

Course Code: **CEng5101**

Credit Hour: **P/F**

Prerequisite: **CEng4114 (Internship Practice**

The course provides students to Prepare a standardized Proposal for their research. The topic for the thesis is assigned by the department and can be connected to any of the major subjects already taught. The subject of the assignment preferably considers the needs of the country.

Course Name: **Final Year Project II (Main Research)**

Course Code: **CEng5102**

Credit Hour: 6 cr. Hr

Prerequisite: CEng5101

A prospective graduate student has to prove the fact that he/she can solve a given problem independently and with little input from his mentor. The prospective graduates could apply their knowledge to solve Ethiopian-related or other problems up to the expectation of his mentor in particular and Department in general. The topic for the thesis is assigned by the department and can be connected to any of the major subjects already taught. The subject of the assignment preferably considers the needs of the country

Course Name: **Reinforced Concrete structures III**

Course Code: **CEng5210**

Credit Hour: 2 cr. Hr

Prerequisite: CEng4108

This course induces the structural design of curved beams, haunched beams, deep beams, and corbels in the ultimate limit state and verify the satisfaction of serviceability limit state requirements.

Course Name: **Highway Engineering III**

Course Code: **CEng5204**

Credit Hour: 2 cr. Hr

Prerequisite: **Highway Engineering II , (CEng4103)**

The course includes the detailed specification of roadway construction, its maintenance and rehabilitation. Introduction to road management system and the economic aspect of road construction are also a part of the course. It also deals with the method of road construction with respect to labor-based method

Course Name: **Water Resource Development**

Course Code: CEng5212

Credit Hour: **2 (4CP)**

Prerequisite: Engineering Hydrology , Hydraulic Structures II (CEng3108) , (CEng4110)

Provide students with the principles of water resources planning and management, principles of integrated water resources development and Legal aspects

Course Name: **Tunnelling**

Course Code: **CEng5206**

Credit Hour: **3**

Prerequisite: Soil Mechanics II (CEng3104)

To know different types of underground opening; elastic and elasto- plastic stresses and displacements around underground openings; ground reaction curve; tunnel ground condition; excavation methods (drill-and-blast and machine); excavation systems for large openings and difficult ground conditions; supports and linings;

determination of rock loads; exploration and instrumentation; case histories

Course Name: **Geographic Information System**

Course Code: **CEng5214**

Credit Hour: **3**

Prerequisite: Engineering Surveying II (CEng3111)

This course is designed to introduce Civil Engineering students to know basic principles of surveying, accurately measure distances and angles using high precision and up-to-date surveying equipment

Course Name: **Building Information Modeling (BIM)**

Course Code: CEng5216

Credit Hour: **3**

Prerequisite: Reinforced Concrete Structures II , Fundamental of Architecture, CEng4108, CEng4115

The course is concerned specifically with the utilization of Building Information Modeling (BIM) technology. The course will be focusing on the processes involved in developing a full 3D design object model, not for the purpose of visualization alone.

Course Name: **Building Information Modeling (BIM)**

Course Code: CEng5216

Credit Hour: **3**

Prerequisite: Reinforced Concrete Structures II , Fundamental of Architecture, CEng4108, CEng4115

3.2 Bachelor of Science Degree in Mining Engineering

3.2.1 Introduction

Mining Engineering is a field of study which embraces all activities related to planning, design, extraction, processing, management and development of mineral resources. The curriculum is based on scientific engineering and geologic fundamentals and the application of these fundamentals to design and operate mines and to construct structures in or on rock to get access to the ores and prepare mine products for the market. Also, the curriculum emphasis on mining technology, ground control and rock mechanics, mine environmental engineering, material transport, methods of working and mine ventilation.

Ethiopia is endowed with substantial number of mineral resources including non-metallic, metal ores, and solid fuel and energy sources. This all resources explored to some extent and reported. However, not upgraded to mining stage and did not contribute the desired economic input to the country. Recently, the Ethiopian government is working heavily in promoting the economic mineral potentials of the country with which both local and foreign Mining investors can get

participated. For that matter, lots of initiative policies and infrastructures have been made. As a result of these, considerable numbers of local and foreign investors are started participating in mining investment the sector.

Keeping in view the growing demand in area of mineral resources and their exploitation, utilization and the realization of its significant role in the national economy, Addis Ababa Science and Technology University, the first public university in Ethiopia launched undergraduate mining engineering program in the country. So far, the Departments has graduated four batch around 100 professionals right now they are working in Mining Companies including Cement industries, companies using Industrial Mineral ,Construction and Dimension Stone Companies, Public Agencies (Local and Federal Government Organizations) and Universities. Generally they are playing vital role in the all rounded developmental activities of the country.

Mining engineering program at AASTU right now has a well-established mine computational lab for mine modeling, planning and design. Other laboratories like mine ventilation, mine machineries, rock mechanics, Ore processing are under development. The Mining engineering curriculum beyond the technical aspects of basic sciences and engineering in mining Engineering, graduates of AASTU also acquire a host of other skills which are essential in today's global through emerging technology and skills during their study at AASTU given by the university course requirement and Mining Engineering department elective courses.

3.2.2 Program Educational Objectives

Mining Engineering Graduates will have expertise to contemporary problem solving, be engaged professionally, and have continued to learn and adapt, and have contributed to their organizations through leadership, teamwork and Entrepreneur. More specifically, the PEO of Bachelor of Science in Mining Engineering is presented in Table 1.

Table: Program Education Objectives (PEO)

PEO	Statement
PEO-1	Competent mining engineers with a broad understanding of extraction, management and development of mineral resources including non-metallic, metal ores, and solid fuel and energy sources.

PEO-2	Entrepreneur mining engineers who can carry out professional work in mineral & energy industries, authority, organizations, consultancy firm and mine research institute.
PEO-3	After five years study, our mining engineering graduates will achieve professional growth in research, education, teamwork and leadership skills.

3.2.3 Program Outcome

The Program Outcome (PO)/Program Learning Outcomes (PLO) of the program are describe or state what mining engineering students shall know, understand, and perform upon completing their course and/or program of study. The Program Outcome (PO) of the mining engineering department prepared by referring (Appendix 3.2) engineering programs.

Table. Program Outcome

PO	Statement
PO-1	Apply knowledge of mathematics, natural science, engineering fundamentals, to the solution of complex mining and underground structure problems.
PO-2	Identify, formulate, research literature and analyze complex mining engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO-3	Design and planning surface and underground mines, selection of their appropriate mining method and machinery. Convert this into actions in sectors such as surface and underground working of mineral deposits, drilling and blasting technology, as well as effective safety, health and environmental management techniques in mining operations.
PO-4	Conduct investigations of complex mining engineering problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information for mineral reserve estimation, feasibility study of mineral deposits, ascertain whether the ore can be extracted economically and decide on the best way to extract it.
PO-5	Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modeling, to complex mining engineering problems, with an understanding of the limitations.

PO-6	Extract and process the mineral resources using the fundamental principles of science and engineering in conjunction with the state-of-the-art tools by consequent responsibilities considering health, safety, legal and cultural issues.
PO-7	Ability to understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development.
PO-8	Apply ethical principles and commit to professional ethics and responsibilities and norms of mining engineering practice.
PO-9	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.
PO-10	Communicate effectively on complex mining engineering activities with the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations and give and receive clear instructions..
PO-11	Demonstrate knowledge and understanding of mine management principles and mineral economic on decision-making and apply these to one's own work as a member and leader in a team, to manage projects and in multi-disciplinary environments.
PO-12	Recognize the need for, and have the preparation and ability to engage in, independent and lifelong learning in the broadest context of technological change.

3.2.4 Graduation Requirement

Graduation requirement for undergraduate Mining Engineering programs should satisfy the following minimum requirements as stated in the university's senate legislation July 2017, Article 109.A student must have passed all prescribed courses earning minimum total **187 credits** except to phase in and phase out program.

- A cumulative grade point average CGPA of 2.00 must be obtained;
- A cumulative grade point average CGPA of 2.00 in Core courses;
- No "F" grade in any course/ taken for undergraduate program;
- Pass the exit exam
- Student who fail to graduate due to less CGPA than the required or due to "F" grade have no more chance to upgrade their CGPA or to remove their "F" may be given certificate of attendance with their transcript stated as "Graduation failure"

- Successfully defended his/her senior project/senior essay

3.2.5 Course Category and list of Courses

Table. Distribution of credit hours for all course categories

Category			Total Cr. hr.	Percentage (%)
1	Core course		149	79.70%
2	Core Elective courses		6	3.20%
3	Common course	National requirement	29	17.11%
		Integrated Engineering Team Project	3	
Total			187	100%

Table. List of Courses

Common courses = 32Cr.hr = 17.11%		
No.	Course Title	Credit-hour
01	Logic and Critical Thinking	3
02	General Psychology	3
03	Communicative English Language Skill-I	3
04	Geography of Ethiopia and The Horn	3
05	Mathematics (For Natural Science)	3
06	Physical fitness	(P/F)
07	Social Anthropology	2
08	Communicative English Language Skill-II	3
09	Moral and Civic Education	2
10	Inclusiveness	2
11	Global Trend	2
12	History of Ethiopia and the horn	3
13	Integrated Engineering Team Project	3

Core/Compulsory Courses = 149Cr.hr = 79.70%		
No.	Course Title	Credit-hour

01	Applied Mathematics IB	4
02	Engineering Mechanics I (Statics)	3
03	Applied Mathematics IIB	4
04	Engineering Drawing	3
05	Introduction to Computer Programing	3
06	General Physics	3
07	Entrepreneurship for Engineers”	2
08	Emerging Technology for Engineers	3
09	Introduction to Economics	3
10	Geology for Mining Engineers I	4
11	Strength of Materials	3
12	General Chemistry	3
13	Fundamentals of Electrical Engineering	3
14	Engineering Mechanics II (Dynamics)	3
15	Introduction to Mining Engineering	2
16	Mine Equipment and Machinery I	2
17	Geology for Mining Engineers II	4
18	Fluid Mechanics	3
19	Introduction to Surveying	2
20	Rock Mechanics	3
21	Probability and Statistics	3
22	Surface Mining Methods	3
23	Mine Hazards and Rescue	3
24	Structural Geology	3
25	Mine Equipment and Machinery II	3
26	Drilling and Blasting	3
27	Rock Engineering	3
28	Mine Ground Control and Instrumentation	3
29	Mine Systems Engineering	3
30	Surface Mine Planning and Design	3
31	Underground Coal Mining Methods	3

32	Mine Surveying	3
33	Computer Aided Mine Planning and Design	3
34	Mine Safety and Health	3
35	Mineral Processing Technology	3
36	Underground Mine Planning and Design	3
37	Underground Metal Mining	2
38	Resource Estimation and Ore Body Modeling	3
39	Environmental Aspects of Mining	3
40	Internship	6
41	Mine Ventilation	3
42	Integrated Mining Design Project	3
43	Rock Excavation Engineering	3
44	Energy Resource Exploitation	3
45	PART-I B.Sc. Thesis (proposal)	P/F
46	Mine Law and Regulations	3
47	Mineral Economics	3
48	Mine Project Management	3
49	PART-II B.Sc. Thesis	6

Core elective courses =6 Cr.hr = 3.20%			
No.	Category	Course Title	Credit-hour
50	Elective I	1. Gemstone Mining 2. Solution mining	3
51	Elective II	1. Dimensional Stone Mining 2. Mine Automation	3

3.2.6 Program Curriculum Structure: Bachelor of Science in Mining Engineering

Year I Semester I

S. No	Course Title	Course Code	Pre-requisite	Cr.Hrs	Lec.	Tut.	Lab. /Prac.
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1	Logic and Critical Thinking	Phil1009	None	3	3	2	0
2	General Psychology	Psyc1011	None	3	3	2	0
4	Geography of Ethiopia and The Horn	GeEs1005	None	3	3	2	0
5	Mathematics (For Natural Science)	Math1007	None	3	3	2	0
7	General Physics	Phys1001	None	3	2	3	0
3	Communicative English Language Skill-I	FLEn1003	None	3	3	2	0
6	Physical fitness	SpSc1013	None	(P/F)	0	0	0
Sub Total				18	14	13	0

Year I Semester II

S.No	Course Title	Course Code	Pre-requisite	Cr.Hrs	Lec	Tut	Lab./Prac.
1	Emerging Technology for Engineers	EmTe1108	None	3	2	0	3
2	Communicative English Language Skill-II	FLEn1004	FLEn1003	3	2	1	2
3	Applied Mathematics IB	Math1014	Math1007	4	3	3	0
4	Moral and Civic Education	MCiE1012	None	2	3	1	0
5	Inclusiveness	Incl1010	None	2	2	0	0
6	Social Anthropology	Anth1002	None	2	3	0	0
7	Entrepreneurship for Engineers	Ent1106	None	2	3	1	0
Sub Total				18	18	6	5

Year II Semester I

S.No	Course Title	Course Code	Pre-requisite	Cr.Hrs	Lec.	Tut.	Lab./Prac.
1	Engineering Drawing	MEng2001	None	3	2	0	3
2	Introduction to Computer	Comp2003	Math1014	3	2	0	3

	Programing						
3	Engineering Mechanics I Statics	CEng2005	Phys1001	3	2	3	0
4	Applied Mathematics IIB	Math2007	Math1014	4	3	3	0
5	Introduction to Economics	Econ2009	None	3	0	0	0
6	Global Trend	GLTr2011	None	2	2	0	0
Sub Total				18	18	8	9

Year II, Semester II

S. No	Course Title	Course Code	Pre-requisite	Cr.Hr	Lec.	Tut.	Lab./Prac.
1	Geology for Mining Engineers I	MnEg2102	None	4	3	0	3
2	General Chemistry	Chem2004	None	3	2	0	3
3	Strength of Materials	CEng2006	CEng2005	3	2	3	0
4	Fundamentals of Electrical Engineering	EEeg2008	None	3	2	3	0
5	Engineering Mechanics II (Dynamics)	MEng2010	CEng2005	3	2	3	0
6	History of Ethiopia and the horn	Hist2002	None	3	3	0	0
Sub Total				19	15	09	06

Year III, Semester I

S. No	Course Title	Course Code	Pre-requisite	Cr.Hrs	Lec.	Tut.	Lab./Prac.
1	Introduction to Mining Engineering	MnEg3101	None	2	2	0	2
2	Mine Equipment and Machinery I	MnEg3103	EEEG2008	2	2	0	2
3	Geology for Mining Engineers II	MnEg3105	MnEg2102	4	3	0	3
4	Fluids Mechanics	MEng3007	None	3	2	1	1
5	Introduction to Surveying	MEng3109	None	2	2	0	2
6	Rock mechanics	MnEg3111	CEng2005	3	2	0	3
7	Probability and Statistics	Stat2091	None	3	3	3	0
	Sub Total			19	13	3	12

Year III, Semester II

S. No	Course Title	Course Code	Pre-requisite	Cr.Hrs	Lec.	Tut.	Lab./Prac.
1	Surface Mining Methods	MnEg3102	MnEg3101	3	2	0	3
2	Mine Hazards and Rescue	MnEg3104	MnEg3112	3	2	0	3
3	Structural geology	MnEg3106	MnEg2102	3	2	0	3
4	Mining Equipment and Machinery II	MnEg 3108	MnEg3101	3	2	0	3
5	Drilling and Blasting	MnEg3110	MnEg3101	3	2	0	3
6	Rock Engineering	MnEg3112	MnEg3109	3	2	2	3
Sub Total				18	12	2	18

Year IV, Semester I

S.No	Course Title	Course Code	Pre-requisite	Cr.Hrs	Lec.	Tut.	Lab./prac.
1	Mine Ground control and Instrumentation	MnEg4101	MnEg3112	3	2	0	3
2	Integrated Engineering Team Project	IETP4115	None	3		3	2
3	Surface Mine Planning and Design	MnEg4105	MnEg3102	3	3	0	3
4	Underground coal Mining Methods	MnEg4107	MnEg2112	3	2	0	2
5	Mine Surveying	MnEg4109	MnEg3107	3	2	0	2
6	Computer Aided Mine Planning and Design	MnEg4111	None	3	0	2	3
Sub Total				18	12	5	12

Year IV, Semester II

S.No	Course Title	Course Code	Pre-requisite	Cr.Hrs	Lec	Tut.	Lab./prac.
1	Mine Safety and Health	MnEg4102	MnEg3104	3	2	0	3
2	Mineral Processing Technology	MnEg4104	MnEg3103	3	2	0	3
3	Underground Mine Planning and Design	MnEg4106	MnEg4109	3	2	0	3
4	Underground Metal Mining	MnEg4108	MnEg3110	2	2	0	2
5	Resource Estimation and Ore Body Modeling	MnEg4110	MnEg3103	3	2	0	3
6	Environmental aspects of Mining	MnEg4112	None	3	2	0	3
Sub total				17	12	0	17

IV, Semster III (Summer)

1	Internship	MnEg4114	None	6	0	0	18
Sub Total				6	0	0	18

Year V, Semester I

S. No	Course Title	Course Code	Pre-requisite	Cr.Hrs	Lec.	Tut.	Lab./Prac.
1	Mine Ventilation	MnEg5101	MnEg4106	3	2	0	3
2	Integrated Mining Design Project	MnEg5103	MnEg4111	3	0	2	3
3	Mine system engineering	MnEg5105	Stat2091	3	2	1	0
4	Rock Excavation Engineering	MnEg5107	MnEg3110	3	2	0	3
2	Mineral Economics	MnEg5109	Econ2009	3	3	2	0
6	Elective I	MnEg5111/ MnEg5113	None	3	2	0	3
7	PART-I B.Sc Thesis (proposal)	MnEg5115	None	P/F	0	0	0
Subtotal Total				18	10	3	15

Year V, Semester II

S. No	Course Title	Course Code	Pre-requisite	Cr.Hrs	Lec.	Tut.	Lab./Prac.
1	Mine law & Regulations	MnEg5102	None	3	2	0	0
5	Energy Resource Exploitation	MnEg5104	None	3	2	0	3
3	Mine Project Management	MnEg5106	None	3	3	0	0
4	PART-II B.Sc. Thesis	MnEg5108	MnEg5103 & MnEg5115	6	0	0	18
5	Elective II	MnEg5210/ MnEg5212	None	3	2	0	3
Sub Total				18	10	2	21

Conversion rate: from Cr.hr to ECTS is given as, 1ECTS = 1.67Cr.hr.

3.2.7 Program Courses Synopsis

Course Code: Math-1014

Course Name: Applied Mathematics IB

Credit Hour: 4

Prerequisite: Math1007

This course covers basic elements of vectors, vector spaces, matrices, determinants, solving systems of linear equations, concepts and applications of differential and integral calculus of one variable

Course Code: CEng2005

Course Name: **Engineering Mechanics I (Statics)**

Credit Hour: 3

Prerequisite: Phys1001

Engineering mechanics (Statics) is the application of mechanics (one of the three branches of Physics) to solve problems involving common engineering elements

Course Code: Math 2007

Course Name: Applied Mathematics IIB

Credit Hour: 4

Prerequisite: Applied Mathematics IB (Math1014)

This course covers basic elements of sequence and series, power series, differential calculus of several variables and multiple integral concepts and their applications.

Course Code: MnEg2001

Course Name: Engineering Drawing

Credit Hour: 3

Prerequisite

Students will learn the Application of Engineering Drawing to sketch multi – view drawings of any given pictorial drawings.

Course Code: Comp 2003

Course Name **Introduction to Computer Programming**

Credit Hour: 3

Prerequisite: Math 2007

This course covering Fundamentals of Object-Oriented Programming: Basic concepts, Objects and classes, Data abstraction and encapsulation, Inheritance, Polymorphism and Dynamic Binding. JAVA Evolution: Java features, Java versus C and C++, Creating, compiling and running a Java program, Constants, Variables, Data types,

Operators and Expressions, Decision making and branching, Decision making and looping, Classes, Objects, and methods, Sample programs.

Course Code: MnEg2102

Course Name **Geology for Mining Engineers I**

Credit Hour: 4

Prerequisite:

This course aims to introducing the students the principles of general geology , petrology, mineralogy and stratigraphy in Mining Engineering

Course Code: CEng2006

Course Name: **Strength of Materials**

Credit Hour: 3

Prerequisite: CEng2005

Exposed the students to the knowledge of analyze and design structural members subjected to tension, compression, and torsion and bending using fundamental concepts of stress, strain, elastic behavior and inelastic behavior. Conduct themselves professionally and with regard to their responsibilities to society, especially with respect to designing structures to prevent failure.

Course Code: **Chem2004**

Course Name: **General Chemistry**

Credit Hour: 3

Prerequisite: No

The aim of engineering chemistry course is to provide students with an introduction to the basic concepts of chemistry.

Course Code: **EEEG2008**

Course Name: **Fundamentals of Electrical Engineering**

Credit Hour: 3

Prerequisite: No

Electromagnetic Phenomenon and Variables, Electric Circuit parameters, Circuit Analysis: Kirchhoff's laws (KVL & KCL), Steady State Single Phase AC Circuit Analysis, Transient Circuit Analysis, Introduction to poly - phase systems, Generation of three phase voltages; star (Y) and delta (Δ) connections; load/power flow method of three phase ac circuit analysis.

Course Code: MEng2010

Course Name: **Engineering Mechanics II (Dynamics)**

Credit Hour: 3

Prerequisite: CEng 2005

The course rationale is to develop the fundamental equation that characterizes the kinematics and Newtonian dynamics of a particle, systems of particles, and rigid bodies.

Course Code: MnEg3101

Course Name: **Introduction to Mining Engineering**

Credit Hour: 2

Prerequisite: No

To impart knowledge of historical role of mining in human civilization. To know the fundamental mining terminologies used in surface, underground mining and in mineral processing. To identify the basic stages of the life of mine, distinguish minerals, ores, and mineral deposits. Have an understanding about the economic significance and structure of the Ethiopian mining industry.

Course Code: MnEg3103

Course Name: **Mine Equipment and Machinery I**

Credit Hour: 2

Prerequisite: MnEg2112 & EEEg2008

To extract and transport the minerals to the required processing unit/ utilization point variety of machines are used in the mining industry. In this course the student gets acquainted with a few machineries including brief details of the machine parts, their working principles, operation and maintenance in addition to the machine installation, commissioning and safety aspects.

Course Code: MnEg3105

Course Name: **Geology for Mining Engineers II**

Credit Hour: 4

Prerequisite: MnEg2102

This course aims at introducing the students to the economic geology and principles and techniques of mineral exploration.

Course Code: MEng 3007

Course Name: **Fluids Mechanics**

Credit Hour: 3

Prerequisite:

The course aims to introduce Mining engineering students to thermo fluids at a more fundamental level and with a more mathematical approach.

Course Code: MnEg3109
Course Name: **Introduction to Surveying**
Credit Hour: 3
Prerequisite:

The course is intended to provide the students with the basic principles of surveying and land information system.

Course Code: MnEg3111
Course Name: Rock Mechanics
Credit Hour: 3
Prerequisite MnEg2102 and CEng 2003

The course makes students familiar with engineering properties of rocks and Soils in relation to open pit and underground mining

Course Code: Stat2091
Course Name: Probability and Statistics
Credit Hour: 3
Prerequisite

This course covers basic elements of statistics, probability, central tendencies, variation, and types of sampling, concepts and applications of estimation and hypothesis testing, regression analysis.

Course Code: MnEg3102
Course Name: **Surface Mining Methods**
Credit Hour: 3
Prerequisite MnEg2112

This course provides an overview of the basic concepts of surface mining, working principles of various excavation and transportation equipment.

Course Code: MnEg3104
Course Name: **Mine Hazards and Rescue**
Credit Hour: 3
Prerequisite MnEg3101

The course aims to introduce Mining Engineering students to the broader issues of mine hazards and the rescue and recovery work

Course Code: MnEg3106
Course Name: **Structural Geology**
Credit Hour: 3
Prerequisite MnEg 2102

The course aims to provide students with the basic concepts of deformation of rocks, and of the mechanisms and causes of deformation; to familiarize students with how to appreciate, describe and record mine geological structures in the field; to enable students how to evaluate, analyze and interpret structural data and incorporate it into synthesis of an area for mining.

Course Code: MnEg3110
Course Name: **Drilling and Blasting**
Credit Hour: 3
Prerequisite MnEg3101

To introduce various technologies of drilling and blasting techniques used in surface and underground mines, various accessories and safety measures.

Course Code: MnEg3108
Course Name: **Mine Equipment and Machinery II**
Credit Hour: 3
Prerequisite MnEg3101

The aim of the course is to introduce students to various mining equipment and their working principles

Course Code: MnEg3112
Course Name: Rock Engineering
Credit Hour: 3
Prerequisite MnEg3109

This course aims to obtain deeper knowledge on rock engineering. Rock and rock mass classification and their applications. And then characteristics of rock mass.

Course Code: MnEg4101
Course Name: Mine Ground Control and Instrumentation
Credit Hour: 3
Prerequisite MnEg3109

The course provides students with a practical understanding of the application of ground control principles in mining from the perspective of improve the design and stability of open pit and underground structures through clear understanding of factors affecting mining Geo techniques.

Course Code: MnEg5105
Course Name: Mine Systems Engineering
Credit Hour: 3
Prerequisite Stat2091

To impart knowledge on various systems and sub-systems and to introduce students to linear programming and project management techniques

Course Code: MnEg4105
Course Name: Surface Mine Planning and Design
Credit Hour: 3
Prerequisite MnEg3110

Factor affecting the selection of open pit mining; Design of Open Pit, Quarry Operations, Strip Mining of Mineral Deposits and Mechanized earth- moving, Analyze the economic indices of surface mine design and apply software packages to surface mine design

Course Code: MnEg4109
Course Name: **Mine Surveying**
Credit Hour: 2
Prerequisite MnEg3107

To introduce various technologies of surveying on the surface and underground mining situations including distance measurements, leveling, contouring, traversing etc along with descriptions of associated instruments.

Course Code: MnEg4107
Course Name: Underground Coal Mining Methods
Credit Hour: 3
Prerequisite MnEg3110

This course provides an overview of the principles and application of the major underground mining methods and equipment, and the conceptual design of the major materials handling and transport systems and support infrastructure.

Course Code: MnEg4111
Course Name: **Computer Aided Mine Planning and Design**
Credit Hour: 2
Prerequisite MnEg2009& MnEg4105 /

MnEg 4106

At the end of the course, the student will be able to plan and design practically for surface and underground mine for given the geologic data and topo files.

Course Code: MnEg4102
Course Name: Mine Safety and Health
Credit Hour: 2
Prerequisite MnEg3104

The course aims to introduce Mining Engineering students to the broader issues of global environmental challenges facing the mining projects.

Course Code: MnEg4104
Course Name: Mineral Processing Technology
Credit Hour: 3
Prerequisite MnEg 3103

This course aims to equip the student with knowledge of mineral processing unit operations normally as a member of a multi-disciplinary team in a mineral Design project or operating mine, Ore handling and classification and mineral liberation techniques.

Course Code: MnEg4106
Course Name: Underground Mine planning and Design
Credit Hour: 3
Prerequisite MnEg 4109

The course aims to provide students' knowledge of design of modes of entry to a deposit and the selection of suitable mining method. Design a ventilation system for an underground mine. Design underground support systems software package applications of mine design.

Course Code: MnEg4108
Course Name: **Underground Metal Mining**
Credit Hour: 3
Prerequisite MnEg 3110

To give the student an introduction to various metal mining techniques and support systems

Course Code: MnEg4110
Course Name: Resource Evaluation and Ore Body Modeling
Credit Hour: 3
Prerequisite Geol3103 and Stat2091

This course aims to introducing the principles of the resources estimation, ore tonnage and grade of a geological deposit from the developed block model; Geo statistical application; principal steps of computer resource estimation; principles of economic analysis of mining interventions and their role in mining decision making.

Course Code: MnEg4112
Course Name: Environmental Aspects of Mining
Credit Hour: 3
Prerequisite No

The course aims to provide students with the knowledge of the effects of mining on the environment and the remedial measures

Course Code: MnEg4114
Course Name: Internship
Credit Hour: 6
Prerequisite

The summer Internships are an exciting way for mining engineering students to gain a practical knowledge in the mining industry.

Course Code: MnEg5101
Course Name: Mine Ventilation
Credit Hour: 3
Prerequisite MnEg 3110 & MnEg4109

This course aims to provide students to detect and know the availability and determination of air quality in mines. Explain characteristics, mechanics and dynamics of fluid and airflows. Describe principles and operations of ventilation network systems, regulators in different types of mine. Define and understand the basic principles and applications of deep mining climate controlling techniques and systems

Course Code: MnEg5103
Course Name: Integrated Mining Design Project
Credit Hour: 3
Prerequisite

To gain experience in reserve estimation and building geologic models; understand ramifications of the reserve characteristics on mine design and production planning using a modern geologic-modeling design software package

Course Code: IETP4115
Course Name: Integrated Engineering Team Project
Credit Hour: 3
Prerequisite

This is a multidisciplinary team integrated engineering project in connection with a special engineering problem and under the guidance of a faculty member. The project synopsis consists of literature review, design, project management, business acumen, multidisciplinary team work, entrepreneurship

Course Code: MnEg5107
Course Name: Rock Excavation Engineering
Credit Hour: 3
Prerequisite MnEg3110

The course aims to provide the fundamentals of power systems in mining, different types of power systems and their applications used in mining works. Define nature of mechanical excavations and working principles of mechanical impact breaking in mining. Describe principles and operations of rock drilling and explosives, Mechanism of Rock breaking and blasting applications. Define and understand the basic principles and applications of underground blasting and surface mining blasting.

Course Code: MnEg5104
Course Name: Energy Resource Exploitation
Credit Hour: 3
Prerequisite

This course introduces students to the fundamental of geothermal energy from geoscience, engineering and prospective.

Course Code: MnEg5102
Course Name: Mine law and regulation
Credit Hour: 2

Prerequisite

The course aims to provide students to know current Ethiopian exploration and mining law, policy, legislation and proclamations

Course Code: MnEg5109
Course Name: Mineral Economics
Credit Hour: 3
Prerequisite Econ 1009 & MnEg3111

Familiarized with basic financial and economic analysis, funding, revenue of mineral resources and markets, financial reporting.

Course Code: MnEg5106
Course Name: Mine Project Management
Credit Hour: 3
Prerequisite MnEg4106

The main objective of this course is to explain the students about the Mine Project Management. Upon successful completion of this course program, students will be prepared to lead mine project requiring.

Course Code: MnEg5108
Course Name: Part II B.Sc. Thesis
Credit Hour: 4
Prerequisite

The course aims to give senior students the opportunity of undertaking an independent research project in Mining Engineering, in an area of interest to the students, under the supervision of a member of staff/s. Students will acquire and interpret mine datasets, using them to solve a mining engineering problem. Students will also present the results of their own scientific research orally and in poster form to an audience of staff and peers.

Course Code: MnEg5111
Course Name: Gemstone Mining
Credit Hour: 3
Prerequisite

This course unit aims to provide students with a general introduction in the field of Gemstone Mining and understanding, investigation and exploitation of these natural resources; understanding on the occurrence of different types of gemstones; basic knowledge on the nature, composition and maturation of gemstones; and skills on the technological applications in the exploration of these resources.

Course Code: MnEg5113
Course Name: Solution mining
Credit Hour: 3
Prerequisite

The main objective of this course to introduces the students about solution mining, methods and as well as laboratory test of solution mining.

Course Code: MnEg5110
Course Name: Dimensional Stone Mining

Credit Hour: 3

Prerequisite

This course introduces students to study about historical uses of natural stones, geology, occurrences, and mining techniques, cutting tools, equipment uses and selections, environmental issues, application and processing of dimensional stones.

Course Code: MnEg5112

Course Name: Mine Automation

Credit Hour: 3

Prerequisite EEeg1202

This course introduces students to the fundamental tools and techniques of automation and robotics as applied to modern mining practice. This course provides an introduction to the basics of systems control, as well as to the fundamentals of sensing and navigation as applied to the design of robotic mobile equipment.

Course Code: MnEg5112

Course Name: Mine Automation

Credit Hour: 3

Prerequisite EEeg1202

3.3 Bachelor of Science Degree in Chemical Engineering

3.3.1 Introduction

Engineering is a field of study that focuses on invention, design, monitoring, and evaluation according to the field of study. Chemical engineering is the branch of engineering that deals with physical and life sciences with mathematics and economics, to the process of converting raw materials or chemicals into more useful or valuable forms.

Chemical Engineering is a profession that principally hub on the conversion of raw materials into variety of products, and deals with the design, construction and operation of plants and equipment to achieve an optimum quality and efficiency. In addition to producing useful materials, Chemical Engineering is also concerned with investigating valuable materials and techniques which contribute to dynamic Research & Development world. The profession focuses on various processes that cause chemical, physical, and biological changes, designing and inventing new process for sustainable development which can be realized by having profitable, efficient and environmentally friendly industries. Additionally, Chemical Engineering has a great contribution in the area of Design, Operations, Control, Optimization and modification of processes and plants.

Chemical Engineers are trained to consider systems as units where the use, findings and manipulations of chemistry, physics and biology with the aid of mathematics, economics and engineering art are utilized to facilitate conversion of raw materials to products/value-added products. Basic principles in chemistry, biology and physics are interpreted to an application in processes and operations where chemical, physical and biological changes/ conversions take place. The profession is needed for the conception, creation (design), operation, control, optimization, and improvement of the process industries.

The chemical engineering profession is the youngest science based engineering profession. In Ethiopia, it has age not more than 30 years and the program were offered only in two universities (BDU and AAU). Currently, due to larger demand of chemical engineering graduates and the plan of the government of the federal republic of Ethiopia to establish industry based economy, more than 18 universities have already launched the program. Hence, AASTU has been launched chemical Engineering program in 2002 EC by considering the need of chemical engineering profession in various industries and for Government plans.

Therefore, it is clear that at present the country expands and gives ambitions to industry developments but lacks technical professionals. Generally there are no enough professionals of chemical engineers in different sectors of industry which the government has given priority such as sugar industries, textiles, garments, leather, cement, food processing and in general process/chemical industries (Growth and Transformation Plan Policy Matrix, Volume II, November 2010, Page 9-11); considered as the critical infrastructure for the country to change the living standards of the people.

The previous Chemical Engineering program curriculum was restructured to consider current state of art and policy of ministry of science and higher education (MoE). The curriculum is restructured to include nationally required courses and AASTU courses. Further, some new courses are included to enhance the competence of the graduates.

3.3.2 Program Education Objective (PEO)

Chemical engineering graduates will have applied their knowledge and skill to contemporary problem solving, be engaged professionally, and have continued to learn and adapt, and have

contributed to their organizations through leadership and teamwork. More specifically, the objectives are expertise, engagement, learning, leadership and teamwork.

Table: Program Education Objectives (PEO)

PEO	Statement
PEO-1	Managerial Position in Industries, Government and Non-Government Organizations
PEO-2	Professional Chemical Engineers
PEO-3	Entrepreneurs
PEO-4	Researchers/Academicians

3.3.3 Program Outcomes (PO)

The programs would be measured through the attainment of the Program Outcome (PO)/Program Learning Outcomes (PLO) of the program and its courses. Hence, the PO was designed in such ways that contribute to achieve the PEO and stated as follow.

Table: Chemical Engineering Program Outcome

PO	Statement
PO-1	Ability to apply knowledge of mathematics, natural science, engineering fundamentals and chemical engineering to the solution of complex chemical engineering problems.
PO-2	Ability to identify, formulate, research literature and analyze complex chemical engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
PO-3	Ability to design solutions for complex chemical engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health, and safety, cultural, societal and environmental considerations.
PO-4	Ability to conduct investigations of complex chemical engineering problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
PO-5	Ability to create, select and apply appropriate techniques, resources and modern Chemicals engineering and IT tools, including prediction and modeling, to complex chemical engineering problems, with an understanding of the limitations.
PO-6	Ability to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex chemical engineering problems.

PO-7	Ability to analyze and evaluate the sustainability and impact of professional chemical engineering work in the solution of complex chemical engineering problems in societal and environmental contexts.
PO-8	Ability to apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.
PO-9	Ability to function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings
PO-10	Ability to communicate effectively on complex engineering activities with the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations and give and receive clear instructions.
PO-11	Ability to identify the need for, and have the preparations and ability to engage in independent and lifelong learning in the broadest context of technological change.
PO-12	Ability to demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work as a member and leader in a team, to manage projects and in multi-disciplinary environments.

3.3.4 Graduation Requirements

Students must take and pass all the required courses to satisfy the requirements for graduation. The total number of credit hours required for graduation with the Degree of Bachelor of Science in Chemical Engineering is 190 Cr hr including 6 Cr. hr internship. Additionally, the following minimum requirements must be fulfilled;

All the required courses/modules and the minimum credit hours set in the program curriculum by the respective academic unit should be satisfied, except to phase in and phase out program.

- A cumulative grade point average CGPA of 2.00 must be obtained;
- A cumulative grade point average CGPA of 2.00 in major area courses;
- Pass the exit exam
- No "F" grade in any course/ module taken for undergraduate program;
- Score pass mark for all courses, which have Pass/Fail grade.

3.3.5 List of courses and category

Table: Distribution of credit hours for all course categories

Category	Total	Percentage (%)
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		Cr. hr	
1	Core Course (major and supportive course)	149	78.42
2	Core Elective/focus area course	3	1.58
3	University requirement (Core)	3	1.58
Core Course Total Cr. Hr		155	81.58
4	National Requirement	35	18.42
Total Cr. Hr		190	100

Table: List of Courses with title and credit hour

Common (National and University) and Supportive Courses		
No.	Course Title	Credit-hour
01	Communicative English Language Skills I	3
02	Communicative English Language Skills II	3
03	General Psychology	3
04	Moral and Civic Education	2
05.	Geography of Ethiopia and the Horn	3
06	Social Anthropology	2
07	History of Ethiopia and The Horn	3
08	General Physics	3
09	Logic and Critical Thinking	3
10	Economics	3
11	Physical Fitness	P/F
12	Inclusiveness	2
13	Global Trend	2
14	Mathematics for Natural Science	3
	Total credit	35

Core supportive courses		
01	Applied Mathematics IB	4
02	Applied Mathematics IIB	4
03	Applied Mathematics IIIB	4
04	Engineering Mechanics I Statics	3
05	Engineering Mechanics II Dynamics	3
06	Basic Electrical Circuit And Introduction to Electrical Machine	3
07	Technical Report Writing	1
08	Engineering Drawing	3
09	Workshop Practice	1
10	Introduction to Computer Programming	3
11	Probability and Statistics	3
12	Strength of Materials	3
	Total credit	35

Core Courses		
No.	Course Title	Credit-hour
01	Applied Inorganic Chemistry	3
02	Applied Organic Chemistry	3
03	Fundamentals of Analytical Chemistry	3
04	Basic Principles of Process Calculations	3
05	Chemical Engineering Thermodynamics I	3
06	Chemical Engineering Thermodynamics II	3
07	Fluid Mechanics	3
08	Transport Phenomena	3
09	Fluid Machines for Chemical Engineers	3
10	Mechanical Unit Operations	3
11	Mechanical Unit Operations Laboratory	1
12	Thermal Unit Operations	3
13	Mass Transfer Unit Operations	4
14	Thermal and Mass Transfer Unit Operations Laboratory.	2

15	Reaction Engineering I-Reaction Kinetics	3
16	Reaction Engineering II-Reactor Design	3
17	Reaction Engineering Laboratory.	1
18	Fundamentals of Biochemical Engineering	3
19	Electro-chemical Engineering	2
20	Material Science and Engineering	3
21	Numerical Methods in Chemical Engineering	3
22	Process Industries	4
23	Internship	6
24	Basic Environmental Engineering	3
25	Process Dynamics and Control	3
26	Process Control Laboratory	1
27	Chemical Engineering Apparatus Design	4
28	Computer Aided Process Design and Simulation	3
29	Plant Design and Economics	3
30	Process Integration and Optimizations	3
31	Sustainable Resources and Energy Technology	3
32	Production and Project Management	3
33	Industrial Safety and Loss Management	3
34	Research Method and Experimental Design	2
35	Integrated Engineering Team Project	3
36	Integrated Plant Design Project	3
37	Emerging Technology for Engineers	3
38	Entrepreneurship for Engineers	3
39	Final Year project: Phase one	P/F
40	Final Year Project: Phase Two	6
		Total credit= 117

Core Elective Courses		
No.	Course Title	Credit-hour
01	Introduction to Food Process Technology	3
01	Polymer Science and Engineering	3
01	Petroleum Refinery Engineering	3
01	Fundamentals of Bioprocessing Engineering	3
01	Fundamentals of Pharmaceutical Industries	3
01	Energy Management and Audit	3
01	Fundamentals of Textile Chemical Processing	3
01	Fundamentals of Nuclear Engineering	3
	Total credit	3

3.3.6 Program Curriculum Structure: Bachelor of Science in Chemical Engineering

Year I, Semester I Course

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
Phil1009	Logic and Critical Thinking	3	3	0	0
Psyc1011	General Psychology	3	3	0	0
FLEn1003	Communicative English Language Skills I	3	2	3	0
GeEs1005	Geography of Ethiopia and The Horn	3	3	0	0
Math1007	Mathematics for Natural Science	3	2	3	0
SpSc1013	Physical Fitness	P/F	0	0	
Phys1001	General Physics	3	2	3	0
Total		18	14	9	

Year I, Semester II

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
EmTe1108	Emerging Technology for Engineers	3	3	0	0
Entr1106	Entrepreneurship for Engineers	3	2	3	0
Anth1002	Social Anthropology	2	2	0	0
FLEn1004	Communicative English Language Skills-II	3	2	3	0
Math1014	Applied Mathematics IB	4	3	3	0
MCiE1012	Moral and Civic Education	2	2	0	0
Incl1010	Inclusiveness	2	2	0	
Total		19	16	6	0

Year II, Semester I

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng2001	Engineering Drawing	3	2	3	0
Comp2003	Introduction to Computer programming	3	2	0	3
CEng2005	Engineering Mechanics I Statics	3	2	3	0
Math2007	Applied Mathematics IIB	4	3	3	0
Econ2009	Economics	3	2	3	0
GLTr2011	Global Trend	2	2	0	0
Total		18	13	12	3

Year II, Semester II

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng2002	Engineering Mechanics II Dynamics	3	2	3	0
Stat2004	Probability and Statistics	3	2	3	0
ChEg2106	Basic Principles of Process Calculations	3	2	3	0
InCh2108	Applied Inorganic Chemistry	3	2	3	0
HiES2002	History of Ethiopia and The Horn	3	3	0	0
Math2002	Applied Mathematics IIIB	4	3	3	0
MEng2014	Workshop practice	1	0	0	3
Total		20	14	15	3

Year III, Semester I

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
ChEg3101	Technical Report Writing	1	1	0	0
InCh3103	Applied Organic Chemistry	3	2	3	0
ChEg3105	Numerical Methods in Chemical Engineering	3	2	3	
ChEg3107	Chemical Engineering Thermodynamics I	3	2	3	0
InCh3109	Fundamentals of Analytical Chemistry	3	2	3	
ChEg3111	Transport Phenomena	3	2	3	0
ChEg3113	Fluid Mechanics	3	2	3	0
Total		19	13	18	

Year III, Semester II

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
ChEg3102	Mechanical Unit Operations	3	2	3	0
ChEg3104	Reaction Engineering I -Reaction Kinetics	3	2	3	0
ChEg3106	Thermal Unit Operations	3	2	3	0
ChEg3108	Material Science and Engineering	3	3	0	0
ChEg3110	Chemical Engineering Thermodynamics II	3	2	3	0
ChEg3112	Fluid Machines for Chemical Engineers	3	2	3	
ChEg3114	Mechanical Unit Operations Laboratory	1	0	0	3
Total		19	13	15	3

Year IV, Semester I

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
ChEg4101	Reaction Engineering II-Reactor Design	3	2	3	0
ChEg4103	Reaction Engineering Laboratory	1	0	0	3
IETP4115	Integrated Engineering Team Project	3			3
ChEg4107	Thermal and Mass Transfer Unit Operations Laboratory	2	0	0	4
EEEg4009	Basic Electrical Circuit and Introduction to Electrical Machine	3	2	0	3
EnEg4111	Basic Environmental Engineering	3	3	0	0
ChEg4113	Mass Transfer Unit Operations	4	3	3	0
Total		19	13	6	13

Year IV, Semester II

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
ChEg4102	Process Dynamics and Control	3	2	3	0
ChEg4104	Process Control Laboratory	1	0	0	3
ChEg4106	Fundamentals of Biochemical Engineering	3	2	3	0
ChEg4108	Process Industries	4	3	0	3
ChEg4110	Chemical Engineering Apparatus Design	4	3	3	0
MEng4012	Strength of Materials	3	2	3	0
Total		18	12	12	6

NB: Three hours (3hr) assigned for lab for the course Process Industries is considered as industry visit in a selected industry.

Year IV, Semester: Summer

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
ChEg4115	Internship	6	0	0	18
Total		6	0	0	18

NB: eighteen hours assigned for lab in the course Internship is considered as industry work in assigned internship place

Year V, Semester I

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab./Pr
ChEg5103	Computer Aided Process Design and Simulation	3	2	3	0
ChEg5117	Integrated Plant Design Project	3			3
ChEg5107	Plant Design and Economics	3	2	3	0
ChEg5109	Sustainable Resources and Energy Technology	3	3	3	0

ChEg5111	Research Method and Experimental Design	2	2	0	0
ChEg5113	Process Integration and optimizations	3	2	3	0
ChEg5115	Final Year Project-Phase One	P/F	0	0	3
Total		17	11	12	6

Year V, Semester II

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
ChEg5102	Electro-chemical Engineering	2	2	0	0
ChEg5104	Production and Project Management	3	3	0	0
ChEg5106	Industrial Safety and Loss Management	3	3	0	0
ChEgxxxx	Elective (ChEgxxxx)	3	2	3	0
ChEg5108	Final Year Project –Phase Two	6	0	0	6
Total		17	10	3	6

NB: Elective course will be selected from list of elective courses based on student interest and available staff for teaching.

Table: List of elective courses

No.	Course Title	Code
01	Introduction to Food Process Technology	ChEg5210
01	Polymer Science and Engineering	ChEg5212
01	Petroleum Refining Engineering	ChEg5214
01	Fundamentals of Bioprocessing Engineering	ChEg5216
01	Fundamentals of Pharmaceutical Technology	ChEg5218
01	Energy Management and Audit	ChEg5220
01	Fundamentals of Textile Chemical Processing	ChEg5222
01	Fundamentals of Nuclear Engineering	ChEg5224

3.3.7 Program Courses Synopsis

3.3.7.1 Core courses

Course Code: InCh2108
Course Name: Applied Inorganic Chemistry
Credit Hour: 3
Prerequisite: None

This course focuses on reinforcement of theory of inorganic chemistry and synthesis of basic inorganic chemical products. The course includes overview of molecular orbital theory, chemistry of transition metals, ligands and isomerism, crystal field theory, ligand field theory and the spectrochemical series, NMR of coordination complexes, color in TM complexes, reactivity of oxides, sulphides, halogenides, hydrides etc. It also intricates about synthesis and structure of inorganic compounds, chemical reactions in solutions. It will also help to develop important problem solving and critical thinking skills.

Course Code: InCh3103
Course Name: Applied Organic Chemistry
Credit Hour: 3
Prerequisite: None

This course will make students who take the course about the basic things in organic chemistry, which are in touch with our day-to-day activities and us also. So, the course makes the students to be familiar with the various biological molecules, natural products, synthetic compounds, polymers and their roles, functions, functional groups, chemical and physical properties so that they will apply to their different fields.

Course Code: InCh3109
Course Name: Fundamentals of Analytical Chemistry
Credit Hour: 3
Prerequisite: None

The course covers ; Introduction to analytical chemistry; Acids bases, and neutralization titration; Complex formation, and complex metric titration: Gravimetric analysis; Chromatographic technique: paper, thin layer, gas chromatography and HPLC; Spectroscopic techniques; UV-Vis, AAS, IR, and Introduction to mass spectroscopy and Nuclear Magnetic Resonance spectroscopy (NMR), fluorescence, and coupled techniques such as HPLC/GC-MS, ICP-MS/OES,

Course Code: ChEg3107
Course Name: Chemical Engineering Thermodynamics I
Credit Hour: 3
Prerequisite: Applied Mathematics IIB (Math2007)

The course has the objective of fundamental principles equipping the student with the knowledge of the laws of thermodynamics and basic principle of thermodynamics in understanding heat and mass transfer and efficiency process units in process industry.

Course Code: ChEg3110
Course Name: Chemical Engineering Thermodynamics II
Credit Hour: 3
Prerequisite: Chemical Engineering Thermodynamics I

The course aims to introduce the principles of solution thermodynamics with understanding of the phase and reaction equilibrium. It focuses on determination of heat of reactions, sensible and latent heats. The course also enables to develop mathematical relationships for thermodynamic properties of ideal non-ideal solutions.

Course Code: ChEg3113
Course Name: Fluid Mechanics
Credit Hour: 3
Prerequisite: None

This course is designed to commence students to have a basic principles and characteristics of fluid flow in different processes (fluids motion and fluids at rest). It will introduce a student about fluid mechanics, the properties of fluid and types of fluids. It will also discuss on forces (pressure) applied by a fluid at rest or in rigid body motion. Using conservation principles (momentum, mass and energy) the student will examine properties of flow parameters on the fluid flow. Finally, this course will introduce dimensional analysis and the concept of similarity between a model and a prototype and also pipe flow and external flows.

Course Code: ChEg3111
Course Name: Transport Phenomena
Credit Hour: 3
Prerequisite: Applied Mathematics IIB (Math2007)/Chemical Engineering Thermodynamics I (ChEg3107) and Fluid Mechanics (ChEg3113)

This course will emphasize on the theoretical aspect of momentum, heat and energy in solving complex transport behavior. Purpose of this knowledge can analyze problems in momentum transfer, heat transfer and mass transfer.

Course Code: ChEg3081
Course Name: Fluid Machines for Chemical Engineers
Credit Hour: 3
Prerequisite: Fluid mechanics (ChEg3113)

This course is designed to commence students to have a basic principles and Characteristic Features and Definition of Fluid Machines, Energy Transfer and Specific Work of Fluid Machines, Performance Characteristics of

Centrifugal Machines, Theory of Positive Displacement Pumps, Theory of Positive Displacement Compressors, Specification of Fluid, Selection and specification of Fluid Machines.

Course Code: ChEg3102
Course Name: Mechanical Unit Operations
Credit Hour: 3
Prerequisite: Fluid Mechanics (ChEg3113)

This course aims to introduce students the subject of powder technology which focuses on processing and handling of particles and powders. It gives coverage of broad areas within powder technology including characterization, particle size reduction and enlargement, fluid-particle separation based on aero- and hydro-classifications, fluid-particle separation using gravity and pressure applications, fluid flow through solid beds and fluidization, and storage and transport of mixture of particles. Application of this knowledge can solve problems relevant to the design of chemical engineering systems, particularly to the mechanical unit operation equipment in process industries. It will also help to develop important problem solving and critical thinking skills. Upon the completion of the course, students will be able to analyse laboratory data and investigate the effects of different process parameters on the performance of powder handling and processing equipment.

Course Code: ChEg3114
Course Name: Mechanical Unit Operations Laboratory
Credit Hour: 3
Prerequisite: Mechanical Unit Operations

This course will provide practical experience on the basic mechanical unit operations ensuring a thorough understanding of principles of unit operation and appropriate theory. The course includes execution of experiments, data analysis and technical report writing and team work. Students will be able to do experiments on size reduction equipment and analysis, particle size distribution using screening, particle movement in a fluid and flow of fluid through a solid bed (Filtration, sedimentation, fluidization etc.)

Course Code: ChEg3106
Course Name: Thermal Unit Operations
Credit Hour: 3
Prerequisite: Transport Phenomena (ChEg3111)

This course will emphasize on the physical principles underlying heat flow by conduction, convection, and radiation mechanisms in heat exchangers, condensers, reboilers, boilers, evaporation processes. Application of this knowledge can solve problems relevant to the design of chemical engineering systems, especially heat exchange equipment. It will also help to develop important problem solving and critical thinking skills.

Course Code: ChEg4113
Course Name: Mass Transfer Unit Operations
Credit Hour: 3
Prerequisite: Thermal Unit Operations (ChEg3106)

Introduces the students with fundamental principles of mass transfer kinetics and chemical equilibrium. It introduces design concept of various separation processes involved in the chemical and other physical processing industries such as drying, distillations, absorption, adsorption, liquid-liquid extraction, crystallization, ion exchange and solid-liquid extraction (leaching).

Course Code: ChEg4107
Course Name: Thermal and Mass Transfer Unit Operations laboratory
Credit Hour: 2
Prerequisite: Thermal Unit Operations, Mass Transfer Unit Operations

This course will emphasize on the experimental setup of thermal and mass transfer equipment to measure effect of different parameters on heat and mass transfer, and also to evaluate performance of transfer equipment.

Course Code: ChEg3104
Course Name: Reaction Engineering I-Reaction Kinetics
Credit Hour: 3
Prerequisite: Applied Mathematics IIB (Math2007)

This course will emphasize on the principles of chemical reaction engineering which focuses on reaction classifications; reaction kinetics (rate & mechanism of the reactions and operating conditions); reactor types; developing the performance equation for various reactors and controlling product distribution for multiple reactions.

Course Code: ChEg4101
Course Name: Reaction Engineering II-Reactor Design
Credit Hour: 3
Prerequisite: Reaction Engineering, I- Reaction Kinetics (ChEg3104)

The course emphasizes on basic concepts of chemical reactor design and evaluate operational performance of ideal, non-ideal, isothermal, non-isothermal, adiabatic, non-adiabatic of batch and flow reactors for homogeneous and heterogeneous reactions.

Course Code: ChEg4103
Course Name: Reaction Engineering Laboratory
Credit Hour: 1
Prerequisite: Reaction Engineering, I-Reaction Kinetic (ChEg3104)/ Reaction Engineering II-Reactor Design (ChEg4101)

This course will acquire the ability to carry out experiments and analyse experimental data of chemical reactors. They should also be able to obtain and interpret experimental data on batch, CSTR, PFR, Cascade CSTR reactors, Catalytic reactors.

Course Code: ChEg4106
Course Name: Fundamentals of Biochemical Engineering
Credit Hour: 3
Prerequisite: Reaction Engineering II- Reactor Design (ChEg4101)

This course covers knowledge about principles of biochemical engineering used in industries, design, operation and control of bioreactors used in bioprocess. Various unit operations employed in bioprocess industries, production of chemicals using fermentation and bioprocesses will also be introduced.

Course Code: ChEg5102
Course Name: Electro-chemical Engineering
Credit Hour: 2
Prerequisite: None

The focus of the course is to give an in-depth understanding in the application areas of electrochemistry:

- Mainly on energy storage devices (fuel cells, batteries),
- Electroplating as a means to form coatings
- Electroforming to form metallic parts

Course Code: ChEg3108
Course Name: Material Science and Engineering
Credit Hour: 3
Prerequisite: None

The course discusses the physical, mechanical, electrical and chemical properties of materials, with an emphasis on the mechanical properties of metals and alloys. The course covers the fundamental understanding of:

- Types of materials (metals, ceramics, polymers, semiconductors)
- Different levels of structure in materials (atomic, microscopic, macroscopic)
- Properties of materials (mechanical, thermal, electrical, optical)

Relation among material processing, structure, properties, and performance

Course Code: ChEg3105
Course Name: Numerical Methods in chemical Engineering
Credit Hour: 3
Prerequisite: Applied Mathematics IIIB (Math2002)

This course focuses on the application of numerical technique for solving various chemical engineering problems. Further, this course elaborates about applying different computational software to solve complex equations and compare different numerical methods to obtain the solution with a minimum error. The course also deals about formulation of mathematical model for a given chemical engineering problem.

Course Code: ChEg4108
Course Name: Process Industries
Credit Hour: 4
Prerequisite: Basic principles of process calculation (**ChEg2106**)

This course will concern on different organic and inorganic chemical process technology. The application of this knowledge is to assess and evaluate chemical process industries. This leads to recommend a novel process (emerging technology) that can be effective in terms of economic, environmental, social and technical issues, if any for process industries.

Course Code: ChEg4115
Course Name: Internship
Credit Hour: 6
Prerequisite: Fundamentals of Biochemical Engineering (ChEg4106), Process Industries (ChEg4108), Chemical Engineering Apparatus Design (ChEg4110), and Process Dynamics and Control (ChEg4102), Mechanical Unit Operations (ChEg3102)

Internship is intended to give students a chance of real time industry experience and exposure, which are application of theoretical knowledge covered in class. During this period students are expected to boost practical hands-on experiences, professional/workplace behavior and knowledge, learn company culture, employer expectations. Internship also helps students to explore their career path and opportunity to professional network.

Course Code: EnEg4111
Course Name: Basic Environmental Engineering
Credit Hour: 3
Prerequisite: None

Ecosystems, environmental pollutants and ecosystem disturbances &, Basic concepts of environmental chemistry and Microbiology, Pollutant transport in environmental systems, Environmental engineering technologies and Environmental management tools

Course Code: ChEg4102
Course Name: Process Dynamics and Control
Credit Hour: 3
Prerequisite: Reaction Engineering II- Reactor Design, Fluid Mechanics

This course is designed to commence students to have a background in theory and practice of chemical process control. This course elaborates and analyzes about the characteristics of a control system and formulates the problems that must be solved during the design of a control system. Further, the aims of the course are to present a general introduction to the control aspects of a chemical process, introduce a student to modeling requirements for process control and analyze static and dynamic behavior of the process system. The course also covers the analysis and design of feedback control systems, which represent the control schemes encountered most often in a chemical plant. Finally, dynamics behavior of process control analysis using MATLAB Simulink Software is covered in this course.

Course Code: ChEg4104
Course Name: Process control Laboratory
Credit Hour: 1
Prerequisite: Process dynamics and control (ChEg4102)

Process Control laboratory is providing knowledge on practical performance of different control instruments and equipment through experimental work. The main objective of process control is to maintain a process at the desired operating conditions safely and efficiently.

Course Code: ChEg4110
Course Name: Chemical Engineering Apparatus Design
Credit Hour: 4
Prerequisite: Reaction Engineering II-Reactor Design (ChEg4101) and Mass Transfer Unit Operations (ChEg4113)

The course aims at giving basic understanding of process equipment design. Criteria and procedures for equipment selection and sizing will be discussed in detail. Equipment design techniques will be applied to the common unit operation in the chemical and process industry: internal and external pressure vessels, tall vessels, high pressure vessels, supports etc...

Course Code: ChEg5103
Course Name: Computer Aided Process Design and Simulation
Credit Hour: 3
Prerequisite: Chemical Process Apparatus Design (ChEg4110)

This course introduces the concept of new design and simulation of existing chemical processes. It deals with process simulation programs, sequential modular approach, and equation-oriented approach. Further, the course focuses on interpretation of different process technology, physical properties methods including reactor, separation, stream recycling, concept of energy integration, and environmental, health, and safety effects. This course elaborates development of process flow diagram (PFD), interpret the PFD with simulator software tools.

Course Code: ChEg5107
Course Name: Plant Design and Economics
Credit Hour: 3
Prerequisite: Chemical Engineering Apparatus Design (ChEg4110)

The course emphasizes the knowledge acquired by the student's in previous courses and applies in the complete design of a Chemical Process Plant. The plant design includes piping and instrumentation, cost estimation, profitability analysis, plant wide control, waste management, plant safety, reliability and operability, plant layout, plant startup. It will also help for critical thinking skills and solve crucial problem.

Course Code: ChEg5117
Course Name: Integrated Plant Design Project
Credit Hour: 3
Prerequisite: Chemical Engineering Apparatus Design (ChEg4110)

This is a group plant design project intended to develop and assess the students' capability to co-ordinate the knowledge gained in previous years of chemical engineering and apply it to the comprehensive design of a process plant. It exposes students to work in teams, develop coordination and communication skills. This capstone design project that intended to enable students to apply fundamental theoretical knowledge gained from previous courses to the conceptual design of a chemical process plant, generate cost effective process options with considerations for operability and safety of the design, screen and evaluate potential design alternatives for feasibility study. It also enables the students conduct material and energy balance, plant location, process flow sheeting and simulation. In this project the students integrate their chemical engineering skills to develop instrumentation and control strategy, look after environmental and safety aspects, perform detailed economic analysis and equipment design for the process plant.

Course Code: ChEg5113
Course Name: Process Integration and Optimization
Credit Hour: 3
Prerequisite: Internship (ChEg4115)

This course will emphasize on the physical principles underlying pinch technology, pinch point analysis and Process Optimization, the application of this knowledge can solve problems relevant to the design of chemical engineering problem and it is a powerful tool for designing and optimizing processes in terms of efficiency use of energy, typical problems in chemical engineering process design or plant operation have many solutions.

Course Code: ChEg5109
Course Name: Sustainable Resources and Energy Technology
Credit Hour: 3

Prerequisite: None

This course is designed to enable students to identify sustainable engineering systems that ensure efficient, safe and environment friendly energy utilization.

Course Code: ChEg5104

Course Name: Production and Project Management

Credit Hour: 3

Prerequisite: None

This course is an introduction to the concepts, principles, problems, and practices of production and project management. Emphasis is on managerial processes in chemical manufacturing organizations to continually improve the systems of production to ensure the most efficient and effective output in turning raw materials into a deliverable product by familiarizing the students with the major operational issues that confront managers.

Course Code: ChEg5106

Course Name: Industrial Safety and Loss Management

Credit Hour: 3

Prerequisite: None

This course is designed to provide essential information for students to acquire knowledge about safety in processing industries and to differentiate the different types of industrial hazard with their control measures.

Course Code: ChEg5111

Course Name: Research Methods and Experimental Design

Credit Hour: 2

Prerequisite: None

This course emphasizes on basic concept of research, design of experiment, collection and data analyses and research result reports. The objective of this course is to enable chemical engineering graduates understand basics of research and final years research project in particular, design of research, data collection and analysis and reporting of research outputs.

Course Code: ChEg4115

Course Name: Integrated Engineering Team Project

Credit Hour: 3

Prerequisite: None

This is a multidisciplinary team integrated engineering project in connection with a special engineering problem and under the guidance of a faculty member. The project synopsis consists of literature review, design, project management, business acumen, multidisciplinary team work, entrepreneurship

Course Code: ChEg5115

Course Name: Final Year Project Phase I

Credit Hour: P/F

Prerequisite: Internship (ChEg4115)/ ChEg5103, ChEg5105, ChEg5107, ChEg5111 and ChEg5113

This is an individual research project in connection with a special engineering problem and under the guidance of a faculty member. The project undertaken may fall under one of the following areas; mathematical analysis, experimental tests, computer simulation, hardware and/or software development, device fabrication. The project work includes literature review, research problem identification, proposal development, preliminary research work and interim report generation.

Course Code: ChEg5108

Course Name: Final Year Project Phase II

Credit Hour: 6

Prerequisite: Final Year Project Phase I

This is an individual research project in connection with a special engineering problem and under the guidance of a faculty member. The project undertaken may fall under one of the following areas; mathematical analysis, experimental tests, computer simulation, hardware and/or software development, device fabrication. The project involves complete literature survey and theoretical analysis; carry out experiments and/or design of experiments, computer modeling in pre-determined methodology; and analyze and interpret the obtained data for optimum solutions using suitable techniques/IT tools.

3.3.7.2 Elective courses

Course Code: ChEg5210

Course Name: Introduction to Food Process Technology

Credit Hour: 3

Prerequisite: None

The course describes fundamentals of techniques involved in processing, preservation, packaging, and processes that are used to transform raw materials into food. Food properties and typical characteristics with respect to extending shelf life and improving quality are described.

Course Code: ChEg5212
Course Name: Polymer Science and Engineering
Credit Hour: 3
Prerequisite: Material Science and Engineering (ChEg3108)

The course provides a broad overview of:

- Production routes of monomers and polymers
- Polymerization mechanisms the structures and properties (chemical, physical, viscoelastic, and mechanical properties) of polymers
- Polymer processing mechanisms: extrusion, injection, film blowing.

Application areas of polymeric materials.

Course Code: ChEg5214
Course Name: Petroleum Refining Engineering
Credit Hour: 3
Prerequisite: None

The course focuses on fundamentals of petroleum refining operations which include composition and properties of crude oil, conversion of crude oil to useful fractions using various technologies like distillation processes, and methods of upgrading the fractions such as catalytic cracking and reforming, and thermal processes

Course Code: ChEg5216
Course Name: Fundamentals of Bioprocess Engineering
Credit Hour: 3
Prerequisite: None

This course will focus on the fundamental concepts in Bioprocess Engineering which include enzyme kinetics, major metabolic pathways, microbial cell growth, bioreactor design and operation, sterilization techniques, recovery and purification of products. Application of enzyme and microbial cell kinetics leads to solve problems pertinent to the design of biochemical processes, particularly bioreactor design and operation. It will also help to develop important problem solving and critical thinking skills

Course Code: ChEg5218
Course Name: Fundamentals of Pharmaceutical Technology
Credit Hour: 3
Prerequisite: None

This course emphasizes on the basic principles of pharmaceutical technologies relevant to chemical engineering students which includes introduction to pharmaceutical industries, pharmacokinetics, pharmacodynamics, principles of manufacturing and drug delivery.

Course Code: ChEg5220
Course Name: Energy Management and Audit
Credit Hour: 3
Prerequisite: None

The course emphasizes on efficient energy management through effective utilization of utilities, conduct energy audit on process industries, analysis of boilers and fired systems for energy conservation options, analysis of steam and condensate systems for energy saving opportunities and waste heat recovery. The course also addresses Heating, Ventilation, and Air Conditioning (HVAC) systems as well as lighting and industrial installation for energy conservation opportunities.

Course Code: ChEg5222
Course Name: Fundamentals of textile chemical processing
Credit Hour: 3
Prerequisite: None

This course mainly deals on the overall chemical processing takes place in the textile industry starting from the fibre stage up to the garment finishes.

Therefore, students will learn the basics of textile chemical processing by incorporating fundamental chemical engineering principles such as transport phenomena, chemical process principles, and chemical engineering analysis.

Course Code: ChEg5224
Course Name: Fundamental of Nuclear Engineering
Credit Hour: 3
Prerequisite: None

This course aims to give basic knowledge about nuclear reactor engineering. It includes basic concepts of nuclear reactions and fission reaction chain, which are needed to understand the principles of nuclear reactors and some important issues in nuclear reactor engineering are also explained, including design concepts of nuclear reactors in practical use and innovative nuclear energy systems.

It will also help to develop important problem solving and critical thinking skills.

3.4 Bachelor of Science Degree in Environmental Engineering

3.4.1 Introduction

Environmental Engineering education leading to B.Sc. degree at Addis Ababa Science and Technology University (AASTU) was launched in 2011/12 under the then ‘School of Energy Resource and Environmental Engineering’ which was one of the eleven schools launched by

AASTU while it was established. Then after, environmental engineering department was grouped under college of biological and chemical engineering as part of the first organizational restructuring AASTU in 2016. AASTU was the first university in Ethiopia to develop and launch pure environmental engineering program at bachelor level to fulfill the growing demands of industries, public institutions and universities for graduates. Considering the importance of environmental issues of these days as a priority area, the department also launched M.Sc. and PhD programs in 2013 and 2017, respectively. Since, its establishment, a total of 204 students is graduated from the department from which 131 (95 male and 36 female) are BSc level and 73 (61 male and 12 female) are MSc level graduates.

Since the inception of the program, the department has strived to provide excellent environmental engineering education that produces competent and creative engineers to meet the challenges of environmental sustainability in this competitive age of globalization; researchers that advances the state of knowledge in the environmental engineering profession; and experts that serves the public and industries. For this reason, the program has been launched for developing well-trained and educated environmental engineers capable of designing, planning, operating, managing and maintaining public utilities, serving the demands of industries and promoting a benign ecological environment to safeguard its valuable resources of water, air and soil, and health.

This curriculum is equipped with most demanding and influential topics that can attract students in the environmental engineering discipline and able to enhance the knowledge, skill and attitude of the graduates. The curriculum is concerned not only about environmental pollution control and technological remediation methods, but also about impact of remediation processes on stakeholders of the environment. As such, it aims to combine scientific and technical knowledge with environmental resources management skills to design, plan, manage, maintain, analyze and operate a system by involving manpower, money, material, equipment, energy, information and all other related resources.

The most distinct feature of the program is its diverse elective areas such as air and water pollutions, water and wastewater treatment, air, water and soil pollution control, solid and hazardous waste management, biological and chemical issues on environmental remediation process with water resources management and sustainable utilization of energy. The department

is also very much concerned about environmental pollution control and remediation technologies. The Environmental Engineering program is, thus, structured to provide the necessary background in mathematics, basic sciences and engineering skills followed by environmental engineering specific courses with the intent of preparing graduates to tackle for the 21st century environment and economic development challenges.

3.4.2 Program Education Objective (PEO)

The program educational objectives describe accomplishments of environmental engineering program graduates are expected to attain within five years after graduation. Graduates are expected to apply their expertise to contemporary environmental problem solving, be engaged professionally, have continued to learn and adapt, and contributed to their organizations through leadership and teamwork. More specifically, the objectives are expertise, engagement, learning, leadership and teamwork as listed in Table 2.1.

Table 3.1: Program Education Objectives (PEO)

PEO	Statement
PEO-1	Apply essential environmental engineering knowledge and skills in designing environmental friendly solution to environmental problems.
PEO-2	Develop entry level managerial and leadership skill to render their expertise in industries
PEO-3	Create entrepreneurial business using their environmental engineering knowledge and skills
PEO-4	Pursue further studies and conduct research in environmental engineering and related fields

3.4.3 Program Outcomes (PO):

Engineering programs are measured through their attainment of the Program Outcome (PO)/Program Learning Outcomes (PLO) of the program and its courses. These program outcomes are statements on what students shall know, understand, and perform upon completing their course and/or program of study. State the Program Outcomes of this environmental engineering Bachelor of Science degree program are stated below (Table 2.3).

Table 3.2: Program Outcome

PO	Statement
PO-1	Apply knowledge of mathematics, science, engineering fundamentals and an environmental engineering specialization to the solution of complex environmental engineering problems.
PO-2	Identify, formulate, research literature and analyze complex environmental engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
PO-3	Design solutions for complex environmental engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
PO-4	Conduct investigation into complex environmental problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.
PO-5	Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modeling, to complex engineering activities, with an understanding of the limitations.
PO-6	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional environmental engineering practice and solution to complex environmental engineering problems.
PO-7	Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development.
PO-8	Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.
PO-9	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.

PO-10	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO-11	Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO-12	Recognize the need for, and have the preparations and ability to engage in independent and life-long learning in the broadest context of technological change

3.4.4 Graduation Requirements

Graduation requirement for environmental engineering undergraduate programs should satisfy the following minimum requirements as stated in the university's senate legislation July 2017, Article 109.

- All the required courses/modules and the minimum credit hours set in the program curriculum by the respective academic unit should be satisfied, except to phase in and phase out program.
- A cumulative grade point average CGPA of 2.00 must be obtained;
- A cumulative grade point average CGPA of 2.00 in major area courses;
- Pass the exit exam
- No "F" grade in any course/ module taken for undergraduate program;
- Score pass mark for all courses which have Pass/Fail grade.

3.4.5 List of Courses and Category

Table: Distribution of credit hours for all course categories

Category		Total Cr. hr	Percentage (%)
1	Core Course (major and supportive course)	145Cr.hr	76.72
2	Core Elective/focus area course	6 Cr.hr.	3.17

3	University requirement (Core)	3 Cr.hr	1.59
Core Course Total Cr. Hr			81.48
4	National Requirement	35 Cr.hr	18.52
Total Cr. Hr		189	100

List of Courses

Table: List of Courses

Common Courses		
No.	Course Title	Credit-hour
01	General Physics	3
02	Communicative English Language Skill – I	3
03	General Psychology	3
04	Geography of Ethiopia and the Horn	3
05	Mathematics (for Natural Science)	3
06	Logic and Critical Thinking	3
07	Physical Fitness	(P/F)
08	Social Anthropology	2
09	Communicative English Language Skill-II	3
10	Inclusiveness	2
11	Moral and Civic Education	2
12	Introduction to Economics	3
13	Global Trend	2
14	History of Ethiopia and the Horn	3
Core/Supportive Courses		
No.	Course Title	Credit-hour
01	Engineering Drawing	3
02	Engineering Mechanics I (Statics)	3
03	Applied Mathematics IB	4
04	Applied mathematics IIB	4
05	Applied Mathematics IIIB	4

06	Engineering Mechanics II (Dynamics)	3
07	Probability and statistics	3
08	Numerical analysis for Environmental Engineering	3
09	Material and energy balance	2
10	Thermodynamics	3
11	Environmental chemistry	3
12	Environmental microbiology	3
13	Strength of Materials	3
14	Fluid Mechanics	3
15	Reaction engineering	3
16	Unit Operation	2
17	Reinforced concrete Design	3
18	Survey	3
19	Transport phenomenon for environmental engineering	3
20	Environmental engineering hydrology	3
21	Fundamentals of soils & pollution Control	2
22	Air pollution engineering I	3
23	Water treatment engineering	3
24	Air pollution engineering II	3
25	Solid Waste Engineering	3
26	Engineering Economics	3
27	Environmental policy and management systems	2
28	Wastewater treatment engineering	3
29	Hazardous and e-waste engineering	2
30	Sewerage and drainage engineering	3
31	GIS and Remote Sensing	3
32	Industrial wastes management technologies	3
33	Waste to energy technology	3
34	Environmental impact assessment	3
35	Climate change Adaptation & Mitigation technologies	3
36	Internship	6

37	Renewable energy engineering	3
38	Soil pollution remediation technologies	3
39	Software application for environmental engineering	3
40	Health, Safety and environment (HSE) engineering	3
41	Final Year Project I	-
42	Final Year Project II	6
43	Energy and the Environment	3
44	Ecological engineering	3
45	Integrated Environmental Engineering Project Design	3
46	Integrated Engineering Team Project	3
47	Introduction to Computer Programming	3
48	Introduction to Emerging Technologies	3
49	Entrepreneurship for engineers	3
Core Elective Courses		
50	Environmental biotechnology	3
51	Cleaner Production and LCA	3
52	Air quality modeling and forecasting	3
53	Sludge treatment technologies	3
54	Ecological risk assessment	3
55	Green building and energy conservation	3
56	Emerging technologies for water and waste water treatment	3

3.4.6 Program Curriculum Structure: Bachelor of Science in Environmental Engineering

Table: Course Breakdown for Regular program

First Year First Semester							
S. No	Course Code	Course Title	Prerequisite	Cr.hr.	Lect.	Tut.	Lab.
1	Phys1001	General Physics	None	3	2	3	0
2	FLEn1011	Communicative English Language Skill – I	None	3	2	3	0
3	GeES1005	Geography of Ethiopia and the Horn	None	3	3	0	0
4	Math1007	Mathematics (for Natural Science)	None	3	2	3	0
5	Phil 1001	Logic and Critical Thinking	None	3	3	0	0
6	Psyc1011	General Psychology	None	3	3	0	0
7	SpSc1013	Physical Fitness	None	(P/F)	2	0	0
Total				18	17	9	0

First Year Second Semester							
S. No	Course Code	Course Title	Prerequisite	Cr.hr.	Lect.	Tut.	Lab.
1	Anth1012	Social Anthropology	None	2	2	0	0
2	FLEn1012	Communicative English Language Skill-II	FLEn1011	3	2	3	0
3	Entr1106	Entrepreneurship for Engineers	None	3	2	3	0
4	EmTe1108	Introduction to Emerging Technologies	None	3	2	3	0
5	Incl 1010	Inclusiveness	None	2	2	0	0
6	MCiE1012	Moral and Civic Education	None	2	2	0	0
7	Math1014	Applied Mathematics IB	Math1007	4	3	3	0
Total				18	15	9	0

Second Year First Semester							
S. No	Course Code	Course Title	Prerequisite	Cr.hr.	Lect.	Tut.	Lab.
1	MEng2001	Engineering Drawing	None	3	2	3	0
2	Comp 2003	Introduction to Computer Programming	None	3	2	0	3
3	CEng2005	Engineering Mechanics I (Statics)	Phys1001	3	2	3	0
4	Math 2001	Applied Mathematics IIB	Math1014	4	3	3	0
5	Econ 2009	Introduction to Economics	None	3	3	0	0
6	GLTr 2011	Global Trend	None	2	2	0	0
Total				18	14	9	3

Second Year Second Semester							
S. No	Course Code	Course Title	Prerequisite	Cr.hr.	Lect.	Tut.	Lab.
1	Math2002	Applied Mathematics IIIB	Math 2001	4	3	3	0
2	EnEg 2102	Environmental Microbiology	None	3	2	1	2
3	MEng2002	Engineering Mechanics II (Dynamics)	CEng2005	3	2	3	0
4	ChEg2004	Thermodynamics	None	3	2	3	0
5	Hist 2002	History of Ethiopia and the Horn	None	3	3	0	0
6	CEng2002	Strength of Materials	CEng2005	3	2	3	0
Total				19	14	15	0

	Third Year First Semester						
S. No	Course Code	Course Title	Prerequisite	Cr.hr.	Lect.	Tut.	Lab.
1	EnEg3103	Environmental Chemistry	None	3	2	1	2
2	ChEg3101	Material and Energy balance	None	2	2	1	0
3	EnEg3105	Environmental Engineering Hydrology	None	3	2	3	0
4	ChEg3103	Fluid Mechanics	None	3	2	3	0
5	EnEg3107	Transport phenomena for environmental engineering	None	3	2	3	0
6	CEng3101	Survey	None	3	2	1	2
7	EnEg3101	Fundamental of Soils and Pollution Control	None	2	2	0	1
	Total			19	14	12	5

	Third Year Second Semester						
S. No	Course Code	Course Title	Prerequisite	Cr.hr.	Lect.	Tut.	Lab.
1	Stat3102	Probability and statistics	Math1007	3	2	3	0
2	EnEg3104	Air pollution Engineering I	EnEg3103	3	2	3	0
3	CEng3102	Reinforced Concrete Design	CEng2002	3	2	3	0
4	EnEg3106	Water Treatment Engineering	EnEg3103 & EnEg 2102	3	2	3	0
5	ChEg3102	Unit operation	None	2	2	1	0
6	ChEg3104	Reaction Engineering	None	3	2	3	0
	Total			17	12	16	0

Fourth Year First Semester							
S. No	Course Code	Course Title	Prerequisite	Cr.hr.	Lect.	Tut.	Lab.
1	EnEg4101	Air pollution engineering II	EnEg3104	3	2	3	0
2	IETP 4115	Integrated Engineering Team Project		3	2	3	0
3	EnEg4105	Engineering economics	Econ 2009	3	2	3	0
4	EnEg4103	Numerical analysis for Environmental Engineering	Math2002	3	2	1	2
5	EnEg4107	Energy and the environment	None	3	2	3	0
6	EnEg4109	Wastewater treatment engineering	EnEg3103& EnEg 2102	3	2	3	0
Total				18	12	16	2

Fourth Year Second Semester							
S. No	Course Code	Course Title	Prerequisite	Cr.hr.	Lect.	Tut.	Lab.
1	EnEg4102	Solid waste engineering	ChEng3101	3	2	3	0
2	EnEg4104	Sewerage and drainage engineering	EnEg3105	3	2	3	0
3	EnEg4106	GIS and remote sensing	None	3	2	0	3
4	EnEg4108	Software applications for Environmental	Comp 2003	3	2	0	3
5	EnEg4110	Waste to Energy Technology	EnEg4107&EnEg4102	3	2	3	0
6	EnEg4112	Environmental Impact Assessment	None	3	2	3	0
7	EnEg4114	Environmental Policy and Management Systems	None	2	2	2	0
Total				19	12	10	6

S. No	Course Code	Course Title	Prerequisite	Cr.hr.	Lect.	Tut.	Lab.
1	EnEg4116	Internship	EnEg4101, EnEg4106 EnEg4109 EnEg4102 EnEg4104 EnEg4110 EnEg4112, EnEg4114	6	0	0	18
	Total			6	0	0	18

Fifth Year First Semester							
S. No	Course Code	Course Title	Prerequisite	Cr.hr.	Lect.	Tut.	Lab.
1	EnEg5111	Integrated Environmental Engineering Project Design	EnEg4114	3	2	3	0
2	EnEg5101	Renewable Energy Engineering	EnEg4107	3	2	2	1
3	EnEg5103	Ecological Engineering	None	3	2	3	0
4	EnEg5105	Hazardous and E-waste engineering	EnEg4102	2	2	1	0
5	EnEg5109	Soil Pollution Remediation Technologies	EnEg3107 &EnEg3101	3	2	3	0
6	EnEg52xx	Elective I		3	2	3	0
7	EnEg5107	Final Year Project I	EnEg4114	P/F	2	0	0
	Total			17	14	18	1

Fifth Year Second Semester							
S. N	Course Code	Course Title	Prerequisite	Cr.hr.	Lect.	Tut.	Lab.
1	EnEg5102	Industrial Waste Management Technologies	EnEg4102 & EnEg4109	3	2	3	0
2	EnEg5104	Health, Safety and Environment (HSE) Engineering	None	3	2	3	0
3	EnEg5106	Final Year Project II	EnEg5107	6	3	5	2
4	EnEg5108	Climate Change, Adaptation & Mitigation Technologies	EnEg4114	3	2	3	0
5	EnEg52xx	Elective II	None	3	2	3	0
Total				18	11	13	6

N.B: Conversion rate from Cr.hr to ECTS is given as 1Cr.hr. =1.67 ECTS

Electives: I

No	Course code	Course name	Prerequisite
1	EnEg5201	Environmental Biotechnology	EnEg 2102
2	EnEg 5203	Cleaner Production and LCA	None
3	EnEg 5205	Air Quality Modeling and Forecasting	EnEg3104
4	EnEg 5207	Sludge Treatment Technologies	EnEg3106&EnEg4109

Electives: II

No	Course code	Course name	Prerequisite
1	EnEg5202	Ecological risk assessment	EnEg5103
2	EnEg 5204	Green building and energy conservation	None
3	EnEg 5206	Emerging technologies for water and wastewater treatment	EmTe1108

3.4.7 Program Courses Synopsis

3.4.7.1 Core supportive and compulsory courses

Course Code: Math 1014

Course Name: Applied Mathematics IB

Credit Hour: 4

Prerequisite: No

The course is designed to develop students' confidence with mathematical concepts, use of mathematical skills and techniques in a wide range of contexts specifically problem solving and abstract thinking. It covers basic elements of vectors, vector spaces, matrices, determinants, solving systems of linear equations, concepts and applications of differential and integral calculus of one variable.

Course Code: Math2007

Course Name: Applied Mathematics IIB

Credit Hour: 4

Prerequisite: Applied Mathematics IB

The course is designed to develop students' confidence with mathematical concepts, use of mathematical skills and techniques in a wide range of contexts specifically problem solving and abstract thinking. It covers basic elements of sequence and series, power series, differential calculus of several variables and multiple integral concepts and their applications.

Course Code: Math2042

Course Name: Applied Mathematics IIIB

Credit Hour: 4

Prerequisite: Applied Mathematics IIB

This course covers basic elements of ordinary differential equations, Laplace and Fourier transforms, vector differential calculus, Line and surface integral and complex analytic functions.

Course Code: CEng2005

Course Name: Engineering Mechanics I Statics

Credit Hour: 3

Prerequisite: None

Engineering mechanics (Statics) is the application of mechanics (one of the three branches of Physics) to solve problems involving common engineering elements.

Course Code: CEng2002

Course Name: Engineering Mechanics II Dynamics

Credit Hour: 3

Prerequisite: Engineering Mechanics I Statics &), Applied Mathematics IB

Basic equations of motion; Kinematics of particles and rigid bodies; Kinetics of particles and rigid bodies.

Course Code: Stat3102

Course Name: Probability and Statistics

Credit Hour: 3

Prerequisite: Math 1007

This course covers basic elements of statistics, probability, central tendencies, variation, and types of sampling, concepts and applications of estimation and hypothesis testing, regression analysis.

Course Code: MEng2001

Course Name: Engineering Drawing

Credit Hour: 3

Prerequisite: None

The course Engineering Drawing covers introduction and importance of engineering drawing, drawing instruments, drawing standards and conventions; theories of projection: types and systems of projections, projections of points and lines; multi-view drawings; pictorial drawing; auxiliary views and sectional view.

Course Code: Comp2003

Course Name: Introduction to Computer programming

Credit Hour: 3

Prerequisite: None

A programming language is a formal language comprising a set of instructions that produce various kinds of output. As programming involves activities such as analysis, developing understanding, generating algorithms, verification of requirements of algorithms including their correctness and resources consumption, and coding of algorithms, this course teaches students to program using a target programming language. As an introduction, this course gives students an overview of the different concepts of programming and problem solving strategies such as Variables, Reserved words, Syntax Diagram, Constant declarations, Data types, Conditional statements, Loops, Error handling and functions. For each topic, some coding guidelines will be discussed.

Course Code: ChEg2004

Course Name: Thermodynamics

Credit Hour: 3

Prerequisite: None

The course has the objective of fundamental principles equipping the student with the knowledge of the laws of thermodynamics and basic principle of thermodynamics in understanding heat and mass transfer and efficiency process units in process industry.

Course Code: EnEg2102
Course Name: Environmental Microbiology
Credit Hour: 3
Prerequisite: None

The course is desired to provide students a general knowledge on microbes, their environment and the diverse roles of microorganisms in natural and artificial environments. It will cover topics including: Biology of microorganisms, microbial ecology, microbial metabolism, Factors affecting the growth of microbes, determination of microorganisms in the environment and the application of microbiology in environmental engineering.

Course Code: CEng 2002
Course Name: Strength of Materials
Credit Hour: 3
Prerequisite: CEng 2005

This course introduces the properties and strength of materials i.e. Flexure, Shear, Torsion, Compound Stress analysis and Buckling of Compression Members.

Course Code: EnEg3103
Course Name: Environmental Chemistry
Credit Hour: 3
Prerequisite: None

This course was designed to deliver a fundamental understanding of chemical and biochemical processes in aquatic, atmospheric, and soil chemistry. It also gives high emphasis on understand, analyze, and interpret toxic and hazardous chemicals in the major segments of the environment under natural and anthropogenic processes. Additionally, the course intended to provide the opportunity to develop experiences in laboratory sample analyses and instrumental operation techniques. The knowledge (skills) of physicochemical and biochemical principles are expected to be acquired which will be applied for pollution prevention, control, and remediation in the discipline of Environmental Engineering.

Course Code: ChEg3101
Course Name: Material and Energy Balances
Credit Hour: 2
Prerequisite: None

This course will emphasize on basic principles of process engineering discipline and to acquaint students with the fundamentals of process engineering calculations such as material and energy balances. It will also help to develop important problem solving and critical thinking skills.

Course Code: EnEg3105
Course Name: Environmental Engineering Hydrology
Credit Hour: 3
Prerequisite: None

This course is designed to enable students understand how observations of the hydrologic cycle are made and how they can be appropriately used, understand how to control non-point sources of pollution and comprehend how to predict risks and reliabilities of flood control systems

Course Code: EnEg3107
Course Name: Transport phenomenon for environmental engineering
Credit Hour: 3
Prerequisite: None

The course is designed to enable the students to apply the basic concepts of chemical transport phenomenon in the environmental systems. It will cover topics including: Introduction to transport phenomenon in the environment, physical mass transport of chemicals in environment and, Different models in environmental mass transportation.

Course Code: ChEg3103
Course Name: Fluid Mechanics
Credit Hour: 3
Prerequisite: None

This course is designed to commence students to have a basic principles and characteristics of fluid flow in different processes (fluids motion and fluids at rest). It will introduce a student about fluid mechanics, the properties of fluid and types of fluids. It will also discuss on forces (pressure) applied by a fluid at rest or in rigid body motion. Using conservation principles (momentum, mass and energy) the student will examine properties of flow parameters on the fluid flow. Finally, this course will introduce dimensional analysis and the concept of similarity between a model and a prototype and also pipe flow and external flows.

Course Code: CEng 3101
Course Name: Survey
Credit Hour: 3
Prerequisite: None

This course is designed to introduce Civil Engineering students to know basic principles of surveying, accurately measure distances and angles using high precision and up-to-date surveying equipment.

Course Code: EnEg3101
Course Name: Fundamental of Soils and Pollution Control
Credit Hour: 2
Prerequisite: None

This course is designed to provide essential information for students to explain the concepts of environmental soil science, definition of environmental soil science, principle and practice of environmental soil science and soil component. In addition to this, the course is designed to classify the factors and process of soil formation, soil profiles and its horizons and to judge physical, chemical and biological properties of soil

Course Code: ChEg3104
Course Name: Reaction Engineering
Credit Hour: 3
Prerequisite: None

This course will emphasize on the principles of chemical reaction engineering which focuses on reaction classifications; reaction kinetics (rate & mechanism of the reactions and operating conditions); reactor types; developing the performance equation for various reactors and controlling product distribution for multiple reactions.

Course Code: CEng3102
Course Name: Reinforced Concrete Design
Credit Hour: 3
Prerequisite: CEng2002

This course induces the structural analysis and design of reinforced concrete structural members subjected to gravity load.

Course Code: EnEg3104
Course Name: Air pollution engineering I
Credit Hour: 3
Prerequisite: EnEg3103

This course aimed to enable the student to explain air pollution source, effect, concentration in the atmosphere and understand the basic air pollution control Technologies. The course also deigned to model the pollutant dispersion modeling in the atmosphere.

Course Code: EnEg3106
Course Name: Water Treatment Engineering
Credit Hour: 3
Prerequisite: EnEg3103 and EnEg2102

The course covers hydrological cycle, source of water quality analysis, water treatment method, convectional water treatment design (screening , aeration, coagulation, flocculation sedimentation, filtration, and disinfection) and advanced technology for water treatments (membrane filtration, reverse osmosis, ion exchange and elector dialysis).

Course Code: ChEg3102
Course Name: Unit Operation
Credit Hour: 2
Prerequisite: None

This course will provide practical experience on the basic mechanical unit operations ensuring a thorough understanding of principles of unit operation and appropriate theory. The course includes execution of experiments, data analysis and technical report writing and team work. Students will be able to do experiments on size reduction equipment and analysis, particle size distribution using screening, particle movement in a fluid and flow of fluid through a solid bed (Filtration, sedimentation, fluidization etc.)

Course Code: EnEg4103
Course Name: Numerical Analysis for environmental Engineering
Credit Hour: 3
Prerequisite: Math2002

The course aims at introducing students in finding numerical solutions to problems for which analytical solutions either do not exist or are not readily or cheaply obtainable. It enables students to apply linear algebra and calculus. It also aims to helps student develop programming skills.

3.4.7.2 Core course

Course Code: ChEg4101
Course Name: Air pollution Engineering II
Credit Hour: 3
Prerequisite: ChEg3104

Air pollution engineering focuses on mitigating or preventing air pollutant emissions. The course introduces the principles and practice of air pollution source control technology for gaseous air pollutants and particulate matter with emphasis on equipment operation and design parameters.

Course Code: EnEg 4105
Course Name: Engineering Economics
Credit Hour: 3
Prerequisite: Econ2009

This course is designed to enable students understand the basic concepts of engineering economics, the time value of money, the concepts behind benefit-cost analyses, the concept of depreciation, and Judge any designed project on the basis of the cost implications when compared with the expected benefits.

Course Code: IETP4115
Course Name: Integrated Engineering Team Project
Credit Hour: 3
Prerequisite: None

This is a multidisciplinary team integrated engineering project in connection with a special engineering problem and under the guidance of a faculty member. The project synopsis consists of literature review, design, project management, business acumen, multidisciplinary team work, entrepreneurship

Course Code: EnEg 4107
Course Name: Energy and the Environment
Credit Hour: 3
Prerequisite: None

Modern industrial societies are characterized by a significant consumption of fossil, renewable and nuclear fuels to fulfill the energy need for the operation of the physical infrastructure (production of food and water, heating, shelter and transportation, etc.). The amount of energy used, especially in urban areas of industrialized countries, has had a significant environmental impact on both local and regional scales. The 20th century and the recent energy crisis in particular, have a remarkable interest in the topics of energy and environment. Therefore the objectives of the

course are enabling a better understanding of the complex interactions between society, energy needs and production and its effect on the environment.

Course Code: EnEg4109
Course Name: Wastewater Treatment Engineering
Credit Hour: 3
Prerequisite: EnEg2102, and EnEg 3103

This course is designed to expose the students to wastewater Engineering. Topics discussed include Collection and conveyance of sewage, Wastewater flow rates, Hydraulic design of sewers, Sewage pumping, Wastewater characteristics, Unit processes and operations for wastewater treatment, Preliminary treatment, Sedimentation and chemical clarification, Biological treatment (Sewage filtration, Activated sludge process, miscellaneous methods), Treatment and disposal of sludge, Advanced wastewater treatment and Industrial wastewater treatment. For design of wastewater treatment systems, it will cover the design of sedimentation tank, Waste Stabilization Pond, Conventional Activated Sludge, Extended Aeration, Aerated Lagoon and Trickling Filters. At the end of the course, the student should be able to apply the knowledge to design a simple unit operation of wastewater treatment systems and also to modify existing unit operation.

Course Code: EnEg 4102
Course Name: Solid waste Engineering
Credit Hour: 3
Prerequisite: ChEng3101

Solid waste engineering is a course designed to enable students to identify, and analyze the economic importance of solid waste, hierarchies of its management and design the engineered way of landfilling. The course covers the following major themes to produce this outcome. The syllabus covers Economic use of solid waste, Characterization and quantification, Engineering principles of solid waste disposal, Solid waste transformation, Energy recovery and use and, Engineering communication.

Course Code: EnEg4104
Course Name: Sewerage and drainage engineering
Credit Hour: 3
Prerequisite: EnEg 3105

The course is designed to enable the students to apply the basic knowledge and concepts of engineering hydrology, fluid mechanics and open channel hydraulics in the design of sewerage and drainage systems. It will cover topics including: Why sewerage and drainage systems are important, components of sewerage and drainage system, Basic engineering hydrology, hydraulics and open channel flows, design and construction of sewerage systems, and design and construction of storm and irrigation drainage systems.

Course Code: EnEg 4106
Course Name: GIS and Remote sensing
Credit Hour: 3
Prerequisite: None

Course Code: EnEg 4110
Course Name: Waste to Energy Technology
Credit Hour: 3
Prerequisite: EnEg 4107 and EnEg 4102

This course is designed to provide an understanding of the various aspects of Waste to Energy. The various sources of waste generation are analyzed with a focus on its potential for energy production. The need for characterization of wastes will be discussed along with the existing norms for waste utilization for alternate energy source. Various Technological options available for the production of energy form waste will delineated along with economics of using alternate sources.

Course Code: EnEg 4108
Course Name: Software applications for Environmental Engineering
Credit Hour: 3
Prerequisite: Comp 2003

The course is designed to enhance the student's skill on software applications and models; it enables the students to integrate theoretical courses with application software's. It also enables them to use application software's and models to design and predict environmental problems. It will cover topics including: Introduction to software applications and mathematical models for environmental engineering, CAD application, Water CAD, Sewer CAD and Storm CAD, SWAT and Air pollution models.

Course Code: EnEg 4112
Course Name: Environmental Impact Assessment
Credit Hour: 3
Prerequisite: None

This course is designed to enable students to assess environmental impacts of development projects and develop mitigation measures for the impacts identified.

Course Code: EnEg4114
Course Name: Environmental policy and Environmental Management Systems
Credit Hour: 2
Prerequisite: None

This course deals with the current practices and policies pertaining to the development, implementation and maintenance of environmental management systems (EMS) in an organization. ISO standards will be examined and case studies will be used to highlight the key factors involved with EMS implementation. The process of assessing environmental impacts and aspects and implementing continuous improvement measures will be addressed. This course also reviews the Procedures and Guidelines to conduct Environmental Audit.

Course Code: EnEg 4116
Course Name: Internship
Credit Hour: 6
Prerequisite: EnEg 4101; [EnEg4106]; [EnEg4109] ; [EnEg4102]; [EnEg4104]; [EnEg4110]; [EnEg4112], [EnEg4114]; [EnEg4108]

This course offers workplace training and experiences that relate to the student's general and technical course of study in preparation for work in the environmental field, including aspects of the duties, skills, and functions of the sponsoring entity. It is intended to give students a chance of real time industry experience and exposure, which are application of theoretical knowledge covered in class. During this period students are expected to boost practical hands-on experiences, professional/workplace behavior and knowledge, learn company culture, employer expectations. Internship also helps students to explore their career path and opportunity to professional network.

Course Code: EnEg 5111
Course Name: Integrated Environmental Engineering Project Design
Credit Hour: 3
Prerequisite: EnEg4114

An integrated project will assist the students in consolidating their knowledge acquired in various previous modules. The students will apply their knowledge in solving practical and real-time projects in due time. Students will perform a comprehensive design project using a team approach requiring interaction with practitioner's development of a team project report and a formal presentation. The main aim of this project work will let students to acquire knowledge and skills of planning and management of Civil Engineering projects. In addition to this, the objective of the course is to equip students with effective report writing skills and research methodologies. Students shall develop solid technical report and paper/thesis writing skills, analysis and data interpretation techniques, and research methodologies

Course Code: EnEg 5107
Course Name: Environmental Policy and Management Systems
Credit Hour: 3
Prerequisite: None

This course deals with the current practices and policies pertaining to the development, implementation and maintenance of environmental management systems (EMS) in an organization. ISO standards will be examined and case studies will be used to highlight the key factors involved with EMS implementation. The process of assessing environmental impacts and aspects and implementing continuous improvement measures will be addressed. This course also reviews the Procedures and Guidelines to conduct Environmental Audit.

Course Code: EnEg 5103
Course Name: Ecological engineering
Credit Hour: 3
Prerequisite: None

The course designed to deliver the principles of ecological engineering, components and characteristics of Freshwater Ecosystems, the energy and flow in the ecosystem for sustainable ecosystems that integrate human society with its natural environment for the benefit of both. Particular emphasis is placed on application of ecological engineering in the restoration of streams and rivers, lakes and reservoirs and wetlands, as well as treatment wetlands and mined land reclamation

Course Code: EnEg 5105
Course Name: Hazardous and E-waste Engineering
Credit Hour: 2
Prerequisite: EnEg4102

The course syllabus for hazardous and electronic waste engineering is focused on the identification, analysis and synthesis of wastes that are hazardous in nature and contain electronic components in them. It also explores the emerging technologies to treat and recycle different parts of these wastes. The courses covers Sources of waste, Classification of waste, Characterization and Emerging treatment and recycling technologies.

Course Code: EnEg 5109
Course Name: Soil pollution and Remediation technologies
Credit Hour: 3
Prerequisite: EnEg3107 and EnEg 3101

This course deals with soil pollution, technologies and approaches that are used for contaminated soil remediation and restoration. It will introduce the subject of soil pollution from a wide range of sources and human risks

associated with contaminated site remediation. Different remediation technologies and strategies to clean up contaminated soil will be covered.

Course Code: EnEg 5101
Course Name: Renewable Energy Engineering
Credit Hour: 3
Prerequisite: EnEg4107

In global wide renewable energy is one of the best solutions for current global climate deterioration and its effect on human being. Bringing this to this course, knowing renewable energy is, one step being the part of the solution for this problem. The course introduces the engineering aspect of the renewable energy, how we are going to use the energy using the tools of engineering. It introduces the growing field of renewable energy i.e. solar energy, wind energy, bio energy and other alternative energy. In this course students will be introducing on the preliminary design of different renewable energy equipment and they will participate in a combination of interactive classroom instruction and tries to construct simple renewable equipment from the resource at the disposal. As it is known all of the primary energy for electricity in Ethiopia is obtain from renewable energy therefore, this course will give also site for students to see those applications and their procedural follow up before implemented to real case scenario.

Course Code: EnEg 5107
Course Name: Final Year Project-I
Credit Hour: P/F
Prerequisite: EnEg 4114

The course provides students to Prepare a standardized Proposal for their project. The topic for the project is assigned by the department and can be connected to any of the major subjects already taught. The subject of the assignment preferably considers the needs of the country.

Course Code: EnEg 5102
Course Name: Industrial waste management technologies
Credit Hour: 3
Prerequisite: EnEg4102 and EnEg4109

The course is designed to provide students with an understanding of technical issues and the management of wastes how to treat and manage waste and control pollution in an integrated and comprehensive manner, permitting evaluation of benefits of waste minimization to industry and of pollution reduction in the environment.

Course Code: EnEg 5106
Course Name: Final Year Project-II (Main Research)
Credit Hour: 6
Prerequisite: EnEg5107

A prospective graduate student has to prove the fact that he/she can solve a given problem independently and with little input from his mentor. The prospective graduates could apply their knowledge to solve Ethiopian-related or other problems up to the expectation of his mentor in particular and department in general.

Course Code: EnEg 5108
Course Name: Climate change, adaptation and mitigation Technologies
Credit Hour: 3
Prerequisite: EnEg4114

Course Code: EnEg 5104
Course Name: Health, Safety and Environment (HSE) Engineering
Credit Hour: 3
Prerequisite: None

This course is designed to enable students describe the fundamental concepts and principles of occupational health, Safety and environment. It enables students to identify and analyses the major work place and environmental hazards and risks as well as assess risks and design mitigation measures for risks and hazards.

3.4.7.3 Elective courses

Course Code: EnEg 5201
Course Name: Environmental Biotechnology
Credit Hour: 3
Prerequisite: EnEg2102

This course is designed to enable students discusses and analyze the principles and applications of biotechnology in solving to environmental problems.

Course Code: EnEg 5203
Course Name: Cleaner Production and Life Cycle Assessment
Credit Hour: 3
Prerequisite: None

This course is designed to improve the environmental performance of industrial sectors that can enhance competitiveness and profitability. This is a win-win situation for both business and the environment while achieving sustainable development goals. Cleaner production is the technique that reduces or eliminates wastes at the source of pollution from the industry. Similarly, the life cycle assessment is also designed to give an essential overview of environmental prevention. It will help to interpret and evaluate LCA studies that can help to reduce or eliminate waste production. This analytical framework and quantitative techniques are used for systematically and holistically evaluation of environmental trade-offs presented by different alternatives to enable more informed decision-making.

Course Code: EnEg 5205
Course Name: Air quality modeling and forecasting
Credit Hour: 3
Prerequisite: EnEg3104

The course provides the fundamentals of air pollution meteorology and numerical modeling of air quality. It covers the air pollution methodology, Air Pollutants Source Distribution Models for the development, application and evaluation of air quality models. It will provide students with hands-on computer practicing. Different computer modeling techniques for solving major atmospheric processes used in current air quality models

Course Code: EnEg 5207
Course Name: Sludge treatment technologies
Credit Hour: 3
Prerequisite: EnEg2106 and EnEg4109

This course is designed to introduce the students to handling, treating, resource and energy recovery and disposing of sludge produced by wastewater treatment plants. It covers sludge pumping, thickening, conditioning, dewatering, digestion, drying storage and final disposal. Various ways of transforming sludge into useful products has also been described in detail. It also discusses loading parameters for gravity thickeners and operational parameters for air flotation thickeners. This course will provide guidelines for chemical and thermal conditioning of sludge. Finally, it illustrates the design parameters for aerobic, standard and high-rate anaerobic sludge digesters. Advanced sludge treatment (AST) options, including pyrolysis and gasification will also be briefly covered. Finally energy recovery alternatives will be analyzed.

Course Code: EnEg 5202
Course Name: Ecological Risk Assessment
Credit Hour: 3
Prerequisite: EnEg5103

This course is designed to enable students describe the fundamental concepts and principles of ecological risk assessment. The course will cover environmental exposure assessment; ecotoxicology at sub-organism, organism, population and community levels. It also includes human toxicology and risk assessment.

Course Code: EnEg 5204
Course Name: Green Building
Credit Hour: 3
Prerequisite: None

Green buildings offer widespread benefits to a number of building industry stakeholders, including building occupants and society as a whole. Knowing this course will be a solution for environmental issues related to the building. This course introduces the key principles of green building and sustainable development for buildings by examining energy usage, water consumption, options for green building materials and resources selections , selecting sustainable sites, and how site and landscape design can maximize natural benefits and indoor environmental quality. In this course students will be introducing on key trends in building design, like the application of net zero buildings and building in high risk zones then put green building in a broader sustainability context by completing small home and community design project that maximize the sustainability and green building concepts, such as green roof, green wall and Green gardening landscape, by using locally available materials.

Course Code: EnEg 5203
Course Name: Cleaner Production and Life Cycle Assessment
Credit Hour: 3
Prerequisite: None

3.5 Bachelor of Science Degree in Electrical and Computer Engineering

3.5.1 Introduction

This curriculum is developed to cater the demand of innovative, highly skilled, practice oriented, entrepreneur, and ethical manpower in the various fields of Electrical and Computer Engineering in five streams, namely, Communications Engineering, Computer Engineering, Control Engineering, Electronics Engineering, and Power Engineering.

Students will have options to choose their stream/focus area from the above five fields of specializations after successfully completing their fourth year second semester. A successful graduate of the program will not only be globally competent but also have active and decisive role in local industries. It will promote industrialization by motivating staff and students and offering consultancy services for prospective investors and entrepreneurs in the stage of industrial setup.

Moreover, this curriculum has special place in the history of the University in that it is designed according to the Washington Accord as an effort to accredit the entire Engineering program running in the University.

3.5.2 Program Education Objective (PEO)

The program education objectives shall describe accomplishments that the five years program graduates are expected to achieve in the first 3 to 5 years after graduation. The Department of Electrical and Computer Engineering has set the following program education objectives:

PEO1: The graduates will become professional engineers.

PEO2: The graduates will establish their own start-up companies.

PEO3: The graduates will be employed in high-ranking companies and universities locally and internationally.

PEO4: The graduates will be involved in research, design, and development works.

3.5.3 Program Outcome (PO)

Program outcomes are statements that express the knowledge, skills and attitudes of graduates of this program. Apparently, they are extensions and summaries of graduate profiles presented above. For this program twelve Program Outcomes are identified as presented below:

PO1: Apply fundamental knowledge of mathematics, physics and various computational methods to analyze and solve electrical and computer engineering related problems.

PO2: Identify, formulate, research literature and analyze complex electrical and computer engineering problems reaching substantiated conclusions using first principles of mathematics, applied physics and engineering sciences.

PO3: Design solutions for complex engineering problems and systems, components and processes that met specified needs, and develop new solutions with appropriate considerations for public, health, safety, cultural, societal and environmental considerations.

PO4: Conduct experiments, analyze and interpret results, and apply research outcomes to develop new technologies and improve existing ones.

PO5: Create, select and apply appropriate techniques, resources and modern engineering and ICT tools, including prediction and modeling, to complex electrical and computer engineering problems, with an understanding of the limitations.

PO6: Apply contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practices and solutions to engineering problems.

PO7: Understand and evaluate the sustainability and impact of professional electrical and computer engineering works while solving complex engineering problems in societal and environmental contexts.

PO8: Ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments which must consider the impact of engineering solutions in global economic, environmental, and societal contexts.

PO9: Ability to work independently and as a team, and as a leader, motivating professionalism in multidisciplinary settings.

PO10: Communicate effectively, in both written and orally, on complex electrical and computer engineering activities with a variety of audiences.

PO11: Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work as a member and leader in a team, to manage projects in multi-disciplinary environments.

PO12: Recognize the personal, national, and global needs for, and the preparation and ability to engage in, independent and life-long learning in the broadest context of technological dynamism.

3.5.4 Graduation Requirement

Graduation requirement for all undergraduate programs should satisfy the following minimum requirements as stated in the University Senate Legislation Article 109.

- All the required courses/modules and the minimum credit hours set in the program curriculum by the respective academic unit should be satisfied, except to phase in and phase out courses.
- A cumulative grade point average CGPA of 2.00 must be obtained overall.
- A cumulative grade point average CGPA of 2.00 in major area courses.
- Pass the exit exam
- No "F" grade in any course/module taken for undergraduate program.
- Student who fails to graduate due to less CGPA than the required or due to "F" grade have no more chance to upgrade their CGPA or to remove their "F". However, they may be given Certificate of Attendance with their transcript stated as "Graduation Failure"
- Successfully defended his/her B.Sc. Thesis.

List of Courses and Category

Table: List of Common Courses

No	Course Code	Course Name	Course Credit
1	Phil1009	Logic and Critical Thinking	3
2	Psyc1011	General Psychology	3
3	FLEn1003	Communicative English Language Skill I	3
4	GeEs1005	Geography of Ethiopia and the Horn	3
5	Math1007	Mathematics for Natural Science	3
6	SpSc1013	Physical Fitness and Conditioning	P/F
7	Phys1001	General Physics	3
8	FLEn1004	Communicative English Language Skill II	3
9	MCiE1012	Moral and Civic Education	2
10	Incl1010	Inclusiveness	2
11	Anth1002	Social Anthropology	2
12	Econ2009	Economics	3
13	GLTr2011	Global Trend	2
14	Hist2002	History of Ethiopia and the Horn	3
		Total Credit Hours	35

List of Core Compulsory Courses

Table: List of Core Courses

No	Course Code	Course Name	Course Credit
1	EmTe1108	Emerging Technology for Engineers	3
2	Entr1106	Entrepreneurship for Engineers	3
3	Math1014	Applied Mathematics I	4
4	MEng2001	Engineering Drawing	3
5	Comp2003	Introduction to Computer Programming	3
6	CEng2005	Engineering Mechanics I - statics	3

7	MEng2102	Engineering Mechanics-II (Dynamics)	3
8	Math2007	Applied Mathematics II	4
9	Math2042	Applied Mathematics-IIIB	4
10	MEng2114	Engineering Thermodynamics	3
11	ECEg2102	Fundamentals of Electrical Engineering	4
12	ECEg2110	Probability and Random Processes	3
13	ECEg3101	Computational Methods	3
14	ECEg3103	Applied Electronics I	4
15	ECEg3105	Signals and Systems Analysis	3
16	ECEg3107	Electromagnetic Fields	3
17	ECEg3109	Object Oriented Programming	3
18	ECEg3111	Research Methods and Presentations	2
19	ECEg3113	Electrical Workshop Practice I	1
20	ECEg3102	Applied Electronics II	3
21	ECEg3104	Digital Logic Design	4
22	ECEg3106	Network Analysis and Synthesis	3
23	ECEg3108	Digital Signal Processing	4
24	ECEg3110	Electrical Machines I	4
25	ECEg3112	Electrical Workshop Practices II	2
26	ECEg4101	Introduction to Communication Systems	3
27	ECEg4103	Computer Architecture and Organization	3
28	ECEg4105	Introduction to Control Systems	3
29	ECEg4107	Electrical Measurement and Instrumentation	3
30	ECEg4109	Power Systems I	3
31	IETP4115	Integrated Engineering Team Project	3
32	ECEg4102	Microprocessors and Interfacing	4
33	ECEg4112	Integrated Design Project	3
34	ECEg4100	Industry Internship	6
35	IEng5104	Industrial Management and Engineering Economy	3
36	ECEg5108	Final year project II	6

37	ECEg5107	Final year project I	P/F
	Total		119

List of Communications Engineering Focus Courses

Table List of Communications Engineering Focus Courses

№	Course Code	Course Name	Course Credit
1	ECEg4304	Digital Communications System	4
2	ECEg4308	EM Waves and Guide Structures	4
3	ECEg5311	Telecommunication Networks	4
4	ECEg5301	Microwave Devices and Systems	3
5	ECEg5303	Fiber Optics Communications	4
6	ECEg5305	Antennas and Radio Wave Propagations	4
7	ECEg5307	Wireless and Mobile Communications	4
8	ECEg5302	Switching and Intelligent networks	3
Total			30

List of Computer Engineering Focus Courses

Table List of Computer Engineering Focus Courses

№	Course Code	Course Name	Course Credit
1	ECEg4406	Data Communications and Computer Networks	4
2	ECEg5410	Advanced Computer Networks	3
3	ECEg4404	Data Structures and Algorithm	4
4	ECEg4410	Data Base Systems	3
5	ECEg5409	Software Engineering	3
6	ECEg5401	Operating Systems	3
7	ECEg5403	Embedded Systems	4
8	ECEg5405	VLSI Design	3
9	ECEg5407	Introduction to Machine Learning	3
10	ECEg5402	New Trends in Computer Engineering	2

11	ECEg5412	Wireless Communications and Mobile Computing	4
Total			36

Control Engineering Courses

Table: List of Control Engineering Focus Engineering

№	Course Code	Course Name	Course Credit
1	ECEg4510	Modern Control Systems	3
2	ECEg4506	Process Control Fundamentals	3
3	ECEg5509	Industrial Automation	4
4	ECEg5507	Digital Control Systems	3
5	ECEg5511	Robotics and Computer Vision	3
6	ECEg5502	Instrumentation Engineering	3
7	ECEg5510	Artificial Intelligence for Control Engineering	3
8	ECEg5503	Embedded Systems for Control Engineering	3
Total			25

Electronics Engineering Courses

Table: List of Electronics Engineering Focus Courses

№	Course Code	Course Name	Course Credit
1	ECEg5606	Analog Systems Design	3
3	ECEg5609	Optoelectronics	3
4	ECEg5602	Digital System Design	3
5	ECEg5604	IC Technology	3
6	ECEg5605	Microelectronic Devices and Circuits	3
7	EEEg5608	Power Electronics	3
Total			18

Power Engineering Courses

Table: List of Power Engineering Courses

№	Course Code	Course Name	Course Credit
1	ECEg4704	Electrical Machines II	4
2	ECEg5709	Power Systems Automation	4
3	ECEg4708	Power System II	4
4	ECEg5703	Energy Conversion and Rural Electrification	4
5	ECEg5711	Power System Protection	3
6	EEEG5701	Power Electronics and Electric Drives	4
7	ECEg5702	Power Systems Operation and Control	4
8	ECEg5705	Electrical Installation	3
Total			30

Credit Hour Distribution

Table: Credit Hour Distribution of Communication Engineering Focus

Category		Total Cr. Hr	Percentage (%)
1	Core Course (major and supportive course)	116	61.38%
2	Core Elective/focus area course	36	18.51%
3	University requirement (Core)	3 Cr.hr	1.59%
Core Course Total Cr. Hr		154	81.48%
4	National Requirement	35 Cr.hr	18.51%
Total Cr. Hr		190	

Table: Credit Hour Distribution of Computer Engineering Focus

Category		Total Cr. hr	Percentage (%)
1	Core Course (major and supportive course)	116	61.05%
2	Core Elective/focus area course	36	18.95%
3	University requirement (Core)	3 Cr.hr	1.58%
Core Course Total Cr. Hr		155	81.58%
4	National Requirement	35 Cr.hr	18.42%
Total Cr. Hr		190	

Table: Credit Hour Distribution of Control Engineering Focus

Category		Total Cr. hr	Percentage (%)
1	Core Course (major and supportive course)	116	61.05%
2	Core Elective/focus area course	36	18.95%
3	University requirement (Core)	3 Cr.hr	1.58%
Core Course Total Cr. Hr		155	81.58%
4	National Requirement	35 Cr.hr	18.42%
Total Cr. Hr		190	

Table: Credit Hour Distribution of Electronics Engineering Focus

Category		Total Cr. hr	Percentage (%)
1	Core Course (major and supportive course)	116	61.05%
2	Core Elective/focus area course	36	18.95%
3	University requirement (Core)	3 Cr.hr	1.58%
Core Course Total Cr. Hr		155	81.58%
4	National Requirement	35 Cr.hr	18.42%
Total Cr. Hr		190	

Table: Credit Hour Distribution of Power Engineering Focus

Category		Total Cr. hr	Percentage (%)
1	Core Course (major and supportive course)	116	61.7%
2	Core Elective/focus area course	36	18.09%
3	University requirement (Core)	3 Cr.hr	1.6%
Core Course Total Cr. Hr		153	81.38%
4	National Requirement	35 Cr.hr	18.62%
Total Cr. Hr		190	

3.5.5 Program Curriculum Structure: Bachelor of Science in Electrical and Computer Engineering

Table 16: Semester Course Breakdown

First Year, First Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
Phil1009	Logic and Critical Thinking	3	2	3	0
Psyc1011	General Psychology	3	2	3	0
FLEn1003	Communicative English Language Skills I	3	2	3	0
GeES1005	Geography of Ethiopia and the Horn	3	2	3	0
Math1007	Mathematics for Natural Science	3	2	3	0
SpSc1013	Physical Fitness	P/F	0	0	2
Phys1001	General Physics	3	2	0	3
Total		18	12	15	5

First Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
EmT1008	Introduction to Emerging Technology	3	2	3	0
FLEn1004	Communicative English Language Skills-II	3	3	0	0
Math1014	Applied Mathematics IB	4	3	3	0
MCiE1012	Moral and Civic Education	2	2	0	0
Incl1010	Inclusiveness	2	2	0	0
Anth1002	Social Anthropology	2	2	0	0
Entr1006	Entrepreneurship	3	2	3	0
Total		19	16	9	0

Second Year, First Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
Comp2003	Introduction to Computer Programming	3	2	0	3
GLTr2011	Global Trend	2	2	0	0
MEng2001	Engineering Drawing	3	2	0	3
CEng2005	Engineering Mechanics I (Statics)	3	2	3	0
Math2007	Applied Mathematics IIB	4	3	3	0
Econ2009	Economics	3	2	3	0
Total		18	13	9	6

Second Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg2102	Fundamentals of Electrical Engineering	4	2	3	3
MEng2102	Engineering Mechanics-II (Dynamics)	3	2	3	0
Math2042	Applied Mathematics IIIB	4	3	3	0
ECEg2110	Probability and Random Processes	3	2	3	0
MEng2114	Engineering Thermodynamics	3	2	3	0
Hist2002	History of Ethiopia and the Horn	3	2	3	0
Total		17	11	12	6

Third Year, First Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg3101	Computational Methods	3	2	0	3
ECEg3103	Applied Electronics I	4	2	3	3
ECEg3105	Signals and System Analysis	3	2	3	0
ECEg3107	Electromagnetic Fields	3	2	3	0
ECEg3109	Object Oriented Programming	3	2	0	3
ECEg3111	Research Methods and Presentation	2	2	0	0
ECEg3113	Electrical Workshop Practices I	1	0	0	3
Total		19	12	9	12

Third Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg3102	Applied Electronics II	3	2	0	3
ECEg3104	Digital Logic Design	4	2	3	3
ECEg3106	Network Analysis and Synthesis	3	2	3	0
ECEg3108	Digital Signal Processing	4	2	3	3
ECEg3110	Electrical Machines I	4	2	3	3
ECEg3112	Electrical Workshop Practices II	2	1	0	3
Total		20	11	12	15

Fourth Year, First Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg4101	Introduction to Communication Systems	3	2	0	3
ECEg4103	Computer Architecture and Organization	3	2	3	0
ECEg4105	Introduction to Control Systems	3	2	0	3
ECEg4107	Electrical Measurement and Instrumentation	3	2	0	3
ECEg4109	Power Systems I	3	2	0	3
IETP4115	Integrated Engineering Team Project	3	0	0	9
Total		18	10	3	21

Specialization Courses

Communication Engineering

Fourth Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg4102	Microprocessors and Interfacing	4	2	3	3
ECEg4304	Digital Communication Systems	3	2	0	3
ECEg4406	Data Communications and Computer Networks	4	2	3	3
ECEg4308	EM waves and Guide Structures	3	2	3	0
ECEg4112	Integrated Design Project	3	0	0	9
Total		17	8	9	18

Fourth Year, Summer

Course Code	Course Title	Cr. Hrs	Lec	Tut.	Lab
ECEg4100	Industry Internship	6	0	0	18
Total		6	0	0	18

Fifth Year, First Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg5301	Microwave Devices and Systems	3	2	0	3
ECEg5303	Fiber Optics Communications	3	2	0	3
ECEg5305	Antennas and Radio Wave Propagations	4	2	3	3
ECEg5307	Wireless and Mobile Communications	4	3	0	3
ECEg5605	Microelectronic Devices and Circuits	3	2	0	3
ECEg5311	Telecommunication Networks	3	2	3	0
ECEg5107	Final year project I	P/F			
Total		20	12	9	15

Fifth Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg5302	Switching and Intelligent Networks	3	2	0	3
ECEg5410	Advanced Computer Networks	3	2	0	3
IEng5104	Industrial Management and Engineering Economy	3	2	3	0
ECEg5108	Final year project II	6	0	0	18
Total		18	9	0	27

Computer Engineering

Fourth Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg4102	Microprocessors and Interfacing	4	2	3	3
ECEg4404	Data Structures and Algorithm	4	2	3	3
ECEg4410	Database Systems	3	2	0	3
ECEg4112	Integrated Design Project	3	0	0	9
ECEg4406	Data Communications and Computer Networks	4	2	3	3
Total		18	8	9	21

Fourth Year, Summer

Course Code	Course Title	Cr. Hrs	Lec.	Tut.	Lab
ECEg4100	Industry Internship	6	0	0	18
Total		6	0	0	18

Fifth Year, First Semester

Course Code	Course Title	Cr. Hrs	Lec.	Tut	Lab
ECEg5409	Software Engineering	3	2	3	0
ECEg5401	Operating Systems	3	2	0	3
ECEg5403	Embedded Systems	4	2	3	3
ECEg5405	VLSI Design	3	2	0	3
ECEg5407	Introduction to Machine Learning	3	2	0	3
ECEg5511	Robotics and Computer Vision	3	2	0	3
ECEg5107	Final year project I	P/F			
Total		19	12	6	15

Fifth Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg5402	New Trends in Computer Engineering	2	2	0	0
ECEg5412	Wireless Communications and Mobile Computing	4	3	0	3
IEng5104	Industrial Management and Engineering Economy	3	2	3	0
ECEg5108	Final year project II	6	0	0	18
Total		18	10	3	21

Control Engineering

Fourth Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg4510	Modern Control Systems	3	2	0	3
ECEg4704	Electrical Machines II	4	2	3	3
ECEg4506	Process Control Fundamentals	3	2	0	3
ECEg4112	Integrated Design Project	3	0	0	9
ECEg4102	Microprocessors and Interfacing	4	2	3	3
Total		17	8	6	21

Fourth Year, Summer

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
EEeg4100	Industry Internship	6	0	0	18
Total		6	0	0	18

Fifth Year, First Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg5701	Power Electronics and Electric Drives	4	2	3	3
ECEg5705	Electrical Installation	3	2	0	3
ECEg5503	Embedded Systems for Control Engineering	3	2	0	3
ECEg5507	Digital Control Systems	3	2	0	3
ECEg5511	Robotics and Computer Vision	3	2	0	3
ECEg5509	Industrial Automation	4	2	3	3
ECEg5107	Final year project I	P/F			
Total		20	12	6	18

Fifth Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg5502	Instrumentation Engineering	3	2	0	3
ECEg5510	Artificial Intelligence for Control Engineering	3	2	0	3
IEng5104	Industrial Management and Engineering Economy	3	2	3	0
ECEg5108	Final year project II	6	0	0	18
Total		18	9	3	24

Electronics Engineering

Fourth Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg4102	Microprocessors and Interfacing	4	2	3	3
ECEg4304	Digital Communication Systems	3	2	0	3
ECEg4308	EM Waves and Guide Structures	3	2	3	0
ECEg4112	Integrated Design Project	3	0	0	9
ECEg5606	Analog System Design	3	2	3	0
ECEg5608	Power Electronics	3	2	0	3
Total		19	8	9	18

Fourth Year, Summer

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg4100	Industry Internship	6	0	0	18
Total		6	0	0	18

Fifth Year, First Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg5301	Microwave Devices and Systems	3	2	0	3
ECEg5307	Wireless and Mobile Communications	4	3	0	3
ECEg5609	Optoelectronics	3	2	3	0
ECEg5605	Microelectronic Devices and Circuits	3	2	0	3
ECEg5405	VLSI Design	3	2	0	3
ECEg5107	Final year project I	P/F			
Total		16	11	12	9

Fifth Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg5602	Digital Systems Design	3	2	3	0
ECEg5604	IC Technology	3	2	3	0
IEng5104	Industrial Management and Engineering Economy	3	2	3	0
ECEg5108	Final year project II	6	0	0	18
Total		18	9	9	18

Power Engineering

Fourth Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg4510	Modern Control Systems	3	2	0	3
ECEg4704	Electrical Machines II	4	2	3	3
ECEg4102	Microprocessors and Interfacing	4	2	3	3
ECEg4112	Integrated Design Project	3	0	0	9
ECEg4708	Power Systems II	4	2	3	3
Total		18	8	9	21

Fourth Year, Summer

Course Code	Course Title	Cr. Hrs	Lec.	Tut.	Lab
ECEg4100	Industry Internship	6	0	0	18
Total		6	0	0	18

Fifth Year, First Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg5703	Energy Conversion and Rural Electrification	4	2	3	3
ECEg5711	Power System Protection	3	2	0	3
ECEg5701	Power Electronics and Electric Drives	4	2	3	3
ECEg5705	Electrical Installation	3	2	0	3
ECEg5709	Power Systems Automation	4	2	3	3
ECEg5107	Final year project I	P/F			
Total		18	10	9	15

Fifth Year, Second Semester

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
ECEg5502	Instrumentation Engineering	3	2	0	3
ECEg5702	Power Systems Operation and Control	4	2	3	3
IEng5104	Industrial Management and Engineering Economy	3	2	3	0
ECEg5108	Final year project II	6	0	0	18
Total		13	4	6	21

3.5.6 Program Courses Synopsis

Course Code: Math-1051

Course Name: Applied Mathematics IB

Credit Hour: 4

Prerequisite: None

This course covers basic elements of vectors, vector spaces, matrices, determinants, solving systems of linear equations, concepts and applications of differential and integral calculus of one variable.

Course Code: MEng 2101

Course Name: Engineering Drawing

Credit Hour: 3 (1 Lecture, 1 Tutor and 5 Practical)

Prerequisite: None

The course Engineering Drawing covers introduction and importance of engineering drawing, drawing instruments, drawing standards and conventions; theories of projection: types and systems of projections, projections of points and lines; multi-view drawings; pictorial drawing; auxiliary views and sectional view.

Course Code: Comp 2003

Course Name **Introduction to Computer Programming**

Credit Hour: 3

Prerequisite: None

This course, after a brief introduction of what programming languages are and their classifications extends fundamental programming principles and concepts. It specifically deals with variables and constants; scope; operators and operator precedence; comment statements; input and output statements; control structures and arrays; functions and pointers; and structures and files.

Course Code: Meng2104

Course Name Engineering Mechanics-I (Statics)

Credit Hour: 3

Prerequisite: General Physics (Phys-1101)

Engineering mechanics (Statics) is the application of mechanics (one of the three branches of Physics) to solve problems involving common engineering elements.

Course Code: Comp 2003

Course Name Engineering Mechanics II (Dynamics)

Credit Hour: 3

Prerequisite: CEng2103: Engineering Mechanics I (Statics)

Math1051: Applied Mathematics-I

Basic equations of motion; Kinematics of particles and rigid bodies; Kinetics of particles and rigid bodies

Course Code: Math-2011

Course Name Applied Mathematics II

Credit Hour: 4

Prerequisite: Applied Mathematics I (Math 1051)

This course covers basic elements of sequence and series, power series, differential calculus of several variables and multiple integral concepts and their applications.

Course Code: Math-2022

Course Name Applied Mathematics IIIB

Credit Hour: 4

Prerequisite: Applied Mathematics II (Math-2011))

This course covers basic elements of ordinary differential equations, Laplace and Fourier transforms, vector differential calculus, Line and surface integral and complex analytic functions.

Course Code: MEng2105

Course Name **Engineering Thermodynamics**

Credit Hour: 3

Prerequisite Applied Mathematics I (Math-1051)

Fundamental concepts; Pure substances; Vapor pressure curves; Steam tables; Phase diagrams of steam; First law of Thermodynamics: closed and open systems, enthalpy; Second law of Thermodynamics: Reversible and irreversible processes; Carnot cycle; Entropy; Ideal gases and their mixtures; gas-steam mixtures; wet air; psychometric charts and air conditioning process; Vapor power and refrigeration cycles; Air standard cycles;

Course Code: Course Code: ECEg2102

Course Name **Fundamentals of Electrical Engineering**

Credit Hour: 4

Prerequisite Math-1014: Applied Mathematics I for Engineering

In this course we discuss about: Review of Electromagnetic Phenomenon and Variables, Electric Circuit parameters, Circuit Analysis, SteadyState Single Phase AC Circuit Analysis, Transient Circuit Analysis, Introduction to poly phase systems.

Course Code: ECEg2110

Course Name Probability and Random Process

Credit Hour: 3

Prerequisite

Course Code: ECEg3101

Course Name Computational Methods

Credit Hour: 4

Prerequisite Applied Engineering Mathematics III (Math-2022) and C++ Programming (ECEg2108)

The course aims at introducing students in finding numerical solutions to problems for which analytical solutions either do not exist or are not readily or cheaply obtainable. It enables students to apply linear algebra and calculus. It also aims to help student develop programming skills.

Course Code: ECEg3103

Course Name Applied Electronics I

Credit Hour: 4(L 2,P 3,T 3)

Prerequisite Fundamentals of Electrical Engineering (ECEg2102)

The course focuses on discussing basic semiconductor theory, semiconductor diodes and their applications, BJT, FET, Amplifier frequency response, Multistage amplifiers, Power amplifier and Tuned amplifiers.

It also introduces the basic measurements on characteristics of electronic devices and circuits, semiconductor diode characteristics, rectification and filtering, bipolar junction transistor characteristics, transistor biasing and operating point stability, transistor amplifier, frequency response, multistage amplifier, power amplifier and Tuned amplifier.

Course Code: ECEg3105

Course Name **Signals and Systems Analysis**

Credit Hour: 3

Prerequisite Math-2022: Applied Mathematics III

The course aims and enables student to understand and apply the representation, classification, characterization and analysis of signals and systems in time and frequency domains.

Course Code: ECEg3107

Course Name **Electromagnetic Fields**

Credit Hour: 3

Prerequisite ECEg2102: Fundamentals of Electrical Engineering

This course lays the foundation for understanding basic electromagnetic concepts, fundamental governing laws and guiding principles which students can apply in subsequent courses. The course covers: Vector Algebra, Scalar and Vector Fields, Coordinate Systems and Transformation, Vector Calculus, Line, Surface and Volume Integrals, Vector Differential Operator, Electrostatic fields, Coulomb's and Gauss's Laws for static electric charge distributions in vacuum and material bodies, Electrostatic Boundary Value Problems, Poisson's and Laplace's equations, Resistance and Capacitance, Magneto static Fields, Biot-Savart's and Ampere's Laws for steady state current source in free and material space, Magnetic Forces, Materials, and Devices, Faraday's Law, Maxwell's Equations for Static and dynamic (time-varying) Electromagnetic Fields

Course Code: ECEg3109

Course Name **Object Oriented Programming**

Credit Hour: 3

Prerequisite C++ Programming (ECEg2108)

This course is designed to provide the fundamental theories and principles and techniques of object-oriented programming paradigms. Topics to be dealt with are: classes, objects, data abstraction, encapsulation and information-hiding, overloading, subclasses and inheritance, polymorphism, class hierarchies and exception handling. This course also gives chances to students to work on object-oriented programming language during practical session to implement; test and experiment an object oriented paradigm of the program design and implementation.

Course Code: ECEg4111

Course Name **Research Methods and Presentation**

Credit Hour: 2

Prerequisite None

This course deals with discussion of research and development methods. Moreover, it addresses current and emerging research topics in the area of computer engineering. Selection of topics for individual projects depends on the instructors and department. Literature review and analysis, intensive readings, discussions and reports will be the main activities and assessment methods.

Course Code: ECEg3112

Course Name **Electrical Workshop Practice I**

Credit Hour: 1

Prerequisite None

In this course we discuss about: General Safety rules, Common types of Electrical hand Tools, Wiring materials and accessories, Termination, splicing of solid and flexible wires, Soldering; Residential installation which includes Lighting, Power and Signal circuits; Fire Alarm installation, Security camera installation, testing and inspecting these electrical installations.

Course Code: ECEg3102

Course Name **Applied Electronics II**

Credit Hour: 3

Prerequisite ECEg3103-Applied Electronics I

The course focuses on discussing advanced electronics circuits and concepts of feedback system. Discusses the working principles of differential and operational amplifiers. Design and simulate the circuits of differential and operational amplifiers using Multisim. Discuss, design and simulate applications of amplifiers.

Introduces to the wave shaping circuits such as oscillators and Multi vibrators, timers and switching circuits.

Course Code: ECEg3104

Course Name **Digital Logic Design**

Credit Hour: 4

Prerequisite Applied Electronics II (ECEg3102)

Digital Logic Design is a comprehensive study of the principles and techniques of modern digital system. It provides an overview of the principles underlying number systems, arithmetic operations, decimal and alphanumeric codes, Boolean algebra, combinational and sequential circuits. Furthermore, analysis and design of combinational sequential logic system is discussed.

Course Code: ECEg3106

Course Name **Network Analysis and Synthesis**

Credit Hour: 3

Prerequisite ECEg3105: Signals and Systems Analysis

The course covers the techniques of modeling, analysis, design and synthesis of N-Port passive and active and passive electrical filters in a classical and a modern approach

Course Code: ECEg3108

Course Name **Digital Signal Processing**

Credit Hour: 4

Prerequisite ECEg3105: Signals and Systems Analysis

The course mainly deals with digital signal processing: sampling and quantization, A/D and D/A converters, Discrete-time systems convolution, Z-transforms and its Implementation, Digital filter realizations Fast Fourier transforms digital filter design, decimation and interpolation, random signals, and some applications. It also introduce FIR and IIR Filters, Filter Design Methods, Interpolation and Decimation, Adaptive Signal Processing and its application, Image and Speech Processing.

Course Code: ECEg4113

Course Name **Electrical Machines I**

Credit Hour: 4

Prerequisite ECEg3107: Electromagnetic Fields

In this course we discuss about: Principle of magnetics, Transformers, Induction Machine, DC Machine and synchronous Machine

Course Code: ECEg4107

Course Name **Electrical Workshop Practice II**

Credit Hour: 2

Co. requisite ECEg4113 Electrical Machine I

In this course we discuss about: Rewinding of transformers and Stator of three phase Squirrel cage Induction motor; starting and control of 3-phase Induction Motor, Explain and perform Computer maintenance and Repair of electrical apparatus coil (Relay coil)

Course Code: ECEg4101

Course Name **Introduction to Communication Systems**

Credit Hour: 3

Pre requisite ECEg3102: Applied Electronics II

ECEg3105: Signals and System Analysis

ECEg2110: Probability and Random Processes

This course is intended to give a strong background on Communication Systems Engineering. It starts by

introducing basic concepts of communication systems, and it contains in-depth study of analog communication systems such as: AM, PM, and FM modulation/demodulation techniques. It also introduces digital modulation techniques and data communication.

Course Code: ECEg4103

Course Name **Computer Architecture and Organization**

Credit Hour: 3

Pre requisite Digital logic design (ECEg3104)

This course introduces students to the organization and architecture of computer systems, beginning with the standard von Neumann model and then moving forward to more recent architectural concepts. **General Introduction:** Organization and architecture, computer evolution, measuring performance, models of a computer system; **The Central Processing Unit:** Computer arithmetic, Instruction sets, Instruction format and addressing modes, RISC and CISC, Pipelining, The Control Unit (Hardwired and Micro-Programd Implementations); assembly/machine language programming; **Memory Systems:** Classification and hierarchy of Memory systems, Main memory, Cache Memory, Secondary Memory, Other types of memory, Memory Management I/O and interrupts; **Input Output Systems:** Input Output devices, modes of transfer, I/O interface, Techniques used for I/O Operations: Programd, Interrupt-driven, Direct Memory Access.

Course Code: ECEg4153

Course Name **Introduction to Control Systems**

Credit Hour: 3

Pre requisite Computational Method, Applied Electronics I

This course mainly provides students with the knowledge and skills required to the basic concept of control systems, mathematical modeling of physical system, characteristics of feedback and non-feedback system, time and frequency response analysis of control system. At last the students will able to design and analyze the stability of linear control systems.

Course Code: ECEg3110

Course Name **Electrical Measurement and Instrumentation**

Credit Hour: 3

Pre requisite Fundamentals of Electrical Engineering (ECEg2102)

This course mainly provides students with the basic operational principles and limitations of electrical and electronic instruments, sensors and transducers with their application, signal conditioning and conversion, output presentation, and instrument transformer.

Course Code: ECEg4109

Course Name **Power Systems I**

Credit Hour: 3

Pre requisite Electrical Machine I (ECEg4113)

This course introduces and explains fundamentals of electrical power systems design and engineering. Phasors and their application to power systems analysis are reviewed. The concept of the per-unit system is introduced and applied to circuit calculations. Transmission line parameters, their calculation, and transmission line modeling are introduced. Steady-state operation of transmission lines is modeled and investigated. Mechanical design aspect of overhead transmission line discussed. Brief introduction of underground cable construction and calculation of its insulation resistance is explained.

Course Code: ECEg4102

Course Name **Microprocessors and Interfacing**

Credit Hour: 4

Pre requisite Computer architecture and Organization (ECEg4103)

This course is designed to impart in-depth knowledge in the design, programming and organization of microcomputers and interfacing circuits. This course completely covers the popular Intel μ P 8086, which would be a stepping-stone for learning the X 86 families of microprocessors. Fundamentals of Microprocessors and Microcontrollers Architecture; Register level organization; Memory organization; Bus systems; Peripheral hardware organization; Instruction Set; Assembly language programming; C programming; Microcontroller programming starting from the architecture to all its peripherals; Different applications of microcontroller in real world, lower level communication protocols for microcontrollers; Interfacing Fundamentals; Programmable Interrupt Controller (PIC); Programmable Peripheral Interface (PPI).

Course Code: ECEg4112

Course Name **Integrated Design Project**

Credit Hour: 3

Pre requisite Senior standing

This Integrated Design Project is either a group-based (up to 3) or individualized engineering design project to evaluate the overall knowledge of the student(s) before going for the BSc Thesis. If it is an individualized project the student will be evaluated on the basis of his/her understanding, application and presentation of basic computing knowledge and skills. If it is a group-based project each student will also be evaluated its role in the group work and its team work dimensions.

Course Code: ECEg4100

Course Name **Industry Internship**

Credit Hour: 6

Pre requisite Senior standing

The Students must be attached with an industry for at least five months in order to get real world experience in their field of study, which compliments their education. The student and the internship coach at the industry have to submit a report on the attachment program, which is evaluated at the departmental committee with pass or fail grade.

Course Code: ECEg5104

Course Name **Industrial Management and Engineering Economy**

Credit Hour: 3

Pre requisite Economics

Basics of management and functions of management; basics of plant layout, ergonomics and industrial safety; defining forecasting and the use of forecasting techniques; inventory management and control; basics of project management and resource allocation; project crashing and project risks; time value of money and cash flow; present and future worth of investments; basic concepts and computations of depreciation; cost comparison of alternative investments; accounting and budgeting fundamentals.

Course Code: ECEg518

Course Name **Final Year Project**

Credit Hour: 6

Pre requisite All senior standing courses

Success of the project is determined in large part by whether students have adequately solved the problem that they are working on.

3.6 Bachelor of Science Degree in Electromechanical Engineering

3.6.1 Introduction

The availability of sufficient skilled manpower plays a key role in realizing national goals and in accelerating the country's growth. Consistent with the national goals, AASTU aspires to surpass in accredited BSc programs in Science, Technology, and Engineering disciplines which will have a principal contribution to the industrial transformation plan of the nation.

The curriculum for BSc program in Electromechanical Engineering is developed, mainly by taking into consideration the specific objectives of Addis Ababa Science and Technology University to train technically sound and professionally competent engineers who can support the transformational activities of the nation, who can solve the pressing problems of the industry, who are competent in the global skilled labor market.

Recently, advancements in Science and Technology become decisive arrays for survival and competitiveness in the global arena. It was due to this fact that the government decided to open two distinctive Science and Technology Universities. AASTU envisions being a center of excellence in Science and Technology of various Engineering disciplines. Therefore, Launching a BSc program in Electromechanical Engineering helps students to grasp multi-domain state-of-the-art knowledge and skills in the field of study which can help them to actively engage in the national industrial transformation. Moreover, the inauguration of this BSc program in Electromechanical Engineering gives an opportunity for staffs and students to be engaged in practical problem-solving researches and technology transfer practices. In this context, the curriculum is expected to provide the students a lurching and strong background in Electromechanical Engineering.

Complex mechatronic systems of today like airplanes, cars, robot, industrial process machines, medical equipment, and spaceships have many embedded mechanical and electronic systems that monitor and control the behavior to avoid catastrophic failure and improve the performance. With its origin in Japan in late 1960s the term mechatronics was in principle coined to define such control and operation systems. The concept has since spread all over the world and a significant international growth has been observed within the last decades. According to technology review

of MIT press, mechatronics is identified as one of the top 10 technologies that will change the future world. The field is in general viewed as the vehicle by which students and professionals are introduced to and made to comprehend the diverse disciplines (Fig.1) such as mechanical engineering, computer science, control theory, and electrical and electronics engineering areas concurrently. By combining diverse fields, mechatronics curriculum provides sufficient background, knowledge, depth and breadth enabling the new breed of graduates to tackle complex engineering problems the world we are living in.

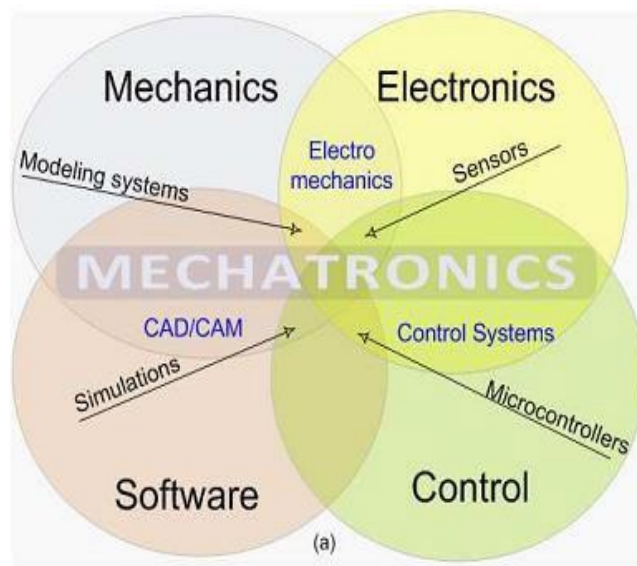


Figure 1. Disciplinary foundations of Mechatronics

In Ethiopia, industry sector is booming and those industries are using state of the art technologies in their production line. Having an industry by itself is nothing unless there is enough skilled manpower with the skill and knowledge they demand. Industries can demand from the basic skill up to engineers who are able to develop and design system level operations. These day's machines and production lines are fully integrated systems (mechanical, electrical and software). In addition to traditional engineers, like other countries around the globe, Ethiopia needs a new breed of engineers and professionals who have multidisciplinary knowledge and skills to tackle complex engineering problems that we are facing these days.

Within the scope of this wide discipline, engineering disciplines closely related to Electromechanical Engineering is recently being offered in few Ethiopian Universities at BSc degree level.

However, with the sweeping development of the manufacturing and construction industrial sectors of the Ethiopian economy and expansion of engineering and technology institutions throughout the country, basic knowledge and applications of Electromechanical Engineering are being highly required. As a stepping-stone towards meeting this growing demand, the College of Electrical and Mechanical Engineering of Addis Ababa Science and Technology have deemed to elevate the level of training to BSc, MSc and PhD degree programs.

3.6.2 Program Educational Objectives (PEO)

Our graduates of the BSc degree in Electromechanical Engineering program are assumed to attain the following career achievements within three to five years of graduation. These objectives are based on the needs of the program's constituencies (government, employers, industries, alumni, students and research and development centers).

Table: Program Education Objectives (PEO)

PEO-1	Specify, design, deploy, implement, troubleshoot and maintain mechatronic systems.
PEO-2	Translate customer requirements and effectively integrate multiple mechanical, electronic, electrical and control systems.
PEO-3	Analyze the produced system and formulate its economic impacts on the overall organization
PEO-4	Communicate effectively in the professional environment in individual and group situations.
PEO-5	Apply appropriate strategies to maintain professional, ethical and social responsibilities in the workplace and include a respect for diversity.
PEO-6	Participate in lifelong learning to stay technically updated in the profession
PEO-7	Work effectively in individual and group-oriented settings by applying safety to all aspects of work.

3.6.3 Program Outcomes (PO)

Upon graduation, our BSc in Electromechanical Engineering program graduates will have the following attributes

PO-1 Engineering Knowledge Select and apply the knowledge, techniques, skills and modern tools in Electromechanical/Mechatronics Engineering

PO-2 Problem analysis Identify, formulate, research literature and analyze complex Electromechanical engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

PO-3 Design /Development of Solutions Design solutions for complex Electromechanical health and safety, cultural, societal, and environmental considerations.

PO-4 Investigation conduct investigations of complex problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

PO-5 Modern Tool usage Ability to create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modeling, to complex engineering activities, with an understanding of the limitations.

PO-6 Environment and sustainability Ability to apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.

PO-7 The engineer and society Ability to understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of and need for sustainable development.

PO-8 Professional Ethics Ability to apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

PO-9 Individual and teamwork Function effectively as an individual and as a member or leader in diverse teams and in multidisciplinary environments engineering problems

and design systems, components or processes that meet specified needs with appropriate consideration for public

PO-10 Communication Ability to communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO-11 Life-long Learning Understand the need for an ability to engage in self-directed continuing professional development

PO-12 Project management and finance Ability to demonstrate knowledge and understanding of engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

3.6.4 Graduation Requirements

- The minimum requirements for successful completion of the program are 188 Cr.Hr. of course work. The student must also complete and successfully defend his/her final year project by providing the originality of the work. The student shall attain a minimum cumulative grade of 2.00 points in major courses as well as in his/her entire study and Pass the exit exam.

3.6.5 List of Courses and Category

Table: Distribution of credit hours for all course composition

Category		Total Cr. hr	Percentage (%)
1	Core Course (major and supportive course)	152	80.42%
2	Core Elective/focus area course	-----	-----
3	University requirement (Core)	3	1.97%
Core Course Total Cr. Hr		155	82%
4	National Requirement	34	18%
Total Cr. Hr		189	

List of Courses

Table: Common Courses/National Requirement

Common Courses		
No.	Course Title	Credit-hour
01	Logic & Critical Thinking	3
02	General Psychology	3
03	Communicative English Skills I	3
04	Geography of Ethiopia and the Horn	3
05	Physical fitness	2
06	Social Anthropology	2
07	Communicative English Skills II	3
08	Moral & Civic education	2
09	Inclusiveness	2
10	Global trend	2
11	Introduction to Economics	3
12	History of Ethiopia and the Horn	3
13	Industrial management & Engineering Economy	3
Total Cr.		34

Table: Core Courses

No.	Course Title	Credit-hour
01	Object oriented programming in python	3
02	Computational Methods	3
03	Introduction to Machine learning	3
04	Mechanism Machinery	3
05	Mechanical Vibrations	3
06	Strength of materials	4
07	Design of Machine Elements I	3
08	Design of Machine Elements II	3
09	Machine drawing with CAD	3
10	Manufacturing Processes & Automation	3
11	Engineering thermodynamics	3
12	Fluid Mechanics	3
13	Hydraulics & Pneumatics	3
14	Fundamental of Electrical Circuits	4
15	Applied Electronics I	3
16	Applied Electronics II	3
17	Electrical Machine	3
18	Embedded systems	3
19	Power Electronics & Drive	4
20	Control systems	3
21	Modern control	3
22	Signal and systems	3
23	Digital signal processing	3
24	Digital Logic Design	3
25	Instrumentation & Measurement	3
26	Virtual instrumentation	3
27	Introduction to robotics	3
28	Introduction to Computer Vision	3
29	Industrial Automation & Process Control	3
30	Design of Mechatronic systems	4
31	Mechanical workshop practice	2
32	Workshop for mechatronics	2

33	Smart Materials & Applications	3
34	Integrated Engineering Team Project	3
35	Industrial internship	6
36	BSc thesis	6
37	Mathematics for Natural Sciences	3
38	General Physics	3
39	Emerging Technology for Engineers	3
40	Applied mathematics I	4
41	Applied mathematics II	4
42	Applied mathematics III	4
43	Probability & Statistics	3
44	Introduction to Computer Programing	3
45	Engineering Drawing	3
46	Engineering Mechanics I :Statics	3
47	Engineering Mechanics II: Dynamics	3
48	Entrepreneurship for Engineers	3
Total Cr.		155

3.6.6 Program Curriculum Structure: Bachelor of Electromechanical engineering

Table: Course Breakdown for Regular Program

Year I, Semester I

Course Code	Course Title	Cr. Hrs	Lec	Tut	Lab
Phil1009	Logic and Critical Thinking	3	2	3	0
Psyc1011	General Psychology	3	2	3	0
FLEn1003	Communicative English Language Skills I	3	2	3	0
GeES1005	Geography of Ethiopia and the Horn	3	2	3	0
Math1007	Mathematics for Natural Science	3	2	3	0
SpSc1013	Physical Fitness	P/F	0	0	2
Phys1001	General Physics	3	2	0	3
Total		18	12	15	5

Year I, Semester II

S.No.	Course code	Course title				
			Cr.hr	Lec.	Tut.	Lab.
1	EmTe2011	Emerging Technology for Engineers	3	3	0	0
2	Anth1002	Social Anthropology	2	2	0	0
3	Entr1106	Entrepreneurship for Engineers	3	3	0	0
4	FLEn1004	Communicative English Skill-II	3	3	0	0
5	Math1014	Applied Mathematics I B	4	3	3	0
6	MCiE1012	Moral and Civic Education	2	2	0	0
7	Incl1010	Inclusiveness	2	2	0	0
Total			19	18	3	-

Year II, Semester I

S.No.	Course code	Course title				
			Cr.hr	Lec.	Tut.	Lab.
1	EMeG2001	Engineering Drawing	3	1	0	4
2	Comp2003	Introduction to Computer Programing	3	2	0	3
3	CEng2005	Engineering Mechanics I :Statics	3	2	3	0
4	Math2007	Applied Mathematics II B	4	3	3	0
5	Econ2009	Economics	3	3	0	0
6	GLTr2011	Global Trend	2	2	0	0
Total			18	13	6	7

Year II, Semester II

S.No.	Course code	Course title				
			Cr.hr	Lec.	Tut.	Lab.
1	MEng2102	Engineering Mechanics II Dynamics	3	2	3	0
2	Math2042	Applied Mathematics IIIB	4	3	3	0
3	Stat2091	Probability& Statistics	3	2	3	0
4	EMeG3107	Fundamental of Electrical Circuits	4	3	3	0
5	Hist2002	History of Ethiopia and the Horn	3	3	0	0
6	EMeG2106	Mechanical Workshop Practice	2	1	0	3
Total			19	13	19	9

Year III, Semester I

S.No.	Course code	Course title				
			Cr.hr	Lec.	Tut.	Lab.
1	EMeg3101	Engineering Thermodynamics	3	2	3	0
2	EMeg2102	Strength of Materials	4	2	3	3
3	EMeg3103	Computational Methods	3	2	0	3
4	MEng3151	Design of Machine Elements I	3	2	3	0
5	EMeg3104	Electrical Machine	3	2	0	3
6	EMeg3105	Workshop for Mechatronics	2	0	0	6
Total			18	11	18	9

Year III, Semester II

S.No.	Course code	Course title				
			Cr.hr	Lec.	Tut.	Lab.
1	EMeg3102	Signals & Systems	3	2	0	3
2	MEng3110	Mechanism of Machinery	3	2	3	0
3	EMeg2104	Object Oriented Programming in Python	3	2	0	3
4	MEng3112	Design of Machine Elements II	3	2	3	0
5	EMeg3106	Applied Electronics I	3	2	0	3
6	EMeg3108	Machine Drawing with CAD	3	1	3	3
Total			19	11	19	15

Year IV, Semester I

S.No.	Course code	Course title				
			Cr.hr	Lec.	Tut.	Lab.
1	MEng4109	Mechanical Vibrations	3	2	3	0
2	EMEG4101	Applied Electronics II	3	2	0	3
3	EMEG4103	Fluid Mechanics	3	2	3	0
4	EMEG4105	Control Systems	3	2	0	3
5	EMEG4107	Digital Signal Processing	3	2	0	3
6	EMEG4108	Smart Materials & Applications	3	2	3	0
Total			18	13	6	12

Year IV, Semester II

S.No.	Course code	Course title				
			Cr.hr	Lec.	Tut.	Lab.
1	EMEG4102	Modern Control System	3	2	0	3
2	EMEG4104	Industrial Automation & Process Control	3	2	0	3
3	IETP4115	Integrated Engineering Team Project	3	0	0	9
4	EMEG4113	Instrumentation & Measurement Systems	3	2	0	3
5	EMEG4110	Power Electronics & Drive	4	3	0	3
6	EMEG4112	Digital Logic Design	3	2	0	3
Total			19	14	0	18

Year IV, Semester Summer

S.No.	Course code	Course title				
			Cr.hr	L	T	P
1	EMeg4111	Industrial Internship	6	0	0	18

Year V, Semester I

S.No.	Course code	Course title				
			Cr.hr	Lec.	Tut.	Lab.
1	EMeg5101	Design of Mechatronic System	4	3	0	3
2	EMeg5103	Virtual Instrumentation	3	2	0	3
3	EMeg5105	Embedded systems	3	2	0	3
4	EMeg5107	Introduction to Robotics	3	2	0	3
5	EMeg5109	Hydraulics & Pneumatics	3	2	0	3
6	EMeg4106	Manufacturing Processes & Automation	3	2	0	3
7	EMeg5113	B.Sc. Thesis Phase I	--	0	0	0
Total			19	11	0	18

Year V, Semester I

S.No.	Course code	Course title				
			Cr.hr	Lec.	Tut.	Lab.
1	EMeg5102	Industrial Management & Engineering Economy	3	2	3	0
2	EMeg5104	Introduction to Machine learning	3	2	0	3
3	EMeg5106	Introduction to Computer Vision	3	2	0	3
4	EMeg5114	B.Sc. Thesis phase II	6	-	-	18
Total			15	6	3	6

3.6.7 Program Courses Synopsis

Course Code: Math-1051

Course Name: **Applied Mathematics IB**

Credit Hour: 4

Prerequisite: None ?

This course covers basic elements of vectors, vector spaces, matrices, determinants, solving systems of linear equations, concepts and applications of differential and integral calculus of one variable.

Course Code: MEng 2101

Course Name: Engineering Drawing

Credit Hour: 3 (1 Lecture, 1 Tutor and 5 Practical)

Prerequisite: None

The course Engineering Drawing covers introduction and importance of engineering drawing, drawing instruments, drawing standards and conventions; theories of projection: types and systems of projections, projections of points and lines; multi-view drawings; pictorial drawing; auxiliary views and sectional view.

Course Code: Comp 2003

Course Name **Introduction to Computer Programming**

Credit Hour: 3

Prerequisite: None

This course, after a brief introduction of what programming languages are and their classifications extends fundamental programming principles and concepts. It specifically deals with variables and constants; scope; operators and operator precedence; comment statements; input and output statements; control structures and arrays; functions and pointers; and structures and files.

Course Code: Meng2104

Course Name Engineering Mechanics-I (Statics)

Credit Hour: 3

Prerequisite: General Physics (Phys-1101)

Engineering mechanics (Statics) is the application of mechanics (one of the three branches of Physics) to solve problems involving common engineering elements.

Course Code: Comp 2003

Course Name Engineering Mechanics II (Dynamics)

Credit Hour: 3

Prerequisite: CEng2103: Engineering Mechanics I (Statics)

Math1051: Applied Mathematics-I

Basic equations of motion; Kinematics of particles and rigid bodies; Kinetics of particles and rigid bodies

Course Code: Math-2011

Course Name **Applied Mathematics II**

Credit Hour: 4

Prerequisite: Applied Mathematics I (Math 1051)

This course covers basic elements of sequence and series, power series, differential calculus of several variables and multiple integral concepts and their applications.

Course Code: EMEg3107

Course Name Fundamental of Electrical Circuits

Credit Hour: 4

Prerequisite: Applied Mathematics III

This course mainly provides students with the knowledge and skills required to identify basic electromagnetic phenomenon, circuit variables and parameters. The students will have the ability to apply the fundamental and derived circuit laws and theorems to the analysis of dc and steady state poly-phase ac circuits.

Course Code: EMEg3101

Course Name **Engineering Thermodynamics**

Credit Hour: 3

Prerequisite Applied Mathematics I

Fundamental concepts; Pure substances; Vapor pressure curves; Steam tables; Phase diagrams of steam; First law of Thermodynamics: closed and open systems, enthalpy; Second law of Thermodynamics: Reversible and irreversible processes; Carnot cycle; Entropy; Ideal gases and their mixtures; gas-steam mixtures; wet air; psychometric charts and air conditioning process; Vapor power and refrigeration cycles; Air standard cycles;

Course Code: EMEg2106

Course Name Mechanical Workshop Practice

Credit Hour: 2

Prerequisite No

Measuring and layout tools; Bench work; Wood work, Metal work, cutting with the aid of conventional machine tools; Operation and maintenance of appliances and machines; Welding, brazing and soldering; Manufacture of parts.

Course Code: EMEg2104

Course Name Object Oriented Programming in Python

Credit Hour: 3

Prerequisite Introduction to Computer Programming

This course will provide a pragmatic and hands-on introduction to the Python programming language, with a focus on practical applications and projects, rather than theoretical topics. Students will design and build software to solve problems from various disciplines. As the course progresses, students will learn to work with packages, data structures, & object-oriented programming.

Course Code: EMEg2102

Course Name Strength of Materials

Credit Hour: 3

Prerequisite Basics of Engineering Mathematics, Engineering mechanics

Concept of stress and strain – stress and strain under axial loading – torsion of circular shafts-shear force and bending moment diagrams – bending stresses in beams – transformation of stress and strain.

1. To know about how a solid (materials, structures) behaves when it is exposed to forces and deformations.
2. To apply the fundamental concepts of principle of superposition, equilibrium, compatibility, force deformation, and stress-strain relationships to the solid and structural mechanics problems.
3. To analyze determinate and indeterminate bars, beams, to determine axial forces, torques, shear forces, and bending moments.

To identify the biaxial stresses in acting in a body or an element. The course also aims to provide practical assignments and projects so as to prepare students for final year graduation project, by enhancing planning and team work skills as well as practical project work and building of prototypes

Course Code: EMEg3103

Course Name Computational Methods

Credit Hour: 3

Prerequisite Applied Mathematics III, Introduction to Computer programming

The course aims at introducing students in finding numerical solutions to problems for which analytical solutions either do not exist or are not readily or cheaply obtainable. It enables students to apply linear algebra and calculus. It also aims to help student develop programming skills.

Course Code: MEng3151

Course Name Design of Machine Elements –I

Credit Hour: 3

Prerequisite Strength of Materials

Introduction to Machine Design, Material properties, Tolerances, Factor of Safety, Design for Static load and Design for Fatigue load, Static Theories of Failure, Fatigue Theories of Failure, Design of Welded, Bolted, Riveted Joints, Design of keys and keyways, Pin Joints, Joints, Design of Springs.

Course Code: MEng3112

Course Name Design of Machine Elements –II

Credit Hour: 3

Prerequisite Design of Machine Elements – I, Strength of Materials

Design of Shafts on the basis of strength and rigidity, Design procedures of couplings, Design of clutches using uniform pressure theory and uniform wear theory, Design of Rolling and sliding contact bearings, Design of Gears drives – Spur, Helical, Bevel and worm, Design of Flat and V belt drives, rope drives and chain drives.

Course Code: EMEg3105

Course Name: Workshop for Mechatronics

Credit Hour: 2

Prerequisite: Introduction to Computer Programming

This course mainly provides students with the knowledge and skills required to perform hand on projects. Students will have the ability to apply practical knowledge to real-world problems.

Course Code: EMEg3105

Course Name Signals & Systems

Credit Hour: 3

Prerequisite Applied mathematics III

The course will focus on how signals can be represented in time domain and how they can be transformed into other domains. The students will also learn about analytical techniques that allow modeling the behavior of the systems and gain an insight into the characteristics of signals. The course will provide skills to model, analyze and design signals and systems in general.

Course Code: MEng3110

Course Name Mechanism of Machinery

Credit Hour: 3

Prerequisite Engineering Mechanics II (Dynamics),Design of Machine Element II

Basics of Kinematic mechanisms – Velocity –Acceleration analysis for different mechanisms – law of gearing – gears- gear trains- balancing of rotating masses- force analysis – vibration analysis – control etc.,

Course Code: EMEg4108

Course Name Smart Materials & applications

Credit Hour: 3

Prerequisite No

The course aims to familiarize students with the structure and physical properties of smart materials used in Mechatronics engineering. New nanostructured active materials are being developed thanks to the advances in the fabrication technologies and to the understanding of frequently non-equilibrium conditions during their synthesis. In this course smart materials properties and structures will be discussed in the broader external conditions for the use with the presentation of selected application technologies and provide the students with the state-of-the-art knowledge on smart materials and smart systems, as well as the techniques used to functionalize common materials. They are now basis of numerous 21st century engineering solutions related to the control of environment, for energy management, for biotechnology, for integrated electronics, for aerospace, and for many others. Particular emphasis will be placed on the application of these materials in the industry. Examples of advanced practical applications will include smart sensors, smart actuators, energy conversion devices, advanced materials for electronics and data storage, etc.

Course Code: EMEg3108

Course Name Machine Drawing with CAD

Credit Hour: 3

Prerequisite Engineering Drawing

Introduction common engineering graphics software, representing of drawing primitives on a computer; hardware and software; Basic commands of drawing and editing, dimensioning and text annotations of a CAD software, Detail and assembly drawings of machine elements, Machine tolerance, allowances and surface texture symbols, 3D mechanical elements and their assemblies, Project works on machine elements and assemblies with CAD software.

Course Code: EMEg3106

Course Name Applied Electronics I

Credit Hour: 3

Prerequisite Fundamentals of Electrical Circuit

The course focuses on discussing basic semiconductor theory, semiconductor diodes and their applications, BJT, FET, Amplifier frequency response, Multistage amplifiers, Power amplifier and Tuned amplifiers.

It also introduces the basic measurements on characteristics of electronic devices and circuits, semiconductor diode characteristics, rectification and filtering, bipolar junction transistor characteristics, transistor biasing and operating point stability, transistor amplifier, frequency response, multistage amplifier, power amplifier and Tuned amplifier.

Course Code: EMEg4101

Course Name Applied Electronics II

Credit Hour: 3

Prerequisite Applied Electronics I

The course focuses on discussing advanced electronics circuits and concepts of feedback system. Discusses the working principles of differential and operational amplifiers. Design and simulate the circuits of differential and operational amplifiers using Multisim. Discuss, design and simulate applications of amplifiers.

Introduces to the wave shaping circuits such as oscillators and Multivibrators, timers and switching circuits.

Course Code: MEng4109

Course Name Mechanical Vibrations

Credit Hour: 3

Prerequisite Mechanisms of Machinery

Introduction to mechanical vibration; Modeling of dynamic systems; Single-degree of freedom System-Two degrees of freedom System-Damped and Undamped Longitudinal Vibrations; Transverse Vibration; Whirling of shafts; Torsional vibrations; Causes of vibrations; Introduction to vibration control and measurements.

Course Code: EMEg4103

Course Name Fluid Mechanics

Credit Hour: 3

Prerequisite Applied mathematics III

This subject aims to enable students to: understand key concepts and fundamental principles, together with the assumptions made in their development, pertaining to fluid behavior, both in static and flowing conditions; deal effectively with practical engineering situations, including the analysis and design of engineering systems and devices involving fluids and flow; appreciate possible applications and links to other disciplines; and engage in further specialized study or research. The subject also aims to enhance interests in fluid phenomena and applications. Topics include: fluid properties and statics; conservation laws of mass, momentum and energy; flow in pipes; external flow (lift and drag); boundary layers; flow measurements; and environmental fluid mechanics.

Course Code: MEng4105

Course Name Control Systems

Credit Hour: 4

Prerequisite Applied Mathematics III, Signals and Systems

This course mainly provides students with the knowledge and skills required to model physical systems in the form of differential equations. Students will have the ability to analyze systems in time and frequency domains and apply those theories to design a compensator and its realization.

Course Code: EMEg4107

Course Name Digital Signal Processing

Credit Hour: 3

Prerequisite Signal and system

This course mainly provides students with the knowledge and skill required to develop mathematical and analytical skills necessary to analyze digital signals both in time and frequency domains. From the system's perspective, the students will have the ability to incorporate extensive design skills which enables

students to develop relevant prototypes with the desired level of accuracy.

Course Code: EMEg4112

Course Name Digital Logic Design

Credit Hour: 3

Prerequisite Applied Electronics II

Digital Logic Design is a comprehensive study of the principles and techniques of modern digital system. It provides an overview of the principles underlying number systems, arithmetic operations, decimal & alphanumerical codes, Boolean algebra, combinational and sequential circuits. Furthermore, analysis and design of combinational sequential logic system is discussed.

Course Code: EMEg4113

Course Name Instrumentation & Measurement Systems

Credit Hour: 3

Prerequisite Applied Electronics II

This course will introduce function, operation, and application of common mechatronics engineering instruments, measurement principles, and statistical analysis applicable for industrial instrumentation as used for troubleshooting, process measurements and process control. Specifically, the course will discuss measurement terminology, differentiating between analog and digital, describe the instrumentation used for electronic and mechanical testing and develop the principles of operation of transducers used for mechatronics system in industrial environment process parameter measurement and control.

The course will deal with the basic knowledge, techniques and components for transduction, conditioning and conversion of measurement signals. Thorough knowledge of elementary measurements principles for selected physical process variables.

The student will select and design simple measurement systems for a given application, based on different physical measurement principles, error analysis, signal conditioning, and data recording.

Course Code: EMEg4102

Course Name Modern Control Systems

Credit Hour: 4

Prerequisite DSP, Control systems

This course mainly provides students with the knowledge and skills required to analyze the mathematical models of simple and complex physical systems. Students will have the ability to find out the solutions and apply different controller design algorithms to real-world problems.

Course Code: EMEg4104

Course Name Industrial Automation and Process Control

Credit Hour: 3

Prerequisite Control systems

This course mainly provides students with the knowledge and skills required to analyze a process in the industry. The students will have the ability to apply different controller types and designing an automated system using PLC and SCADA.

Course Code: EMEg3104

Course Name Electrical Machine

Credit Hour: 3

Prerequisite Fundamentals of Electrical Circuit

In this course we discuss about: Principle of magnetics, Transformers, Induction Machine, DC Machine and synchronous Machine

Course Code: EMEg4110

Course Name Power Electronics & Drive

Credit Hour: 4

Prerequisite Applied Electronics II

This course mainly provides students with the knowledge and skills required to analyze various types of power electronics devices and their application to control circuits. The students will have the ability to apply power electronics to design motor drive circuits to implement feedback control mechanism.

Course Code: EMEg5109

Course Name Hydraulics & Pneumatics

Credit Hour: 3

Prerequisite Fluid Mechanics

Basics of fluid power systems-power generators hydraulic pumps-hydraulic actuators – control components-hydraulic circuits- working principles of pneumatic power system- trouble shooting-design of hydraulic circuits

1. Applying the working principles of fluid power systems and hydraulic pumps.
2. Applying the working principles of hydraulic actuators and control components.
3. Designing and develop hydraulic circuits and systems.
4. Applying the working principles of pneumatic power system and its components.
- 5 Solving problems and troubles in fluid power systems.

Course Code: EMEg4111

Course Name Industrial Internship

Credit Hour: 6

Prerequisite All Core Courses

The purpose of Industrial internship training is to expose students to real work of environment experience and at the same time, to gain the knowledge through hands on observation and job execution. From the industrial training, the students will also develop skills in work ethics, communication, management and others. Moreover, the practical training program allows students to relate theoretical knowledge with its application in manufacturing industry.

Course Code: EMEg5101

Course Name Design of Mechatronic Systems

Credit Hour: 4

Prerequisite Instrumentation measurement systems, Control systems, Modern control , Applied Electronics I , Embedded systems

The course provides the student with general overview of Mechatronic systems, their main components, actuator & transducer selection, development of control system algorithms, selection of physical controllers, and the design procedure for Mechatronic systems.

It mainly allows students to integrate their knowledge of measurement systems, control, electronics, programming and mechanics into designing comprehensive Mechatronic systems.

The students will have the ability to apply unified/integrated design approach to real-world problems of

Mechatronic system design and synthesis.

The course also aims students to provide practical assignments and the project work to prepare students for the final year graduation project, by enhancing planning and team work skills as well as practical project work and the building of prototypes

Course Code: EMEg5105

Course Name Embedded Systems

Credit Hour: 3

Prerequisite Applied Electronics II

This course mainly provides students with the knowledge and skills required to prototype any Industrial automation systems using microcontrollers. The students will master the techniques to interface different peripherals with microcontrollers and use various communication interfaces to build control automation systems from remote location.

Course Code: EMEg5103

Course Name Virtual Instrumentation

Credit Hour: 3

Prerequisite Instrumentation & Measurement

This course mainly provides students with the knowledge and programming skills required to interface instruments and acquire the data from sensors in Industrial mechatronic systems. The students will master the art of Visual and virtual instrumentation techniques.

Course Code: EMEg5107

Course Name Introduction to Robotics

Credit Hour: 3

Prerequisite control system

This course mainly provides students with the knowledge and skills required to analyze different robot functions. The students will have the ability to Use robot for industrial product effectiveness and simplification of human burden.

Course Code: EMEg4106

Course Name Manufacturing Processes and Automation

Credit Hour: 3

Prerequisite Mechanical Workshop Practice

This course provides the students an insight into different manufacturing processes used in the industry. It will provide the student with an introduction to the concepts and technologies of the principal manufacturing processes utilized by industry. Discussion subjects include the manufacturing system and its operating principles, casting, forming, material removal, welding, and advanced manufacturing processes. In addition, basic manufacturing automation concepts like, CNC Programming, Machining Codes, and Computer Assisted Programming.

Course Code: EMEg5106

Course Name Introduction to Computer Vision

Credit Hour: 3

Prerequisite DSP

This course mainly provides students with the knowledge and skills required to analyze digital images and video to extract vital information buried in them. The students will have the ability to apply theories and algorithms for real life applications such as face recognition, video surveillance and visual traffic management

Course Code: IETP4115

Course Name Integrated Engineering Team Project

Credit Hour: 3

Prerequisite

To expose the students to complex engineering problems by way of designing an engineering product/process/system/solution within such constraints as sustainability, cost-effectiveness, safety and environmental impact; to enhance students' skill in project management.

Course Code: EMEg5102

Course Name Industrial Management & Engineering Economy

Credit Hour: 3

Prerequisite

Basic management concepts and industrial organization; Work environment; Plant design; Materials

management; Forecasting techniques; Basics of accounting principles; Project management; Financial evaluation.

Course Code: EMEG5104

Course Name Introduction to Machine Learning

Credit Hour: 3

Prerequisite Object oriented Programming in Python , Probability & Statistics

Machine learning uses interdisciplinary techniques such as statistics, linear algebra, optimization, and computer programming to create automated systems that can sift through large volumes of data at high speed to make predictions or decisions without human intervention. Machine learning as a field is now incredibly pervasive, with applications spanning from business intelligence to security, from analyzing biochemical interactions to structural monitoring of aging bridges, and from emissions to astrophysics, etc. This class will familiarize students with a broad cross-section of models and algorithms for machine learning, and prepare students for research or industry application of machine learning techniques.

The course provides students with an in-depth introduction to two main areas of Machine Learning: supervised and unsupervised. Main models and algorithms for regression, classification & clustering and will be covered. Topics will include linear and logistic regression, probabilistic (Bayesian) inference, SVMs and kernel methods, ANNs, clustering, and dimensionality reduction. To be eligible for this course, prior knowledge of python programming, familiarity with linear algebra, probability theory has to be first fulfilled.

Course Code: EMEg5114

Course Name BSc Thesis

Credit Hour: 6

Prerequisite All courses in the curriculum

The purpose of a thesis is to enable the student to develop deeper knowledge, understanding, capabilities and attitudes in the context of the program of study.

It offers the opportunity to delve more deeply into and synthesize knowledge acquired in previous studies.

It also aims to create awareness on how to use the principles of different courses in a coordinated way and how to apply them in a product design, modification, adaptation, reengineering and development.

3.7 Bachelor of Science Degree in Mechanical Engineering

3.7.1 Introduction

Mechanical Engineering Department of AASTU has been established in 2006 E.C, with the objective of responding to the need for rapid industrialization and the changing societal needs of the country for sustainable development. The department believes in cultivating the full potential of students, and the advancement of all forms of knowledge keeping in pace with international standards of academic quality, including the high skilled employment needs presented by a growing economy operating in global environment.

3.7.2 Program Education Objective (PEO)

The program educational objectives describe accomplishments that graduates of the program are expected to attain within five years after graduation. Graduates are expected to apply their expertise to contemporary problem solving, be engaged professionally, and develop the skill of independent and continuous learning, and will contribute to their organizations through leadership and teamwork. More specifically, the objectives are expertise, engagement, learning, leadership and teamwork.

Table: Program Education Objectives (PEO)

PEO	Statement
PEO-1	Our graduate will be able to pursue further study
PEO-2	Our graduate will be able to solve technical problems of the industry and participate in research and consultation service
PEO-3	Our graduate will be able to acquire industry and project leadership position
PEO-4	Our graduate will be engineering entrepreneur.

3.7.3 Program Outcomes (PO):

Engineering knowledge

PO1: Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization to the solution of complex mechanical engineering problems.

Problem analysis

PO2: Identify, formulate, research literature and analyze complex mechanical engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.

Design/ development of solutions

PO3: Design solutions for complex mechanical engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health, and safety, cultural, societal and environmental considerations.

Investigation

PO4: Conduct investigations of complex mechanical engineering problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

Modern tool usage

PO5: Create, select and apply appropriate techniques, resources and modern engineering and IT tools, including prediction and modeling, to complex mechanical engineering problems, with an understanding of the limitations.

The engineer and society

PO6: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional mechanical engineering practice and solutions to complex mechanical engineering problems.

Environment and sustainability

PO7: Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex mechanical engineering problems in societal and environmental contexts.

Ethics

PO8: Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

Individual and teamwork

PO9: Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.

Communication

PO10: Communicate effectively on complex mechanical engineering activities with the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations and give and receive clear instructions.

Project management and finance

PO11: Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work as a member and leader in a team, to manage projects and in multi-disciplinary environments.

Lifelong learning

PO12: Recognize the need for, and have the preparation and ability to engage in, independent and life-long learning in the broadest context of technological change.

3.7.4 Graduation Requirements

Graduation requirement for all undergraduate programs should satisfy the following minimum requirements as stated in the university's senate legislation July 2017, Article 109.

- All the required courses/modules and the minimum credit hours set in the program curriculum by the respective academic unit should be satisfied, except for phase in and phase out program.
- A minimum cumulative grade point average CGPA of 2.00 must be obtained;
- A minimum cumulative grade point average CGPA of 2.00 in major area courses;
- Pass the exit exam
- No "F" grade in any course taken for undergraduate program;
- Score pass mark for all courses which have Pass/Fail grade.

3.7.5 List of Courses and Category

Table: Distribution of credit hours for all course categories

Category		Total Cr. hr	Percentage (%)
1	Core Course (major and supportive course)	137	73.65
2	Core Elective/focus area course	12	6.45
3	University requirement (Core)	3 Cr.hr	1.61
Core Course Total Cr. Hr		152	81.72
4	National Requirement	34 Cr.hr	18.27
Total Cr. Hr		186	100

Table: List of Courses

Common Courses		
No.	Course Title	Credit-hour
01	Logic and Critical Thinking	3
02	General Psychology	3
03	Communicative English Language Skill – I	3
04	Geography of Ethiopia and The Horn	3
05	Mathematics (For Natural Science)	3
06	General Physics	3
07	Physical Fitness	0
08	Social Anthropology	2
09	Communicative English Language Skill-II	3
10	Moral and Civic Education	2
11	Inclusiveness	2
12	Economics	3
13	Global Trend	2
14	History of Ethiopia and the Horn	3
Total		34

Core Courses		
No.	Course Title	Credit-hour
01	Engineering Mechanics II	3
02	Strength of Materials I	3
03	Engineering Thermodynamics I	3
04	Basic Workshop Practice	2
05	Machine Drawing with CAD	4
06	Engineering Thermodynamics II	3
07	Strength of Materials II	3
08	Basic Electricity and Electronics	3
09	Engineering Materials	4
10	Fluid Mechanics I	3
11	IC Engines	3
12	Introduction to Numerical Methods and FEM	3
13	Machine Elements I	3
14	Manufacturing Process	3
15	Introduction to Mechatronics	3
16	Fluid Mechanics II	3
17	Machine Elements II	3
18	Heat Transfer	3
19	Fluid Power Systems	3
20	Electrical Power and Machines	3
21	Mechanisms of Machinery	3
22	University Requirement Course	3
23	Material Handling Equipment	3
24	Motor Vehicles Engineering	3
25	Refrigeration and Air Conditioning	4
26	Machine Design Project	3
27	Industrial Internship	6
28	Power Plant Engineering	3

29	Mechanical Vibration	3
30	Maintenance of Machinery	3
31	Integrated Design Project	3
32	Final Year Project Phase I	--
33	Industrial Management and Engineering Economy	3
34	Measurement and Instrumentation	3
35	Control Systems Engineering	3
36	Final Year Project Phase II	6
36	Turbomachinery	4
37	Introduction to Emerging Technology	3
38	Entrepreneurship	2
39	Applied Mathematics I	4
40	Engineering Drawing	3
41	Engineering Mechanics I	3
42	Applied Mathematics II	4
43	Applied Mathematics III	4
44	Integrated Engineering Team Project	3
45	Total Quality Management	3
46	Introduction to computer programing	3
47	Probability and Statistics for Engineers	3
Total		152

Automotive Stream Courses		
No.	Course Title	Credit-hour
1	Automotive Electrical Systems	3
2	Automotive Electronics Systems	3
3	Fleet and Transport Management	2
4	Heavy Duty and Construction Equipment	2
5	IC Engines and Motor Vehicles Lab	1
6	Automotive Maintenance	3
Total		14

Manufacturing Stream Courses		
No.	Course Title	Credit-hour
1	Computer Integrated Manufacturing	3
2	Metal Casting Technology	3
3	Machining Technology	3
4	Welding Technology	3
5	Tool and Die Design and Manufacturing	3
6	Engineering Measurement and Metrology	2
7	Modern Manufacturing Technology	3
8	Metal Forming Technology	3
Total		23

Design Stream Courses		
No.	Course Title	Credit-hour
1	Product Design and Development	3
2	Introduction to Tribology	3
3	Rotor Dynamics	3
4	Introduction to Robotics and Automation	3
Total		12

Thermal Stream Courses		
No.	Course Title	Credit-hour
1	Turbomachinery I	3
2	Renewable Energy Systems	3
3	Thermo-Fluid System Design	3
4	Turbomachinery II	2
5	Energy Auditing and Management	3
6	Gas Turbines and Jet Propulsion	3
Total		17

3.7.6 Program Curriculum Structure: Bachelor of Science in Mechanical Engineering

Year I, Semester I

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
Phil 1012	Logic and Critical Thinking	3	3	0	0
Psyc 1011	General Psychology	3	3	0	0
FLEn 1011	Communicative English Language Skill – I	3	3	0	0
GeES 1005	Geography of Ethiopia and The Horn	3	3	0	0
Math 1007	Mathematics (For Natural Science)	3	3	0	0
Phys1001	General Physics	3	3	0	0
SpSc 1013	Physical Fitness	0	0	0	2
Total		18	18	0	2

Year I, Semester II

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
EmTe 1112	Introduction to Emerging Technology	3	3	0	0
Entr.1106	Entrepreneurship	2	2	0	0
Anth1011	Social Anthropology	2	2	0	0
FLEn 1012	Communicative English Language Skill-II	3	3	0	0
Math1102	Applied Mathematics I	4	3	3	0
MCiE1012	Moral and Civic Education	2	2	0	0
INCL1010	Inclusiveness	2	2	0	0
Total		18	17	3	0

Year II, Semester I

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng2101	Engineering Drawing	3	1	1	5
Comp2103	Introduction to computer programming	3	2	0	3
CEng2105	Engineering Mechanics I	3	2	3	0
Math2101	Applied Mathematics II	4	3	3	0
Econ2009	Economics	3	3	0	0
GLTr2011	Global Trend	2	2	0	0
Total		18	13	7	8

Year II, Semester II

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng2102	Engineering Mechanics II	3	2	3	0
MEng2104	Strength of Materials I	3	2	3	0
Math2142	Applied Mathematics III	4	3	3	0
MEng2106	Engineering Thermodynamics I	3	2	3	0
MEng2108	Basic Workshop Practice	2	1	0	3
MEng2110	Machine Drawing with CAD	4	1	3	6
Total		19	11	15	9

Year III, Semester I

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng3101	Engineering Thermodynamics II	3	2	3	0
MEng3103	Strength of Materials II	3	2	3	0
ECEg3101	Basic Electricity and Electronics	3	2	0	3
MEng3105	Engineering Materials	4	3	0	3
MEng3107	Fluid Mechanics I	3	2	3	0
Hist2002	History of Ethiopia and the Horn	3	3	0	0
Total		19	14	9	6

Year III, Semester II

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng3102	IC Engines	3	2	0	3
MEng3104	Introduction to Numerical methods and FEM	3	2	2	1
MEng3106	Machine Elements I	3	2	3	0
MEng3108	Manufacturing Processes	3	2	3	0
MCEg3110	Introduction to Mechatronics	3	2	3	0
MEng3112	Fluid Mechanics II	3	2	2	1
Total		18	12	13	5

Year IV, Semester I

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng4101	Machine Elements II	3	2	3	0
MEng4103	Heat Transfer	3	2	3	0
MEng4105	Fluid Power Systems	3	2	1	2
ECEg4107	Electrical Power and Machines	3	2	0	3
MEng4109	Mechanisms of Machinery	3	2	3	0
IETP4115	Integrated Engineering Team Project	3	1	0	6
Total		18	12	10	8

Mechanical Core Stream

Year IV, Semester II (Mechanical Core Stream)

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng4102	Material Handling Equipment	3	2	3	0
MEng4404	Computer Integrated Manufacturing	3	2	1	2
MEng4106	Motor Vehicles Engineering	3	2	1	2
MEng4606	Turbomachinery I	3	2	2	1
MEng4110	Refrigeration and Air Conditioning	4	2	3	3
MEng4112	Machine Design Project	3	0	0	9
Total		19	10	11	16

Year IV, Summer Semester (Term III)

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng4114	Industrial Internship	6	0	0	18
Total		6	0	0	18

Year V, Semester I (Mechanical Core Stream)

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng5101	Power Plant Engineering	3	2	3	0
MEng5103	Mechanical Vibration	3	2	3	0
MEng5105	Maintenance of Machinery	3	2	3	0
MEng5607	Turbomachinery II	2	1	3	0
MEng5309	Heavy Duty and Construction Equipment	2	2	0	0
MEng5111	Integrated Design Project	3	0	0	9
MEng5113	Final Year Project Phase I	--	0	3	0
Stat5115	Probability and Statistics for Engineers	3	2	3	0
Total		19	11	18	10

Year V, Semester II (Mechanical Core Stream)

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
IEng5102	Industrial Management and Engineering Economy	3	3	0	0
MEng5104	Measurement and Instrumentation	3	2	3	0
MEng5106	Control Systems Engineering	3	2	3	0
MEng5108	Final Year Project Phase II	6	0	0	18
Total		15	07	06	18

Automotive Stream

Year IV, Semester II (Automotive Stream)

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng4102	Material Handling Equipment	3	2	3	0
MEng4106	Motor Vehicles Engineering	3	2	1	2
MEng4108	Turbomachinery	4	3	2	1
MEng4110	Refrigeration and Air Conditioning	4	2	3	3
MEng4310	Automotive Electrical Systems	3	2	0	3
MEng4112	Machine Design Project	3	0	0	9
Total		20	11	09	18

Year IV, Summer Semester (Term III)

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng4114	Industrial Internship	6	0	0	18
Total		6	0	0	18

Year V, Semester I (Automotive Stream)

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
Stat 5115	Probability and Statistics for Engineers	3	2	3	0
MEng5103	Mechanical Vibration	3	2	3	0
MEng5305	Automotive Electronics Systems	3	2	0	3
MEng5307	Fleet and Transport Management	2	2	0	0
MEng5309	Heavy Duty and Construction Equipment	2	2	0	0
MEng5311	IC Engines and Motor Vehicles Lab	1	0	0	3
MEng5111	Integrated Design Project	3	0	0	9
MEng5113	Final Year Project Phase I	--	0	3	0
Total		17	10	06	15

Year V, Semester II (Automotive Stream)

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng5102	Industrial Management and Engineering Economy	3	3	0	0
MEng5104	Measurement and Instrumentation	3	2	3	0
MEng5106	Control Systems Engineering	3	2	3	0
MEng5308	Automotive Maintenance	3	2	0	3
MEng5108	Final Year Project Phase II	6	0	0	18
Total		18	09	06	21

Manufacturing Stream

Year IV, Semester II (Manufacturing Stream)

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng4102	Material Handling Equipment	3	2	3	0
MEng4404	Computer Integrated Manufacturing	3	2	1	2
MEng4406	Metal Casting Technology	3	2	0	3
MEng4402	Machining Technology	3	2	0	3
Stat5115	Probability and Statistics for Engineers	3	2	3	0
MEng4112	Machine Design Project	3	0	0	9
Total		18	10	07	17

Year IV, Summer Semester (Term III)

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng4114	Industrial Internship	6	0	0	18
Total		6	0	0	18

Year V, Semester I (Manufacturing Engineering Stream)

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng5411	Engineering Measurement and Metrology	2	1	0	3
MEng5403	Welding Technology	3	2	0	3
MEng5105	Maintenance of Machinery	3	2	3	0
MEng5103	Mechanical Vibration	3	2	3	0
MEng5405	Tool and Die Design and Manufacturing	3	2	2	1
MEng5111	Integrated Design Project	3	0	0	9
MEng5113	Final Year Project Phase I	--	0	3	0
Total		17	09	08	16

Year V, Semester II (Manufacturing Engineering Stream)

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng5102	Industrial Management and Engineering Economy	3	3	0	0
MEng5404	Modern Manufacturing Technology	3	2	3	0
MEng5106	Control Systems Engineering	3	2	3	0
MEng5408	Metal Forming Technology	3	2	3	0
MEng5108	Final Year Project Phase II	6	0	0	18
Total		18	09	8	19

Mechanical Design Stream

Year IV, Semester II (Mechanical Design Stream)

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng4102	Material Handling Equipment	3	2	3	0
Stat5115	Probability and Statistics for Engineer	3	2	3	0
MEng4106	Motor Vehicles Engineering	3	2	1	2
MEng4108	Turbomachinery	4	3	2	1
IEng4110	Total Quality Management	3	2	3	0
MEng4112	Machine Design Project	3	0	0	9
Total		19	11	09	14

Year IV, Summer Semester (Term III)

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng4114	Industrial Internship	6	0	0	18
Total		6	0	0	18

Year V, Semester I (Mechanical Design Stream)

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng5501	Product Design and Development	3	2	3	0
MEng5103	Mechanical Vibration	3	2	3	0
MEng5105	Maintenance of Machinery	3	2	0	3
MEng5503	Introduction to Tribology	3	2	3	0
MEng5505	Rotor Dynamics	3	2	3	0
MEng5111	Integrated Design Project	3	0	0	9
MEng5113	Final Year Project Phase I	--	0	3	0
Total		18	10	12	09

Year V, Semester II (Mechanical Design Stream)

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng5102	Industrial Management and Engineering Economy	3	3	0	0
MEng5504	Introduction to Robotics and Automation	3	2	3	0
MEng5106	Control Systems Engineering	3	2	3	0
MEng5108	Final Year Project Phase II	6	0	0	9
Total		15	10	12	09

Thermal Stream

Year IV, Semester II (Thermal Stream)

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng4102	Material Handling Equipment	3	2	3	0
MEng4106	Motor Vehicles Engineering	3	2	1	2
MEng4110	Refrigeration and Air Conditioning	3	2	3	1
MEng4606	Turbomachinery I	3	2	2	1
MEng4602	Renewable Energy Systems	3	2	3	0
Stat5115	Probability and Statistics for Engineer	3	2	3	0
Total		18	12	15	4

Year IV, Summer Semester (Term III)

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng4114	Industrial Internship	6	0	0	18
Total		6	0	0	18

Year V, Semester I (Thermal Stream)

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
MEng5101	Power Plant Engineering	3	2	3	0
MEng5603	Thermo-Fluid System Design	3	2	0	3
MEng5607	Turbomachinery II	2	1	3	0
MEng5601	Energy Auditing and Management	3	2	3	0
MEng5111	Integrated Design Project	3	0	0	9
MEng5103	Mechanical Vibration	3	2	3	0
MEng5105	Maintenance of Machinery	3	2	3	0
MEng5113	Final Year Project Phase I	-	0	3	0
Total		20	11	15	12

3.7.7 Program Courses Synopsis

Course Code: Math-1051

Course Name: Applied Mathematics IB

Credit Hour: 4

Prerequisite: None ?

This course covers basic elements of vectors, vector spaces, matrices, determinants, solving systems of linear equations, concepts and applications of differential and integral calculus of one variable.

Course Code: MEng 2101

Course Name: Engineering Drawing

Credit Hour: 3 (1 Lecture, 1 Tutor and 5 Practical)

Prerequisite: None

The course Engineering Drawing covers introduction and importance of engineering drawing, drawing instruments, drawing standards and conventions; theories of projection: types and systems of projections, projections of points and lines; multi-view drawings; pictorial drawing; auxiliary views and sectional view.

Course Code: Comp 2003

Course Name Introduction to Computer Programming

Credit Hour: 3

Prerequisite: None

This course, after a brief introduction of what programming languages are and their classifications extends fundamental programming principles and concepts. It specifically deals with variables and constants; scope; operators and operator precedence; comment statements; input and output statements; control structures and arrays; functions and pointers; and structures and files.

Course Code: Meng2104

Course Name Engineering Mechanics-I (Statics)

Credit Hour: 3

Prerequisite: General Physics (Phys-1101)

Engineering mechanics (Statics) is the application of mechanics (one of the three branches of Physics) to solve problems involving common engineering elements.

Course Code: Comp 2003

Course Name Engineering Mechanics II (Dynamics)

Credit Hour: 3

Prerequisite: CEng2103: Engineering Mechanics I (Statics)

Math1051: Applied Mathematics-I

Basic equations of motion; Kinematics of particles and rigid bodies; Kinetics of particles and rigid bodies

Course Code: Math-2011

Course Name Applied Mathematics II

Credit Hour: 4

Prerequisite: Applied Mathematics I (Math 1051)

This course covers basic elements of sequence and series, power series, differential calculus of several variables and multiple integral concepts and their applications.

Course Code: MEng2104

Course Name Strength of Materials I

Credit Hour: 3

Prerequisite: Applied Mathematics I (Math 1102)

Concept of stress and strain – stress and strain under axial loading – torsion of circular shafts-shear force and bending moment diagrams – bending stresses in beams – transformation of stress and strain.

Course Code: Math2142

Course Name Applied Mathematics IIIB

Credit Hour: 4

Prerequisite: Math2101 - Applied mathematics IIB

This course covers basic elements of ordinary differential equations, Laplace and Fourier transforms, vector differential calculus, Line and surface integral and complex analytic functions

Course Code: Math2142

Course Name Engineering Thermodynamics I

Credit Hour: 3

Prerequisite: Applied Mathematics I (Math 1102)

Thermodynamic notions and systems; Fundamental concepts; Pure substances; Vapor pressure curves; Steam tables; Phase diagrams of steam; Work and Heat; First law of Thermodynamics: closed and open systems, enthalpy; Second law of Thermodynamics: Reversible and irreversible processes; Carnot cycle; Entropy; Availability; Irreversibility

Course Code: MEng2108

Course Name Basic Workshop Practice

Credit Hour: 2

Prerequisite: Applied Mathematics I (Math 1102)

Measuring and layout tools; Bench work; Wood work, Metal work, cutting with the aid of conventional machine tools; Operation and maintenance of appliances and machines; Welding, brazing and soldering; Manufacture of parts.

Course Code: MEng2110

Course Name Machine Drawing with CAD

Credit Hour: 4

Prerequisite: MEng2101 -Engineering Drawing

Introduction common engineering graphics software, representing of drawing primitives on a computer; hardware and software; Basic commands of drawing and editing, dimensioning and text annotations of a CAD software, Detail and assembly drawings of machine elements, Machine tolerance, allowances and surface texture symbols, 3D mechanical elements and their assemblies, Project works on machine components and assemblies with CAD software.

Course Code: MEng3101

Course Name Engineering Thermodynamics II

Credit Hour: 3

Prerequisite: Math1102 - Applied Mathematics I

MEng2106 - Engineering Thermodynamics

Fundamental concepts like Ideal gases and their mixtures, gas-steam mixtures, wet air, psychometric charts and air conditioning process. Vapor power and refrigeration cycles. Air standard cycles. Thermodynamic relations. Combustion. Phase equilibrium.

Course Code: MEng3103

Course Name Strength of Materials II

Credit Hour: 3

Prerequisite MEng2104 – Strength of Materials I

Concept of stress and strain – stress and strain under axial loading – torsion of circular shafts-shear force and bending moment diagrams – bending stresses in beams – transformation of stress and strain.

Course Code: ECEg3101

Course Name Basic Electricity and Electronics

Credit Hour: 3

Prerequisite

Course Code: MEng3105

Course Name Engineering Materials

Credit Hour: 4

Prerequisite

Fundamental concepts like Basic methods of iron and steel production, Properties and applications of steels and alloyed steels, Heat treatment process, Crystalline Structure and Defect in Crystals, Mechanical Properties, Properties and applications of different cast irons

Causes of corrosion and their protection, Properties and applications of polymer

Course Code: MEng3107

Course Name Fluid Mechanics I

Credit Hour: 3

Prerequisite Math2142 - Applied Mathematics III

Introduction to Fluid Mechanics; Hydrostatics pressure in Fluids; Flow Classification; Properties of flows; Reynolds Transport theorem, Differential Relations for fluid flow, Viscous fluid flows, Newtonian flows; Boundary Conditions, Stream functions, Vortices

Course Code: MEng3102

Course Name IC Engines

Credit Hour: 3

Prerequisite MEng3101 - Engineering Thermodynamics II

To impart the basic engine types, working principle, emission and alternative fuels skills to the learners such that there is seamless transition to advanced concepts

Course Code: MEng4106

Course Name Motor Vehicle Engineering

Credit Hour: 3

Prerequisite

Introduction; Pneumatic tires and wheel; Suspension systems;

Vehicle stability; Power train; Vehicle road performance; Braking system; Steering system; Vehicle Frame Construction; Vehicle body construction and detail; Vehicle body aerodynamics

Course Code: MEng3104

Course Name Introduction to Numerical Methods and FEM

Credit Hour: 3

Prerequisite Comp2103 - Introduction to computer programing,

Math2142 -Applied Mathematics III,

This course provides an introduction to Numerical Methods and Finite Element Methodsto solution of equations, interpolation and data fitting, numerical differentiation and integration, solution of ordinary differential equations, Introduction to FEM, comparison with other discretization methods, Global stiffness matrix, shape function, boundary conditions, plane truss problem, two-dimensional elements, Isoparametric elements, review of elasticity, beams and frames, dynamic problems, steady state heat transfer, potential flow problem, torsion of shafts.

Course Code: MEng3106

Course Name Design of Machine Elements I

Credit Hour: 3

Prerequisite MEng3103 - Strength of Materials

Introduction to Machine Design, Material properties, Tolerances, Factor of Safety, Design for Static load and Design for Fatigue load, Static Theories of Failure, Fatigue Theories of Failure, Design of Welded, Bolted, Riveted Joints, Design of keys and keyways, Pin Joints, Joints, Design of Springs.

Course Code: MEng3108

Course Name Manufacturing Processes

Credit Hour: 3

Prerequisite MEng2108 - Basic workshop Practice

Introduction and classification of manufacturing processes; Metal Casting Processes: Introduction to Metal Casting, Fundamentals of metal casting, Patterns: Types and various Pattern Materials, Molding processes, Casting Processes and Design, Defects in casting; Metal forming processes: Fundamentals of metal-forming process: Bulk metal forming processes, fundamentals of sheet metal processes; Metal removal Processes: Machining, Fundamentals of Metal Cutting, machining processes: Turning, Drilling, Milling and Shaping, Abrasive machining process, Machinability and Economics of metal cutting; Joining processes: Fundamentals of welding, Arc welding, Oxyacetylene welding and Solid state welding; Non-traditional machining processes

Course Code: MCng3110

Course Name Introduction to Mechatronics

Credit Hour: 3

Prerequisite Math2142 - Applied Mathematics III

ECEg3101 - Basic Electricity and Electronics

Introduction, Sensors and Transducers, Actuation Systems, Digital and Control systems, Recent advances

Course Code: MEng3112

Course Name Fluid Mechanics II

Credit Hour: 3

Prerequisite MEng3107 - Fluid Mechanics I

Dimensional analysis, Similitude, Boundary layer, shock waves, speed of sound, Normal shock, duct flow with friction, plane potential flow, aerofoil theory

Course Code: MEng4101

Course Name Machine Elements II

Credit Hour: 3

Prerequisite MEng3106 - Machine Elements I,

MEng3103 - Strength of Materials II

Design of Shafts on the basis of strength and rigidity, Design procedures of couplings, Design of clutches using uniform pressure theory and uniform wear theory, Design of Rolling and sliding contact bearings, Design of Gears drives – Spur, Helical, Bevel and worm, Design of Flat and V belt drives, rope drives and chain drives.

Course Code: MEng4103

Course Name Heat Transfer

Credit Hour: 3

Prerequisite MEng3101-Engineering Thermodynamics II

Math2142- Applied Mathematics III

Steady heat conduction: One and two dimensional applications; Analytical and numerical solutions; One dimensional transient heat conduction: Analytical, numerical and graphical solutions; Convective heat transfer: Forced and natural with laminar and turbulent flows; Boiling and condensation heat transfer coefficients; Dimensionless parameters; Radiation: Basic definitions; Black body radiation; Radiation of technical surfaces in the presence of absorbing and emitting gases; Heat exchangers: parallel, counter and cross flow.

Course Code: MEng4402

Course Name Machining Technology

Credit Hour: 3

Prerequisite MEng2108 – Basic Workshop Practice

MEng3108 - Manufacturing Processes

This course covers introduction to machining technology, metal cutting saws, drilling machines and operation, lathe machines and operations, shaper machines and operations, milling machines and operations, grinding machines, planning and slotting machines, boring machines and operations, broaching machines and operations & metal cutting experiments.

Course Code: MEng4105

Course Name Fluid Power Systems

Credit Hour: 3

Prerequisite MEng3112-Fluid Mechanics II &

MEng2102-Engineering Mechanics II

Introduction to Principles of Hydraulics and Pneumatics; Components and Design of Hydraulic and Pneumatic Systems; Electrical and Electronic Control Devices

Course Code: MEng4606

Course Name Turbomachinery I

Credit Hour: 3

Prerequisite MEng3112 - Fluid Mechanics II

MEng3101 - Engineering Thermodynamics II

This course will cover the basics of turbomachines for use with liquid (incompressible fluid). It will lead the student from the basic knowledge of fluid mechanics and thermodynamics to being able to perform the preliminary design of turbomachines, understand their wide applications in households, process industry and energy generation. Types and design of turbomachines, thermodynamics and fluid mechanics considerations, performance of turbomachines, velocity triangles, Euler turbomachinery equation, degree of reaction and its effect on efficiency, as well as topics on cavitation and water hammering, energy cascade will be discussed.

Course Code: ECEg4107

Course Name Electrical Power and Machines

Credit Hour: 3

Prerequisite ECEg3101 - Basic Electricity and Electronics

In this course we discuss about: Principle of magnetics, Transformers, Induction Machine, DC Machine and synchronous Machine

Course Code: IETP4115

Course Name Integrated Engineering Team Project

Credit Hour: 3

Prerequisite

This is a multidisciplinary team integrated engineering project in connection with a special engineering problem and under the guidance of a faculty member. The project synopsis consists of literature review, design, project management, business acumen, multidisciplinary team work, entrepreneurship

Course Code: MEng4109

Course Name Mechanisms of Machinery

Prerequisite MEng3101- Engineering Thermodynamics II

MEng3112 - Fluid Mechanics II

Credit Hour: 3

Prerequisite MEng2102 - Engineering Mechanics II

Basics of Kinematic mechanisms – Velocity –Acceleration analysis for different mechanisms, Cam Design –Gear Trains– gears- gear trains- balancing of rotating masses- force analysis, gyroscopes, Introduction to Computer Methods

Course Code: MEng5411

Course Name Engineering Measurement and Metrology

Credit Hour: 2

Prerequisite

This course includes introduction to metrology; error measurement and calibrations for various measuring instruments; limit fits and tolerances surface measurement; screw thread measurements, interferometry, Comparator, computer aided metrology and co-ordinate measuring machine.

Course Code: MEng4602

Course Name Renewable Energy Systems

Credit Hour: 3

Prerequisite MEng4103 - Heat Transfer

MEng3101- Engineering Thermodynamics II

MEng3112 - Fluid Mechanics II

Introduction to Renewable Energy Technology, Solar Energy, Solar Thermal Energy applications, Photovoltaic, Biomass Energy, biomass characterization Biomass Conversion, Wind Power, Hydropower Energy, Small and medium scale hydropower plants, Geothermal Energy

Course Code: MEng4102

Course Name Material Handling Equipment

Credit Hour: 3

Prerequisite MEng3101- Engineering Thermodynamics II

MEng3112 - Fluid Mechanics II

Prerequisite

Introduction; Main types and major classifications of material handling equipment; Basic principles of materials handling; Unit load concept; Basic considerations for design and selection of material handling equipment, Design of conveyers;

Course Code: MEng5601

Course Name Energy Auditing and Management

Credit Hour: 3

Prerequisite MEng3101 - Engineering Thermodynamics II,

MEng4103 - Heat Transfer

This course covers theoretical and empirical topics related to energy demand and consumption, energy production and supply, energy markets, public energy policies and regulation affecting (international) energy markets - with a strong emphasis on environmental effects and a focus on sustainable (alternative) energy sources. The course examines energy taxation, price regulation, energy efficiency and policies for controlling emissions. Energy efficiency and energy conservation opportunities and strategies of different sectors: energy auditing. Supply and demand side energy management. Special attention will be given to factors affecting the development of more sustainable energy sources.

Course Code: MEng4404

Prerequisite MEng3101- Engineering Thermodynamics II

MEng3112 - Fluid Mechanics II

Course Name Computer Integrated Manufacturing

Credit Hour: 3

Prerequisite MEng2110 - Machine Drawing with CAD

This course includes Introduction to CIM, automation and production types, Manufacturing Systems: Types of Manufacturing systems, Single and multi-station manufacturing systems, lines and assemblies; Product design and CAD/CAPP and CAM and their integrations; Group Technology and FMS: Part families, Part classification and coding systems: OPITZ, PFA, FFA, Applications of group technology, Component of FMS, Analysis of FMS, Production Planning and Control System: Production planning and schedule, Material requirements planning, Capacity planning, Shop floor control, Inventory control, manufacturing resource planning; Introduction to Robotics: Robot anatomy and related attributes, Robot control, Applications of industrial robots, Robot programming.

Course Code: MEng5311

Course Name IC Engines and Motor Vehicles Lab

Credit Hour: 1

Prerequisite MEng3102 - IC Engines,

MEng4106 - Motor Vehicles Engineering

Valve timing and port timing diagrams using timing diagram and dial gauge;

Determination of rotation and firing order with the help of valve overlap; Valve clearance adjustment and Influence of valve clearance to valve timing and engine performance; Study of hydraulic, electrical and eddy current dynamometers; Engine performance testing and emission analysis; Determination of fuel properties; Determination of the center of gravity location and brake force distribution; Performing vehicle inspection based on the regulation Ethiopian transport authority.

Course Code: MEng4406

Course Name Metal Casting Technology

Credit Hour: 3

Prerequisite MEng3105 - Engineering Materials

Fundamentals of casting: Pattern making, mold making, core making, solidification of metals, molding and core materials, design of castings, location of gates and risers, allowances for casting design and finishing of casting.

Course Code: MEng4108

Course Name Turbomachinery

Credit Hour: 4

Prerequisite MEng3112 - Fluid Mechanics II,

MEng3101 - Thermodynamics II

This course will cover the basics of turbomachines, both for use with liquid (incompressible) as well as gaseous (compressible) fluids. It will lead the student from the basic knowledge of fluid mechanics and thermodynamics to being able to perform the preliminary design of turbomachines, understand their wide applications in households, process industry, air transport and energy generation. Types and design of turbomachines, thermodynamics and fluid mechanics considerations, performance of turbomachines, velocity triangles, Euler turbomachinery equation, degree of reaction and its effect on efficiency, as well as topics on cavitation and water hammering, energy cascade will be discussed.

Course Code: MEng5607

Course Name Turbomachinery II

Credit Hour: 2

Prerequisite Turbomachinery I

This course will cover the basics of turbomachines for use with gaseous (compressible) fluids. It will lead the student from the basic knowledge of fluid mechanics and thermodynamics to being able to perform the preliminary design of turbomachines, understand their wide applications in households, process industry, air transport and energy generation. Types and design of turbomachines, thermodynamics and fluid mechanics considerations, performance of turbomachines, velocity triangles, Euler turbomachinery equation, degree of reaction and its effect on efficiency, as well as topics on surge and choking in compressors will be discussed.

Course Code: MEng5305

Course Name Automotive Electronic systems

Credit Hour: 3

Prerequisite MEng4310 - Automotive Electrical Systems

Fundamentals of Automotive Electronics: Sensors Actuators and Microprocessor: Si Engine Management and digital engine control system: Ci Engine Management and Digital Engine Control System: Vehicle Management Systems: Introduction to Automotive Networking.

Course Code: MEng4310

Course Name Automotive Electrical systems

Credit Hour: 3

Prerequisite ECEg3101 - Basic Electricity and Electronics,

ECEg4107 - Electrical Power Machines,

MEng3102 - IC Engines

Batteries, starting system and Charging system, ignition, Lighting systems and accessories

Course Code: MEng4110

Course Name Refrigeration and Air Conditioning

Credit Hour: 4

Prerequisite MEng3110 - Engineering Thermodynamics II

MEng4103 - Heat Transfer

The aim of the course is to introduce and explain the basic fundamental principles of refrigeration and air conditioning systems. This course will introduce students with basic analysis, design and selection of refrigeration and air conditioning systems and equipment's applicable for different purpose. The course will provide students with a working knowledge of computer-aided calculations of thermal loads and their use in design of RAC systems

Course Code: MEng4114

Course Name Industrial Internship

Credit Hour: 6

Prerequisite MEng4101 - Machine Elements II

MEng4103 - Heat Transfer

This course focuses on providing students with an understanding of various operations and activities within the industry. Moreover, the course intended to create exposure to various engineering activities within the industry including operation and handling of different equipment, process and production methods, design and quality control & maintenance etc. The course will also equip students with hands -on-experience in handling various instruments.

Course Code: MEng4112

Course Name Machine Design Project

Credit Hour: 3

Prerequisite MEng4101 -Machine Elements II

Conceptual Design; Embodiment Design. Design procedures and special calculation methods related to the design projects; Practical design of typical machine assemblies; Simple machine units and elements; Design project: Unfired pressure vessels and jacks (Bottle, Scissor, Fiat Type, Service, etc.)

Course Code: MEng5101

Course Name Power Plant Engineering

Credit Hour: 3

Prerequisite Thermodynamics II, Fluid Mechanics or Fluid Mechanics II, Heat Transfer

Analysis of steam cycles; Fuels and combustion; Steam generators (Boilers); Combustion mechanisms, Combustion equipment and Firing methods; Steam turbines; Steam condensers, Condensate-feed-water and circulating water systems; Internal combustion power plants; Miscellaneous topics; Engineering economy

Course Code: MEng5501

Course Name Product Design and Development

Credit Hour: 3

Prerequisite MEng4112 - Machine Design Project

Product Design and Development is a project-based course that covers modern tools and methods for product design and development. The cornerstone is a project in which teams of management, engineering, and industrial design students conceive, design and prototype a physical product. Topics include identifying customer needs, concept generation, product architecture, industrial design, and design-for-manufacturing.

Course Code: MEng5503

Course Name Mechanical Vibration

Credit Hour: 3

Prerequisite MEng4109 – Mechanisms of Machinery

Introduction to mechanical vibration; Modeling of dynamic systems; Single-degree of freedom System-Two degrees of freedom system-Damped and Undamped Longitudinal Vibrations; Transverse Vibration; Whirling of shafts; Torsional vibrations; Causes of vibrations; Introduction to vibration control and measurements.

Course Code: MEng5505

Course Name Maintenance of Machinery

Credit Hour: 3

Prerequisite MEng4101 - Machine Elements II

MEng2108 - Basic Workshop Practice

Maintenance of machinery and installation is a comprehensive course designed to equip students with knowledge and skill of installation and commissioning of industrial machineries, damages of typical machine components and their causes; determination of the state of damage of equipment; elements of maintenance technology; how to maintain machineries and equipment; reliability, maintainability and availability; maintenance planning and organization; inventory management in maintenance.

Course Code: MEng5309

Course Name Heavy Duty and Construction Equipment

Credit Hour: 2

Prerequisite MEng3102 - IC Engines

MEng4106 - Motor Vehicles

Construction Equipment classification. Construction Equipment design, operation and specifications: Motor Grader, Scrapers, Dozers Excavators, Crushers, Asphalt plant, Concrete mixer.

Course Code: MEng5309

Course Name Final Year Project Phase I

Credit Hour:

Prerequisite MEng4112 - Machine Design Project

A project based subject in which students are required to undertake several group projects covering both the conceptual and detailed aspects of design. It involves different areas of the mechanical engineering discipline.

Course Code: MEng5309

Course Name Final Year Project Phase I

Credit Hour:

Prerequisite

3.8 Bachelor of Science Degree in Software Engineering

3.8.1 Introduction

Software Engineering is essential for modern industries in the 21st century. Broadly, software engineering can be defined as the application of a systematic, disciplined, quantifiable approach to the development, operation and maintenance of software. Software engineers are responsible for the specification, design, and development of software programs that support computer and computer-controlled systems or products. In other words, software engineering is an ideal field for creative, detail-oriented individuals who enjoy solving-problems by using technology. This undergraduate program in software engineering is aimed at equipping students with the major principles and concepts in software production, management and design. The students will learn how to develop, create, and troubleshoot computer software, programs, and applications, mastering the skills needed to work in software development and other information technology fields.

Although the field of Software Engineering was established more than 30 years ago, Bachelor degree programs in Software Engineering in Ethiopia came into existence in the last few years. According to the GTP (Growth and Transformation Plan) goals and strategies set by the Ministry of Science and Technology (MoST, the ministry focuses on creating an enabling environment for ICT professionals to be engaged in digital innovation, creativity, and research activities. The Ministry had been supporting GTP reform to ensure sustainable development, poverty eradication, human resource development and capacity building through the application of well-designed information communication technology. Taking these facts into account, the Department of Computer Science and Information Technology under the School of Electrical Engineering and Computing had opened a new program in Software Engineering that was aimed at producing well-qualified and skilled graduates in the field of Software Engineering to satisfy the growing demand of manpower in the country.

To this end, the former Department of Computer Science and Information Technology (CS&IT) established at AASTU in 2011 as undergraduate program within the School of Electrical Engineering and Computing. Five years later, the undergraduate program in Software Engineering is launched within Department of Computer Science and Information Technology. In other words, the undergraduate program in Software Engineering had been launched during

the 2014-2015 academic year by admitting students in the regular program under the College of Electrical and Mechanical Engineering. Though this undergraduate program was initially four-year program, it becomes five-years program in September 2019 as a result of 15 new courses are prearranged from MOE as a common course to be delivered in all higher institutions in Ethiopia.

The Software Engineering program of the Department is structured in such a way that supports the study of both theoretical and practical aspects of software design and development. The program tries to find the balance between breadth and depth to provide a solid foundation in principles of software development methodologies on one hand, and comprehensive exposure to software applications, professionalism, and leadership skills on the other hand.

Students are guided to pursue their interest in software engineering by studying the fundamental principles and application of software construction including basic concepts, techniques, and methods of software development through the core required courses such as data structures, discrete mathematics, algorithms, object-oriented programming, fundamental of software engineering, etc. followed by tailored individual education through the selection of elective courses. Moreover, every senior student is required to undertake a research project under the guidance of a senior academic advisor. Students are encouraged to participate in multidisciplinary research projects at the Excellence Centers of the University.

3.8.2 Program Education Objective (PEO)

The program education objective of the Software Engineering program is shown in table 2.

Table 2: Program Education Objectives (PEO)

PEO	Statement
PEO-1	To produce large scale software developers, project managers and leaders who has an ability to address complex and large-scale problems.
PEO-2	To produce digital innovators and entrepreneurs.

PEO-3	To nurture professionally qualified software engineers with the potential to become consultant, trainer, software requirement engineer, software architect, system analyst, software quality assurance and tester.
PEO-4	To produce software engineers who incorporate standards and security in software development with appropriate consideration of socio-cultural and environmental safety.
PEO-5	To produce researchers and software engineers who will develop dynamic and smart software solutions using emerging technologies to address complex national and global challenges and demonstrate lifelong learning and continuing professional development.

3.8.3 Program Outcomes (PO):

Graduates of the Software Engineering program expected up on their graduation to demonstrate the ability to:

PO-1 Apply knowledge of mathematics, computing science, engineering fundamentals and software engineering specialization to address complex and large-scale problems.

PO-2 Identify, formulate, research literature and analyze complex problems, reaching substantiated (justifiable, validated) conclusions using software development principles, methodologies and tools.

PO-3 Design integrated and efficient software solutions, component or processes to address complex problems and implement them to meet the specified needs with appropriate consideration for public health and safety, cultural, societal (local and international) and environmental considerations.

PO-4 Conduct investigation of complex problems using research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

PO-5 Adopt, innovate, select and apply appropriate state of the art software design and development tools, methodologies, resources and emerging technologies including simulation,

prediction and modeling for complex business process with understanding of their potential capabilities, limitations and associated risks.

PO-6 Apply logical reasoning and informed decision to assess societal, health, safety, privacy, security, legal and cultural issues and the consequent responsibilities relevant to software engineering application.

PO-7 Understand the impact of software applications in societal and environmental context and apply the principle of green computing for sustainable development.

PO-8 Apply ethical principles and commit to professional ethics and responsibilities and norms of software engineering practice.

PO-9 Function effectively as an independent software engineer, as a member or as a leader in a diverse team and multidisciplinary settings.

PO-10 Communicate effectively on complex business and system activities with the software engineering community and with a society at large; such as being able to comprehend and write effective reports, articles and design documentations using various tools, make effective presentations and give and receive clear instructions.

PO-11 Demonstrate knowledge and skill on software project management principles and apply these to manage software development processes and its resource requirements, costs in multidisciplinary projects.

PO-12 Recognize the need for and have the preparation as well as ability to engage in an independent and lifelong learning in the information and knowledge economy

3.8.4 Graduation Requirements

Graduation requirement for all undergraduate programs should satisfy the following minimum requirements as stated in the university senate legislation July 2017, Article 109.

- All the required courses (a total of 185 credit hrs.) set in the program should be completed.
- A cumulative grade point average CGPA of 2.00 must be obtained;

- A cumulative grade point average CGPA of 2.00 in major area courses;
- Pass the exit exam
- No "F" grade in any course taken for undergraduate program;
- Successfully defended his/her senior project/senior essay

3.8.5 List of Courses and Category

Table: Distribution of credit hours for all course categories

Category	Total Cr. hr.	Percentage (%)
1 Core course (Major and Supportive courses)	139 Cr. hrs.	75.54%
2 Core elective/focus areas/ course (if any)	6 Cr. hrs.	3.26%
3 University requirement (Core course)	3 Cr. hrs.	1.63%
4 National requirements	36 Cr. hrs.	19.56
Total	184	100%

List of Courses

Table: List of core courses

No	Course Code	Course Title	Credithour	Prerequisite
1	SWEG2101	Introduction to Software Engineering and Computing	4	None
2	SWEG2103	Fundamentals of Programming I	3	None
3	SWEG2102	Fundamentals of Programming II	3	SWEG2103
4	SWEG2105	Discrete Mathematics for Software Engineering	3	Math1012
5	SWEG2106	Data Communication and Computer Networks	4	SWEG2101
6	SWEG2108	Database Systems	4	None
7	SWEG3101	Object Oriented Programming	3	SWEG2102
8	SWEG3103	Data Structure and Algorithms	4	SWEG2102
9	SWEG3105	Computer Organization and Architecture	4	EEng2004
10	SWEG3109	System Analysis and Modeling	4	SWEG2101
11	SWEG3104	Software Requirements Engineering	3	SWEG3109

12	SWEG3106	Operating Systems	4	SWEG3105
13	SWEG3107	Internet Programming I	3	SWEG2106
14	SWEG3102	Internet Programming II	3	SWEG3107
15	SWEG3108	Advanced Programming	4	SWEG3101
16	SWEG4102	Embedded Systems	3	SWEG3101, SWEG3106
17	SWEG3110	Formal Language and Automata Theory	3	SWEG2105, SWEG3103
18	SWEG4101	Principles of Compiler Design	3	SWEG3110, SWEG2102
19	SWEG4103	Mobile Computing and Programming	3	SWEG2106, SWEG3108
20	SWEG4105	Software Design and Architecture	3	SWEG3109
21	SWEG4106	Software Quality Assurance and Testing	3	SWEG3104, SWEG4105
22	SWEG4104	Software Project Management	3	SWEG3104
23	SWEG4109	Computer Graphics	3	SWEG2102, SWEG3103
24	SWEG4110	Human Computer Interaction	3	SWEG4109
25	SWEG4117	Introduction to Artificial Intelligence	3	SWEG3103, SWEG2105
26	SWEG4108	Research Methods in Software Engineering	2	SWEG4105
27	SWEG4112	Introduction to Machine Learning	3	SWEG4103
28	SWEG4114	Industrial Internship	6	SWEG4112
29	SWEG5101	Senior Research Project Phase I	3	SWEG4114
30	SWEG5102	Senior Research Project Phase II	3	SWEG5101
31	SWEG5111	Distributed Systems	4	SWEG3101, SWEG2106
32	SWEG5105	Computer System Security	3	SWEG2106
33	SWEG5106	Software Evolution and Maintenance	3	SWEG4106,

				SWEG5103
34	SWEG5107	Software Component Design	3	SWEG4105
35	SWEG5108	Software Defined Systems	3	SWEG2106, SWEG3101, SWEG5107
36	SWEG5109	Open-Source Software Paradigms	3	SWEG4104
37	SWEG5103	Software Configuration Management	3	SWEG4104
38	EmTe1108	Emerging Technology for Engineers	3	None
39	Entr1106	Entrepreneurship for Engineers	2	None
40	IETP4115	Integrated Engineering Team Project	3	All courses that given up to 3 rd Year
41	SWEG5110	Selected Topics in Software Engineering	3	All core courses (Excluding 5 th Year 2 nd semester courses)
Total			133	

Table 9: List of elective courses

Elective Courses 1				
No	CourseCode	Course Title	Credit hour	Prerequisite
1	SWEG5201	Introduction to Big Data Analytics	3	Stat2091, SWEG2108
2	SWEG5203	Data Mining and Data Warehousing	3	Stat2091, SWEG2108
	SWEG5205	Simulation and Modeling	3	Stat2091, SWEG2105
Total			9	

Elective Courses 2				
No	CourseCode	Course Title	Credit hour	Prerequisite
1	SWEG5202	Introduction to Cloud Computing	3	SWEG2106, SWEG5111

2	SWEG5204	High Performance Computing	3	SWEG5111
3	SWEG5206	Multimedia Systems	3	SWEG2106
Total			9	

Table: List of common courses and supportive courses

No.	CourseCode	Course Title	Credit hour	Prerequisite
1	LoCT1011	Logic and Critical Thinking	3	None
2	Psch1011	General Psychology	3	None
3	FLEn1011	Communicative English Language Skill – I	3	None
4	GeES1011	Geography of Ethiopia and The Horn	3	None
5	Math1011	Mathematics (For Natural Science)	3	None
6	SpSc1011	Physical Fitness	-	None
7	Phys1011	General Physics	3(2+1)	None
8	Anth1012	Social Anthropology	2	None
9	FLEn1012	Communicative English Language Skill-II	3	FLEn1011
10	Math1012	Applied Mathematics IB	4	None
11	MCiE1012	Moral and Civic Education	2	None
12	Incl1012	Inclusiveness	2	None
13	Econ1101	Introduction to Economics	3	None
14	GLaf1002	Global Trend	2	None
15	HiES2010	History of Ethiopia and the horn	3	None
	Total		36	

Supportive Courses				
19	EEng2004	Digital Logic Design	3	Math1012
20	Stat2091	Probability and Statistics	3	None
	Total		9	

3.8.6 Program Curriculum Structure: Bachelor of Science in Software Engineering

Year I, Semester I

No	Course Code	Course Title	Cr. Hr.
1.	LoCT1011	Logic and Critical Thinking	3
2.	Psch1011	General Psychology	3
3.	FLEn1011	Communicative English Language Skill – I	3
4.	GeES1011	Geography of Ethiopia and The Horn	3
5.	Math1011	Mathematics (For Natural Science)	3
6.	SpSc1011	Physical Fitness	(P/F)
7.	Phys1011	General Physics	3(2+1)
Total			18

Year I, Semester II

No	Course Code	Course Title	Cr. Hr.
1.	EmTe1108	Emerging Technology for Engineers	3
2.	Entr1106	Entrepreneurship for Engineers	2
3.	Anth1012	Social Anthropology	2
4.	FLEn1012	Communicative English Language Skill-II	3
5.	Math1012	Applied Mathematics IB	4
6.	MCiE1012	Moral and Civic Education	2
7.	Incl1012	Inclusiveness	2
Total			18

Year II, Semester I

No	Course Code	Course Title	Cr. Hr.
1.	SWEG2101	Introduction to Software Engineering and Computing	4
2.	SWEG2103	Fundamentals of Programming I	3
3.	SWEG2105	Discrete Mathematics for Software Engineering	3
4.	HiES2010	History of Ethiopia and the horn	3
5.	Econ1101	Introduction to Economics	3
6.	GLaf1002	Global Trend	2
Total			18

Year II, Semester II

No	Course Code	Course Title	Cr. Hr.
1.	SWEG2102	Fundamentals of Programming II	3
2.	EEng2004	Digital Logic Design	3
3.	SWEG2106	Data Communication and Computer Networks	4
4.	SWEG2108	Database Systems	4
5.	Stat2091	Probability and Statistics	3
Total			17

Year III, Semester I

No	Course Code	Course Title	Cr. Hr.
1.	SWEG3101	Object Oriented Programming	3
2.	SWEG3103	Data Structure and Algorithms	4
3.	SWEG3105	Computer Organization and Architecture	4
4.	SWEG3107	Internet Programming I	3
5.	SWEG3109	System Analysis and Modeling	4
Total			18

Year III, Semester II

No	Course Code	Course Title	Cr. Hr.
1.	SWEG3102	Internet Programming II	3
2.	SWEG3104	Software Requirements Engineering	3
3.	SWEG3106	Operating Systems	4
4.	SWEG3108	Advanced Programming	4
5.	SWEG3110	Formal Language and Automata Theory	3
Total			17

Year IV, Semester I

No	Course Code	Course Title	Cr. Hr.
1.	SWEG4101	Principles of Compiler Design	3
2.	SWEG4103	Mobile Computing and Programming	3
3.	SWEG4105	Software Design and Architecture	3
4.	SWEG4117	Introduction to Artificial Intelligence	3
5.	SWEG4109	Computer Graphics	3
6.	IETP4115	Integrated Engineering team project	3
Total			18

Year IV, Semester II

No	Course Code	Course Title	Cr. Hr.
1.	SWEG4102	Embedded Systems	3
2.	SWEG4104	Software Project Management	3
3.	SWEG4106	Software Quality Assurance and Testing	3
4.	SWEG4108	Research Methods in Software Engineering	2
5.	SWEG4110	Human Computer Interaction	3
6.	SWEG4112	Introduction to Machine learning	3
Total			17

Year IV, Semester Summer

No	Course Code	Course Title	Cr. Hr.
1	SWEG4114	Industrial Internship	6
Total			6

Year V, Semester I

No	Course Code	Course Title	Cr. Hr.
1.	SWEG5101	Senior Research Project Phase I	P/F
2.	SWEG5103	Software Configuration Management	3
3.	SWEG52XX	Elective I	3
4.	SWEG5105	Computer System Security	3
5.	SWEG5107	Software Component Design	3
6.	SWEG5109	Open Source Software Paradigms	3
7.	SWEG5111	Distributed Systems	4
Total			19

Year V, Semester II

No	Course Code	Course Title	Cr. Hr.
1.	SWEG5102	Senior Research Project II	6
2.	SWEG52XX	Elective II	3
3.	SWEG5106	Software Evolution and Maintenance	3
4.	SWEG5108	Software Defined Systems	3
5.	SWEG5110	Selected Topics in Software Engineering	3
Total			18

3.8.7 Program Courses Synopsis

Common Course Synopsis

Course Code: SWEG2101

Course Name: **Introduction to Software Engineering and Computing**

Credit Hour: 4

Prerequisite: No

Software engineering is a discipline that allows us to apply engineering and computer science concepts in the development and maintenance of reliable, usable, and dependable software. The purpose of this course is to present a general introduction to software engineering and computing. The course is designed to equip students with theoretical and practical aspects of fundamentals of computing and software engineering by furnishing them with a broad oversight of software engineering concepts and principles in parallel with the software development life cycle.

The course begins with an introduction to computing, giving definition, characteristics, types, application of computer system, components of computing system. The students then will able to make use of computer system, learn data representation in computer system followed by numbering system, binary arithmetic, and digital logic gate and computer system architecture. The students will go on to learn about software engineering evolution, principles, applications, impact of software on economic, societal and environmental safety and Software Engineering Professional Practice. The students will also learn about Software Development Life Cycle (SDLC), methodologies of software engineering, software development paradigms, trends and CASE tools. Moreover, the students will learn about data communication and computer networks followed by computer system security to introduce students with potential security threats to computer system and methods to safeguard it.

Course Code: SWEG2103

Course Name: **Fundamentals of Programming I**

Credit Hour: 3

Prerequisite: No

Computer programming involves various activities such as understanding and analysis of problems, algorithm design, verification of requirements of algorithms including their correctness and resources consumption, and coding of algorithms. This course is designed to allow students to acquire the three basic skills of programming: problem solving skill (understand how they solve problems), algorithmic skill (knowing how to write the steps of their proposed solutions to specific problems), language skill (learning how to put the solutions in to code and communicate it to the machine).

More specifically as an introduction, this course designed to introduce students to an overview of the different concepts of programming and problem solving strategies. Moreover, the learners will be able to identify and demonstrate the basic programming constructors like Variables, Reserved words, Syntax diagram, Constant declarations, Data types, and also the concepts of various programming techniques such as, Conditional statements, Loops, array, strings, pointers (static and dynamic memory management), Error types and error debugging techniques. For each topic, some coding guidelines will be discussed.

Course Code: SWEG2105

Course Name: **Discrete Mathematics for Software Engineering**

Credit Hour: 3

Prerequisite: No

This course covers the basic concepts of discrete mathematics used in computer science and other disciplines that involve formal reasoning. The topics include logic, proof, counting, discrete probability,

relations, graphs, trees, and Boolean algebra.

Course Code: EEng2004

Course Name: **Digital Logic Design**

Credit Hour: 4

Prerequisite: Math1012

Digital Logic Design is a comprehensive study of the principles and techniques of modern digital system. It provides an overview of the principles underlying number systems, arithmetic operations, decimal & alphanumerical codes, Boolean algebra, combinational and sequential circuits. Furthermore, analysis and design of combinational sequential logic system is discussed

Course Code: SWEG2105

Course Name: **Fundamentals of Programming II**

Credit Hour: 3

Prerequisite: SWEG2103

This course is a continuation of “Fundamentals of Programming I”: It mainly covers the concepts of modular programming, user-defined data types (structure, union, and enumeration). In addition, further application of programming techniques including file manipulation (processing of text files and binary files) and other advanced techniques like templates and vectors will also be discussed. This could help the students to enhance the three basic

skills of programming: problem solving skill (understand how they solve problems), algorithmic skill (knowing how to write the steps of their proposed solutions to specific problems), language skill (learning how to put the solutions in to code and communicate it to the machine). More specifically, it empower the learners in performing projects and solving different real world problems using the target programming language.

Course Code: SWEG2106

Course Name: **Data Communication and Computer Networks**

Credit Hour: 4

Prerequisite: SWEG2101

This Course covers a foundation in the study of data communications and computer networking. Topics covered will include fundamentals of data communications, layered model, and TCP/IP and OSI Reference models including common protocol and services.

Course Code: SWEG2108

Course Name: Database Systems

Credit Hour: 4

Prerequisite: No

This course is intended enable students to implement different concepts from the very basic to advance SQL on a variety of DBMS. It starts by dealing with introduction to SQL like SQL Data Definition, SQL Data Manipulation, Basic Structure of SQL Queries, Nested Subqueries and Additional Basic Operations. Then it goes to working with Intermediate SQL like Join Expressions, Views, Integrity Constraints, Transactions, Authorization. Lastly, the advanced SQL like Trigger, Concurrency Control Techniques, Database Recovery Techniques, Database Security and Authorization Enhanced Data Models for Advanced Applications will be covered. Finally, besides to the concept of SQL Systems, the NoSQL systems will be briefly introduced.

Course Code: SWEG3101

Course Name: Object Oriented Programming

Credit Hour: 3

Prerequisite: Fundamental Programming II

The goal of the course is to introduces object-oriented programming (OOP) using the Java programming language. Students will learn how to program in Java and using OOP principles:- classes and objects, encapsulation, polymorphism, inheritance; exception handling; introduction to GUI design concepts, introduction to Event handling; Special importance will be given to

the object-oriented nature with Hands-on labs and exercises.

Course Code: SWEG3103

Course Name: **Data Structures and Algorithms**

Credit Hour: 4

Prerequisite: SWEG2102

This course focuses on the study of data structures, algorithms and program efficiency. Topics include: analysis of time and space requirements of algorithms; program efficiency improving techniques, abstract data types such as linked lists, stacks, queues, trees (traversal, implementations); simple searching algorithms (linear search, binary search, ...), simple sorting algorithms (bubble sort, insertion sort, selection sort, ...), advanced sorting algorithms (merge sort, quick sort, heap sort ...)

Course Code: SWEG3105

Course Name: **Computer Organization and Architecture**

Credit Hour: 4

Prerequisite: DLD

Computer Organization and Architecture, Computer function and Structure, Computer Evolution and Performance, Top level view of Computer function and interconnection, Cache Memory, Internal and External Memory, CPU instruction sets and Assembly programming language will be discussed.

Course Code: SWEG3107

Course Name: **Internet programming I**

Credit Hour: 3

Prerequisite SWEG2106

Overview of Internet and World Wide Web Technology: Client-Server architecture, HTTP Protocol, Web page design and development, static pages, client-side programming using hypertext and scripting languages (HTML, JavaScript) also applying different styles will be discussed in this course.

Course Code: SWEG3109

Course Name: **System Analysis and Modeling**

Credit Hour: 4

Prerequisite SWEG2101

This course provides concepts on how to capture software requirements and Handle difficult situations ingathering data to build systems. Special emphasis is given to working with clients and to learning about the needs of users

who interact with a system. The course addresses elicitation, specification, and Management of software system requirements.

The primary function of an analyst in the requirements capture process is to analyze and organize informal requirement statements into a form that can be Verified by a user and used as input to design. Requirement's specifications must also provide the sole criteria for the validation of the end software product. In achieving these goals an effective analyst must bring to a project the ability to listen to users' needs and create a complete, correct, consistent and unambiguous Requirement specification. Analysis models describing functional and time dependent behavior and data relationships are therefore an essential component of the analyst's tool box. Useful models also act as an aid to understanding complex systems, representing the essence of a system in graphical form and Downplaying excessive detail.

Course Code: SWEG3102

Course Name: Internet programming II

Credit Hour: 3

Prerequisite SWEG3107

This course is a continuation of Internet Programming one. It deals with web application development using an enterprise application framework focusing on server-side scripting. Topics offered include Server-Side Scripting Basics, HTML Forms and Server-Side Scripting, Files handling and directories, Connecting and working with the databases, Session and Cookie tracking, Introduction to content management system and framework will be discussed in this course.

Course Code: SWEG3104

Course Name: Software Requirements Engineering

Credit Hour: 3

Prerequisite System Analysis and Modeling

Software requirement engineering is the process of determining what is to be produced in a software system. It provides concepts on how to capture software requirements and handle difficult situations in gathering data to build systems. Special emphasis is given to working with clients and to learning about the needs of users who interact with a system. The course addresses elicitation, analysis, specification, validation and management of software system requirements.

The primary function of an analyst in the requirements capture process is to analyze and organize informal requirement statements into a form that can be verified by a user and used as input to design. Requirements specifications must also provide the sole criteria for the validation of the end software product. In achieving these

goals an effective analyst must bring to a project the ability to listen to users' needs and create a complete, correct, consistent and unambiguous requirement specification. Ultimately the techniques presented reduce project risk, improve product quality, and allow for effective control of requirements volatility – all measures for increasing the likelihood of a successful software project.

Course Code: SWEG3106

Course Name: Operating Systems

Credit Hour: 4

Prerequisite SWEG3105

This course is a general introduction to the design and implementation of modern operating

Systems. Historical development of operating systems, synchronization, scheduling, deadlocks, paging, virtual memory, input/output devices, and files will be discussed.

Course Code: SWEG3108

Course Name: **Advanced Programming**

Credit Hour: 4

Prerequisite SWEG3101

Introductory Programming courses. Topics like multithreading, socket programming, object serialization, and Event objects are crucial components of this course. Including the above-mentioned advanced programming concepts other important features of the technologies like socket and web programming are also discussed.

Basics of high-level programming concepts and threads will be given priority to pave the way for developing advanced programming skills. The course then advances to some of the most popular APIs in the language: Remote Method Invocation (RMI), Database connection and GUI, a specification for developing a software component.

Course Code: SWEG3101

Course Name: **Formal Language and Automata Theory**

Credit Hour: 3

Prerequisite SWEG2105, SWEG3103

Formal languages and automata theory is based on mathematical computations. These computations are used to represent various mathematical models. Automata theory is a theory of models. The working of every process can be represented by means of models. The model can be a theoretical or mathematical model. The model helps in representing the concept of every activity. In this Basics of all fundamental concepts of automata theory and those are strings, languages, operations on the languages will be discussed. Formal languages including grammar, finite automaton, regular expression, pushdown automaton, Context-free Grammar, Context-free Languages, Backus-Naur, Chomsky Normal Form, and the Turing machine also will be discussed. Not only do they form basic models of computation, but they are also the foundation of many branches of computer science, e.g. compilers, software engineering, and concurrent systems.

Course Code: IETP4115

Course Name: Integrated Engineering Team Project

Credit Hour: 3

Prerequisite

This is a multidisciplinary team integrated engineering project in connection with a special engineering problem and under the guidance of a faculty member. The project synopsis consists of literature review, design, project management, business acumen, multidisciplinary team work, entrepreneurship

Course Code: SWEG4101

Course Name: Principles of Compiler Design

Credit Hour: 3

Prerequisite SWEG3110, SWEG2102

This course is intended to teach the basic principles involved in compiler design. These include lectures and experiments on compiler design, implementation techniques and tools. The course will provide deeper insights into the fundamental aspects of lexical analysis, syntax analysis, semantic analysis, optimization, code generation, symbol table handling, and error handling. Finally, students will design and implement a mini language compiler.

Course Code: SWEG4103

Course Name: **Mobile Computing and Programming**

Credit Hour: 3

Prerequisite SWEG2106, SWEG3108

This course will mainly introduce enrollees of the program to mobile network, mobile computing and mobile application development. Mobile computing will be discussed from three perspectives: mobile technology, application development, and user interaction. The course will priority sum-up various mobile computing technologies and introduce to appropriate mobile application frameworks and development environments to fortify concepts covered in lectures and use it to implements assignments, course project and FYP as well.

In addition to this, each enrollee will have a seminar on the current mobile computing technologies so as to deeply understand, internalize, explain and reveal different issues on the computing technologies

Course Code: SWEG4105

Course Name: Software Design and Architecture

Credit Hour: 3

Prerequisite SWEG3109

Software design is a foundational layout for a software system. This course covers the fundamental design principles and strategies for software architecture and design, including the structural view of software components, and their characteristics and interrelationships, at a high level of abstraction. Topics included are design concepts, architectural structures and views, software architectural design process, architectural styles and patterns, software design quality attributes, middleware architectures, design patterns, design principles and

architectural documentation and evaluation. The course uses software design tools to reinforce design processes and representations.

Course Code: SWEG4109

Course Name: **Computer Graphics**

Credit Hour: 3

Prerequisite SWEG2102, SWEG3103

Basics of Computer Graphics: Overview of Computer Graphics, Computer Graphics Application and Software, Video display devices :(Color CRT Monitors, Raster scan & Random Scan Display Systems, DVST), Flat Panel Display: (LCD, LED, Plasma display) 3D Viewing devices, Input & Output Devices. **Graphics Primitives:** Scan Conversion, Rasterization, Line drawing algorithms, Circle drawing algorithms, Ellipse drawing algorithms, Character generation, Color Models & color application, Filled area algorithms, Aliasing & anti-aliasing. **2D Transformation & Views:** Basic transformations: Translation, Rotation, Scaling, Matrix representation and Homogeneous Coordinates, Composite transformations, Other Transformations: Reflection & Shear, Line Clipping algorithm, Viewing transformation, Polygon. **3D Concepts and Object Representation:** 3D Display Methods, Polygon Surfaces, Polygon Table, Plane Equation, Polygon Meshes, Spline Representation, Cubic Spline Interpolation Methods, Bazier Curves and Surfaces. **3D Transformation & Views:** 3D Translation, 3D Rotation, 3D Scaling, Matrix representation and Homogeneous Coordinates, Composite transformations, Other Transformations: Reflection & Shear, View Pipeline, Viewing Co-ordinates, Projection, View Volume and General Transformation Projection. **Advanced Topics in Computer Graphics:** Classification of Visible-Surface Detection Algorithms, Back-Face Detection, Depth Buffer Method/ Z Buffer Method, Light Source, Basic Illumination Models, Ambient Light, Diffuse Reflection, Properties of Light and Color Model semester. Hence, they will be advised to repeat the semester with the next batch of students.

Course Code: SWEG4117

Course Name: **Introduction to Artificial Intelligence**

Credit Hour: 3

Prerequisite SWEG3103, SWEG2105

Artificial Intelligence (AI) is concerned with the design and analysis of autonomous agents. These are software systems and/or physical machines, with sensors and actuators, embodied for example within a robot or an autonomous spacecraft. An intelligent system has to perceive its environment, to act rationally towards its assigned tasks, to interact with other agents and with human beings. It rely on a broad set of general and specialized knowledge representations and reasoning mechanisms, on problem solving and search algorithms, and on machine learning techniques. Hence, the focus of this course is to deal with Fundamental issues in intelligent systems; Search and constraint satisfaction; Knowledge representation and reasoning.

Course Code: SWEG4102

Course Name: Embedded System

Credit Hour: 3

Prerequisite SWEG3101, SWEG2106

The aim of this course is to enable students develop theoretical and practical knowledge about embedded systems hardware as well as acquire skills in programming embedded processors.

Course Code: SWEG4104

Course Name: Software Project Management

Credit Hour: 3

Prerequisite SWEG3104

This course discusses about the fundamental concepts, roles and purpose of software project management in software engineering. The course covers software project management fundamentals, software project planning, software project scheduling, software project estimation, project estimation approaches, risk management, configuration and change management, software quality management, and staffing and team management skill.

Course Code: SWEG4106

Course Name: **Software Quality Assurance and Testing**

Credit Hour: 3

Prerequisite SWEG3104, SWEG4105

This course focuses on techniques for ensuring software quality. Here, quality assurance is viewed as an activity that runs through the entire development process: understanding the needs of clients and users; analyzing and documenting requirements; verifying and Validating solutions through testing.

It encompasses activities and related techniques to ensure the implementation of appropriate functionality that satisfy the requirements/needs of its targeted client/users for the intended software system, product, or service as the case may be, both correctly and efficiently.

Course Code: SWEG4108

Course Name: Research Methods in Software Engineering

Credit Hour: 2

Prerequisite No

Basic reviews of the nature of research and the methods which are used to undertake a variety of investigations relevant to a wide range of subjects, specifically Software Engineering, prepare to carry out independent research endeavor, in the form of projects or theses. Research methods are a range of tools that are used for different types of enquiry, just as a variety of tools are used for doing different practical jobs. Hence, it is necessary to understand what the correct tools are for doing the job, and how to use them to best effect. The course provides students with the basic notions and the tools used in research, the situations in which they are applied and indicates briefly how they are used.

Course Code: SWEG4110

Course Name: **Human Computer Interaction**

Credit Hour 3

Prerequisite: SWEG4109

HCI is an interdisciplinary field that integrates theories and methodologies from computer science, cognitive psychology, design, and many other areas. The course will span current theory and practice in interface specification, design, implementation and evaluation of a user interface. The students will be incrementally led through the phases of ethnographic study and requirements analysis, scenario-based design, paper prototyping, computer prototyping, and several methods of usability analysis and evaluation. The course involves also an introduction to software architectures used in modern graphical user interfaces, including the implementation of a few simple interfaces using different toolkit.

Course Code: SWEG4112

Course Name: **Introduction to Machine Learning**

Credit Hour: 3

Prerequisite: SWEG4117

Machine learning is an exciting and fast-moving field of Software Engineering with many recent consumer applications (e.g., Microsoft Kinect, Google Translate, Iphone's Siri, digital camera face detection, Netflix recommendations, Google news) and applications within the sciences and medicine (e.g., predicting protein-protein interactions, species modeling, detecting tumors, personalized medicine). This course introduces undergraduate Software Engineering students to the field of machine learning. Students learn about the theoretical foundations of machine learning and how to apply machine learning to solve new problems. Assuming no prior knowledge in machine learning, the course focuses on two major paradigms in machine learning which are supervised and unsupervised learning. In supervised learning, we learn various methods for classification and regression. Dimensionality reduction and clustering are discussed in the case of unsupervised learning

Course Code: SWEG4114

Course Name: **Industrial Internship**

Credit Hour: 6

Prerequisite: All core courses (Excluding 5th Year courses)

This course focuses on providing students with an understanding of various operations and activities within the industry. Moreover, the course intended to create exposure to various engineering activities within the industry including operation and handling of different equipment, process and production methods, design and quality control & maintenance etc. The course will also equip students with hands-on-experience in handling various instruments

Course Code: SWEG5101

Course Name: **Senior Research project I**

Credit Hour: 3

Prerequisite: SWEG4114

The student will practice to define a research problem and define a research to solve problem. The student will also

explore the existing systems and literatures to find the gaps and the connections to his/her defined research problem. The student also will present his/her research idea for a research community and make sound defense for any academic critics.

Course Code: SWEG5102

Course Name: Senior Research Project II

Credit Hour: 3

Prerequisite: SWEG5101

The student will practice to design model for a problem and implement a designed solution model. The student will successfully complete a major project working in a team with demonstrated ability to partition a project among multiple people

The student also will evaluate the solution implemented and present his/her research solution for a research community and make sound defence for any academic critics.

Course Code: SWEG5103

Course Name: **Software Configuration Management**

Credit Hour: 3

Prerequisite: SWEG4104

In Software Engineering, Software development is not a straightforward process and often requires multiple iterations between the development phases. To cope with this dynamic nature of software development, software engineers need to control the changes in the development process. *Software Configuration Management (SCM)* is the activity that helps us control the evolution of a software project. *Software Configuration Management (SCM)* is a process to systematically manage, organize, and control the changes in the documents, codes, and other entities during the Software Development Life Cycle. The primary goal is to increase productivity with minimal mistakes. SCM is part of cross-disciplinary field of configuration management and it can accurately determine who made which revision.

This course is a comprehensive review of SCM as a software-engineering discipline supporting all life-cycle phases by relying on tools and techniques to manage changes in software, including documentation, code, interfaces, and databases.

Course Code: SWEG5111

Course Name: **Distributed Systems**

Credit Hour: 3

Prerequisite: SWEG2106 SWEG3101

This course studies the key design principles of distributed systems, which are collections of independent networked computers that function as single Coherent systems. Covered topics include communication protocols, processes and threads, naming, synchronization, consistency and replication, fault tolerance and security.

We also examine some specific real-world distributed systems case studies, ranging from the Internet to file

systems. Class discussions based on readings from the textbook and research papers. The goals of this course are to understand how large-scale, distributed computational systems are built, and to provide you with the tools necessary to evaluate new technologies after the course end

Course Code: SWEG5105

Course Name: Computer System Security

Credit Hour: 3

Prerequisite: SWEG2106

Learn the goals of computer security and explore the threats and vulnerabilities which might compromise the goals. Examine security architectures and cryptographic systems. Study security protocols and standards to take countermeasures.

Course Code: SWEG5107

Course Name: Software Component Design

Credit Hour: 3

Prerequisite: Software Design and Architecture

The course focuses on an approach to software component design based on extensive use of pre-existing standard (or customizable) components. It describes the principles and practices of systematically engineering software systems from components. It also illustrates how a repository of reusable candidate components can be integrated into a typical evolutionary process model. The course will introduce various models that can be used to describe an object-oriented analysis and design models suitable for specifying Components-Based Software using Unified Modeling Language (UML).

Course Code: SWEG5109

Course Name: **Open Source Software Paradigms**

Credit Hour: 3

Prerequisite: SWEG4104

This course introducing the principles and practices of open source software development & communities of practice: history, intellectual property rights, economics & business models, development models, and community standards. The course begins with a historical study of the emergence and maturity of open source software from a variety of influences: the computer industry, thought leaders, and cultural movements. From there the course then covers copyright/patents, /trademarks/licensing and basic concepts of contracts. The impact of open source development and adoption on economics & various business models, in particular clarifying the apparent contradiction between free flow of information and the funding & compensation of participants, will be explored. The operational and organizational aspects of open source communities are covered to help understand the motivations that drive authentic participation: community, collaboration, contribution and co-creation. Throughout the course, hands-on activities using open source technologies are performed in order to develop skills on the practices of participation.

Course Code: SWEG5106

Course Name: Software Evolution and Maintenance

Credit Hour: 3

Prerequisite: SWEG4106, SWEG5103

Once a software product is delivered, it must evolve or change over time. Evolution and Maintenance of software is concerned with continuously correcting, adapting, and perfecting the software. While the Maintenance phase of the life cycle generally begins following a post-implementation period, actual maintenance activities may occur earlier. It is covering the fundamental developments in software evolution and common maintenance practices in the industry. This course begins with an examination of the fundamentals, the nature and need for maintenance and maintenance and evolution models. From there, maintenance proceeds through key issues; the maintenance process and techniques for maintenance; reengineering, refactoring, reuse and finally deals with the cost of maintenance.

Course Code: SWEG5108

Course Name: **Software Defined Systems**

Credit Hour: 3

Prerequisite: SWEG2106, SWEG3101, SWEG5107

The concept of software defined is all the rage these days when it comes to Information Technology Infrastructure. This is true for systems also. You can have software defined systems that are virtual systems or that are cloud based systems.

Software Defined Systems include Software Defined Networking (SDN), Software Defined Storage, Software Defined Servers (Virtualization), Software Defined Datacenters (SDD), Software Defined Security (SDSec), and ultimately Software Defined Clouds (SDCloud) to name a few possibilities. Individual solutions and seamless integration of these abstractions remains in many respects a challenge

The main objective of this course is to understand Software defined Networks, Storage, Communication, and Radios and Understand virtualization concepts like Hypervisor and how it helps in Software Defined Systems.

Course Code: SWEG5110

Course Name: **Selected Topics in Software Engineering**

Credit Hour: 3

Prerequisite: All core courses (Excluding 5th Year 2nd semester courses)

Selected Topics in Software Engineering studies the state of the art of and main research challenges of selected topics in software engineering. It covers current research topics in software engineering such as major challenges/issues, best practices and research directions in various phases of software development life cycle, including requirements analysis and specification, software design, software implementation, software maintenance, and software process. It also covers emerging topics in software engineering. Students are expected to review particular topics and present their findings. The course will also prepare students to course will also

prepare students to create motivation to conduct research in software engineering and to apply that knowledge in their future research and practice.

Course Code: SWEG5025

Course Name: **High Performance Computing**

Credit Hour: 3

Prerequisite:

Introduction to HPC and scientific computing (review of computational complexity, performance: metrics and measurements, basic methods for parallel programming), GPU computing and Cluster analysis, Grid and cluster computing, High-performance parallel algorithms, Distributed and parallel computing, memory hierarchies and network models, multi-core processors, shared-memory symmetric multiprocessors, supercomputers and Peta scale systems, reconfigurable computing), Measuring HPC performance (identifying performance bottlenecks, restructuring and partitioning applications for heterogeneous resources).

Course Code: SWEG5201

Course Name: **Introduction to Big Data Analytics**

Credit Hour: 3

Prerequisite: Introduction to Statistics and Probability

Database Systems

With rapid advances in information and communication technologies (ICTs), modern computing and communication technologies have allowed us to capture, process, store and communicate vast amounts of data. As a result, more and more organizations these days are interested in using their data as decision supporting tool and to build data-intensive products and services. This course provides an introduction to big data technologies and business analytics, which include the use of data, statistical and quantitative analysis, exploratory and predictive models, and evidence-based methods to inform business decisions and actions. The main objective of this course is to enable students to grasp BDA concepts, components and tools that are commonly used to address critical business problems. The course begins with a basic introduction to BDA and explores what the analysis of massive amounts of data entails, as well as associated technical, conceptual and ethical challenges. The unprecedented opportunities and challenges presented by big data are discussed in depth using real-world examples. Students will be then engaged in case studies and team projects in which each group of students develop and present a customized big data solution. The course focuses on practical exercises and hands-on experience in analyzing large and complex datasets on Hadoop Eco-System. The course is designed as an introductory for senior students who are interested in attaining a basic understanding of Big Data and its applications.

4. College of Natural and Applied Sciences

- Bachelor of Science degree in Food Science and applied nutrition
- Bachelor of Science degree in Geology
- Bachelor of Science degree in Biotechnology
- Bachelor of Science degree in Industrial Chemistry

4.1 Bachelor of Science Degree in Food Sciences and Applied Nutrition

4.1.1 Introduction

Mission of teaching-learning, research and community services is mainly supporting transformation of the Ethiopian agriculture and industry sectors. In line with this, being one of the highly demanded sectors in Ethiopia, the Food and Nutrition sector and seeking accreditation. Accordingly, the staffs of the Department of Food Science and Applied Nutrition, AASTU, prepared this curriculum.

In Ethiopia the estimated yearly production of food materials and food products is significantly low to feed the growing population of the country. Though, a remarkable increase rate is recorded in recent years in food products, it is below the increasing per capital food need of the country.

The food industry which originally provided only primary products for final preparation in home; find itself responding to market demands for more refined, sophisticated, and convenient products. The demand to prepare fortified, nutritionally sound safe convenience foods poses major scientific challenges which cannot be met without highly trained human resource capable of understanding the complex chemistry/biochemistry, new product development of food systems and knowledge of how to process, preserve and package them. The increased tendency of society on foods has led to greater responsibility for processors in terms of quality, safety and nutritional value. Thus the current need for Food Science and Nutrition experts at various central and regional government institutions and organizations in the country can be considered highly significant. On the other hand malnutrition is one of the major health problems in Ethiopia affecting primarily children and women of the child bearing age. Protein-energy malnutrition,

vitamin A deficiency (VAD), iron deficiency anaemia (IDA), iodine deficiency disorders (IDD) are the forefront nutritional problems prevailing in the country.

Addis Ababa Science and Technology University in turn, thus, took the initiative to develop a curriculum to produce professionals equipped with all the technical knowledge and professional competence with readiness to assist the society (in various areas of food science, nutrition, and dietetics). Based on the situation in Ethiopia, the graduates are supposed to support farmers and consumers in using modern and traditional techniques of food processing and preservation and thereby alleviate the problem of nutrition and food security issues as well as problems due to unsafe food handling practices, also creating graduates with both the knowledge of food science and nutrition who have the ability to appraise the food industry problems and can find the solution for them to make them competent in the national and international market.

4.1.2 Program Education Objective (PEO)

The program educational objectives describe accomplishments that a four years program graduates are expected to attain within four years after graduation. Graduates will have applied their expertise to contemporary problem solving, be engaged professionally, and have continued to learn and adapt, and have contributed to their organizations through leadership and teamwork. More specifically, the objectives are expertise, engagement, learning, leadership and teamwork.

Table 3. Program Education Objectives (PEO)

PEO-1	Engage and succeed in their professional careers through teamwork, ethical behavior, proactive involvement, and effective communication
PEO-2	Develop an understanding of the importance of life-long learning through professional development, practical training, and specialized certifications; in addition, pursue postgraduate studies and succeed in academic and research careers.
PEO-3	Build progressively managerial, leading, and influential roles in their organizations and communities.

4.1.3 Program Outcomes (PO)

Applied science programs are measured through their attainment of the Program Outcome (PO)/ Program Learning Outcomes (PLO) of the program and its courses. These program outcomes are statements on what students shall know, understand, and perform upon completing their course and/or program of study.

Table : Program Outcome

PLO-1	An ability to identify, formulate, and solve broadly defined technical or scientific problems by applying knowledge of Food Science and Nutrition and/or technical topics to areas relevant to the discipline.
PLO-2	An ability to formulate or design a product, system, process, procedure or program to meet desired needs in Food Science and Nutrition.
PLO-3	An ability to develop and conduct experiments or test hypotheses, analyze and interpret data and use scientific judgment to draw conclusions in the area of Food Science and Nutrition.
PLO-4	An ability to communicate effectively with a range of audiences to provide Food Science and Nutrition Knowledge.
PLO-5	An ability to understand ethical and professional responsibilities in food regulation and the impact of technical and/or scientific solutions in global, economic, environmental, and societal contexts in the area of Food Science and Nutrition
PLO-6	An ability to function effectively on teams that establish goals, plan tasks, meet deadlines, and analyze risk and uncertainty in the sector of Food Science and Nutrition.
PLO-7	An ability to contribute in building Ethiopia, for the peaceful coexistence of peoples, healthy and productive citizen.

4.1.4 Graduation Requirements

Graduation requirement for all undergraduate programs should satisfy the following minimum requirements as stated in the university's senate legislation July 2017, Article 109.

- All the required courses/modules and the minimum credit hours set in the program curriculum by the respective academic unit should be satisfied, except to phase in and phase out program.
- A cumulative grade point average CGPA of 2.00 must be obtained;
- A cumulative grade point average CGPA of 2.00 in major area courses;
- Pass the exit exam
- No "F" grade in any course/ module taken for undergraduate program;
- Score pass mark for all courses which have Pass/Fail grade.

4.1.5 List of Courses and Category

Table: Distribution of credit hours for all course categories

Category		Total Cr. hr.	Percentage (%)
1	Core Course (major and supportive course)	119	78.29
2	Core Elective/focus area course	-	0
3	University requirement (Core)	3	1.97
Core Course Total Cr. Hr		122	80.26
4	National requirement	30	19.74
Total		152	100

List of Courses

Table: List of Courses

0. Common Course		
No.	Course Title	Credit-hour
1	Logic and Critical Thinking	3
2	General Psychology	3
3	Communicative English Language Skill-I	3
4	Geography of Ethiopia and The Horn	3
5	Physical Fitness	P/F
6	Social Anthropology	2
7	Communicative English Language Skill-II	3
8	Moral and Civic Education	2

9	Inclusiveness	2
10	History of Ethiopia and the horn	3
11	Global Trend	2
12	Applied Mathematics IA	4

1. Core/Compulsory Courses		
No.	Course Title	Credit-hour
1	Fundamentals of Analytical Chemistry	3
2	Introduction to Food Science and Nutrition	3
3	Human Anatomy and Physiology	3
4	Food Chemistry	3
5	Principle of Food Processing and Preservation	3
6	Biochemistry	3
7	Food Microbiology	3
8	Food Toxicology	3
9	Food Analysis and Instrumentation	3
10	Unit Operation in Food Processing	3
11	Indigenous food processing and Biotechnology	2
12	Biostatistics	3
13	Sport and Exercise Nutrition	2
14	Fruit and Vegetable Science and Technology	3
15	Dairy Science and Technology	3
16	Meat, Poultry and Fish Science and Technology	3
17	Coffee, Tea, Spice and herbs Science and Technology	3
18	Nutrition Assessment	2
19	Beverage Science and Technology	3
20	Community Nutrition	3
21	Food Safety, Quality Assurance and Legislation	3
22	Food Economics, Marketing and Distribution	2
23	Cereal and Pulse Science and Technology	3
24	Fat and oil Science and Technology	2
25	Practical Attachment	3
26	Sugar, Honey and Confectionery Science and Technology	3
27	Food Fortification and Functional Foods	3

28	Research Method in Food Science and Nutrition	3
29	Food and Nutrition Security	2
30	Nutrition Education and Counseling	2
31	Senior Seminar	1
32	Food and Nutrition Policy and program	3
33	Nutrition throughout the life cycle	3
34	Nutritional Epidemiology	3
35	Diet Therapy	2
36	Food Product Development and Sensory Evaluation	3
37	Food Packaging Technology	3
38	Entrepreneurship	2
39	Mathematics (For Natural Science)	3
40	Economics	3
41	Introduction to Emerging Technology	3
42	General Chemistry	3
42	AASTU (Integrated Science Team Project)	3
43	Final Year Project I	P/F
44	Final Year Project II	4

4.1.6 Program Curriculum Structure: Bachelor of FSAN

Year I, Semester I

Course Code	Course Title	Cr.Hrs	Lect.	Tut.	Lab
Phil1009	Logic and Critical Thinking	3			
Psyc1011	General Psychology	3			
ELEn1013	Communicative English Language Skill-I	3			
GeEs1005	Geography of Ethiopia and The Horn	3			
Math1001	Mathematics (For Natural Science)	3			
SpSc1013	Physical Fitness	P/F			
Chem1006	General Chemistry	3			
	Total	18			

Year I, Semester II

Course Code	Course Title	Cr.Hrs	Lect.	Tut.	Lab
EmTe2011	Introduction to Emerging technology	3			
Entr1108	Entrepreneurship	2			
Anth1002	Social Anthropology	2			
ELEn1004	Communicative English Language Skill-II	3			
Math1014	Applied Mathematics IA	4			
MCiE1012	Moral and Civic Education	2			
Incl 1010	Inclusiveness	2			
	Total	18			

Year II, Semester I

Course Code	Course Title	Cr.Hrs	Lect.	Tut.	Lab
Econ2009	Economics	3			
GLTr 2001	Global Trend	2			
Inch2205	Fundamentals of Analytical Chemistry	3	2		1
FSAN2101	Introduction to Food Science and Nutrition	3	2	1	
FSAN2103	Human Anatomy and Physiology	3	2		1
FSAN2105	Food Chemistry	3	2	1	
FSAN2107	Principle of Food Processing and Preservation	3	2	1	
Total		20			

Year II, Semester II

Course Code	Course Title	Cr.Hrs	Lect.	Tut.	Lab
Hist 2002	History of Ethiopia and the horn	3			
Inch2503	Biochemistry	3	2	1	
FSAN2102	Food Microbiology	3	2		1
FSAN2104	Food Toxicology	3	2		1
FSAN2106	Food Analysis and Instrumentation	3	2		1
FSAN2108	Unit Operation in Food Processing	3	2		1
FSAN2110	Indigenous food processing and Biotechnology	2	1		1
Total		20			

Year III, Semester I

Course Code	Course Title	Cr.Hrs	Lect.	Tut.	Lab
FSAN3109	Biostatistics	3	2	1	
FSAN3111	Community Nutrition	3	2	1	
FSAN3113	Fruit and Vegetable Science and Technology	3	2		1
FSAN3115	Dairy Science and Technology	3	2		1
FSAN3117	Cereal and Pulse Science and Technology	3	2		1
FSAN3119	Coffee, Tea, Spice and herbs Science and Technology	3	2		1
Total		18			

Year III, Semester II

Course Code	Course Title	Cr. Hrs	Lect.	Tut.	Lab
ISTP3116	Integrated Science Team Project	3			
FSAN3112	Sport and Exercise Nutrition	2	1	1	
FSAN3114	Beverage Science and Technology	3	2		1
FSAN3118	Food Fortification and Functional Foods	3	2		1
FSAN3120	Research Method in Food Science and Nutrition	3	2	1	
FSAN3122	Nutrition Assessment	2	1		1
FSAN3124	Meat, Poultry and Fish Science and Technology	3	2		1
Total		19			

Year III, Summer Semester

Course Code	Course Title	Cr.Hrs	Lect.	Tut.	Lab
FSAN3126	Practical Attachment	3	2	1	
Total		3			

Year IV, Semester I

Course Code	Course Title	Cr.Hrs	Lect.	Tut.	Lab
FSAN4121	Food Economics, Marketing and Distribution	2	1	1	
FSAN4123	Honey, sugar and confectionery Science and Technology	3	2		1
FSAN4125	Fat and oil Science and Technology	2	1		1
FSAN4127	Food Product Development and Sensory Evaluation	3	2		1
FSAN4129	Food and Nutrition Policy and program	3	2	1	
FSAN4131	Nutrition Throughout the life cycle	3	2	1	
FSAN4133	Senior Seminar	1		1	
FSAN4135	Final Year Project I	P/F			
Total		17			

Year IV, Semester II

Course Code	Course Title	Cr.Hrs	Lect.	Tut.	Lab
FSAN4128	Food and Nutrition Security	2	1	1	
FSAN4130	Nutrition Education and Counseling	2	1	1	
FSAN4132	Nutritional Epidemiology	3	2	1	
FSAN4134	Diet Therapy	2	1	1	
FSAN4136	Food Safety, Quality Assurance and Legislation	3	2	1	
FSAN4138	Food Packaging Technology	3	2		1
FSAN4140	Final Year Project II	4	3	1	
Total		19			

Conversion rate: from Cr.hr to ECTS is given as, 1 Cr.hr. =1.67ECTS

4.1.7 Program Courses Synopsis

Course Code: Math 1007

Course Name: Mathematics for natural science

Credit Hour: 3

Prerequisite: No

This course covers the basic concepts of logic and set theory, the real and complex number systems, Mathematical induction, least upper bound and greatest lower bound, functions and types of functions, polynomial and rational functions, logarithmic and exponential functions, trigonometric functions, hyperbolic functions and their graphs and analytic geometry.

4.1.7.1 Supportive courses

Course Code: InCh 2205

Course Name: Fundamentals of Analytical Chemistry

Credit Hour: 3

Prerequisite: None

The course covers ; Introduction to analytical chemistry; Acids and bases, neutralization titration; Complex formation, and complexometric titration: Gravimetric analysis; Chromatographic technique: paper, thin layer, gas chromatography and HPLC; Spectroscopic techniques; UV-Vis, AAS, IR, and introduction to mass spectroscopy

and Nuclear Magnetic Resonance spectroscopy (NMR), fluorescence, and coupled techniques such as HPLC/GC-MS, ICP-MS/OES,

Course Code: Math1002

Course Name: Applied Mathematics IA

Credit Hour: 4

Prerequisite: None

This course is designed to develop students' confidence with mathematical concepts and relationships and use of mathematical skills and techniques in a range of contexts specifically problem solving and abstract thinking. Topics covered are basic elements of vectors, vector spaces, matrices, determinants, solving systems of linear equations, concepts and applications of differential and integral calculus of one variable.

Core course Course Code: Ent1108

Course Name: Entrepreneurship for Science

Credit Hour: 2

Prerequisite: No

This course is designed to encourage students to start their own business and to acquaint them with the peculiar challenges and management decisions faced by owners of small business. It will develop such entrepreneurial skills as: identifying business opportunities; initiating, financing, and developing new venture business plans. It also addresses issues of small business, such as: legal aspects, financing, costing, locating, personnel, marketing, competition, sources of funding, and constituency services.

Course Code: EmTe 2011

Course Name: Introduction to Emerging technologies

Credit Hour: 3

Prerequisite: No

This course will enable students to explore current breakthrough technologies in the areas of Artificial Intelligence, Internet of Things and Augmented Reality that have emerged over the past few years. Besides helping learners become literate in emerging technologies, the course will prepare them to use technology in their respective professional preparations.

Course Code: InCh1011

Course Name: General Chemistry

Credit Hour: 3

Prerequisite: No

The course General chemistry is designed to make students more prepared to the all chemistry courses by refreshing and summarizing the previous preparatory chemistry concepts before tackling the advanced chemistry courses. The course covers units and measurement, the composition of matter, chemical reactions, reactions stoichiometry, atomic structure and the periodic table, the chemical bond, structure of molecules, the properties of solutions, chemical equilibrium, introduction to functional groups and their typical reactions.

Course Code: InCh2508

Course Name: Biochemistry

Credit Hour: 3

Prerequisite: No

The course, Biochemistry is designed to make our students familiar with the different types of biological molecules, their structure and its application in industries. So that they will understand the applications of biochemistry in life and in industry. Moreover, the students will understand the different metabolic reactions and pathways in different kinds of living things.

Course Code: FSAN2101

Course Name: Introduction to food science and nutrition

Credit Hour: 3

Prerequisite: None

Introduction to food science and nutrition course covers the major food components (water, fats, carbohydrates and protein) in terms of their nature and role in foods. It also covers the nutritional principles of fat carbohydrate and protein and how dietary macronutrient requirement values are derived.

Course Code: FSAN2103

Course Name: Human Anatomy and Physiology

Credit Hour: 3

Prerequisite: None

Human anatomy and physiology course covers the structure and function of the tissues, the skeletal system, the nervous system, the endocrine system, and muscle function from the level of the cell to the level of the organism.

Course Code: FSAN2105

Course Name: Food Chemistry

Credit Hour: 3

Prerequisite: None

Food chemistry deals with the chemical, physical and functional properties of food constituents and the chemical changes these constituents undergo during handling, processing and storage including those that limit food shelf life. Food chemistry is a major aspect of Food science, that is an interdisciplinary subject in which the engineering, biological, and physical sciences are used to study the nature of foods, the causes of deterioration, the principles underlying food processing, and the improvement of foods from a consumer and sustainability perspective. The aim

of the course is to provide the students with a deep understanding of how food components contributes to overall quality of foods; and to enable students to evaluate and explain how the highly complex nature of food may result in a multitude of desired and undesired reactions which are controlled by a variety of parameters.

Course Code: FSAN2107

Course Name: Principle of Food Processing and Preservation

Credit Hour: 3

Prerequisite: None

Food processing and preservation focus on the importance of food processing and preservation. Classification of foods on the basis of shelf life, pH, origin; Different types of food spoilage viz. microbiological, bio-chemical, chemical, physical and their effects on food quality

Course Code: FSAN2102

Course Name: Food microbiology

Credit Hour: 3

Prerequisite: None

Food microbiology focuses on the general biology of the microorganisms that are found in foods including: their growth characteristics, identification, and pathogenesis. Specifically, areas of interest which concern food microbiology are food poisoning, food spoilage, food preservation, and food legislation

Course Code: FSAN2104

Course Name: Food toxicology

Credit Hour: 3

Prerequisite: None

Food Toxicology covers various aspects of food safety and toxicology, including the study of the nature, properties, effects, and detection of toxic substances in food and their disease manifestations in humans. It will also include other aspects of consumer product safety.

Course Code: FSAN2106

Course Name: Food Analysis and Instrumentation

Credit Hour: 3

Prerequisite: None

Food analysis and instrumentation course focused on the application of qualitative and quantitative techniques used in the physical, chemical, sensory and instrumental examination of food products.

Course Code: FSAN2108

Course Name: Unit operation in food processing

Credit Hour: 3

Prerequisite: None

Unit operation in food processing focuses on different unit operations like mixing, pumping, heating, cooling, concentration, drying, filtration, evaporation, distillation, extraction, and many others.

Course Code: FSAN3109

Course Name: Biostatistics

Credit Hour: 3

Prerequisite: None

The field of Biostatistics allows students to use statistical principles to find patterns in public health, maintain consistent clinical trials, or conduct research in other areas of human health where pattern finding can lead to breakthroughs. Data is so crucial in the research field and processing that data spurs innovations in new treatments and medicines. It also helps us better predict the outcomes of public health issues and could even teach us how to see the beginnings of problems before they become a public health crisis. Biostatisticians are at the forefront of using big data to the advantage of human health.

Course Code: FSAN3111

Course Name: Community Nutrition

Credit Hour: 3

Prerequisite: None

This course will examine the role of nutrition in promoting, maintain and improving health in the community. It investigates traditional aspects of the emerging health delivery system, as well as entrepreneurial ventures. Students will study the financial, legislative, political, sociological, and scientific aspects of public and community health nutrition

Course Code: FSAN3113

Course Name: Fruit and vegetable science and technology

Credit Hour: 3

Prerequisite: None

The goal of this course is to introduce the students to the sciences of fruit and vegetable with the overall objective of introducing to the student to the theories, principles and practices utilized in production. It is the objective of this course to increase the capacity of the student for intelligent decision-making and problem solving based upon scientific knowledge and resources of fruits and vegetables.

Course Code: FSAN3115

Course Name: Dairy science and technology

Credit Hour: 3

Prerequisite: Successful completion of Food Processing and Preservation FSAN2041

Dairy science and technology is a field that encompasses the production and manufacturing of all dairy products as well as the machinery and methods used in the dairy industry. The largest part of the food supply chain is, by far, the dairy industry. This industry is an integral part of our food economy that not only supplies consumers with many ready-to-eat products such as milk, butter, and cheese but also produces many of the ingredients like milk powder and condensed milk that are found in processed foods. Milk itself has also become a key ingredient for the deployment of probiotics and the development of functional food products designed to improve consumer health. As such, dairy products have become an area of accelerated research and innovation, particularly in the areas of processing, sustainability, and health, and marketing strategy.

Course Code: FSAN3124

Course Name: Meat, Poultry and Fish Science and Technology

Credit Hour: 3

Prerequisite: Successful completion of Food Processing and Preservation

This course is included in the curriculum to make students familiar with the basic knowledge and skills of various technologies and equipment used for production of raw as well as processed meat, fish and poultry products. It will also familiarize students with chemical composition and microscopic structure of meat. It will make an in depth examination of the effect of feed, breed and management on meat production and quality; slaughtering of animals and poultry, inspection and grading of meat. Moreover, it will discuss the structure, composition, nutritive value and functional properties of eggs and its preservation by different methods; factors affecting egg quality and measures of egg quality. Types of fish, composition, structure, post-mortem changes in fish; handling of fresh water fish; canning, smoking, freezing and dehydration of fish; fish sausage and home making will also be dealt with.

Course Code: FSAN3119

Course Name: Coffee, Tea, Spice, and herbs Science and Technology

Credit Hour: 3

Prerequisite: Principles of Food Processing and Preservation

The course aims to introduce students to the traditional as well as state of the art industrial processing and marketing of coffee, tea and spices. It deals with the importance of coffee in the Ethiopian economy and the potential of tea and spices as a foreign currency earning commodity. It enables students to develop the skills in wet and dry processing of coffee and tea

Course Code: FSAN4016

Course Name: Integrated Science Team Project

Credit Hour: 3

Prerequisite: None

This is a multidisciplinary integrated science team project for special science computational problems under the guidance of a faculty member and a computer Programr. The course introduces the student with computer programing basics using python language to help students deal with scientific problems by drawing methods and concepts from biology, chemistry, physics and mathematics. The course deals with binary computation, problem-solving methods and algorithm development. The course also covers data types, control structures, functions, parameter passing, library functions, arrays, inheritance and graph plotting using python Matplotlib. It is aimed to aid multidisciplinary team based science problem solving and product development.

Course Code: FSAN3112

Course Name: Sport and Exercise Nutrition

Credit Hour: 2

Prerequisite: None

This class will focus on the nutritional needs and requirement of different exercises. First, the student will recognize and establish sound nutrition principles and the nutrients that play a role in determining these principles. Nutrients and other food constituents are integrated into the human body. Nutrition plays an integral part of our daily lives. It becomes even more important for athletes, who push the physical limits of their body. Providing adequate fuel is essential and can make a difference between success and failure. Yet most athletes and coaches lack basic nutrition knowledge important for enhancing strength, speed and endurance.

Course Code: FSAN3122

Course Name: Nutritional Assessment

Credit Hour: 2

Prerequisite: None

The goal of the course is to introduce the concepts of nutritional assessment and the practical application of these concepts in the nutritional care of clients in clinical, community, and research settings. It focuses on methods of human nutritional assessment, including anthropometric (A), biochemical (B), clinical (C), and dietary (D) approaches (the A, B, C, Ds). Students will learn how to apply nutrition assessment methods across various settings ranging from the individual level (e.g., clinic) to the population level (e.g., community). Students will also develop knowledge and skills in the Nutrition Care Process, which includes nutrition screening, assessment, diagnosis, and preparing and writing nutrition assessments.

Course Code: FSAN3114

Course Name: Beverage Science and Technology

Credit Hour: 3

Prerequisite: Successful completion of Food Processing and Preservation

Being graduates of food science and applied nutrition, students are required to have a basic skill and theoretical knowledge of beverage processing and technology. With the increasing consumption of beverages all over the world, the production needs to be managed and supervised by those who are capable of producing safe and quality products.

Course Code: FSAN3118

Course Name: Food Fortification and functional food

Credit Hour: 3

Prerequisite: None

Food fortification is the enrichment of food by adding micronutrients. The course is designed to cover food fortification, types of fortification, physical characteristics, selection and use with specific food vehicle: minerals and vitamins, diets in specific disease, formulation of diets for target groups along the lifecycle, functional food, antioxidant, and different foods as functional food, anti-nutritional factors.

Course Code: FSAN3120

Course Name: Research Method in Food Science and Nutrition

Credit Hour: 3

Prerequisite: None

Research Method in Food Science and Nutrition offers an overview of research methodology and the basic concepts employed in quantitative and qualitative research methods. The course covers a brief description on the types of research, research and experiment, fundamentals of sampling and data collection, analysis of experiments, fundamentals of designs of experiments, basic principles of design of experiments, summarizing data sets using statistics, measures of data variability, analysis of variance (ANOVA), completely randomized design (CRD), randomized complete block designs (RCBD) Latin square designs (LSD), designs and analysis of multiple factors experiments, two, three and fractional factorial experiments, regression and correlation analysis, nonlinear regression analysis, selection of research topics, research proposal writing, scientific report writing in relation to food science and nutrition.

Course Code: FSAN3117

Course Name: Cereal and pulse Science and Technology

Credit Hour: 3

Prerequisite: None

To acquaint with production trends, structure, composition, quality evaluation and processing technologies for product development and value addition of various cereals and pulses.

Course Code: FSAN4125

Course Name: Food economics, marketing and distribution

Credit Hour: 2

Prerequisite: None

This course is based on a systems approach to the study of food production, processing, marketing, and consumption. The importance of technology and economic relationships is emphasized. Students will learn a variety of concepts, including analytical models and how to use them in assessment of the real-world marketing strategies, consumer behavior, and market performance.

Course Code: FSAN4127

Course Name: Honey, Sugar and Confectionery Science and Technology

Credit Hour: 3

Prerequisite: Principle of Food Processing and Preservation

The course is designed to provide students with sugar, honey confectionery sciences and technological applications. Sugar, Honey and Confectionery Science and Technology covers sugar technology, analysis of quality of raw materials, extraction and refining of sugar, honey production in Ethiopia, the process of honey synthesis and ripening, botanical origin, physicochemical and biofunctional property of honey, honey processing and quality attributes, sugar and chocolate confectioneries.

Course Code: FSAN4132

Course Name: Food safety, quality assurance and legislation

Credit Hour:

Prerequisite: None

In this course, students will acquire the knowledge necessary for the assurance of food quality and safety. After passing through this course, they will be able to orient themselves in legislation relevant for the performance of official controls. Basing on practical skills and knowledge acquired when elaborating HACCP plans they will be able to prevent risks jeopardizing people either directly or indirectly (environmentally) and to eliminate them or, at least, to reduce them to an acceptable level.

Course Code: FSAN4133

Course Name: Food and Nutrition Security

Credit Hour: 2

Prerequisite: None

Human Nutrition provides an integrated overview of the physiological requirements and functions of different macro and micronutrients, which are determinants of health and diseases in human populations. The course cover food security and its dimensions, levels of analysis of food security, the conceptual framework of malnutrition, instruments for assessment and analysis of food and nutrition security (FNS) situation at macro level, indicators & instruments for assessment & analysis of FNS situation at micro & meso level, designing a plan for assessing and analysing the FNS situation, interventions to improve FNS at macro level, interventions to improve FNS at micro and meso level, developing appropriate strategies for FNS.

Course Code: FSAN4135

Course Name: Nutrition Throughout the Life Cycle

Credit Hour: 3

Prerequisite: None

Nutrition needs throughout the stages of the life cycle, which include pregnancy and lactation, infancy, adolescence and aging. Nutrition throughout the Life Cycle encompasses the socioeconomic, cultural and psychological influences on food and nutrition behavior. The course covers macronutrients and micro nutrients, the lifespan approach to nutrition, malnutrition and associated diseases, assessment of nutritional status of individuals and populations, preconception nutrition, nutrition during pregnancy, nutritional requirements during lactation, infant and child nutrition, school age nutrition, adolescent nutrition, adult nutrition and nutrition and aging.

Course Code: FSAN4137

Course Name: Senior Seminar

Credit Hour: 1

Prerequisite: Successful completion of practical attachment

Senior Seminar is a course prepared by the student for public lecture in support of their instructor, which concentrates on delivering information on the pertinent area of Food Science and Nutrition. The students are engaged to prepare a written document using a scientific way of writing, which consequently presented for students, staffs and interested groups.

Course Code: FSAN4139

Course Name: Final Year Project I

Credit Hour: P/F

Prerequisite: Research methods in Food Science and Nutrition (FSAN3118)

This is an individual research project in connection with Food Science and applied Nutrition cases and under the guidance of a faculty member. The project undertaken may fall under one of the following areas; nutritional composition analysis, sensory analysis, nutritional assessment, product development. The project work includes literature review, research problem identification, proposal development, preliminary research work and interim report generation.

Course Code: FSAN4124

Course Name: Food and Nutrition Policy and program

Credit Hour: 3

Prerequisite: None

The purpose of this course is to familiarize and engage the student in the steps and dynamics of policy making processes that address nutrition problems and issues. Where ever nutrition problems exist, policy and program options address the problems. Food and Nutrition Policy and program course covers the definitions of policies, strategies, and related terms, the concept nutrition policy and food strategy, the process of policy and strategy development, dietary standards, food guides and dietary goals, national nutrition policies, strategies of nutrition improvement, approaches/ strategies for improving nutrition and health status of the community, selecting policy options and developing policy strategy, drafting public health legislation, reviewing proposed public health legislation, legislative oversight, developing policies, strategies and public health programs.

Course Code: FSAN4126

Course Name: Nutrition Education and Counselling

Credit Hour: 2

Prerequisite: None

This course aims to provide students with the theories and skills necessary to design and implement nutrition education programs. It shows how nutrition education can facilitate dietary changes in a group of people. Specific educational counselling of customers with specific health conditions is also emphasized. The course cover the concept of nutrition education and counselling, dietary counselling and communication for behavior change, development of a plan for nutrition education Programs in community, designing a plan for assessing and analysing the FNS situation, selecting the right counselling method, nutrition education in emergencies.

Course Code: FSAN4128

Course Name: Nutrition Epidemiology

Credit Hour: 3

Prerequisite: Biostatistics

The course directed students to key concepts and methods in Nutrition Epidemiology in order to equip them with the tools needed to design, analyse, and critically evaluate population-based nutrition research. Nutrition Epidemiology covers evolution of epidemiology, important terminologies and their definitions, types and purpose of epidemiology, basic epidemiologic assumptions, natural history of disease, epidemiological approach in disease causation, frequency measures used in epidemiology, epidemiological designs, nutritional assessment methods, surveillance and outbreak investigation, introduction to nutrition screening and surveillance, outbreak investigations.

Course Code: FSAN4130

Course Name: Diet Therapy

Credit Hour: 2

Prerequisite: Human Physiology and Anatomy

This *course* explores the various nutrients, their sources, digestion, absorption, metabolism, interaction, storage, and excretion with relating to human health. The course Diet Therapy cover diet and therapeutic nutrition, food service in health care institutions, principles and objective of diet therapy: screening, assessment, monitoring, and counselling and diet manuals, methods of nutrient delivery: oral delivery, enteral tube feeding and parenteral feeding, interactions: food and nutrients versus medications and supplements, diet in renal disease, diet in diabetes mellitus and hypoglycaemia, other health disorders.

Course Code: FSAN4131

Course Name: Food Product Development and Sensory Evaluation

Credit Hour: 3

Prerequisite: None

This course is designed to provide students with a basic ability of product development and sensory evaluation. Food product development and sensory evaluation is designed to successfully produce food prototypes or food concepts, formulate products by preparing laboratory samples and sourcing raw materials, develop formulations to meet cost targets, design effective food safety plans (HACCP), run different sensory test methods and statistically analyse all data from sensory tests. The course include product development processes, developing strategies, stages of product development, the consumers in product development, shelf life and safety issues, sensory physiology, sensory attributes and ways of perception, consumer- and product- oriented testing, factors influencing sensory judgment, basic description and application of sensory test methods, sensory data analysis and interpretation, computer laboratory and desk discussion.

Course Code: FSAN4136

Course Name: Final Year Project II

Credit Hour: 4

Prerequisite: Final Year Project I

Final year project is a BSc Thesis work in the area of Food Science and Applied Nutrition, which is prepared by graduating students with the support of their assigned advisors. The course is the continuation of final project one. The proposal developed in the previous semester will be used for the project. Designed to develop a BSc Thesis based on the scientific methods of writing. The thesis will be defended and finally submitted to the Department of Food Science and Applied Nutrition.

4.2 Bachelor of Science Degree in Geology

4.2.1 Introduction

Geology teaching and research in Ethiopia started in 1956 GC when the Department of Geology was established in the Addis Ababa University. Through the years, there have been significant curriculum revisions which have taken into account new ideas and advances in the field. Then another geology Department named Department of Applied Geology has been established at the Mekelle University in 1997. Following the higher education expansion program of the Government of Ethiopia in early 2000s, several other geology institutions have been established or are being established in existing or new universities in the country. Among them Department of Applied Geology of Arba Minch University and Department of Earth Sciences of Wollega University have started teaching and research activities in 2005 and 2007 respectively. The Departments have been conducting training and research in the wide areas of geosciences in collaboration with local and international partners and institutions. These have led to significant contribution in the development of human resources in geosciences and in understanding the geology and geologic resources of Ethiopia. Notable contributions include generation of knowledge in the geodynamics of the East African Rift, the petrogenesis of various rocks of Ethiopia, the reconstruction of paleo-environmental setting of Ethiopia, the inventory/baseline survey of earth materials and earth resources of Ethiopia. Researchers in the Departments and graduates from the Departments have been contributing in sectors such as geological surveying, groundwater, construction, energy, mining, and environment.

4.2.2 Program Education Objective (PEO)

The program educational objectives describe accomplishments those four years geology program graduates are expected to attain within four years after graduation. Graduates will have applied their expertise to contemporary problem solving, be engaged professionally, and have continued to learn and adapt, and have contributed to their organizations through leadership and teamwork. More specifically, the objectives are expertise, engagement, learning, leadership and teamwork.

Table: Program Education Objectives (PEO)

PEO	Statement
PEO-1	Apply and evaluate data, models, hypotheses, and conclusions published in scientific literatures.
PEO-2	Identify problems, design projects and conduct an independent geoscience investigation
PEO-3	Communicate the findings of the geoscientific investigations in the form of presentations, reports and publications.
PEO-4	Investigate geoscientific issues that affect the society and to find scientifically viable solutions
PEO-5	Collaborate with others, including multidisciplinary groups to solve scientific problems in accordance with professional and ethical standards
PEO-6	Develop the abilities to engage in lifelong learning

4.2.3 Program Outcomes (PO):

Applied science programs are measured through their attainment of the Program Outcome (PO)/Program Learning Outcomes (PLO) of the program and its courses. These program outcomes are statements on what students shall know, understand, and perform upon completing their course and/or program of study. State the Program Out comes of the department by referring (Appendix 3.2).

Table: Program Outcome

PO	Statement
PO-1	Ability to integrate the concept of natural and numerical sciences in solving complex geological problems
PO-2	Ability to identify, describe, measure and record geological data using conventional and modern technologies in order to produce geoscience maps with accompanying report
PO-3	Ability to analyze, synthesize and apply fundamental geoscience knowledge into actionable practices in sustainable earth resources evaluation and development.

PO-4	Ability to conduct basic and applied geoscience researches and be able to communicate with stakeholders (public, scientific community and industries)
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4.2.4 Graduation Requirements

Graduation requirement for all undergraduate programs should satisfy the following minimum requirements as stated in the university's senate legislation July 2017, Article 109.

- All the required courses/modules and the minimum credit hours set in the program curriculum by the respective academic unit should be satisfied, except to phase in and phase out program.
- A cumulative grade point average CGPA of 2.00 must be obtained;
- A cumulative grade point average CGPA of 2.00 in major area courses;
- Pass the exit exam
- No "F" grade in any course/ module taken for undergraduate program;
- Score pass mark for all courses which have Pass/Fail grade.

4.2.5 List of Courses and Category

Table: Distribution of credit hours for all course categories

Category		Total Cr. hr	Percentage (%)
1	Core Course (major and supportive course)	113 Cr.hr	80.8 %
2	Core Elective/focus area course	6 Cr.hr	
3	University requirement (Core)	3 Cr.hr	
Core Course Total Cr. hr			
4	National Requirement	29 Cr.hr	19.2%
Total Cr. hr		151%	100%

List of Courses

Table: List of Courses

Common Courses		
No.	Course Title	Credit-hour
01	Fundamentals of Programming (Comp3003)	3
02	Logic and Critical Thinking (Phil1009)	3
03	General Psychology (<i>Psyc 1011</i>)	3
04	Communicative English Skill – I (EnLa 1003)	3
05	Physical Fitness & Conditioning (SpSc 1013)	P/F
06	Introduction to Anthropology (Anth1002)	2
07	Communicative English Language Skill-II (EnLa1004)	3
08	Moral and Civic Education (MCiE1012)	2
09	Inclusiveness (Incl010)	2
10	History of Ethiopia and the horn (Hist2002)	3
11	Economics (Econ 2009)	3
12	Global Trend (GLTr2001)	2

Core Courses			
No.	Course Title	Credit-hour	Pre-requisite
01	General Physics (Phys2003)	3	
02	Introduction to statistics (Stat2006)	3	
03	Physical Chemistry (InCh2004)	3	
04	Geography of Ethiopia and The Horn (GeES 1005)	3	
05	Mathematics (For Natural Science) (Math1007)	3	
06	Applied Mathematics IA (Math1014)	4	
07	General Chemistry (InCh1001)	3	
08	Emerging Technology for Science (EmTe	3	

	2011)		
09	Entrepreneurship for Science (Entr1108)	3	
10	General Geology (Geo2101)	4	
11	Paleontology (Geol2103)	3	
12	Crystallography and Mineral Optics (Geol2105)	3	
13	Mineralogy (Geol2108)	3	Geol2103
14	Stratigraphy and Earth History (Geol2102)	3	Geol2101
15	Plate Tectonics (Geol 2104)	2	
16	Sedimentary Petrology (Geol2106)	3	Geol2102
17	Igneous Petrology (Geol2110)	3	
18	Structural Geology (Geol3101)	4	Geol2101
19	Geological Mapping and Report Writing (Geol3103)	3	
20	Remote Sensing and GIS (Geol3105)	3	
21	Mapping Sedimentary Tarrain (Geol3107)	3	Geol2106, Geol3103
22	Mapping Igneous Tarrain (Geol3102)	3	Geol2108, Geol3103
23	Geophysics (Geol3104)	4	
24	Fundamentals of Soil and Rock Mechanics (Geol3106)	3	Geol3101
25	Principles of Hydrogeology (Geol3108)	3	
26	Metamorphic Petrology (Geol3110)	3	
27	Geochemistry (Geol3112)	3	
28	Groundwater Exploration and Development (Geol4101)	3	Geol3108
29	Mapping Metamorphic Terrain (Geol4103)	3	Geol3110, Geol3103
30	Internship (Geol4113)	3	

31	Economic Geology (Geol4105)	3	Geol2103
32	Engineering Geology (Geol4107)	3	Geol3106
33	Mineral Exploration and Resource Evaluation (Geol4109)	2	
34	Research Methods in Geosciences (Geol4111)	2	
35	Mining Geology(Geol4108)	2	Geol4105
36	Energy Resource of the Earth (Geol4102)	3	
37	Geology and Geologic Resource of Ethiopia (Geol4104)	3	
38	Environmental Geology (Geol4106)	2	
39	Fynal Year project (Geol4110)	3	Geol4111

Core Elective Courses		
No.	Course Title	Credit-hour
1	Industrial Minerals and Rocks (Geol4201)	3
2	Introduction to Gemology (Geol4203)	3
3	Introduction to Geohazard Assesment (Geol4204)	3
4	Environmental Impact assessment (Geol4206)	3
5	Groundwater Resources Management (Geol4202)	3
6	Quaternary Geology (Geol4205)	3

4.2.6 Program Curriculum Structure: Bachelor of geology

Table Course Breakdown for Regular Program

Year I, Semester I

Course Code	Course Title	Cr.Hr.	Lect.	Tut.	Lab.
Chem1001	General Chemistry	3	2	3	
FLEn1003	Communicative English Language Skills I	3	2	3	
GeEs1005	Geography of Ethiopian and the horn	3	2	3	

Math1007	Mathematics for Natural Science	3	2	3	
Phil1009	Logic and Critical Thinking	3	2	3	
Psyc1011	General Psychology	3	2	3	
SpSc1013	Physical Fitness	P/F	2		
Total		18			

Year I, Semester II

Course Code	Course Title	Cr.Hr.	Lect.	Tut.	Lab.
Anth1002	Social Anthropology	2	2		
FLEn1004	Communicative English Language Skills II	3	2	3	
Entr1108	Entrepreneurship for Science	3	2	3	
EmTe2011	Emerging Technology for Science	3	2		
Incl1010	Inclusiveness	2	2		
MCiE1012	Moral and Civic Education	2	2		
Math1014	Applied Mathematics IA	4	3	3	
Total		19			

Year 2, Semester I

Course Code	Course Title	Cr.Hr.	Lect.	Tut.	Lab.
Geo2101	General Geology	4	3		3
Geol 2103	Paleontology	3	2		3
Geol2105	Crystallography and Mineral Optics	3	2		3
GLTr2001	Global Trend	2	2		
Phys2003	General Physics	3	2	3	-
Econ2009	Economics	3	2	3	
Total		18			

Year 2; Semester II

Course Code	Course Title	Cr.Hr.	Lect.	Tut.	Lab.
InCh2004	Physical Chemistry	3	2	3	-
Hist2002	History of Ethiopia and the horn	3	2	3	-
Stat2006	Introduction to Statistics	3	2	3	-
Geol2102	Stratigraphy and Earth History	3	2	3	-
Geol2104	Plate Tectonics	2	2	2	-
Geol2106	Sedimentary Petrology	3	2	-	3
Geol2108	Mineralogy	3	2		3
Total		20			

Year 3, Semester I

Course Code	Course Title	Cr.Hr.	Lect.	Tut.	Lab.	Field work
Comp3003	Fundamentals of Programming	3	2	-	3	
Geol3101	Structural Geology	4	3	-	3	
Geol3103	Geological mapping and Report Writing	3	2	-	3	
Geol3105	Remote Sensing and GIS	3	2	-	3	
Geol 3107	Mapping Sedimentary Terrain	3	-	-	3	3
Geol3109	Igneous Petrology	3	2	-	3	
Total		19				

Year 3, Semester II

Course Code	Course Title	Cr.Hr.	Lect.	Tut.	Lab.	Field work
Geol3102	Mapping Igneous Terrain	3	-	-	3	3
Geol3104	Geophysics	4	3	3	-	

Geol3106	Fundamentals of Soil and Rock Mechanics	3	2	-	3	
Geol3108	Principles of Hydrogeology	3	2	3	-	
Geol3110	Metamorphic Petrology	3	2	-	3	
Geol3112	Geochemistry	3	2	-	3	
Total		19				

Course Code	Course Title	Cr.Hr.	Lect.	Tut.	Lab.	Field work
Geol4113	Internship	3	-	-	-	Summer
Total		3				

Year 4, Semester I

Course Code	Course Title	Cr.Hr.	Lect.	Tut.	Lab.	Fieldwork
Geol4101	Groundwater Exploration and Development	3	2	3		
Geol4103	Mapping Metamorphic Terrain	3	-	-	-	3
Geol4105	Economic Geology	3	2	-	3	
Geol4107	Engineering Geology	3	2	3		
Geol4109	Mineral Exploration and Resource Evaluation	2	2	-		
Geol4111	Research Methods in Geoscience	2	2	-		
Geol4115	Final Year I project Proposal preparation	P/F				
Total		16				

Year 4; Semester II

Course Code	Course Title	Cr.Hr.	Lect.	Tut.	Lab./Field
Geol4102	Energy resource of the Earth	3	2	3	
IGTP4016	Integrated Geological Team	3			3

	Project				
Geol4104	Geology and Geologic Resources of Ethiopia	3	2	3	
Geol4106	Environmental Geology	2	2	3	
Geol4108	Mining Geology	2	2	3	
Geol4110	Final year project	3			3
	Elective	3	2	3	
Total		19			

Conversion rate: from Cr.hr to ECTS is given as, 1Cr.hr= 1.67 ECTS

4.2.7 Program Courses Synopsis

4.2.7.1 Core course Synopsis

Course Code: Math 1007

Course Name: Mathematics for natural science

Credit Hour: 3

Prerequisite: No

This course covers the basic concepts of logic and set theory, the real and complex number systems, Mathematical induction, least upper bound and greatest lower bound, functions and types of functions, polynomial and rational functions, logarithmic and exponential functions, trigonometric functions, hyperbolic functions and their graphs and analytic geometry.

Course Code: EmTe 2011

Course Name: Introduction to Emerging technologies

Credit Hour: 3

Prerequisite: No

This course will enable students to explore current breakthrough technologies in the areas of Artificial Intelligence, Internet of Things and Augmented Reality that have emerged over the past few years. Besides helping learners become literate in emerging technologies, the course will prepare them to use technology in their respective professional preparations.

Course Code: InCh1101

Course Name: General Chemistry

Credit Hour: 3

Prerequisite: No

The course General chemistry is designed to make students more prepared to the all-chemistry courses by refreshing and summarizing the previous preparatory chemistry concepts before tackling the advanced chemistry courses. The course covers units and measurement, the composition of matter, chemical reactions, reactions stoichiometry, atomic structure and the periodic table, the chemical bond, structure of molecules, the properties of solutions, chemical equilibrium, introduction to functional groups and their typical reactions.

Course Code: Math 1114

Course Name: Applied Mathematics I A

Credit Hour: 4

Prerequisite: No

This course is designed to develop students' confidence with mathematical concepts and relationships and use of mathematical skills and techniques in a range of contexts specifically problem solving and abstract thinking. Topics covered are basic elements of vectors, vector spaces, matrices, determinants, solving systems of linear equations, concepts and applications of differential and integral calculus of one variable.

Course Code: Geol2101 General Geology

Course Name: General Geology

Credit Hour: 4

Prerequisite: No

The course aims to provide students with a general introduction to the geological science, the planet Earth, the Earth materials, as well as the physical and historical processes of the Earth. It also introduces students with basic concept of simplified topographic and geologic maps and their cross-sections; and with varieties of representative mineral, rock and fossil specimens. The course will serve as a basis for many of the geology courses. Also introduce the students with basic knowledge of the different landforms and geomorphological processes that are involved in the formation and reshaping of these landform features of the earth.

Course Code: Geol2103

Course Name: Palaeontology

Credit Hour: 3

Prerequisite: None

To introduce the following: the history of life on Earth; the preservation of animals and plants as fossils and methods of studying them; the morphology and classification of the main groups of invertebrate animals with emphasis on those of geological importance; the science of palaeo botany; the stratigraphical importance of fossils including microfossils; comparison of living organisms with their fossil relatives. To provide students with the aptitude to the classification and to the systematic recognition of some of the main groups of fossil organisms.

Course Code: Geol2105 Crystallography and Mineral Optics

Course Name: Crystallography and Mineral Optics

Credit Hour: 3

Prerequisite: Geol1011

The course covers the following topics: Historical background of Crystallography. Introduction to morphological relations of crystals: Molecular, structure, crystal structure and amorphous structure. Introduction to mineralogy: definition, history of mineralogy and its early use, mineral deposits, ore, rocks, genesis of minerals, mineral nomenclature, mineral classification. Crystal Symmetry: Symmetry elements, symmetry operations, crystal classes (point groups), crystal systems, Steno's and Haüy's laws, crystal forms, Miller indices, Stereographic projections. Real crystals: crystal growth, crystal aggregates. Internal structure of crystals: crystal lattice and its properties, lattice types, Bravais lattices, Space groups. Introduction to X-ray diffraction (XRD) in crystals

Course Code: Geol2108

Course Name: Mineralogy

Credit Hour: 3

Prerequisite: Geol2105

The course covers the following topics: Introduction to mineralogy: definition, history of mineralogy and its early use, mineral deposits, ore, rocks, genesis of minerals, mineral nomenclature, mineral classification; Crystal property: definition, chemistry, structure types, polymorphism, pseudomorphism, isomorphism and mineraloids; Introduction to X-ray diffraction (XRD); Physical properties related to symmetry: cleavage, piezoelectricity, pyroelectricity; Crystal optics: light and its properties; Examination of crystals in plane-polarized light: opacity, color, refractive index, pleochroism; Examination of crystals under crossed polars: isotropic and anisotropic crystals, interference colors, birefringence; Examination of crystals in convergent light: uniaxial and biaxial crystals, extinction, cleavage, twinning, interference figures, optic angle, optic sign; Miscellaneous optical properties; Mineral identification: Physical characteristics of minerals: Polarizing microscope (use of transmitted and polarized light, reflected light); Systematic mineralogy: native elements, oxides, silicates, sulfides and sulfates, phosphates, chromates, tungstates, molybdates, arsenates, vanadite. Carbonates, halides: Sampling methods of mineralogical materials: Fractionation of mineral particles.

Course Code: Entr1108

Course Name: Entrepreneurship for science

Credit Hour: 2

Prerequisite: None

This course is designed to encourage students to start their own business and to acquaint them with the peculiar challenges and management decisions faced by owners of small business. It will develop such entrepreneurial skills as: identifying business opportunities; initiating, financing, and developing new venture business plans. It also addresses issues of small business, such as: legal aspects, financing, costing, locating, personnel, marketing, competition, sources of funding, and constituency services.

Course Code: InCh2010

Course Name: Fundamentals of Physical Chemistry

Credit Hour: 3

Prerequisite: None

The Fundamentals of Physical Chemistry course has been designed to provide for the basics of Physical Chemistry to non-chemistry students. It helps the learners to acquire a deep understanding on their area of specialization.

Course Code: Geol2102

Course Name: Stratigraphy and Earth History

Credit Hour: 3

Prerequisite: Geol 1111, Geol 2121

The course aims to provide students with a general introduction to the basic principles of stratigraphy and broad information on Earth's history; to provide the basic concepts of the spatial and temporal relationships among rock formations, stratigraphic correlations, depositional environment and evolution; to introduce students with the evolution of the Earth's lithosphere, atmosphere, hydrosphere and biosphere; to briefly introduce students with Ethiopian stratigraphic succession.

Course Code: Geol2104

Course Name: Plate Tectonics

Credit Hour: 2

Prerequisite: None

The course is to help students understand the present configuration of lithospheric plates and the characteristic features and processes at their boundaries, know the dynamics of mantle plume and plate motion; introduce students to the basic methods of plate reconstruction of the past.

Course Code: Geol2106

Course Name: Sedimentary Petrology

Credit Hour: 3

Prerequisite: Stratigraphy and Earth History (Geol 2102)

The course covers the following topics: Sedimentary grain textures and rock fabrics; Sedimentary structures; Siliciclastic sedimentary grain compositions; Weathering; Sedimentary environments rivers, lakes, deserts, deltas, shallow-marine siliciclastic and carbonate settings; facies concepts and Stratigraphy; Turbidity currents on continental margins; Glaciations; The Earth's climate system - present day and a geological perspective.

Course Code: Phys2003

Course Name: General Physics

Credit Hour: 3

Prerequisite: None

This algebra based introductory course is designed to enable students to learn the basic concepts, principles, and applications of elementary physics. The topics include vectors, Kinematics & Dynamics of Particles, Fluids Mechanics, Heat and Thermodynamics, Oscillations, Waves and optics, Electromagnetism & Electronics, Cross-Cutting Applications of Physics.

Course Code: Geol2110

Course Name: Igneous Petrology

Credit Hour: 3

Prerequisite: None

The integration of descriptive and interpretative petrology is the main aim of igneous petrology work. This course is intended to give students a broader understanding of igneous rock types/classifications and their origin and mode of emplacements as well as the tectonic environments that enables the formation of various types of igneous rocks. And the lab session introduces igneous rocks (plutonic and volcanic) in hand specimen and thin section. It further elucidates the different physical parameters of igneous rocks in a lab set up. Practical session includes: Review of rock forming minerals; classification of igneous rocks; (based on mineralogical, texture and modal classifications).

Course Code: Stat2006

Course Name: Introduction to statistics

Credit Hour: 3

Prerequisite: None

This course covers basic elements of statistics, probability, central tendencies, variation, and types of sampling

Course Code: Geol3101

Course Name: Structural Geology

Credit Hour: 4

Prerequisite: None

The course aims to provide students with the basic concepts of deformation of rocks, and of the mechanisms and causes of deformation; to familiarize students with how to appreciate, describe and record geological structures in the field; to enable students how to evaluate, analyze and interpret structural data and incorporate it into regional synthesis of an area.

Course Code: Geol3103

Course Name: Geological mapping and Report Writing

Credit Hour: 3

Prerequisite: None

The course aims to introduce students with the concepts and components of topographic and geologic maps and cross-sections as well as the basic techniques of geological mapping, field data collection, report writing and presentation techniques. The course introduces students with the aerial photography and its interpretation, and basics of drawing. The course introduce students with the basic skills of fieldwork planning, surveying and sampling techniques, field data recording, communication through field report writing and presentation.

Course Code: Geol3105

Course Name: Remote Sensing and GIS

Credit Hour: 3

Prerequisite: None

Course Code: Geol3107

Course Name: Mapping Sedimentary Terrain

Credit Hour: 3

Prerequisite: Geol2106

The course aims to provide students with the necessary knowledge, skill and attitude to plan and carry out independent geological mapping of a sedimentary area including outcrop description, sample collection and study, measurement/ evaluation/ interpretation of structures, preparation of maps and cross sections, reconstruction of geologic history as well as production of a geological report.

Course Code: Geol3102

Course Name: Mapping Igneous Terrain

Credit Hour: 3

Prerequisite: Geol2108, Geol3103 and Geol3101

The course aims to provide students with the necessary knowledge, skill and attitude to plan and carry out independent geological mapping of an igneous terrain including outcrop description, sample collection and study, measurement/ evaluation/ interpretation of structures, preparation of maps and crosssections, reconstruction of geologic history as well as production of a geological report.

Course Code: Geol4105

Course Name: Economic Geology

Credit Hour: 3

Prerequisite: None

The course introduces the students with the basic principles of economic geology and origin of mineral deposits; mode of formation of the deposits and mineral and country rock association. It briefly discusses the different ore deposits models, it highlights the comparison between similar deposits globally and plate tectonics and associated mineral deposits.

Course Code: Geol3106

Course Name: Fundamentals of Soil and Rock Mechanics

Credit Hour: 3

Prerequisite: Geol3101

Course Code: Geol3108

Course Name: Principles of hydrogeology

Credit Hour: 3

Prerequisite: None

Course Code: Geol3103

Course Name: Geological mapping and Report Writing

Credit Hour: 3

Prerequisite: None

The course is designed to address basic principles and concepts of hydrogeology. What the students must learn from this course has outlined in six learning outcomes (CLO) outlined here under. The course adequately addresses the principles and basic concepts regarding the hydrologic cycle, Surface water hydrology & water balance, groundwater recharge and discharge, groundwater flow, Aquifer characterization, Pumping tests, fundamentals of groundwater flow modeling and basics of groundwater geochemistry and water quality.

Course Code: Geol3110

Course Name: Metamorphic Petrology

Credit Hour: 3

Prerequisite: None

The course aims to provide students with the fundamental concepts on metamorphic processes and mineral reactions, metamorphic rocks, their textures and mineral assemblages; introduce students with the metamorphism of sedimentary and igneous rocks at different P-T conditions.

Course Code: Geol3112

Course Name: Geochemistry

Credit Hour: 3

Prerequisite: Igneous Petrology (Geol2110)

Geochemistry has become a very interdisciplinary science, and learning geochemistry provides the student an opportunity to integrate and synthesize knowledge learned in many different science courses into a holistic approach to the study of earth. This course aims to trace the geochemical development of the Earth from the origins of the solar system to the present planetary system and to cover the fundamentals of geochemistry in order to understand the natural system.

Course Code: Geol4101

Course Name: Groundwater Exploration and Development

Credit Hour: 3

Prerequisite: None

Groundwater occurrences in different rocks, Groundwater exploration techniques, Application of hydrological, geological, hydrogeological, geophysical, remote sensing and GIS in groundwater exploration. Drilling methods, Drilling fluids, well design and well constructions, Pumping test, Groundwater resource evaluation & assessment, Sustainable yield of groundwater aquifers, Groundwaters of Ethiopia.

Course Code: Geol4103

Course Name: Mapping Metamorphic Terrain

Credit Hour: 3

Prerequisite: Geol3110,3103

The course aims to provide students with the necessary knowledge, skill and attitude to plan and carry out independent geological mapping of an igneous terrain including outcrop description, sample collection and study, measurement/ evaluation/ interpretation of structures, preparation of maps and cross sections, reconstruction of geologic history as well as production of a geological report.

Course Code: Geol3104

Course Name: Geophysics

Credit Hour: 4

Prerequisite: None

Course Code: Geol4107

Course Name: Engineering Geology

Credit Hour: 3

Prerequisite: Geol3106

Course Code: Geol4109

Course Name: Mineral Exploration and Resource Evaluation

Credit Hour: 2

Prerequisite: None

The course aims to provide the students with basic knowledge and skills on the prospecting and exploration of mineral resources; and with the basic understanding of the occurrence of mineral deposits in space and time.

Course Code: Geol4102

Course Name: Energy Resource of the Earth

Credit Hour: 3

Prerequisite: Geol2106,

This course unit aims to provide students with a general introduction in the field of petroleum and coal geology and understanding, investigation and exploitation of these natural resources; understanding on the occurrence of different types of fossil fuel deposits in Ethiopia and the world; basic knowledge on the nature, composition and maturation of coals and oils; and skills on the technological applications in the exploration of hydrocarbon resource

Course Code: Geol4104

Course Name: Geology and Geologic Resources of Ethiopia

Credit Hour: 3

Prerequisite: None

Proper exploration, exploitation and utilization of the mineral oil, natural gas and water resources of Ethiopia require the knowledge of the geological set up of the country. The course is designed to introduce students the different geological formations and associated resources of the country

Course Code: Geol4106

Course Name: Environmental Geology

Credit Hour: 2

Prerequisite: None

The course aims to introduce geology students to the broader issues of global environmental challenges facing the human society. The students will get opportunity of Introduction to geologic hazards affecting humankind; emphasis on earthquakes, volcanism, floods and landslides

Course Code: Geol4111

Course Name: Research Methods in Geosciences

Credit Hour: 2

Prerequisite: None

To gear students towards undertaking substantial independent research projects in areas of geosciences and produce research proposals of their own.

Course Code: Geol4108

Course Name: Senior Project

Credit Hour: 3

Prerequisite: Geol4111,

The course aims to give senior students the opportunity of undertaking an independent research project in Earth Sciences, in an area of interest to the students, under the supervision of a member of staff. Students will acquire and interpret datasets, using them to solve an Earth Science problem. Students will also present the results of their own scientific research orally and in poster form to an audience of staff and peers.

Course Code: Geol4110

Course Name: Mining Geology

Credit Hour: 2

Prerequisite: None

Proper exploitation and utilization of natural resources is one of the important needs of the Society. It demands for proper acquaintance with the procedures of exploitation. The course introduces the students to the basic knowledge in the application of various mining methods: production and auxiliary activities in mining and problems associated with them, and on comparison of exploration and mining methods used in different countries.

Course Code: Geol4016

Course Name: Integrated Geological Team Project

Credit Hour: 2

Prerequisite: None

This is a multidisciplinary team integrated geological project in connection with a special geological problem and under the guidance of a faculty member. The project synopsis consists of literature review, design, project management, business acumen, multidisciplinary team work, entrepreneurship

4.2.7.2 Elective courses

Course Code: Geol4201

Course Name: Industrial Minerals, Rocks and Gems

Credit Hour: 3

Prerequisite: Economic Geology (Geol 4105)

The course covers the following topics: Industrial minerals and rocks composition, structure, physical properties, and origins, will be covered along with topics of particular interest in gemology, such as mechanisms of color, history and lore of gems, and uses of gems. Occurrence, processing, economics and applications of commodities (nonmetallic minerals and rocks) will be focused such as: rock aggregates, bentonites, datomites, salt, soda ash & chlor-alkali; magnesite, brucite & magnesite; glass; bauxite & alumina; pigments & fillers; high-tech ceramics; limestone & dolomite; dimension stone; cement; gypsum; wollastonite; rare earths; phosphates; borates and gemstone: opal, sapphires, rubies, olivine, garnet.

Course Code: Geol 4202

Course Name: Groundwater Resource Management

Credit Hour: 3

Prerequisite: Geol 3108, Geol 4101

The volume of groundwater stored in an aquifer at any instant of time depends on the dynamic relationship between recharge inputs, and outflow by natural discharge and abstraction. Aquifer storage provides a buffer between highly variable, climatically driven recharge processes and the variable outflow that supports surface water ecology. Uncontrolled abstraction of groundwater both for drinking and irrigation, alters the dynamic balance between “natural”

recharge and the state of surface waters. Both quantity and quality of groundwater can be altered by over-pumping of aquifers. Thus, the groundwater resource management objective is to determine the regime of abstraction without causing unacceptable environmental effects. This course aims at enhancing student's skills and knowledge to use tools, methods and approaches used in groundwater resources management to ensure the development of groundwater in a way that ensures long-term sustainable use for future generations.

Course Code: Geol4205

Course Name: Quaternary Geology

Credit Hour: 3

Prerequisite: None

The course aims to introduce students with the Quaternary history of the Earth in general and that of Ethiopia in particular. Emphasis is given to the evaluation/ interpretation of Quaternary paleoenvironment and paleoclimate.

Course Code: Geol4206

Course Name: Environmental Impact Assessment (EIA)

Credit Hour: 3

Prerequisite: None

The course introduces basic foundation of EIA and its importance for project planning and design, find ways and means to reduce adverse impacts, shape projects to suit the local environment and present the predictions and options to decision-makers. Specific topics can include Definition, Basic Concepts, Principles, and philosophical, methodological and institutional approaches of EIA, environmental risk assessment, steps in EIA, methods of conducting EIA.

Course Code: Geol4203

Course Name: Fundamentals of Gemology

Credit Hour: 3

Prerequisite: None

This course deals with basic gemological knowledge that are important in jewellery design, metal smiting, gem-setting along with instrumentation basic for gem identification in the industrial practice. Gemologists are expected to be able to identify gemstones and detect a number of gems in the market including treatments, synthetics and

imitations. The fundamental course is not an end by itself but requires a series of other courses like Advanced Gemology and Gem Identification practical lab exercises to be qualified being competent in the industry.

Course Code: Geol42045

Course Name: Introduction to Geohazard Assessment

Credit Hour: 3

Prerequisite: Geol4107

After successful completion of the course, students will acquire sufficient knowledge on hazardous earth processes influencing engineering works such as Landslides, Earthquakes, volcanoes, Subsidence, expansive soils; flood and other earth related problems and their impacts on different engineering structures like buildings, dams, reservoirs, roads, bridges and underground excavations. We will investigate hazardous phenomena, their prediction and mitigation. Topics to include: earthquakes, volcanoes, tsunamis and weather/ climate. Provides a foundation in basic geological hazards related to science, suitable for use in teaching, communications, policy, and emergency management careers.

4.3 Bachelor of Science Degree in Biotechnology

4.3.1 Introduction

According to the United Nations Convention on Biological Diversity, Biotechnology is defined as “any technological application that uses Biological systems, living organisms, or derivatives thereof, to make or modify products or processes for specific use such as the production of new knowledge, products, processes and services”. Biotechnology is a rapidly growing and a cross-cutting field that uses a combination of biology and technology to develop technologies and products which help improve our health, agriculture, industry and environmental management. The tremendous agricultural, environmental, industrial and health challenges for Ethiopia which have existed for a long time, remained the difficult to achieve improvement in the standard of living of the population in the face of rapid population growth. Providing food and quality health services to current yet growing population of over 100 million people could not be accomplished using the technologies of the 1940’s. Biotechnology helps to design and produce new molecules, plants, animals and microorganisms with improved characteristics. It offers seemingly unlimited opportunities to combine genes from related or unrelated species to produce useful organisms with desirable properties that were not previously found in nature.

To realize the full potential of this field as a frontline area of research and development with an overwhelming impact on the societal and national economic development, higher education institutions of Ethiopia are expected to produce competent professionals that can make the country beneficiary of the modern applications in the area. To this effect, AASTU has established Department of Biotechnology under the College of Biological and Chemical Engineering. The department runs undergraduate and post-graduate biotechnology programs and has developed four-year structured courses for its undergraduate program. The B.Sc. biotechnology curriculum, in addition to offering basic theoretical knowledge, practical core set of skills and techniques in biotechnology, and entrepreneurial skills, is designed to include fundamental concepts that span across other basic life and social sciences, chemistry, physics, and mathematics. The courses are uniquely selected from basic sciences as well as from major application areas of the discipline allowing the graduate to compete nationally and internationally. Moreover, the department employs an outcome-based learning approach, and

assessment is generally to be made following multiples assessment methods and tools for attaining the knowledge/skill described herein.

4.3.2 Program Education Objectives (PEO)

The Biotechnology program at Addis Ababa Science and Technology University produces graduates with a strong foundation of scientific and technical knowledge and who are equipped with problem solving, team work, and communication skills that will serve them throughout their careers consistent with the following program educational objective. The program prepares students for career in Biotechnology and its related fields. It also prepares students to further their education in graduate schools.

Table: Program Education Objectives (PEO)

PEO	Statement
PEO-1	To prepare graduates who demonstrate knowledge and technical competency and leadership commitments for careers of productive services.
PEO-2	To produce biotechnologists who can sustain the development of the bio-economy for the betterment of the society and the nation.
PEO-3	To prepare graduates for advanced studies in various disciplines of biotechnology and related fields.

4.3.3 Program Learning Outcomes (PO):

The biotechnology program is measured through its attainment of the Program Outcome (PO) of the program and its courses.

Table: Program Learning Outcome

PO	Statement
PO-1	Each graduate will have an ability to identify, formulate, and solve broadly defined technical or scientific problems by applying principles and practices of biotechnology, sciences, and mathematics.

PO-2	Each graduate will have an ability to formulate or design a system, process, procedure or program to meet desired needs of biotechnological applications in agriculture, industry, environment and health development.
PO-3	Each graduate will have an ability to develop and conduct appropriate experimentation, analyze and interpret data and use scientific judgment to draw conclusions.
PO-4	Each graduate will have an ability to communicate effectively with a range of audiences.
PO-5	Each graduate will have an ability to understand ethical and professional responsibilities and make informed judgment, which must consider the impact of biotechnological solutions in global, economic, environmental, and societal contexts.
PO-6	Each graduate will have an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, meet objectives and deadlines, and analyze risk and uncertainty.
PO-7	Each graduate will have an ability to develop an entrepreneurial mindset and understand the dynamics of bioscience industry to translate scientific innovations into bio-based products, processes, services and viable business opportunities
PO-8	Each graduate will have an ability to acquire and apply new knowledge and technologies in the bioeconomy development.

4.3.4 Graduation Requirements

- The graduation requirement for all undergraduate programs should satisfy the following minimum requirements as stated in article 109, July 2017 university's senate legislation. All the required courses/modules and the minimum credit hours set in the program curriculum by the respective academic unit should be satisfied, except phase-in and phase-out programs. For a student to graduate, total credit hours of 151 shall be taken, with a cumulative grade point average (CGPA) of 2.00 in all and major area courses; with No "F" grade in any course taken and shall score pass mark for all courses which have Pass/Fail grade and pass the exit exam

4.3.5 Course Category and list of Courses

Table: Distribution of credit hours for all course categories

Category		Total Cr. Hr	Percentage (%)
1	Core Course (major and supportive course)	117	77.48

2	Core Elective	2	1.3
3	University requirement (Core)	3	1.98
Core Course Total Cr. hr		122	80.79
4	National Requirement	29	19.20
Total Cr. Hr		151	100

List of Courses

Table: List of Courses

Common Courses		
No.	Course Title	Credit-hour
01	Logic and Critical Thinking	3
02	General Psychology	3
03	Communicative English Language skill – I	3
04	Geography of Ethiopia and the Horn	3
05	Social Anthropology	2
06	Communicative English Language skill-II	3
07	Physical Fitness	P/F
08	Moral and Civic Education	2
09	History of Ethiopia and the Horn	3
10	Economics	3
11	Inclusiveness	2
12	Global Trend	2
	Total	29

Core Supportive course		
No.	Course Title	Credit-hour
13	Project Planning and Management	2
14	Introduction to Computer Science	2
	Total	4

Core Courses		
No.	Course Title	Credit-hour
01	Biostatistics and Experimental Design	3
02	Animal Anatomy and Physiology	2
03	Plant Anatomy and Physiology	2
04	Cell biology	3
05	Principles of Genetics	3
27	General Microbiology	3
07	Molecular Biology	2
08	Techniques in Molecular biology	1
09	Microbial Genetics	2
10	Fundamentals of Biochemistry	4
11	Immunology and Immunotechnology	3
12	Biophysics	2
13	Microbial Biotechnology	3
14	Fundamentals of Bioprocessing Engineering	2
15	Genetic Engineering	3
16	Plant Biotechnology	2
17	Molecular Diagnostics	1
18	Animal Biotechnology	3
19	Introduction to Omics science	2
20	Medical Biotechnology	3
21	Cancer Biology	2
22	Biotechnology Internship	3
23	Research Methodology and Academic Writing	2
24	Plant Cell and Tissue Culture	2
25	Enzyme Technology	3
26	Industrial Biotechnology	3
27	Environmental Biotechnology	3
28	Food Biotechnology	2

29	Final year Project I &II	4
30	Pharmaceutical Biotechnology	2
31	Nano biotechnology	2
32	Introduction to Bioinformatics	3
25	Biosafety and IPR in Biotechnology	2
26	Fungal Biotechnology	2
27	Introductory System and Synthetic biology	1
28	Principles of Animal Breeding	2
29	Principles of Plant Breeding	2
30	Aquatic Biotechnology	2
31	Integrated Science Team Project	3
32	Emerging Technologies for Science	3
33	Entrepreneurship for Sciences	3
34	Organic Chemistry	3
35	Analytical chemistry and Instrumentation	3
36	Applied Mathematics IA	4
37	Mathematics for Natural Science	3
38	General Chemistry	3
	Total	116
Core Elective Courses		
No.	Course Title	Credit-hour
01	Energy Biotechnology	2
02	Biodiversity and Conservation of Genetic Resources	2
03	Forensic Biotechnology	2
04	Animal cell and Tissue Culture	2

4.3.6 Course Breakdown for Bachelor of Science in Biotechnology

Table: Year- and Semester-based Course Breakdown

Year I Semester I

Course Code	Course Title	Cr.hr.	Lect.	Lab.
Phil1009	Logic and Critical Thinking	3	3	
Psyc1011	General Psychology	3	3	

FLEn1003	Communicative English Language Skills I	3	3	
GeEs1005	Geography of Ethiopia and the Horn	3	3	
Math1107	Mathematics for Natural Science	3	3	
InCh1101	General Chemistry	3	2	1
SpSc1013	Physical Fitness	P/F		
Total		18	17	1

Year I Semester II

Course Code	Course Title	Cr.hr.	Lect.	Lab.
EmTe1106	Emerging Technologies for Science	3	3	
Entr1108	Entrepreneurship for Science	3	3	
Anth1002	Social Anthropology	2	2	
FLEn1004	Communicative English Language Skills II	3	3	
Math1114	Applied Mathematics IA	4	4	
Incl1010	Inclusiveness	2	2	
MCiE1012	Moral and Civic Education	2	2	
Total		19	19	

Year II Semester I

Course Code	Course Title	Cr.hr.	Lect.	Lab.
GLTr2001	Global Trends	2	2	
SWENG2109	Introduction to Computer Science	2	1	1
Econ2003	Economics	3	3	
Biot2101	General Microbiology	3	2	1
InCh2103	Organic Chemistry	3	2	1
Biot2105	Cell Biology	3	2	1
Biot2107	Principles of Genetics	3	2	1
Total		19	14	5

Year II Semester II

Course Code	Course Title	Cr.hr.	Lect.	Lab.
Biot2102	Molecular Biology	2	2	
Biot2104	Techniques in Molecular biology	1		1
Hist 2002	History of Ethiopia and the Horn	3	3	
Biot2110	Microbial Genetics	2	2	
Biot2112	Fundamentals of Biochemistry	4	3	1
Biot2108	Animal Anatomy and Physiology	2	2	
InCh2114	Analytical Chemistry and Instrumentation	3	2	1
Biot2106	Plant Anatomy and Physiology	2	2	
Total		19	16	3

Year III Semester I

Course Code	Course Title	Cr.hr.	Lect.	Lab.
Biot3111	Principles of Plant Breeding	2	2	
Biot3101	Immunology and Immunotechnology	3	2	1
Biot3103	Principles of Animal breeding	2	2	
Biot3105	Microbial Biotechnology	3	2	1
Biot3107	Fundamentals of Bioprocessing Engineering	2	2	
Biot3109	Biophysics	2	2	
Biot3115	Genetic Engineering	3	2	1
Biot3113	Biostatistics and Experimental design	3	3	
Total		20	17	3

Year III Semester II

Course Code	Course Title	Cr.hr.	Lect.	Lab.
Biot3102	Plant Biotechnology	2	2	
Biot3104	Molecular Diagnostics	1		1
Biot3110	Fungal Biotechnology	2	1	1
Biot3106	Plant Cell and Tissue Culture	2	1	1
Biot3108	Medical Biotechnology	3	2	1
ISTP3116	Integrated Science Team Project	3	3	
Biot3118	Aquatic Biotechnology	2	1	1
Biot3114	Introduction to Bioinformatics	3	2	1
Biot3116	Research Methodology and Academic Writing	2	2	
Total		20	14	6

Year III Summer I

Course Code	Course Title	Cr.hr.	Lect.	Lab.
Biot3120	Biotechnology internship	3		
Total		3		

Year IV Semester I

Course Code	Course Title	Cr.hr.	Lect.	Lab.
Biot4101	Enzyme Technology	3	2	1
Biot4103	Industrial Biotechnology	3	2	1
Biot4105	Environmental Biotechnology	3	2	1
Biot4107	Introduction to Omics science	2	2	
Biot4111	Final year Project I	P/F		
Biot4115	Animal Biotechnology	3	2	1
Biot4113	Nanobiotechnology	2	2	
Total		16	12	4

Year IV Semester II

Course Code	Course Title	Cr.hr.	Lect.	Lab.
Biot4112	Final Year Project II	4		4
Biot4102	Cancer Biology	2	2	
Biot4104	Food Biotechnology	2	2	
Biot4106	Pharmaceutical Biotechnology	2	2	
Biot4108	Biosafety and Intellectual Property Right in Biotechnology	2	2	
Biot4110	Introductory System and Synthetic biology	1	1	
BMgt4116	Project Planning and Management	2	2	
Biot xx	Elective	2	2	
Total		17	13	4

4.3.5 Program Courses Synopsis

4.3.5.1 Core course Synopsis

Course Code: Math 1107

Course Name: Mathematics for natural science

Credit Hour: 3

Prerequisite: No

This course covers the basic concepts of logic and set theory, the real and complex number systems, Mathematical induction, least upper bound and greatest lower bound, functions and types of functions, polynomial and rational functions, logarithmic and exponential functions, trigonometric functions, hyperbolic functions and their graphs and analytic geometry.

Course Code: Ent1108

Course Name: Entrepreneurship for Science

Credit Hour: 2

Prerequisite: No

This course is designed to encourage students to start their own business and to acquaint them with the peculiar challenges and management decisions faced by owners of small business. It will develop such entrepreneurial skills as: identifying business opportunities; initiating, financing, and developing new venture business plans. It also addresses issues of small business, such as: legal aspects, financing, costing, locating, personnel, marketing, competition, sources of funding, and constituency services.

Course Code: InCh1101

Course Name: General Chemistry

Credit Hour: 3

Prerequisite: No

The course General chemistry is designed to make students more prepared to the all-chemistry courses by refreshing and summarizing the previous preparatory chemistry concepts before tackling the advanced chemistry courses. The course covers units and measurement, the composition of matter, chemical reactions, reactions stoichiometry, atomic structure and the periodic table, the chemical bond, structure of molecules, the properties of solutions, chemical equilibrium, introduction to functional groups and their typical reactions.

Course Code: Math 1114

Course Name: Applied Mathematics I A

Credit Hour: 4

Prerequisite: No

This course is designed to develop students' confidence with mathematical concepts and relationships and use of mathematical skills and techniques in a range of contexts specifically problem solving and abstract thinking. Topics covered are basic elements of vectors, vector spaces, matrices, determinants, solving systems of linear equations, concepts and applications of differential and integral calculus of one variable.

Course Code: SWENG2109

Course Name: Introduction to computer science

Credit Hour: 2

Prerequisite: No

This course introduces the student with the use of a PC in an MS Windows environment. It introduces the two main application packages in that context – MS Excel and MS Word. In addition, it introduces the students with the world wide web for e-mail and information gathering.

Course Code: EmTe1106

Course Name: Emerging technologies for Science

Credit Hour: 3

Prerequisite: None

Course Code: InCh2103

Course Name: Organic Chemistry

Credit Hour: 3

Prerequisite: None

This course will make students who take the course about the basic things in organic chemistry, which are in touch with our day-to-day activities and us also. So the course makes the students to be familiar with the various biological molecules, natural products, synthetic compounds, polymers and their roles, functions, functional groups, chemical

and physical properties so that they will apply to their different fields of Biotechnology. This course will complement molecules and molecule-interaction about as theoretical Background and will create basic knowledge for next fundamental biotechnology courses.

Course Code: Biot2005

Course Name: Cell Biology

Credit Hour: 3

Prerequisite: None

This course deals with level of organization of cells; Cell structure of Prokaryotes, Eukaryotic cell, Viruses, viroids; function and structure of Cellular organelles which is a basic knowledge for biotechnological and genetic studies. The course also introduces cell membrane and different transports systems across cell membrane; Cell division; Regulation of cellular activity with enzymes.

Course Code: Biot2007

Course Name: Principles of Genetics

Credit Hour: 3

Prerequisite: None

The course introduces the students about pre-Mendelian views of heredity; Mendel's works, results and deductions; cytological basis of heredity; chromosome banding, crosses involving different levels of hybrids and the concomitant genotypic and phenotypic ratios; probability and chi-square testing in genetics; allelic (dominance) and non-allelic gene interactions and modifications of phenotypic ratios; effect of external and internal environments on gene expression; multiple allelic inheritance; linkage, crossover and gene mapping; extra-nuclear (cytoplasmic) inheritance; mechanisms of sex determination and inheritances related to sex; experiments that led to the identification of DNA as the genetic material; structure of the DNA. Information flow from DNA (replication, transcription, translation); how genes determine phenotypes through the intermediary of proteins; mutations and how they change codons and protein structure and function; brief introduction to genotypic and phenotypic frequencies in populations.

Course Code: Biot2102

Course Name: Molecular Biology

Credit Hour: 2

Prerequisite: Principles of Genetics

The course deals with basic molecular biology concepts such as constituents, structures & functions of DNAs, polymerization and base pairing; denaturation & renaturation of DNA, various forms of DNA, structure of RNA, DNA replication, replication of linear and circular DNA molecules, proofreading activity during DNA replication, structure of genes; transcription and translation process in prokaryotes and eukaryotes; regulation of gene expression in prokaryotes and eukaryotes; protein targeting, mutation and repair mechanism.

Course Code: Biot2104

Course Name: Techniques in Molecular Biology

Credit Hour: 1

Prerequisite: Principles of Genetics

The course deals with basic techniques in molecular biology such as clean handling, solution preparations, sterility (laboratory safety), DNA isolation (Bacterial genomic and plasmid DNA, Eukaryotic DNA); Agarose Gel electrophoresis, Elution of the DNA fragment from the gel, Restriction digestion, PCR amplification of DNA, cloning techniques, Competent cell formation, transformation, protein isolation, RNA isolation, UV spectroscopy techniques for DNA and protein estimation; SDS-PAGE separation of proteins: blotting techniques (Southern, Northern and Western blotting)

Course Code: Biot2110

Course Name: Microbial Genetics

Credit Hour: 2

Prerequisite: None

The course focuses primarily on gene structure and function of prokaryotic systems as a paradigm for processes in eukaryotes. The course also delves extensively into the intricacies of bacterial virus (bacteriophage) molecular genetics. Bacterial genome and its elements, extra-chromosomal elements, plasmids and their properties, DNA replication, recombination in bacteria, gene structure of prokaryotes, operons, control of gene expression, transformation, conjugation, transduction, transposition, genetic fine structure and co-linearity, mapping methods, transposable elements, virulent and avirulent phages, mutation, DNA repair.

Course Code: Biot2112

Course Name: Fundamentals of Biochemistry

Credit Hour: 4

Prerequisite: Cell biology

The students will learn about theory and practically Structure, classification and function of Carbohydrates, Lipids, Amino acids, Proteins, Nucleic acids, Vitamins and Enzymes. And to develop understanding and provide scientific basics of the life processes at the molecular level and explain the structure, function and inter-relationships of biomolecules and their deviation from normal and their consequences for interpreting and solving Industrial problems.

Course Code: InCh2114

Course Name: Analytical Chemistry and instrumentation

Credit Hour: 3

Prerequisite: None

The course deals with Introduction to analytical chemistry; Acids bases, neutralization, titration, Complex formation, and complex metric titration, measurement of BOD and COD, gravimetric analysis, Chromatographic

and spectroscopic techniques. The students will learn about through analytical chemistry and instrumentation subject to knowing the appropriate applications of various analytical techniques used in biotechnology, make the experiment easier and let the researcher obtain a better result which also reduces the time period of experiment by eliminating the trial-and-error methods. Analytical equipment's find its application not only in the research works but also at the industry level for the quality analysis and quantities analysis of the final products.

Course Code: Biot2108

Course Name: Animal Anatomy and Physiology

Credit Hour: 2

Prerequisite: Cell Biology

Anatomy and Physiology is a basic and mandatory course for biologists, medical professionals and Biotechnologies as it gives a clear understanding to make use of biological products and organisms for human use. Through the course students will be exposed to general introduction of animal anatomy and reference terminologies to study anatomy and physiology, ; Cell, Basic structural levels (Cell, Tissue, Organ, Organ system, Different); Homeostasis (Mechanisms of Homeostasis, Transport of material across membrane); Joints (Types of Joints, Classification of joint, Different joints of the body); Muscular system (Types of muscle, Composition of muscle fiber, Muscles found on different body parts); Nervous system (Central Nervous system (CNS), Peripheral Nervous system (PNS), Neuron, Nerve message, Synapses, Neurotransmitter), Cardiovascular system (Heart, Blood vessel, Blood); Respiratory system (Parts of the respiratory system, Mechanism of pulmonary ventilation (Breathing), Muscles of breathing, Pulmonary volumes and capacities, Respiratory pigment, Exchange of air between alveoli and capillaries, Regulation of breathing); Lymphatic system; Digestive system; Urinary system; Reproductive system (Male reproductive organs, Spermatogenesis, Female reproductive organs, Oogenesis, Mammary gland); Endocrine system. After the completion of the course, students interested in the animal biotechnology stream will benefit a lot for their advanced animal biotechnology courses and applications in their careers.

Course Code: Biot2006

Course Name: Plant Anatomy and Physiology

Credit Hour: 2

Prerequisite: None

This course integrates the plant structures with cell biology and allows the student conceptually integrate the ultra-cell structure with its physiological functions and developmental structures and reproductive biology. The course impart a comprehensive knowledge and exposure related to plant anatomy and physiology with the focused topics include the ultra-cell structure, root and stem anatomy and modifications, transport of water and solute translocation, inorganic nutrition, plant metabolisms, regulation of growth and development, plant hormones, control of flowering and stress physiology.

Course Code: Biot3111

Course Name: Principles of Plant Breeding

Credit Hour: 2

Prerequisite: None

The course concedes the aims and objectives of Plant Breeding; Modes of reproduction, sexual, asexual, apomixis and their classification, significance in plant breeding. Modes of pollination, genetic consequences, differences between self and cross-pollinated crops. The course addresses types of breeding: introduction and acclimatization, selection, and hybridization. Methods of handling of segregating generations – pedigree method, bulk method, back cross method and various modified methods considered. Heterosis, inbreeding depression, various theories of heterosis, exploitation of hybrid vigour-development of inbred lines, single cross and double cross hybrids will be covered. During the course, population improvement Programs, recurrent selection, synthetics and composites, methods of breeding for vegetative propagated crops will be focused. The significance of different breeding techniques like clonal selection, mutation breeding, Ploidy breeding, Wide hybridization will be also covered in the course.

Course Code: Biot3103

Course Name: Principles of Animal Breeding

Credit Hour: 2

Prerequisite: Animal Anatomy and Physiology

Through the course students will be exposed to the different techniques of improving the genetic potentials of farm animals with great emphasis on selection and the different mating systems. Students will be familiar with: the core concept and principles of animal genetics that helps students to identify types of farm animal traits and their mode of inheritance, and other related laws governing principles of genetics in a given farm animal population with emphasis on quantitative traits inheritance, populations genetics, genetic parameters and possible genetic improvement strategies (breeding programs) of selection and cross breeding with possible response to animal breeding programs. The genetic parameters including heritability, genetic correlation and repeatability of farm animals: selection and mating systems (Cross breeding and Inbreeding); hybrid vigour and inbreeding depression; relationship characterization of indigenous livestock and principles and methods of conservation of animal genetic resources; breeding schemes, application and use of biotechnology in animal breeding.

Course Code: Biot3101

Course Name: Immunology and immunotechnology

Credit Hour: 3

Prerequisite: Cell Biology

Immunology is the study of an organism immune system. It deals with the defense mechanism including all physical chemical and biological properties of the organisms that help it to combat its susceptibility to foreign organism. It can be defined as the branch of biology that is concerned with the structure and function of the immune system, the bodily distinction of self from non self, and the use of antibody based laboratory techniques or immunoassays.

Immunology deals with a range of immune cells, tissues and chemicals that interact to overcome infection, repair tissue damage and maintain the integrity of the body. It also studies the, environmental, genetic, neurological and psychological factors that affect the immune response of the organism. Immunotechnology is an important arm of biotechnology, constituting industrial scale application of immunological procurers to produce vaccine, monoclonal antibody and other cells and molecules important for treatment, diagnosis and immunization of the organism.

Course Code: Biot3105

Course Name: Microbial Biotechnology

Credit Hour: 3

Prerequisite: General Microbiology

This course entails ecosystem services provided by microorganisms in biogeochemical cycle and their roles in industrial, agricultural and environmental microbial biotechnology applications. It deals with the significance of microbes in wide production of biotechnologically important industrial products that include primary and secondary metabolites such as enzymes, organic acids alcohols, antibiotics biochemical, biopolymers, food ingredients, and pharmaceuticals etc. It also portrays the impacts of microorganisms in using bio fertilizers, bio controls, biocides, and the production of foods and animal feeds to develop agricultural yields and productivity. The course also encompasses the application of microbial cells and products for environmental safety and health. The practical aspects of this course inculcate students with biotechnological applications of microbial screening, isolation, characterization, and preservation as well as manipulation (using relevant molecular biology and gene expression technologies) for the development of potential isolates/strains/ for various products and applications in industries, agriculture, environmental protection and medicine.

Course Code: Biot3107

Course Name: Fundamentals of BioProcessing Engineering

Credit Hour: 2

Prerequisite: None

The students will learn about through bioprocessing engineering subject to operation modes based upon the nature of bioproducts and cell lines and other process criteria. Make to apply modeling and simulation of bioprocesses, so as to reduce costs and to enhance the quality of products and systems. To understand the requirements for successful operations of downstream processing. Future Plan a research career or to work in the biotechnology industry with strong foundation about bioreactor design and scale-up. Integrate research lab and Industry; identify problems and seek practical solutions for large scale implementation of Biotechnology. Industrial visit is needed for 2 times to different industries to learn the production process and safety regulations followed in Food/ Beverage industries as part of the course

Course Code: Biot3109

Course Name: Biophysics

Credit Hour: 2

Prerequisite: None

Biophysics deals with the application of physics to biological systems. The concepts and techniques of biophysics find applications in bio-electronics, medicine/health, and population dynamics and are closely related to statistical mechanics and transport processes. Specifically, this course provides an introduction to the physics of many-body systems, diffusion models, polymeric materials, transport phenomena of neutral and electrically charged particles, thermodynamics, medical applications of electricity, and most biological systems.

Course Code: Biot3115

Course Name: Genetic Engineering

Credit Hour: 3

Prerequisite: Molecular Biology

The course deals with basic molecular biology processes involving gene manipulation, cloning strategies such as cDNA and genomic libraries cloning host such as eukaryotic and prokaryotic hosts. In addition, the course also deals with screening strategies to identify desirable clones from DNA libraries, mutagenesis, gene mapping and applications of recombinant DNA technology in medicine, agriculture, industries and environment.

Course Code: Biot3113

Course Name: Biostatistics and Experimental design

Credit Hour: 3

Prerequisite: None

This course covers and focus on methods of data collection and presentations, basic principles of experimental design, error estimation and control in experimentation, complete randomization, randomized complete block, Latin square, factorial and confounded factorial, incomplete block (lattice square) designs, transformation of abnormal data, Covariance analysis and regression analysis

Course Code: Biot3104

Course Name: Molecular Diagnostic

Credit Hour: 1

Prerequisite: Techniques in molecular biology

Molecular Diagnostics is the process of identifying a disease by understanding the molecules, such as proteins, DNA, and RNA, in a tissue or fluid, which forms the markers of the diseases directly or indirectly. Molecular diagnostics is not confined to human diseases but can be used in animals and plants also. Molecular diagnostics can also be used to identify foodstuffs, vegetables, meat types and food processing methods etc., and can be also used in environmental monitoring, detect the presence of specific microorganisms in various samples including food materials. The course will provides the theory and use of molecular techniques in the diagnostic techniques, with

more importance on nucleic acids based techniques. Common analytical techniques and molecular techniques related to the development and use of diagnostics such as variants of polymerase chain reaction (PCR), quantitative real time PCR (qRT-PCR), immunohisto-compatibility test(IHC) and ELISA tests, chromosomal techniques such as hybridization techniques(FISH), SNP marker analysis and DNA sequencing will be emphasized in this course.

Course Code: Biot3102

Course Name: Plant Biotechnology

Credit Hour: 2

Prerequisite: None

This course to explore the techniques of plant Biotechnology to create genetic variation in plants and to recognize how genes contribute at the molecular level for the phenotypic expression of genotypes that leads to morphological variation. This knowledge is the key aspect to improve the ability to modify the plant responses towards global food security and commercial exploitation of biotechnology in agricultural, industrial and environmental sectors. The course outlines the techniques used in plant biotechnology and the application of these techniques in science and society. The molecular marker techniques for detecting the genetic variation and other applications will be dealt. The key concepts of the course will be integrated in a cohesive mode through a series of case studies of plant biotechnology.

Course Code: Biot3108

Course Name: Medical Biotechnology

Credit Hour: 3

Prerequisite: Immunology and Immunotechnology

The course integrates genetics, genomics, and molecular biology approaches to elucidate the pathogenesis of human diseases, and is intended to encourage students to unravel novel disease mechanism for improving management of human diseases. It deals with omics and structural biology for diseases identification as well as druggable target identification, the mucosal and immune system and the related diseases, neurological diseases, infectious diseases and viral replication in human cell and animal model for human disease study. Finally, new approaches in diseases treatment will be discussed.

Course Code: Biot3110

Course Name: Fungal Biotechnology

Credit Hour: 2

Prerequisite: None

The course is designed to provide basic concepts about kingdom fungi (general characteristics of fungi, fungal evolution and classification) and primarily focus on the role of fungi in agricultural Biotechnology, industrial Biotechnology, medical biotechnology, food Biotechnology and environmental Biotechnology. It extensively discusses about fungal originated biotechnological products (enzymes, vitamins, polysaccharides, alcoholic

beverages, organic acids, pigments, lipids, glycolipids, antibiotics, etc.,) and the role of fungi in solving the major problems the world faced to day (shortage of food, diminishing quality of health and environmental pollution).

Course Code: Biot3112

Course Name: Animal Biotechnology

Credit Hour: 3

Prerequisite: None

Introduction to animal biotechnology, historical perspectives and scope of animal biotechnology, fertilization, embryogenesis, animal cell culture and its manipulation, in vitro fertilization, embryo transfer, genetic engineering in animals, transgenic animals, hybridoma technology, molecular markers for the animal diseases, stem cells and its applications, laboratory practices of animal cell culture and its genetic engineering (experiments details are included in the course content).

Course Code: Biot3118

Course Name: aquatic biotechnology

Credit Hour: 2

Prerequisite: None

The course deals with Introduction to Aquatic biotechnology; Molecular Genetics of Aquatic organisms: Discovery & cloning of novel genes, Genetic manipulations of finfish & shellfish; Medical Applications of aquatic biotechnology: Drugs & medicines from the sea, Environmental applications of Aquatic biotechnology: Antifouling Agents, Biosensors, aquatic environmental remediation; Marine biotechnology: Fish & marine invertebrate, transgenic fish technology & its application in fish production; The uses of macro & micro algae, fish feed technology.

Course Code: Biot3114

Course Name: Introduction to Bioinformatics

Credit Hour: 3

Prerequisite: Molecular Biology

Bioinformatics is an emerging discipline that covers a wide range of topics including but not limited to the storage and retrieval of biological data such as sequence data of DNA, RNA and protein molecules, mathematical modeling of biological sequences, analysis of possible mechanisms behind complex biological processes, understanding and modeling of the evolutionary history of life. The course provides a theoretical and practical introduction to bioinformatics methods with topics that deal with biological databases, sequence similarity analysis, DNA sequencing, phylogenetic analysis, protein structure prediction and homology modeling, restriction mapping and PCR primer designing with sufficient description and coverage on each and every topic.

Course Code: Biot3106

Course Name: Plant cell and Tissue culture

Credit Hour: 2

Prerequisite: None

The course deals with a review of the historical importance of plant tissue culture. This is followed by studying the different media compositions for in vitro techniques to amplify the expertise of in vitro procedures for micro propagation, haploid production through anther and ovule culture, cryopreservation and developing disease free planting materials through meristem tip culture. The course brings out the growth hormone regulation of in vitro techniques and the acclimatization process to harden the tissue cultured plants.

Course Code: Biot3116

Course Name: Research Methodology and Academic writing

Credit Hour: 2

Prerequisite: None

The course deals with basic of researching such as sources of knowledge, types of research, research problem identification, research topic selection, developing a hypothesis, proposal writing; power point preparation and oral presentation; the principles of scientific writing and the use of bibliographic index; preparation of a review paper, correct use and preparation of tables and figures and research ethics.

Course Code: Biot4101

Course Name: Enzyme Technology

Credit Hour: 3

Prerequisite: None

This course elaborates historical background of enzymes/biocatalysts and their advantage over non-biological catalysts, nontraditional enzymes, and enzymes from extremophiles, properties of enzymes, classification of enzymes, enzyme production sources and method, immobilization of enzymes, enzyme engineering, and application of enzymes, current use and potential of enzyme technology for the production of goods to meet various human needs.

Course Code: Biot4103

Course Name: Industrial Biotechnology

Credit Hour: 3

Prerequisite: General Microbiology

Industrial biotechnology is the application of biotechnology for industrial purposes, including the practice of using cell (biology) or components of cells like enzymes to generate industry useful products. The course deals with basic principles of fermentation technology, industrially important microorganisms, handling and development of inocula for various fermentation processes; types of fermentation, process development, production of microbial products, etc. The focus of the techniques taught in this program is the use of biological resources as input to biobased

processes which are economically and environmentally sustainable, to equip students with skills on recent methodologies and practice in industrial biotechnology and to expose students to relevant and recent advances in industrial biotechnology.

Course Code: Biot4105

Course Name: Environmental Biotechnology

Credit Hour: 3

Prerequisite: None

The course deals with the environmental problems, its legislation and biotechnological methods for wastewater treatment and its advancements, solid waste management, sewage sludge disposal and utilization (Composting), bioremediation and its types and strategy, biodegradation of xenobiotic compounds, recalcitrant, hazardous wastes, genetic engineering approach for biodegradation, vermitechnology, sustainable technology (biofertilizers, biopesticides, biosensors, bioleaching, biofuels) and environmental monitoring, removal of heavy metal pollutants and treatment of various industrial waste water.

Course Code: Biot4107

Course Name: Introduction to Omics science

Credit Hour: 2

Prerequisite: Genetic Engineering

This course deals with physical aspects of the genome; construction and study of various types of genome maps, transcript mapping and large scale sequencing. The human genome project and the plant genome program, structural and functional genomics to address gene discovery, isolation, localization and characterization, protein structure and function, discovery and finding genetic markers for plant breeding purposes will also be dealt out. Moreover, strategy for developing diagnostic tests for plant, animal and human diseases and therapeutics techniques will be covered in addition to gene and protein expression and the external and cellular factors that influence their expression. It gives sufficient description and coverage on each and every topic with learning objectives stated under the section, Course Learning Outcome.

Course Code: Biot4109

Course Name: Biotechnology internship

Credit Hour: 3

Prerequisite: None

This course encompasses a pre-arranged 3 credit hour work focusing on practical job/research experience. Attachment of the students to a host organization (private, public, non-profit organizations or/and research centers), engaged in fundamental and/or applied biotechnology related endeavors will be arranged for two months during the summer break, post completion of the third year- second semester. During the practical attachment, the students will remain under the guidance of an identified Industry/research lab-based supervisor. Besides providing an avenue to have hands-on-experience in an industry/research setting, the major assignments of this course are to write, submit

and present the expectations, various experiences the student gains during his/her respective internship, information on the background and focus of the institution, methods or protocols used/learned, instruments handled or product produced, new skills acquired, the plausible challenges or limitations encountering the institution and the attached student and finally recommendations/scope for improvement. Mentored by a departmental advisor, each student will submit a report on his/her attachment, and will make a report presentation at the end of the attachment.

Course Code: Biot4111

Course Name: Final Year Project I

Credit Hour: 3

Prerequisite: Research methods and Academic writing

The course deals with introducing the students to seminar presentation, proposal writing and research problems identification connected to department research divisions (Microbial biotechnology, Agricultural biotechnology, Plant breeding, Industrial biotechnology, Environmental biotechnology and Medical biotechnology). It is emphasized that the student shall learn to identify research problems, present research plans and initiate his/her work individually within the chosen topic, and also learn how to obtain detailed knowledge through literature studies and practical work. The student will have mandatory seminar presentation and progress report submission for their initiated research project at the end of the semester. The laboratory project work is carried under the supervision of an adviser.

Course Code: Biot4113

Course Name: Nanobiotechnology

Credit Hour: 2

Prerequisite: None

This course gears off with a general introduction to nanotechnology, the basic principles of nanoscience and the scope of applications in the domain of biotechnology. The various preparative strategies, tools and techniques for characterization and bio physicochemical attributes of various nanobiomaterials are embraced. With inclusion of topics like nanobiosensing, nanobiomaterials for tissue engineering, micro/nanofluidics, artificial cells, nanodiagnostics and nanotherapeutics, as well as bio macromolecular (DNA, RNA, protein, lipid and carbohydrate based) nanotechnology, the course is streamlined to highlight the various advanced and feasible applications in diverse realms of health care, environment, agriculture, food industry, forensic science etc. Post perusal of various plausible nano-bio interfacial actions and reactions *vis-à-vis* nanotoxicology, the course concludes with analysis of plausible future trend in the realm.

Course Code: Biot4112
Course Name: Final Year Project II
Credit Hour: 3
Prerequisite: Final Year Project I

The course deals with introducing the students to research problems connected to Department research divisions (Microbial biotechnology, Agricultural biotechnology, Plant breeding, Industrial biotechnology, Environmental biotechnology and Medical biotechnology). It is emphasized that the student shall learn to individually within the chosen topic, and also learn how to obtain detailed knowledge through literature studies and practical work. The students will have mandatory report submission and oral presentation of the project at the end of the semester. The laboratory project work is carried out individually under the supervision of an adviser.

Course Code: Biot4106
Course Name: Cancer Biology
Credit Hour: 2
Prerequisite: Molecular biology

The course aims to introduce students with a basic understanding of the molecular and cellular mechanisms that lead to cancer. It primarily focuses on the role of growth factors, oncogenes, tumor suppressor genes, angiogenesis, and signal transduction mechanisms in tumor formation. In addition, students gain knowledge of cancer-related mutagens and pathways, and cancer therapy.

Course Code: Biot4104
Course Name: Food Biotechnology
Credit Hour: 2
Prerequisite: Microbial biotechnology

The food Biotechnology course deals with the application of biotechnology in food science to the selection, preservation, processing, packaging, distribution, and use of safe, nutritious, and wholesome food. Food biotechnology covers some of the main areas of application of biotechnology such as: the production of new kinds of foods and drinks, both by modern developments of conventional techniques and by genetically modifying the products themselves, or producing them using genetically modified organisms or their products. This course makes students learn about current trends in food biotechnology concerning modern biotechnological and biochemical interventions aiming towards food safety and food security with safety guidelines.

Course Code: Biot4106

Course Name: Pharmaceutical Biotechnology

Credit Hour: 2

Prerequisite: None

The course aims to give strong foundation and advanced information on biopharmaceuticals manufacturing techniques, drug discovery & development strategies including different phases of clinical trials. This course also provides core responsibilities for the development and monitoring of the drug and the preparation of medicines according to the standard operating procedures / GMP norms. To gain knowledge in physicochemical properties, pharmacokinetics and pharmacodynamics, and the formulation of commonly used biopharmaceuticals, packaging materials and recent techniques used, etc.

Course Code: Biot4108

Course Name: Biosafety and Intellectual property rights in biotechnology

Credit Hour: 2

Prerequisite: None

The course gives a brief outlook about the social and environmental impacts of the new developments in biotechnology in the areas of agriculture, medicine, and human reproduction. Students will gain a basic scientific/technical literacy pertaining to the biological concepts associated with bioethics, biosafety and intellectual property rights in the field of biotechnology. In addition, principles used to deal with concerns of biotechnology applications, regulatory frameworks in biosafety and IPR will also be dealt in details. There will be discussions and critical analyses of the contributions of science to public policy and the role of values in science.

Course Code: Biot4110

Course Name: Introduction to Systems and Synthetic Biology

Credit Hour: 1

Prerequisite: Introduction to omics Science

Biological systems are complex. System level analysis of interaction of components helps in solving many biological problems and for the reconstruction of artificial gene circuits. This course tries to introduce concepts in systems and synthetic biology. It deals with biological networks, mathematical models for networks, gene circuits and synthesis and assembly approaches for artificial gene circuits. And finally, application of synthetic biology and its safety considerations will be dealt in details.

Course Code: ISTEP3116

Course Name: Integrated Science Team Project

Credit Hour: 3

Prerequisite: None

This is a multidisciplinary integrated science team project for special science computational problems under the guidance of a faculty member and a computer Programmer. The course introduces the student with computer programming basics using python language to help students deal with scientific problems by drawing methods and concepts from biology, chemistry, physics and mathematics. The course deals with binary computation, problem-solving methods and algorithm development. The course also covers data types, control structures, functions, parameter passing, library functions, arrays, inheritance and graph plotting using python Matplotlib. It is aimed to aid multidisciplinary team based science problem solving and product development.

Supportive Module

Course Code: BMgt4116

Course Name: Project Planning and Management

Credit Hour: 2

Prerequisite: None

The course deals with the different stages of project management. Topics covered include: project planning and national planning, project preparation, the project cycle, identification and quantification of costs and benefits, investment criteria, important financial ratios, risk and sensitivity analysis and applications on project planning and management.

4.3.5.2 Elective

Course Code: Biot4218

Course Name: Biodiversity and Conservation of Genetic Resources

Credit Hour: 2

Prerequisite: None

The course of biodiversity is designed to introduce the students to get knowledge about the meaning of biodiversity and its sub components of biosystematics, species diversity, species loss, habitats and ecosystems. Biodiversity is teaching the students the uses and values of biodiversity, as well as conservation and management of biodiversity at national, regional and international levels. The diversity of living organisms like animals, plants and microorganisms. The concept of extinction, threatened organisms is also demonstrated. This course will provide an introduction to the basic principles of conservation and biodiversity. The field of conservation biology is interdisciplinary. Protecting our remaining biological diversity will involve all fields of science, from the humanities to natural sciences.

Course Code: Biot4220

Course Name: Animal Cell and tissue culture

Credit Hour: 2

Prerequisite: Animal Biotechnology

Introduction to cell culture, design of the cell culture laboratory, the culture environment: media preparation and formulations, primary cell culture, maintaining cells: contamination and storage, and laboratory practices of animal cell culture.

Course Code: Biot4222

Course Name: Forensic Biotechnology

Credit Hour: 2

Prerequisite: Molecular biology

This course is designed to introduce the student to the hands-on techniques and opportunities in the field of biotechnology for the forensic field. The course will cover topics including introduction to forensic science, DNA applications in forensic investigation such as DNA fingerprinting. The course will cover various techniques used in biotechnology (very significant for forensics) such as PCR, and DNA diagnostics. There will be field visits, case studies, and group discussions about the latest events in the field of forensic biotechnology.

4.4 Bachelor of Science Degree in Industrial Chemistry

4.4.1 Introduction

Industrial Chemistry deals with the commercial production of chemicals and related products from raw materials and their derivatives. It enables humanity to experience the benefits of Chemistry when applied on the exploitation of available materials and energy. Industrial Chemistry as a branch of Chemistry, therefore, comes in very useful application of scientific knowledge to a range of industrial endeavors. The pharmaceutical, cement, leather, plastic, fertilizer, textile and sugar industries are just some of the endeavors where Industrial Chemistry is applied. Structured training of the students in order to equip them with sound knowledge of the chemical principles and laboratory practices will make them capable and competent to solve problems in such industries. By being able to design alternate and cheaper routes to discover and produce, using readily available raw materials, the trained chemist would catalyze the creation of national wealth and contribute to the well-being of the Ethiopian citizens.

Most of the current public universities train pure chemists which results in shortage of industrial chemists to work on the growing number of industries. Hence, the industrial chemistry program at the Addis Ababa Science and Technology University is opened to address the above-stated problem specifically. The Industrial Chemistry department strives to deliver high-quality education, as per the standard of the country, and produce qualified professionals in order to satisfy the needs of industrial chemists of the country.

The preparation of this curriculum takes the above fact into account and also takes lessons from the past seven years' experience, since the inception of the department. The department continually creates strong collaboration with the nearby industries and research institutes to give students experience with the current industry trend through industrial visits and a two-month compulsory industrial attachment. All the courses have been structured in such a way that the subject matter reflects strong theoretical knowledge as well as essential practical skills. This encourages the students to acquire these skills and to utilize the same in the dynamic industrial chemistry world.

Hence, this four-year industrial chemistry training curriculum will equip the students with the required knowledge and skills to work in different units of various manufacturing industries, as

quality control (QC) chemists, research and development (R&D) chemists, sales representatives and product specialists and production supervisors, for various scientific institutions. A section of the graduates would definitely pursue the postgraduate education at home and abroad, as encouraged by the numerous Faculty serving in the department. The training will also help the students to choose a path of entrepreneurship.

4.4.2 Program Education Objectives (PEO)

In general, the graduates of B. Sc. in Industrial Chemistry in the next 4-5 years of their graduation are expected to have applied their expertise in problem solving, be engaged professionally, and have continued to learn and adapt, and have contributed to their organizations through leadership and teamwork. Specifically, in 4-5 years of their graduation, the graduates will meet the following objectives as shown in Table 2.1.

Table : Program Education Objectives (PEO)

PEO	Statement
PEO-1	Become experts in manufacturing industries (R&D group leader, QC manager).
PEO-2	Develop their career (pursuing postgraduate studies, adapting to new and state-of-the-art technology)
PEO-3	Become entrepreneurs (startups, consultancy)

4.4.3 Program outcome (PO)

Upon successful completion of the four-year program, the B. Sc. Industrial Chemistry graduates will develop the following skills, knowledge, and attributes.

Table : Program Objectives (PO)

PO	Statement
PO-1	An ability to identify, formulate, and solve broadly defined technical and scientific

	problems by applying knowledge of mathematics and science and/or technical topics to Industrial Chemistry.
PO-2	Ability to formulate or design a system, process, procedure or program in chemical industries to meet desired needs.
PO-3	An ability to develop and conduct experiments, or test hypotheses, analyze and interpret data and use scientific judgment to draw conclusions.
PO-4	An ability to operate advanced scientific equipment properly.
PO-5	An ability to communicate effectively with a range of audiences.
PO-6	An ability to understand ethical and professional responsibilities and the impacts of technical and scientific solutions in global, economic, environmental, and societal contexts.
PO-7	An ability to fulfill a multiplicity of roles as research scientists, development chemists, and technical representatives and as lab/unit/plant/company managers.
PO-8	An ability to function effectively in teams together with other professionals that establish goals, plan tasks, meet deadlines, and analyze the risks and uncertainty.

4.4.4 Graduation Requirements

The graduation requirement for the program is based on the university's senate legislation of July 2017, Article 109. To graduate, a student needs to satisfy the following minimum requirements:

- All the required courses and the minimum credit hours set in the program curriculum should be satisfied,
- A cumulative grade point average CGPA of 2.00 must be obtained;
- A cumulative grade point average CGPA of 2.00 in major area courses;
- Pass the exit exam
- No "F" grade in any course taken for the undergraduate program;

The score pass mark for all courses which have Pass/Fail grade

List of Courses and Category

Table: Distribution of credit hours for by course categories

Category	Total Cr. hr.	Percentage (%)
Core courses (Major and supportive)	118	78.66
University Requirement (Core)	3	2.00
Core courses Total Cr hr	121	80.66
National Requirements	29	19.08
Total	150	100%

List of courses

Table : List of Courses

Common Courses		
No.	Course title	Credit-hour
1	Communicative English Skills I	3
2	Communicative English Skills II	3
3	General Psychology	3
4	Moral and Civics Education	2
5	Economics	3
6	Geography of Ethiopia & the Horn	3
8	Inclusiveness	2
9	Global Trends	2
10	Logic and Critical Thinking	3
11	History of Ethiopia & the Horn	3
12	Social Anthropology	2
13	Physical Fitness and conditioning *	P/F
Total		29

Core supportive courses		
No.	Course Title	Credit-hour
14	Applied Mathematics IA for	4
15	Physics for Industrial Chemists	3
16	Unit Operation for Industrial Chemists	3
17	Industrial Microbiology for Chemists	3
18	Statistics for Industrial Chemists	3
Total		15

Major courses		
Course Title		
19	Analytical Chemistry	3
20	Practical Analytical Chemistry	1
21	Instrumental Analysis I	3
22	Practical Instrumental Analysis I	1
23	Instrumental Analysis II	3
24	Practical Instrumental Analysis II	1
25	Real Sample Analysis	1
26	Inorganic Chemistry I	3
27	Inorganic Chemistry II	4
28	Practical Inorganic Qualitative Analysis	1
29	Practical Inorganic Synthesis	1
30	Organic Chemistry I	3
31	Practical Organic Chemistry I	1
32	Organic Chemistry II	3
33	Practical Organic Chemistry I	1
34	Systematic Identification of organic compounds	2
35	Physical Chemistry I	3

36	Practical Physical Chemistry I	1
35	Physical Chemistry II	3
36	Practical Physical Chemistry II	1
37	Physical Chemistry III	3
38	Industrial Inorganic products	3
39	Industrial organic and consumer products	3
40	Chemistry of paper, pigments and natural products	3
41	Chemistry of Cement, glass and ceramics	3
42	Leather Processing and Production	3
43	Sugar Processing and Production	3
44	Chemistry of Materials	3
45	Industrial Pharmacy	3
46	Pharmaceutical Analysis	2
47	Biochemistry	3
48	Environmental Chemistry and Waste Management	3
49	Quality Assurance and Management systems	2
50	Industrial Safety and loss prevention	2
51	Industrial Attachment	3
52	Student project	4
53	Research methods and scientific writing	2
54	Computer Applications for Industrial Chemists	2
55	General Chemistry	3
56	Introduction to Emerging Technologies	3
57	Entrepreneurship	3
58	Mathematics for Natural Sciences	3
59	Integrated Science Team Project	3
	Total	105

4.4.5 Program Curriculum Structure: Bachelor of Industrial Chemistry

Table : Course Breakdown for Regular Program

Year I, Semester I

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
Phil1009	Logic and Critical Thinking	3			
Psyc1011	General Psychology	3			
FLEn1003	Communicative English Language Skills I	3			
GeEs1005	Geography of Ethiopia and the Horn	3			
Math1107	Mathematics for Natural Science	3	2	3	
Chem1101	General Chemistry	3	2	3	
SpSc1013	Physical Fitness and conditioning	P/F			
Total		18			

Year I Semester II

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
EmTe1104	Emerging Technologies for Science	3			
FLEn1004	Communicative English Language Skills II	3			
Math1114	Applied Mathematics IA	4	3	3	
MCiE1012	Moral and Civic Education	2			
Incl1010	Inclusiveness	2			
Ant1002	Social Anthropology	2			
Entr1108	Entrepreneurship for Sciences	3	3		
Total		19			

Year II, Semester I

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
GLTr2001	Global trends	2	2	-	-
InCh2101	Analytical Chemistry	3	2	3	
InCh2105	Practical Analytical Chemistry	1	-	-	3
Stat2103	Statistics for Industrial Chemists	3	2	3	
InCh2107	Inorganic chemistry I	3	2	3	
Econ2003	Economics	3	2	3	
InCh2109	Organic Chemistry I	3	2	3	
InCh2111	Practical Organic Chemistry I	1	-	-	3
Total		19			

Year II, Semester II

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
InCh2102	Instrumental Analysis I	3	2	3	
InCh2104	Practical Instrumental Analysis I	1			3
Hist2002	History of Ethiopia and the Horn	3	3	-	-
InCh2106	Physical Chemistry I	3	2	3	
InCh2108	Practical Inorganic Qualitative Analysis	1	-	-	3
Phys2114	Physics for Industrial Chemists	3	2	3	
InCh2110	Organic Chemistry II	3	2	3	
InCh2112	Practical Organic Chemistry II	1	-	-	3
Total		18			

Year III, Semester I

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
InCh3101	Instrumental Analysis II	3	2	3	
InCh3103	Practical Instrumental Analysis II	1	-	-	3
InCh3105	Industrial organic and consumer Products	3	2	3	-

InCh3107	Physical Chemistry II	3	2	3	-
InCh3109	Practical Physical Chemistry I	1	-	-	3
InCh3111	Inorganic Chemistry II	4	3	3	
InCh3113	Practical Inorganic Synthesis	1	-	-	3
InCh3115	Computer Applications for Chemists	2			6
Total		18			

Year III, Semester II

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
InCh3102	Physical Chemistry III	3	2	3	
InCh3104	Practical Physical Chemistry II	1			3
ChEg3106	Unit operations for Industrial Chemists	3	2	3	
Biot3108	Industrial Microbiology for Chemists	3	2		3
InCh3110	Systematic Identification of organic compounds	2			6
InCh3112	Industrial Inorganic Products	3	2	3	
InCh3114	Research methods and scientific writing	2	2		
ISTP3116	Integrated Science Team Project	3	2		3
Total		20			

Year III, Semester III

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
InCh3117	Industrial Attachment*	3			3
Total		3			

Year IV, Semester I

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
InCh4101	Sugar Processing and Production	3	2	3	
InCh4103	Leather Processing and Production	3	2	3	

InCh4105	Industrial Pharmacy	3	2	3	
InCh4107	Chemistry of Materials	3	3	3	
InCh4109	Environmental Chemistry and Waste management	3	2	3	
InCh4111	Biochemistry	3	2	3	
InCh4114	Student Project*	P/F			
Total		18			

*= pass/fail based on proposal evaluations

Year IV, Semester II

Course Code	Course Title	Cr.hr.	Lect.	Tut.	Lab.
InCh4102	Real Sample Analysis	1			3
InCh4104	Pharmaceutical Analysis	2	1	3	
InCh4106	Quality Assurance and Management systems	2	2	-	-
InCh4108	Chemistry of paper, pigments and natural products	3	2	3	
InCh4110	Industrial safety and loss prevention	2	2		
InCh4112	Cement, glass and ceramics	3	2	3	
InCh4114	Student Project	4			12
Total		17			

4.4.6 Program Courses Synopsis

4.4.6.1 Supportive courses

Course Code: Stat2103

Course Name: Statistics for Industrial Chemists

Credit Hour: 3

Prerequisite: No

This course covers basic elements of statistics, probability, central tendencies, variation, and types of sampling

4.4.6.2 Core course Synopsis

Course Code: Chem1106

Course Name: General Chemistry

Credit Hour: 3

Prerequisite: No

The course General chemistry is designed to make students more prepared to the all-chemistry courses by refreshing and summarizing the previous preparatory chemistry concepts before tackling the advanced chemistry courses. The course covers units and measurement, the composition of matter, chemical reactions, reactions stoichiometry, atomic structure and the periodic table, the chemical bond, structure of molecules, the properties of solutions, chemical equilibrium, introduction to functional groups and their typical reactions.

Course Code: Phys2114

Course Name: Physics for Industrial Chemists

Credit Hour: 3

Prerequisite: None

This course introduces Coulomb's Law, Electric Field, Gauss' Law, Electric Potential, Electric Potential Energy, Capacitors and Dielectric, Electric Circuits, Magnetic Field, Biot-Savart's Law, Ampere's Law, Electromagnetic Induction, Inductance, Circuits with time dependent currents, and Maxwell's equations

Course Code: ChEg3106

Course Name: Unit operation

Credit Hour: 3

Prerequisite: No

This course will give basic introduction on unit operations that are available in chemical process industries. Mainly focuses on mechanical unit operations like particle technology including size reduction and enlargement, mechanical hydromechanical aero mechanical separations, and mixing. Students will have ability to analyze heat transfer operations like heat exchange, condensation and evaporation in detail.

Course Code: Biot3108

Course Name: Industrial Microbiology

Credit Hour: 3

Prerequisite: Organic Chemistry, Basic Biochemistry and Basic knowledge in microbiology

Industrial Microbiology is the application of Microbial biotechnology for industrial purposes, including the practice of using cell (biology) or components of cells like enzymes to generate industry useful products. The course deals with basic principles of Microbiology, fermentation technology, industrially important microorganisms, handling and development of inocula for various fermentation processes; types of fermentation, Process development,

production of microbial products, etc. The focus of the techniques taught in this program is the use of biological resources as input to bio based processes which are economically and environmentally sustainable, to equip students with skills on recent methodologies and practice in industrial biotechnology and to expose students to relevant and recent advances in industrial Microbiology. Industrial visit is needed for 2 times to different industries to learn the fermenter types, operational conditions and production process of fermented products in Beverage or Food industries as part of the course.

Course Code: InCh2101

Course Name: Analytical Chemistry

Credit Hour: 3

Prerequisite: High school chemistry

The course is designed to introduce students to the fundamentals of chemical analysis, and enable them develop competencies on chemical metrology; measurement in chemistry; traceability in chemical analysis; method validation, inter-laboratory test reference materials; calibration and standardization, Presenting analytical data; Evaluation and statistical data treatment; Errors in chemical analysis; estimation of uncertainty; combined standard uncertainty, the ISO GUM and the Nordtest approach; Ionic equilibrium; Titrimetric methods of analysis; as well as Gravimetric analysis.

Course Code: InCh2105

Course Name: Practical Analytical Chemistry

Credit Hour: 3

Prerequisite: No

The course is designed to enable students acquire the basic skills in calibration and standardization. Moreover, the course enables students to develop skills on different types of titrations and determination of end points. The course familiarizes the students with the skills in gravimetric analysis. As a result the students, after completion of the course, will develop the practical competency to carry out chemical analysis in various fields such as chemical industry, agriculture, environmental chemistry, clinical chemistry, medicine, pharmaceutical industries and others. The experiments to be conducted include: Calibration of burettes, pipettes, determination of standard and expanded uncertainty, neutralization, precipitation, complex formation, redox titrations and gravimetric analysis.

Course Code: Inch2102

Course Name: Instrumental Analysis I

Credit Hour: 3

Prerequisite: InCh2201

This course is designed for undergraduate students to develop basic theoretical competency in separation sciences and electro analytical techniques. The course helps students to familiarize students with analytical separation techniques and electro analytical techniques methods. The first segment of this course covers principles,

instrumentations and application of modern chromatographic techniques like Gas chromatography (GC), high performance liquid chromatography (HPLC), hyphenated techniques; GC-MS and LC-MS, supercritical fluid chromatography, size exclusion chromatography, ion exchange chromatography and electrophoresis. The second segment of this course encompasses the basics of electroanalytical techniques such as conductometry, potentiometry, coulometry, electro-gravimetry and voltammetry and their application areas such as toxicology, environmental science, pharmaceuticals, quality controlling, chemical industry, clinical chemistry, medicine etc. is included.

Course Code: InCh2104

Course Name: Practical Instrumental Analysis I

Credit Hour: 1

Prerequisite: InCh2203

The course is designed to provide practical skills and develop competency in carrying out chemical analysis using modern chromatographic techniques and electroanalytical techniques. The course includes practical experiments of chromatography (TLC, CC, GC, HPLC), and Electroanalytical methods (potentiometry, voltammetry, Conductometry, coulometry, electrogravimetry).

Course Code: InCh3101

Course Name: Instrumental Analysis II

Credit Hour: 3

Prerequisite: InCh2102

The course is designed to make the students develop the theoretical competency in using spectroscopic techniques for analytical purposes. The course familiarizes the students with the theoretical background of the principles of spectroscopic instruments like atomic absorption, atomic emission, ultraviolet-visible and infrared spectroscopy; nuclear magnetic resonance and mass spectrometers, as well as hyphenated techniques. The course also addresses the instrumentations of the various techniques and their application in various fields like, toxicology, environmental science, pharmaceuticals, quality controlling, chemical industry, clinical chemistry, medicine and the like.

Course Code: InCh3103

Course Name: Practical Instrumental Analysis II

Credit Hour: 1

Prerequisite: InCh2204

The course is designed to make the students develop the practical competency in using spectroscopic techniques for analytical purposes. The course familiarizes the students with the practical skills of operating spectroscopic instruments for qualitative and quantitative analysis. Practical experiments include atomic spectroscopy (AAS, AES), UV-Vis molecular absorption spectroscopy, IR spectroscopy and refractive index.

Course Code: InCh4102

Course Name: Real sample analysis

Credit Hour: 1

Prerequisite: InCh3103

Systematic analysis of real samples: sampling, preservation and preparation of samples for the determination of the major and/or trace elements, inorganic compounds (speciation) and organic compounds from real samples: biological samples; food and beverages; water and waste water samples; soils and related samples.

Course Code: InCh3111

Course Name: Inorganic Chemistry II

Credit Hour: 4

Prerequisite: InCh2101

This course is designed for undergraduate Industrial chemistry students to introduce the concept of Symmetry and point groups. Most importantly it introduces Coordination chemistry, where formation, stability, reactions, catalytic activity and the electronic spectra of coordination compounds are widely discussed; Industrial applications of selected coordination compounds are also explained. Furthermore, chemistry of Main General Chemistry group and Transition Metal Organometallics: Nomenclature, bonding, and stability, reactivity as well as their reaction mechanisms are discussed. Finally, this course introduces Bioinorganic chemistry that motivates the students to study further on the function of metal ions in biological system.

Course Code: InCh2108

Course Name: Practical Inorganic Qualitative Analysis

Credit Hour: 1

Prerequisite: InCh2101

The course is designed to give student's competency in chemistry of transition elements mainly titanium, vanadium, chromium, manganese, cobalt, nickel, Platinum, copper, silver, cadmium, zinc, and mercury. The course familiarizes the students with the different oxidation states and reactions of transition elements and their compounds. The students will be able to prepare compounds of transition metals and study their behaviors in different media (acidic, basic, and neutral). The course equips the students with the required competency to work in areas that require the competency such as the chemical industry, agriculture, environmental chemistry, Geology, Biology and others.

Course Code: InCh3113

Course Name: Practical Inorganic Synthesis

Credit Hour: 1

Prerequisite: InCh3111

Practical Inorganic Synthesis is designed to give the students competency in analyzing inorganic compounds/species both in laboratory and in real samples. At the end of this course the students have expected to equip with the practical knowledge of synthesis, isolation and characterization of coordination compounds.

Course Code: InCh2109

Course Name: Organic Chemistry I

Credit Hour: 3

Prerequisite: None

This course is primarily designed to offer basic understanding of structures, reactivities and synthesis of simple organic compounds and the relationships between structure and properties. Although the course follows mechanistic approach to reactions of organic compounds (substitution, elimination, addition, rearrangement reactions), a chapter is devoted to brief discussion of functional groups, their typical reactions and synthesis. This will enable the students to understand the twin strategies of studying chemistry of the millions of organic compounds by either classifying them according to the reaction types they undergo (mechanistic approach) or according to their functional groups (functional group approach). The course also introduces the concept of stereochemistry and stereoisomerism (configurational and conformational isomerism) and its importance in organic reactions. This enables the students to appreciate the more subtle types of isomerism than the obvious structural (constitutional) isomerism. This course

will complement practical organic chemistry-I course as theoretical background and will create basic knowledge for next organic chemistry courses.

Course Code: InCh2111

Course Name: Practical Organic Chemistry I

Credit Hour: 1

Prerequisite: InCh2401

This course designed to make the students aware of basic organic laboratory activities such as simple recrystallization, melting point determination, simple, steam and fractional distillation, and chromatography techniques. In addition to this student will prepare simple organic compounds like soap, aspirin in laboratory scale. Student will learn the laboratory safely and regulation rules of organic laboratory. The course will give basic knowledge and skill on experimental organic chemistry because organic chemistry is everywhere, from the delicate smell of violets to the paper these words printed on. It is in the laboratory where the advances of science made. Without laboratory work, science would be just a poetic fabrication.

Course Code: InCh2110

Course Name: Organic Chemistry II

Credit Hour: 3

Prerequisite: InCh2109

This course designed to make students aware of organic reactions in detail and depth. It will elaborate chemistry of aromatic, amine, carbonyl compounds, carboxylic acid, and oxidation– reduction reactions. In addition, biological molecules such as carbohydrates, amino acids, peptides, lipid, and nucleic acids are introduced to address basic concepts about natural product chemistry.

Course Code: InCh2112

Course Name: Practical Organic Chemistry II

Credit Hour: 1

Prerequisite: InCh2111

This course designed to integrate the theoretical organic reaction with small-scale laboratory practice. The course enable students to understand organic reactions such as Esterification reactions; dehydration, acetylation, oxidation, aldol condensation, Friedel-Crafts reaction; and the Diels-Alder reaction. Extraction technique is very helpful in

organic research. Thus, under this course extraction of limonene from citrus fruit and isolation of caffeine from tea are included to introduce basic extraction skills. In addition to this, students will understand dying process.

Course Code: InCh2110

Course Name: Systematic Identification of Organic Compounds

Credit Hour: 2

Prerequisite: InCh2112

This course designed to apply different physio-chemical and spectroscopic techniques. The course enables students to identify different functional groups of organic compounds by applying chemical tests. It also enables students to design and synthesize derivatives for functional groups in organic compounds. The course will help students to apply and exercise the practical as well as theoretical knowledge for the identification of unknown organic compounds.

Course Code: InCh2106

Course Name: Physical Chemistry I

Credit Hour: 3

Prerequisite: None

This course is designed for students of Industrial chemistry as it makes them to have good understanding of bulk properties of system (thermodynamics) and enable them describe and predict chemical and physical changes mathematically by computing the change in properties of the system during the change and Predict criteria for any change to take place..

Course Code: InCh3107

Course Name: Practical Physical Chemistry I

Credit Hour: 1

Prerequisite: Physical Chemistry I, Physical Chemistry II

The Practical Physical Chemistry I course is delivered to the students of Chemistry in order to supplement the compulsory Theoretical Physical Chemistry courses which the students attend in the classroom. The laboratory course enables the students to develop skills of experimental data generation, interpretation and analysis of the data. Students are also motivated to do independent work through exploration on various physical properties of substances.

Course Code: InCh3107

Course Name: Physical Chemistry II

Credit Hour: 3

Prerequisite: Physical Chemistry I (InCh2302)

This course is designed to enhance and extend students' ability to understand rate of chemical reactions, electrochemistry and surface chemistry through learning theoretical laws and principles. Additionally, they will be ready to conduct laboratory experiments, making observations and analyzing results, designing and analyzing products, and formulating and testing hypotheses based on evidence so that they are ready for making environmental and chemical analysis.

Course Code: InCh3104

Course Name: Practical Physical Chemistry II

Credit Hour: 1

Prerequisite: Physical Chemistry II, Physical Chemistry III

This practical course is designed to familiarize students with mechanisms of by which rate of reaction is determined. It also enable students develop skill of analysis of compounds using electrochemical and optical techniques. This practical course familiarizes students with the research tool and computational software used by chemists.

Course Code: InCh3102

Course Name: Physical Chemistry III

Credit Hour: 3

Prerequisite: Physical Chemistry II (InCh3301)

The Physical Chemistry III (Quantum Chemistry) course has been designed for the students to develop the knowledge of matter at the submicroscopic level. It also enriches students with the knowledge of computing properties of atomic and molecular species, which can be compared with the experimental results, so that the learners develop full confidence on the applications of quantum chemistry.

Course Code: InCh3112

Course Name: Industrial Inorganic Products

Credit Hour: 3

Prerequisite: None

This course is designed to introduce concepts of industrial processes and process variables, sources of water and its treatment for chemical industries, metallurgical processes and other chemical industries. It is designed to give deep understanding of manufacture of nitrogen containing compounds, synthesis and transportation means of some selected gases. It also emphasizes on a basic foundation of the manufacturing process of mineral acids and other concept that turns out to be a fundamental basis for many chemical processes.

Course Code: InCh3115

Course Name: Industrial Organic chemicals and Consumer products

Credit Hour: 3

Prerequisite: None

This course is designed to introduces undergraduate Industrial chemistry students about - the coal and petroleum, their value added products, Lubricants, their Classification mechanism of lubricants and their industrial application, synthesis and composition of soaps, detergents, shampoo , their biodegradation and environmental effects, formulation and composition of cosmetics products, chemistry of oil and fat, their function and their industrial applications, formulation and evaluation of Perfume, Industrial organic synthesis and their application

Course Code: InCh4112

Course Name: Chemistry of Cement, Glass and Ceramics

Credit Hour: 3

Prerequisite: None

This course is designed to introduces undergraduate Industrial chemistry students about Introduction to Cements; cements production, types of cements, binders (organic and inorganic), clinker production process and hardening of cements; Glass and its properties, type of glasses, constituents of glasses, Production processes of glasses, fuels needed for glass production, furnaces to be used for glass manufacture; Ceramics classification, properties and manufacturing process.

Course Code: InCh4103

Course Name: Leather Processing and Production

Credit Hour: 3

Prerequisite: None

Course is designed to make students familiar with the process and production of leather from skin histology to last product through different processes so that they will understand the application of chemistry in leather industry. Moreover, the students will understand the different methods of increasing the quality and quantity of skin and leather. The course covers Skin histology and structure: microorganisms: curing: beam house: tanning: combination tanning: dyeing and fat liquoring.

Course Code: InCh4107

Course Name: Chemistry of Materials

Credit Hour: 3

Prerequisite: None

The course, Chemistry of Materials is designed to make students familiar with the different types of solids (crystalline and amorphous solid), solid state reactions and special applications of solid state materials (magnetic materials, liquid crystals, ionic conductors and special materials) so that they will understand the applications of chemistry in real life. Moreover, the students will learn about polymer synthesis and characterization; polymer structure and properties, and their industrial fabrication.

Course Code: InCh4105

Course Name: Industrial pharmacy

Credit Hour: 3

Prerequisite: None

This course is designed to introduce undergraduate Industrial Chemistry students about the concept of formulation, manufacturing, packaging, storage and quality control of pharmaceutical dosage forms (oral liquids, capsules, conventional tablets, coated tablets, and sterile products (parenteral, ophthalmic, irrigating solutions)). In addition students will be familiar with the equipment and machinery used for granulation, production and quality control of pharmaceutical dosage forms. The course will also give overall concepts about quality assurance and good manufacturing practices aspects of these dosage forms.

Course Code: InCh4108

Course Name: Chemistry of paper, pigments and natural products

Credit Hour: 3

Prerequisite: None

This course is designed to make students familiar with production, analysis and utilization of dyes and pigments, and pulp and paper. The students also understand the importance and industrial applications of natural products. The course covers pigments and paints, types of pigments, formulation, composition and related properties of dyes, paper industry, manufacture of paper, and manufacture of pulp by different processes, classes of natural product, effect of chemical structures on physiological activities of natural products and importance of natural products in health care and commerce.

Course Code: InCh4104

Course Name: Pharmaceutical Analysis

Credit Hour: 2

Prerequisite: Industrial Pharmacy

This course designed to make students familiar with the basic concepts of Pharmaceutical Analysis. The main points to be covered include; concepts in Pharmaceutical analysis, Pharmaceutical quality control and quality assurance: Pharmaceutical sample preparation for analysis: Limit tests: Pharmaceutical applications: Biological methods of analysis: Quality control: Current good laboratory practices (GLP). They will also able to propose suitable analytical technique for a sample, carry out analysis for different pharmaceuticals as well as handle, interpret and report data obtained from the analysis. The course also deals with the applications of important instrumental analytical techniques and gives biological assay methods.

Course Code: InCh4111

Course Name: Biochemistry

Credit Hour: 3

Prerequisite: None

The course, Biochemistry is designed to make our students familiar with the different types of biological molecules, their structure and its application in industries. So that they will understand the applications of biochemistry chemistry in life and in industry. Moreover, the students will understand the different metabolic reactions and pathways in different kinds of living things.

Course Code: InCh4109

Course Name: Environmental Chemistry and Waste management

Credit Hour: 3

Prerequisite: General Chemistry

This is an undergraduate level course designed to introduce students with contemporary concepts in environmental chemistry and Waste management. The course covers the chemical composition of different segment of the environment, relationship between different segment of the environment, environmental pollution, sources, transport and fate of pollutants atmospheric chemistry and air pollutants; aquatic chemistry; water quality assessment, toxicology, industrial waste classification, soil chemistry and pollution, waste management; waste treatment, cycling, resource recovery; Green chemistry and sustainability.

Course Code: InCh4106

Course Name: Quality Assurance and Management systems

Credit Hour: 2

Prerequisite: None

This course is designed to introduce students about the concept of quality and contemporary quality management concepts, tools and models specifically in a manufacturing industry set up and prepare them for job market. The concepts, aspects and development of quality; Laboratory quality assurance and control; models and standards for quality management systems etc. will be covered in detail in this course.

Course Code: InCh4110

Course Name: Industrial Safety and Loss prevention

Credit Hour: 2

Prerequisite: None

This course is designed to introduce concepts of industrial safety system and loss prevention. It describes the general safety requirements for the workplace as well as the general safety equipment, facilities and controls. The course presents the basic principles of workplace safety and health protection, as well as the constituents of Safety management program that establish control of the hazards and risks of work. Determination and evaluation of hazardous properties of substances. Materials handling, fire control, Loss prevention working with machinery and apparatus, It explains the possible workplace industrial accidents and loss statistics, nature of the accidents/hazardous processes as well as controlling strategies from its possible dangerous effect. It also deals with personal safety devices and ways of mitigating health and physical hazards. .

Course Code: InCh3117

Course Name: Industrial attachment

Credit Hour: 3

Prerequisite: None

At the end of their third year, during summer vacations (for 2 months) the students are assigned to the various industries to get practical skills and exposure to the real industry working conditions. During this time students will get mentoring and supervision both from the industry and a faculty member from the department. After completion of their practice, students are expected to submit a report and present it to the evaluators in an arranged session.

Course Code: InCh4114

Course Name: Student Project

Credit Hour: 4

Prerequisite: Research Methods and Scientific Writing (InCh 3601)

During their final year, students will be allowed to solve a particular industrial chemistry problem following appropriate research methodology and write a scientific report which will be defended orally. The department assign suitable supervisor based on the scientific problem chosen. The course will help students develop, scientific process and writing skills as well as hands on experience.

Course Code: InCh3114

Course Name: Introduction to Research Methods and Scientific writing

Credit Hour: 2

Prerequisite: None

This course will introduce the meaning of research and its objectives, types of research and research process, proposal writing, writing research reports, and poster and oral presentations. It also introduces to literature review including the source of literature and the information that could be get from them. They will be acquainted with literature reviewing, proposal writing, seminar and poster presentations through projects.

Course Code: InCh3115

Course Name: Computer Application for Industrial Chemists

Credit Hour: 2

Prerequisite: None

The course covers the applications of computers in chemistry including computer fundamentals, data collection, data analysis and presentation. Construction, visualization and geometry optimization of simple molecules would encourage the student to shape his/her career as a good chemist in the future.

Course Code: InCh4101

Course Name: Sugar Processing and Production

Credit Hour: 3

Prerequisite: None

History of Sugar Industry, Cane Sugar Processing, Cane preparation, Sugar Beets; Cultivation, Harvesting, Preparation; Production of Juice; Juice Clarifications and Methods; Production of Sugar from Thick Juice; Separation of Sugar from Masseccuite; Preparation of Refined and White Sugar.

Course Code: InCh3116

Course Name: Integrated Science Team Project

Credit Hour: 3

Prerequisite: None

This is a multidisciplinary integrated science team project for special science computational problems under the guidance of a faculty member and a computer Programmer. The course introduces the student with computer programming basics using python language to help students deal with scientific problems by drawing methods and concepts from biology, chemistry, physics and mathematics. The course deals with binary computation, problem-solving methods and algorithm development. The course also covers data types, control structures, functions, parameter passing, library functions, arrays, inheritance and graph plotting using python Matplotlib. It is aimed to aid multidisciplinary team based science problem solving and product development.

